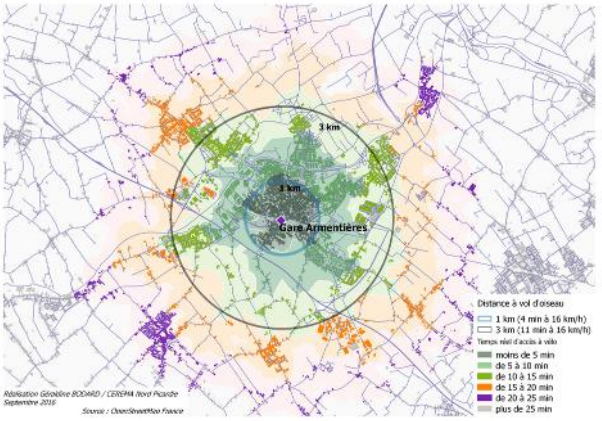
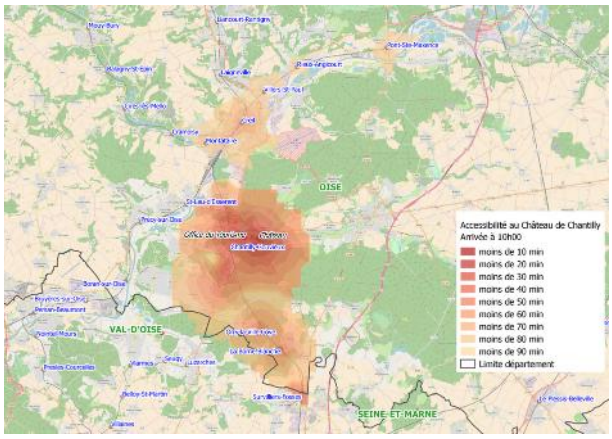
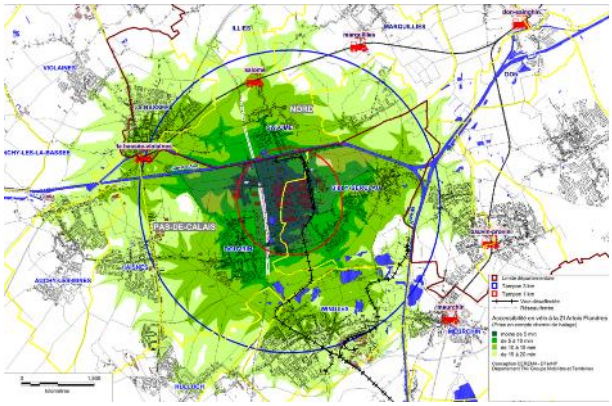
Cerema

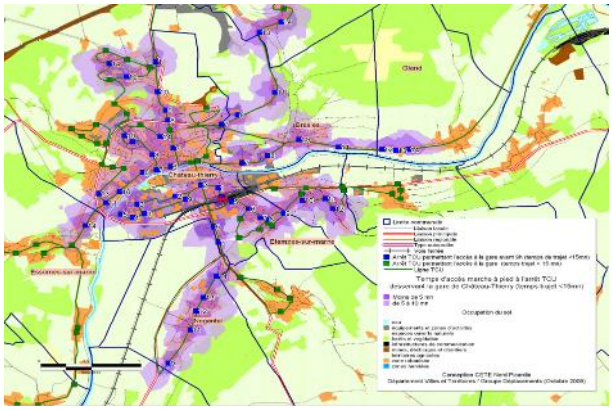
Cerema Nord-Picardie

Multimodal transport analysis with Qgis plugin « networks »









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A- Create an accessibility map (walking, cycling, car or public transport) from a multimodal network

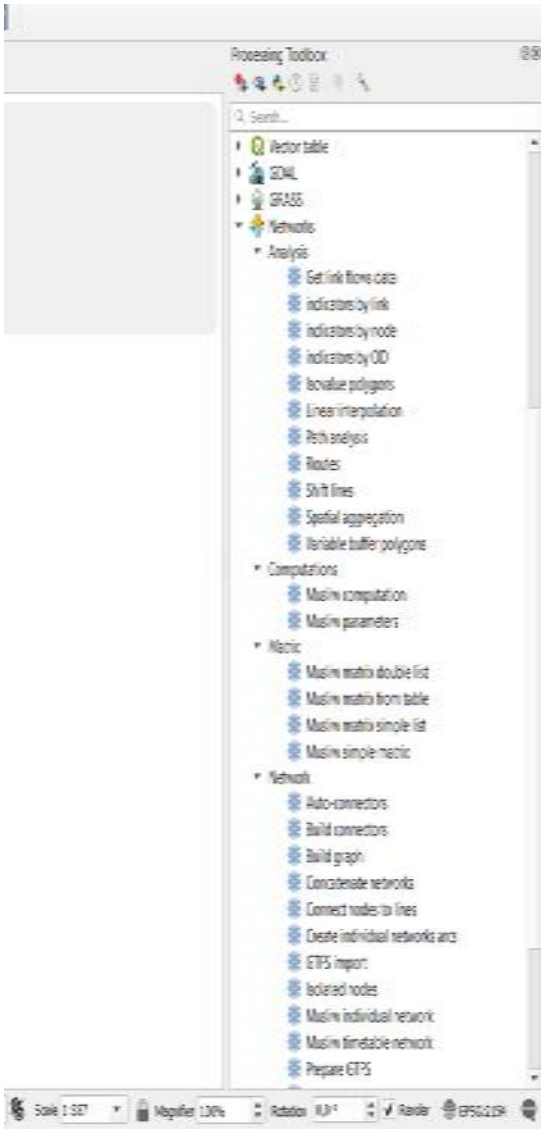
Warning

Do not put blank or accent for directory and / or table names. Example: Armentieres / voirie_decoupe_2senss

Most scripts used for:

- modify the different road layers, nodes, ...
- create different networks (cycling, walking, transportation)

can be found in the processing toolbox, under the Networks tab.

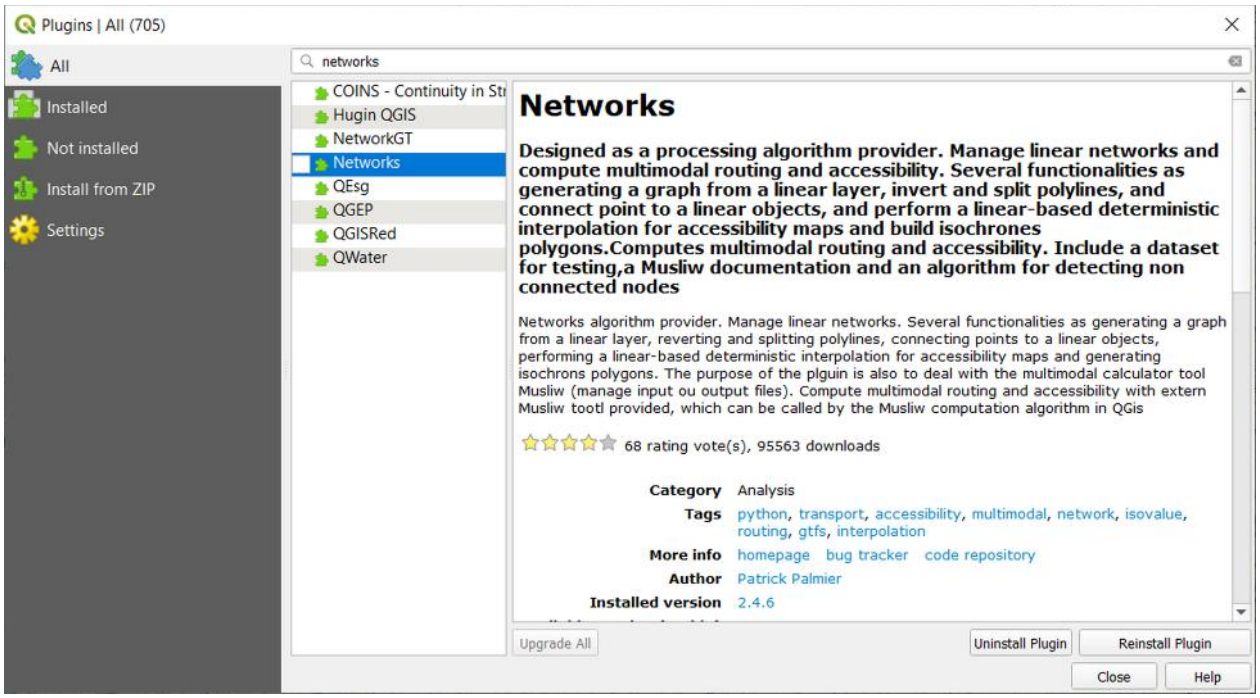


The plugin networks is installed from the Qgis « Plugins/Manage and install,,, » Menu

1. Install « Networks » plugin

In Qgis, **download and activate the "networks"** plugin which add a new algorithm provider in your processing toolbox.

In Plugins/Manage and install Plugins menu, select Networks and install it.



The "networks" algorithm provider is added to your processing toolbox

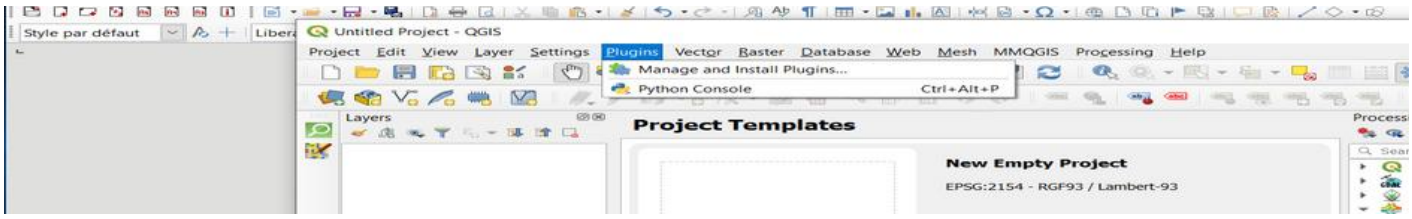


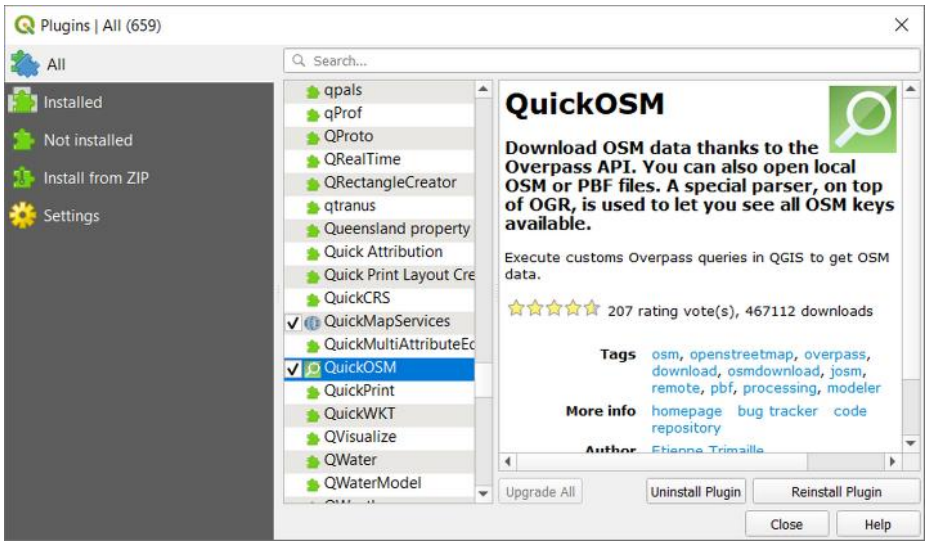
Part 1 : Data preparation – Multimodal network development

chapters 1 and 2 are only for users who do not have a road network and who wish to generate it from OSM. Otherwise you can go directly to 3

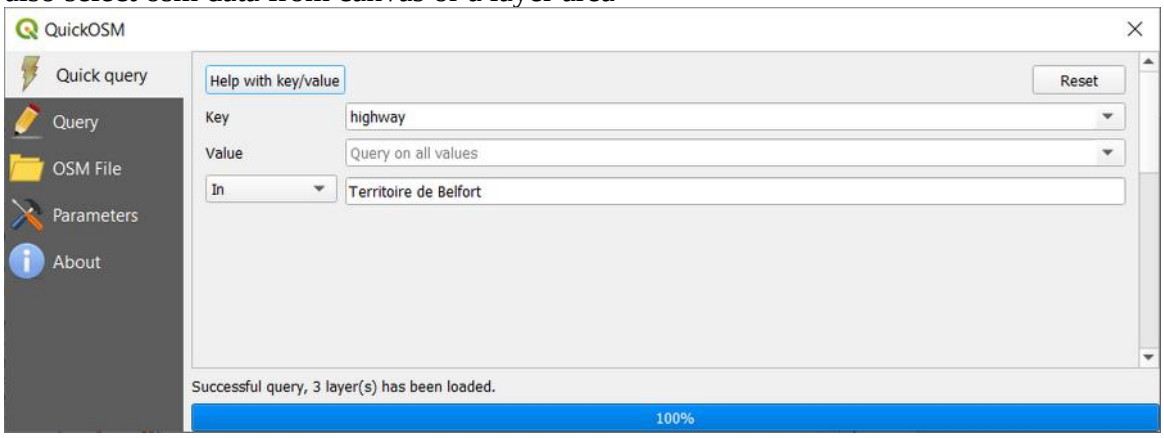
2. Get network data from OSM

First, you can install the QuickOsm plugin
Menu : **Plugins** / Manage and Install Plugins : **Install QuickOSM**

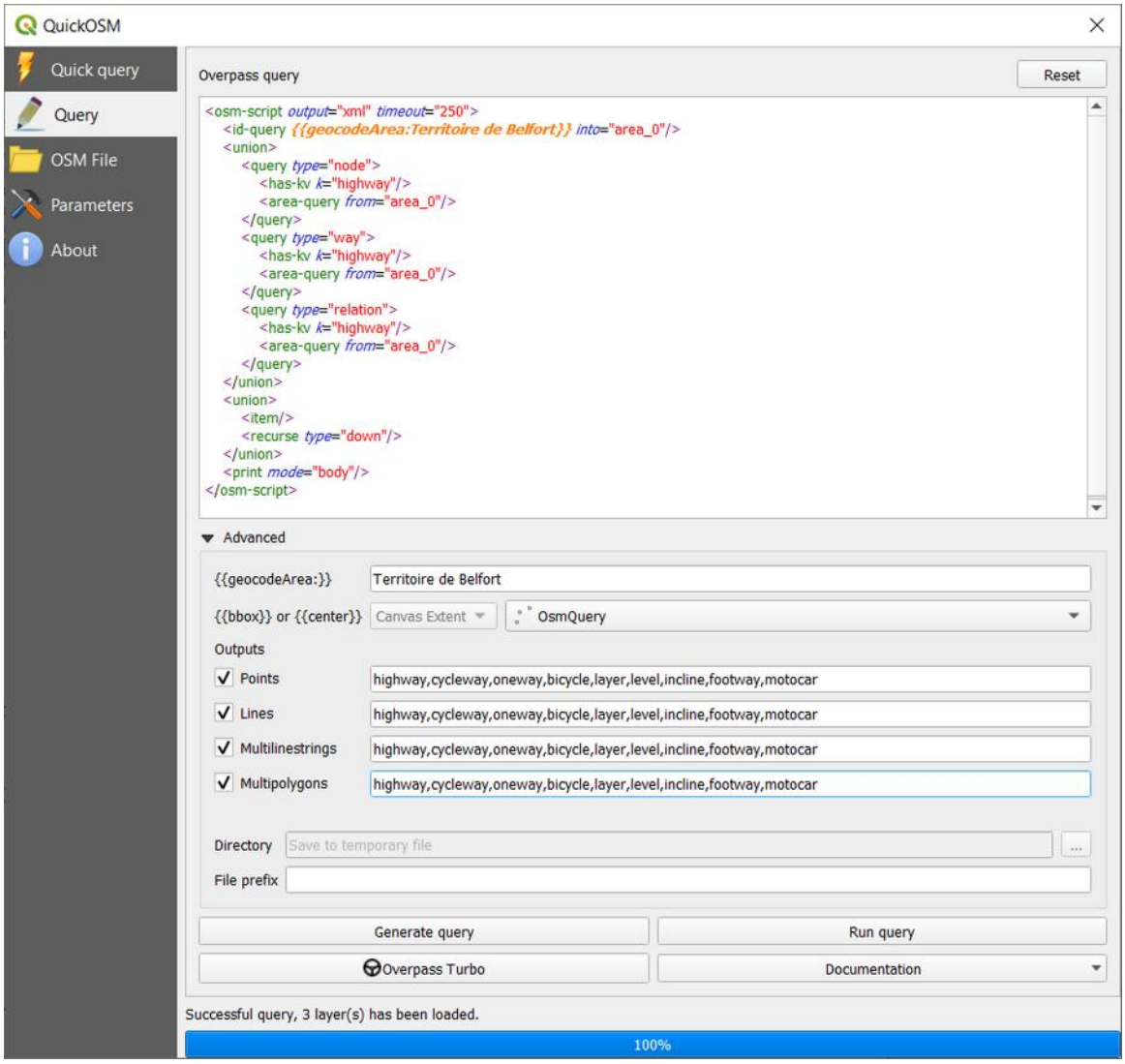




For road network, you can select highway feature, and in this example data from Territory of Belfort, You can also select osm data from canvas or a layer area



You can also select in « Advanced » only « outputs » OSM tags you need to have in your layers, Otherwise you'll have as many fields as different tags in your OSM query data,



You can use either XML query language or OQL query as below. With the OQL query below, you can select roads that are at the border of the area too,
Don’t forget to increase « timeout » and « maxsize » for huge files

Query example to obtain data for Belfort Territory :

```
[timeout:2500]
[maxsize:2000000000];
area[admin_level='6'][ref='90']->.area;
rel(pivot.area)->.rel;
(
node[highway](area.area);
node[highway](r.rel);
way[highway](area.area);
way[highway](r.rel);
rel[highway](area.area);
rel[highway](r.rel);
);
(._;>;);
out;
```

Then :

Run Query
Close
Save « Osmquery » :**Change SCR (ex :EPSG 2154 for France)**
Example : DEPT90_voirie

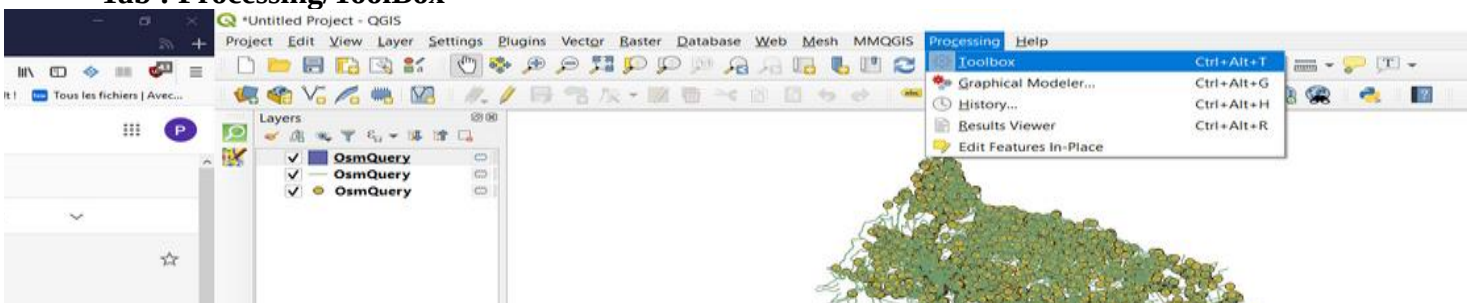
Save the query to load other networks later.

3. Clean/cut OSM data for routing

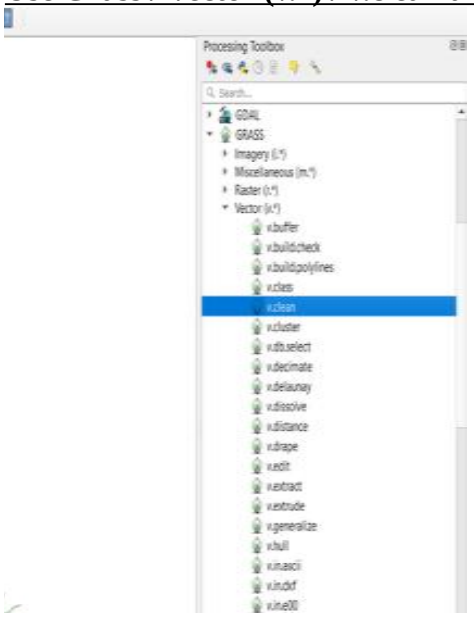
In order to use OSM data for routing you need to do pre-processing operations. For example you need to cut arcs at intersections.
For such tasks, you can use Grass Processing Provider

Show the processing toolbox (if not shown)

Tab : Processing/ToolBox

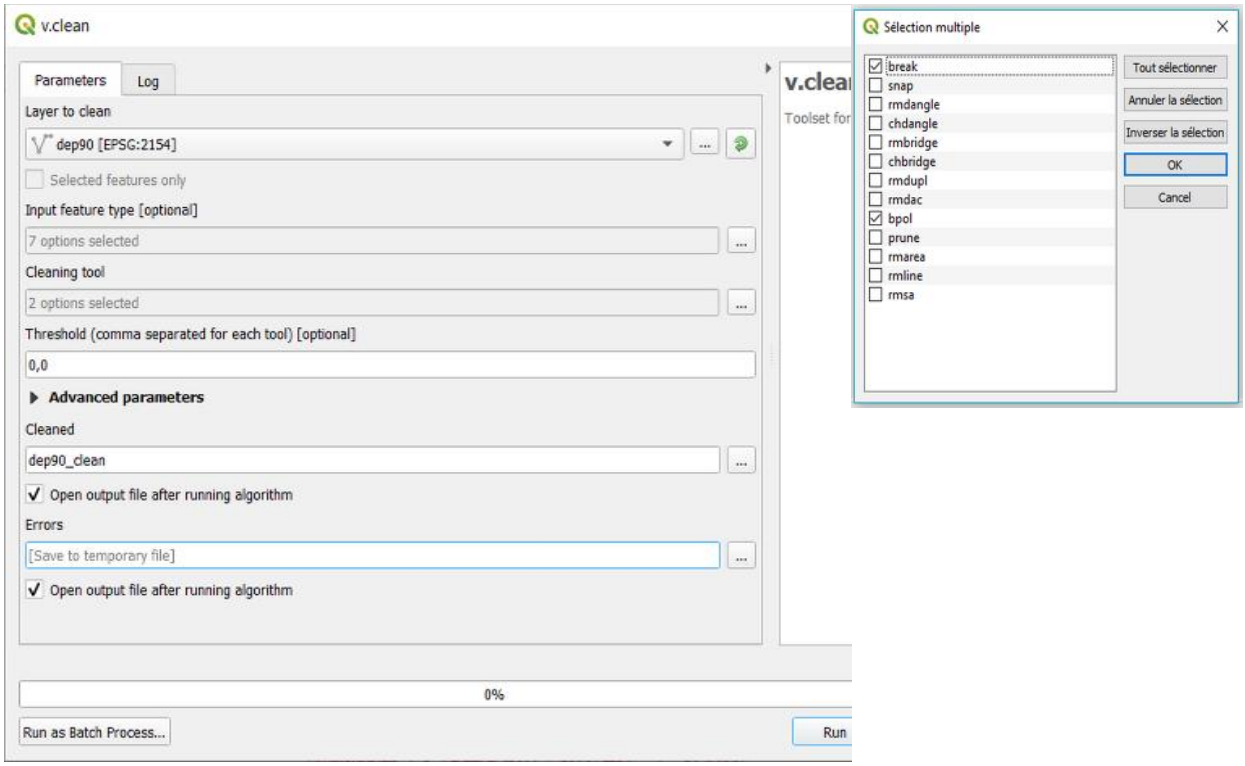


Use Grass / Vector (v.*) / v.clean algorithm



Grass / Vector (v.*) / v.clean Commande :

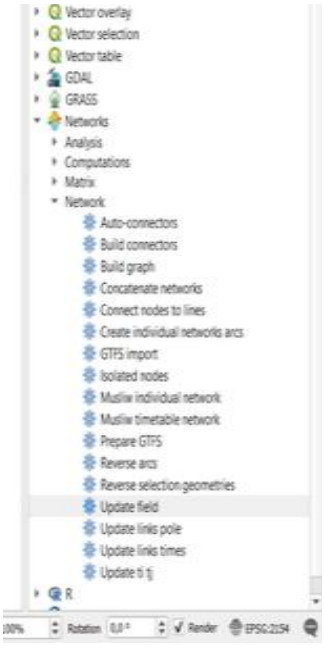
Layer to clean : select the layer to clean : **DEPT01_Voirie**
Cleaning tools : choose the tools : **break**
Threshold (comma separated for each tool) : **0**
Cleaned : save resut in a layer : **dep90_clean**
Errors : save errors file :
Run



Save the clean table (cleaned road layer) and add in Qgis : **change SCR (ex : EPSG 2154)**
Exemple : dep90_clean

4. Add and update fields to the cleaned road layer

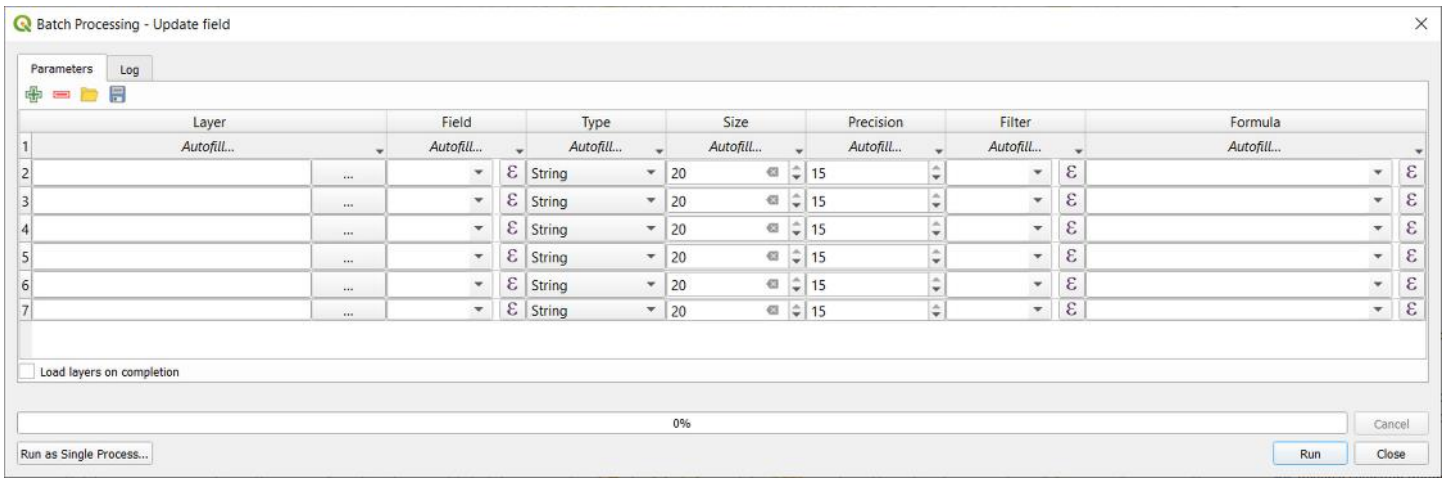
From now, all algorithms used are those from the « networks » provider



Algorithm *:* ***Update field***

Update field algorithm is like « field calculator » in the Standard toolbox, but you can also put a filter for update,

As we need to create and update several fields, we can run the script in « batch mode »
Right Click on « update field » algorithm and select « Execute as batch process » (save query if you want to use it again later)



Layer : select the road layer to update

Fields to create and/or update :

Field : **Sens**
Type : **String**
Size : **1**
Precision : **0**
Filter :
Formula : **‘1’**

Field : **Longueur**
Type : **Double**:
Size : **15**
Precision : **5**
Filter :
Formula : **$length**

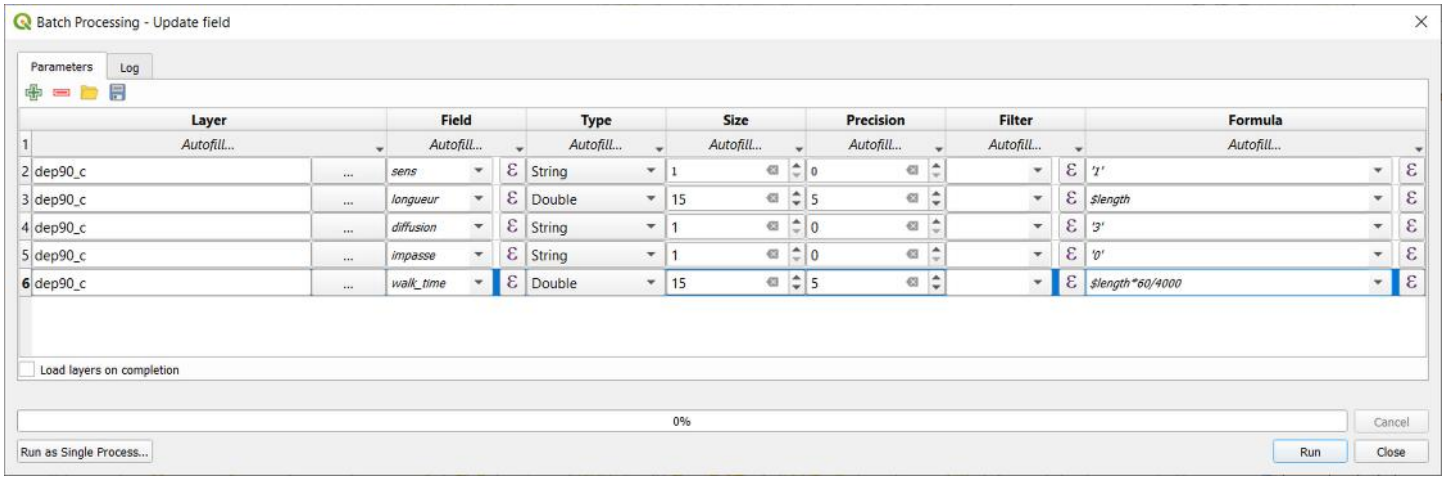
Field : **Diffusion**
Type : **String**
Size : **1**
Precision : **0**
Filter :
Formule : **‘3’** (allow spread in both direction)

Champ : **Impasse**
Type : **String**
Taille: **1**
Précision : **0**
Filtre :
Formula : **‘0’** (passable)

Field : **Cycling_time or Walk_time**
Type : **Double**
Taille : **15**

Precision : **5**
Filter :
Formula : **$length*60/16000** (cycling : 16 km/h) ou **$length*60/4000** (walking : 4 km/h)

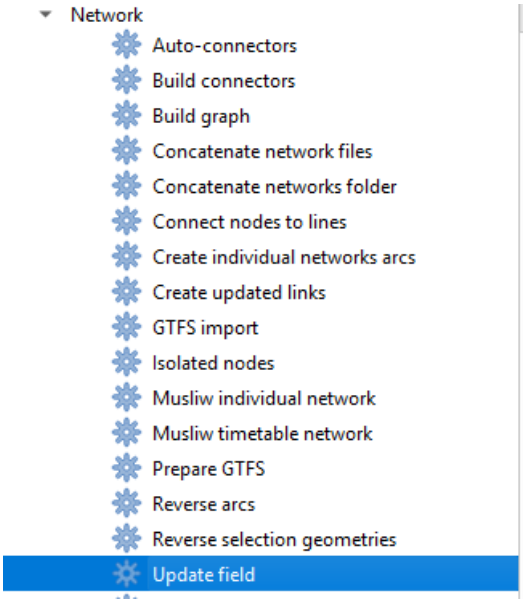
Résultat :



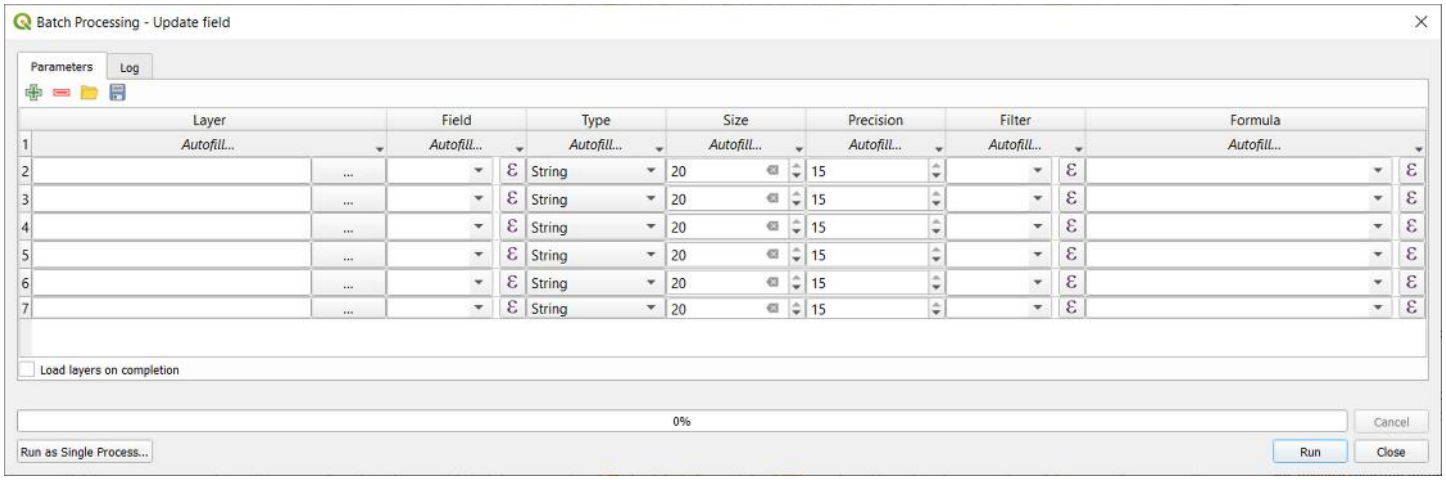
The field « **filter** » is used to update a subset of an existing field

5. Make motorways impassable

SCRIPT *:* ***Update field***



Right Click on « update field » algorithm and select « Execute as batch process » (save query if you want to use it again later)



Fields to update
Field : **Sens (choose a field name for flow direction)**
Type : **keep default value**
Size **: keep default value**
Precision : **keep default value**
Filter : **highway in ('motorway','motorway_link','trunk','trunk_link')**
Formula : ‘0’

Field : **Diffusion**
Type : **laisser le champ affiché par défaut**
Size **: laisser le champ affiché par défaut**
Precision : **laisser le champ affiché par défaut**
Filter : **highway in ('motorway','motorway_link','trunk','trunk_link')**
Formula : ‘0’

Field : **Impasse**
Type : **keep default value**
Size **: keep default value**
Precision : **keep default value**
Filter : **highway in ('motorway','motorway_link','trunk','trunk_link') and "layer" is Null**
Formula : ‘3’ (allowed from both sides of the motorway)

Field : **Impasse**
Type : **keep default value**
Size **: keep default value**
Precision : **keep default value**
Filter : **highway in ('motorway','motorway_link','trunk','trunk_link') and "layer" in ('0')**
Formula : ‘3’

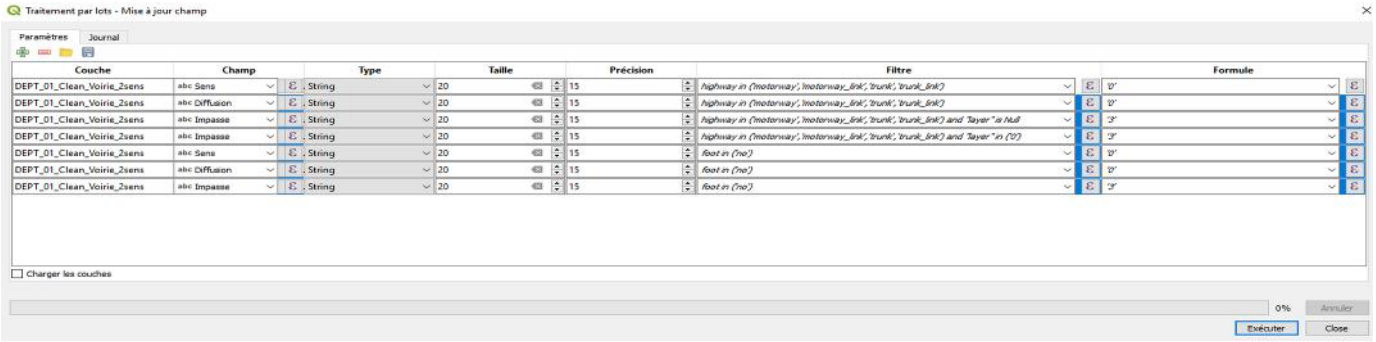
Field : **Sens**
Type : **keep default value**
Size **: keep default value**
Precision : **keep default value**
Filter : **foot in ('no')**
Formula : ‘0’

Field : **Diffusion**
Type : **keep default value**
Size **: keep default value**
Precision : **keep default value**

Filter : **foot in ('no')**
Formula : ‘0’

Field : **Impasse**
Type : **keep default value**
Size **: keep default value**
Precision : **keep default value**
Filter : **foot in ('no')**
Formula : ‘3’

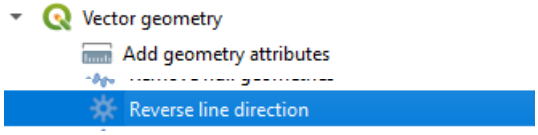
Result :



6. Opposite direction preparation

It is better for modeling to have only one-way arcs, as travel times and characteristics could be different on each direction. For example travel times for cycling are very different downhill and uphill. For cars drivers, many streets are only one-way. To do this, it is necessary to reverse and duplicate arcs

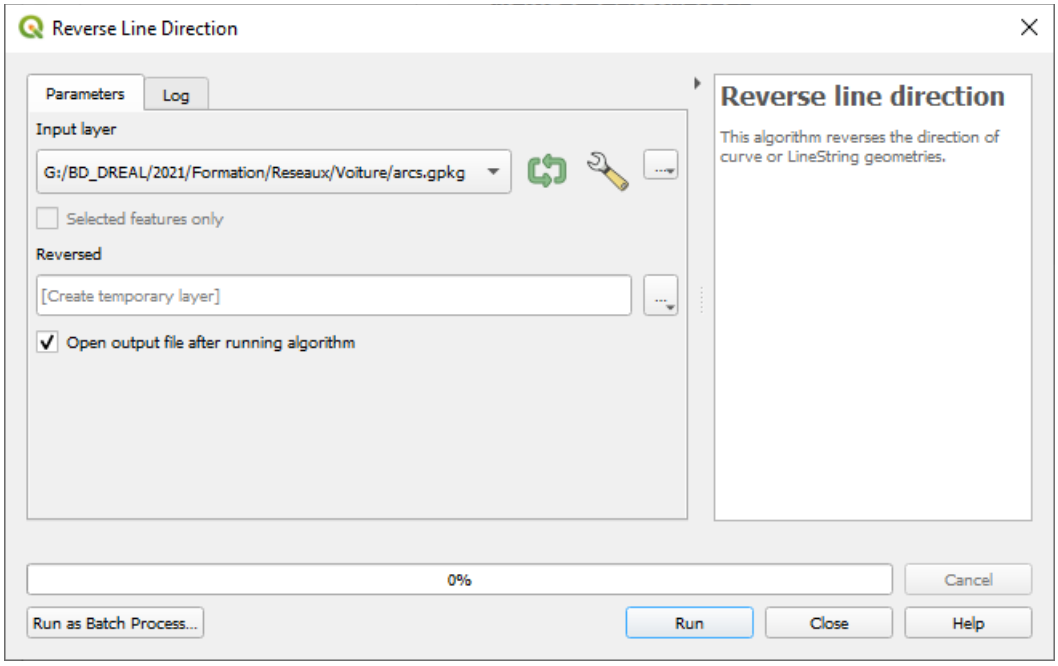
SCRIPT *:* ***Reverse arcs***



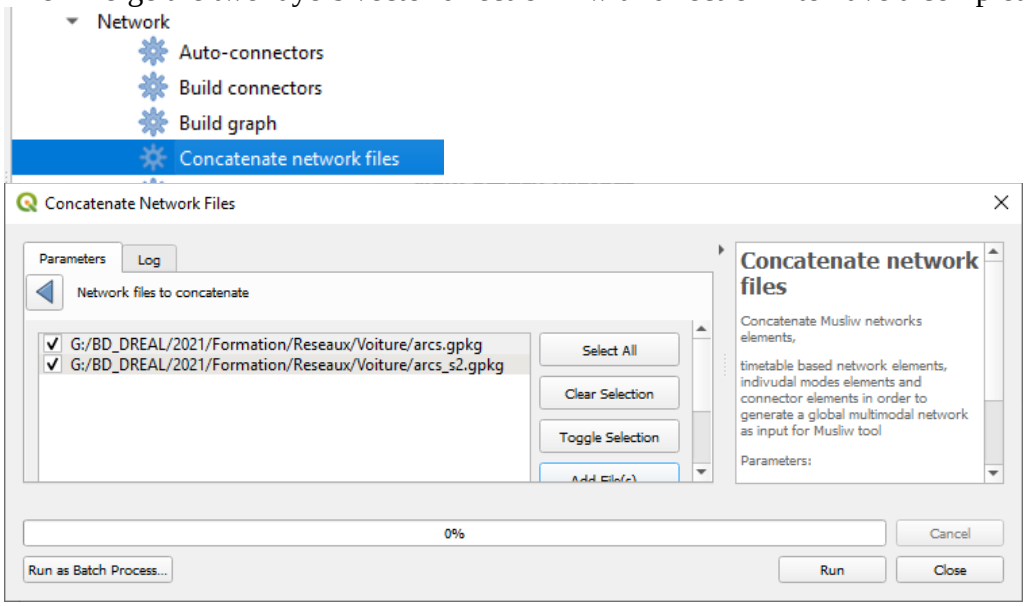
network : ***Open the network layer to reverse***
Add to network ***(if checked : reverse arcs are added directly to the initial layer)***
reverted network: name of the reverse arcs layer
Open output file after running algorithm Qgis.

I advise users to create in first two separate layers, one for original arcs and another for the reserve arcs layer, in order to manage easily flow directions with "oneway" tag for cars and "cycleway" tag for bikes in particular.
For example, an arc with oneway='yes' in the original layer should have "direction"='1' (allowed) in the original layer and "direction"='0' (prohibited) in the reverse layer
Please look at https://wiki.openstreetmap.org/wiki/Key:oneway for more details on one wayslinks and https://wiki.openstreetmap.org/wiki/Key:cycleway for cycling facilities.

Result :



Then merge the two layers vector direction 1 with direction 2 to have a complete network



Only for bike network« **need to be adapted depending on territories »**

oneway= ‘yes’ and highway in (‘primary’,’secondary’,’tertiary’,’primary link’,’secondary link’,’tertiary link’) and cycleway is Null

update field “ Sens” (or direction) : **‘0’**

From update fields :

example for reverting cycling slopes.

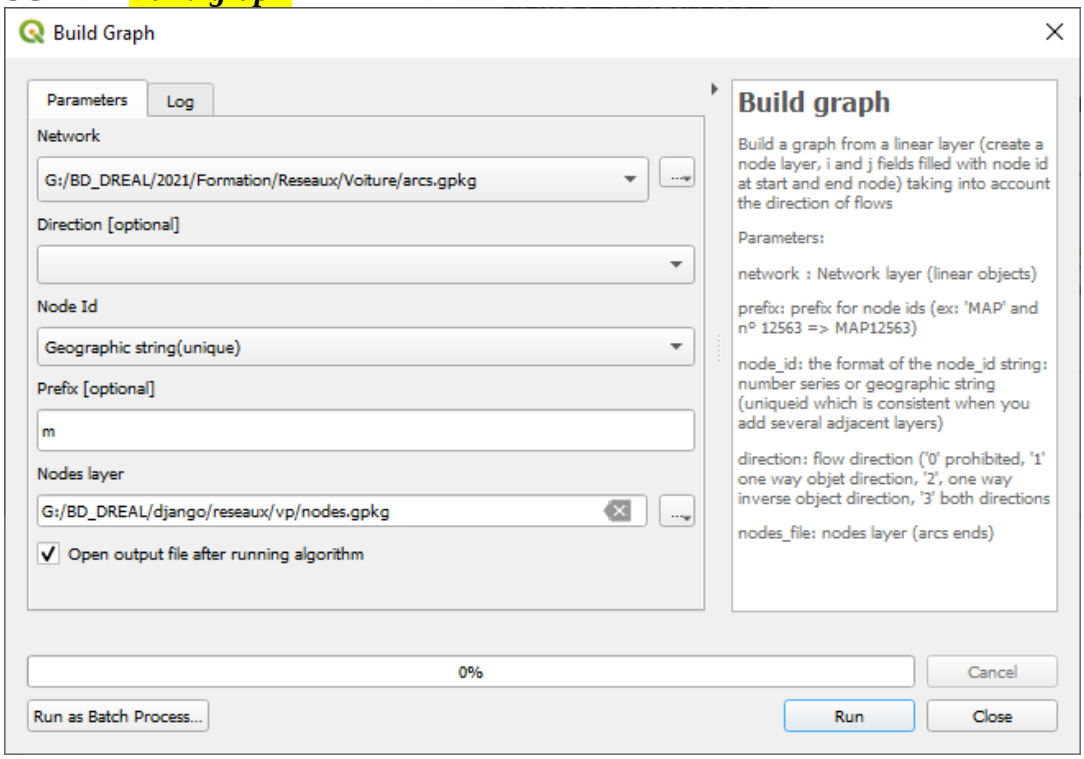
Update field « incline » to revert slopes *(values to adapt depending on slopes values)*

Case when « incline »= ‘up’ then ‘down’
when « incline »= ‘down’ then ‘up’
when « incline »= ‘-8%’ then ‘8%’
when « incline »= ‘10%’ then ‘-10%’
when « incline »= ‘8%’ then ‘-8%’
when « incline »= ‘-10%’ then ‘10%’
else Null
End

7. Build road graph

For shortest paths computing, we need to build a road graph from the network layer. The graph generation consists in the generation a of node layer and the addition for each arc of a “from node” and a “to node” field corresponding to the nodeId of the node layer

SCRIPT *Build graph*



Network : **Network layer (arc layer original + reverse arcs layer)**

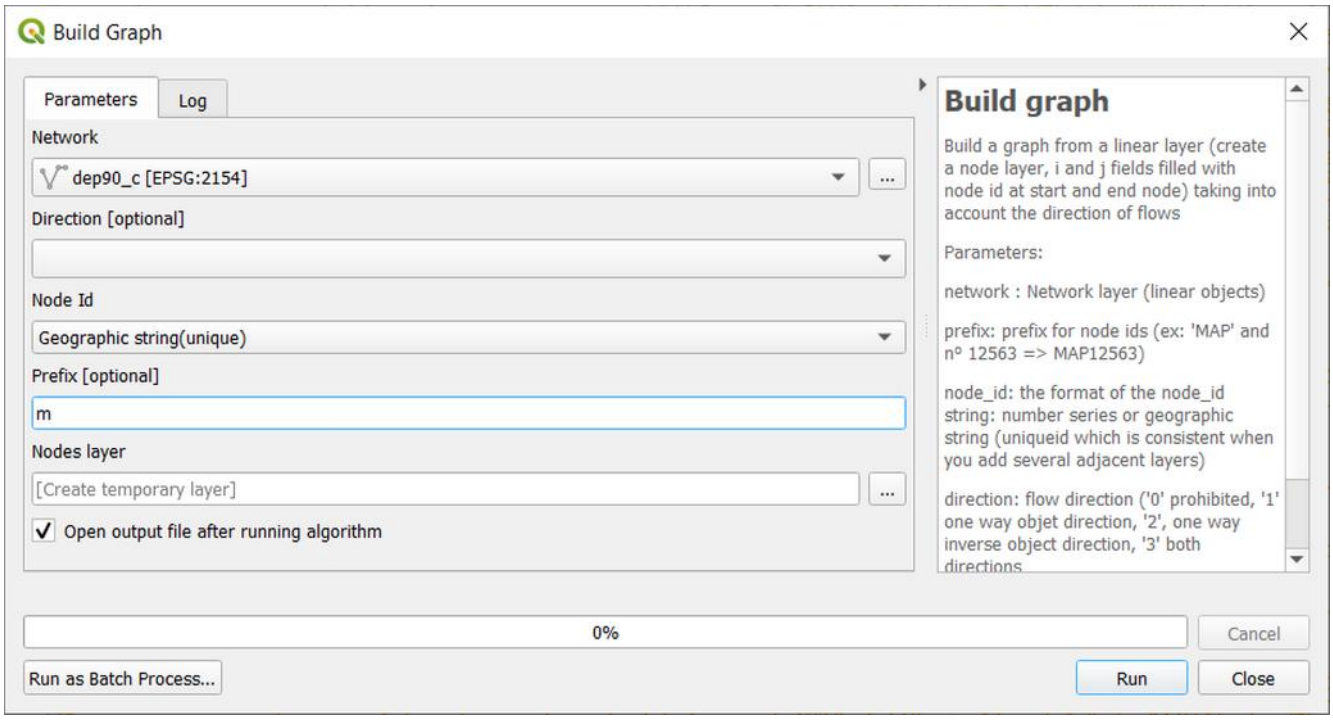
Node id : **geographic string(unique)**

Prefix : **m** (in order to differentiate arcs from for different modes)

Nodes layer : **Name of the generated nodes layer**

Run

Result :

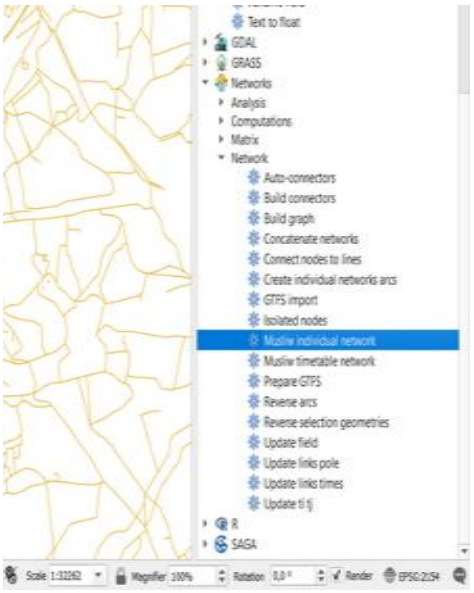


8. Generation of the Musliw individual network file

The shortest path algorithm is working with networks and matrices ascii files in the Musliw file format (semi-column separated text files).

SCRIPT *Musliw individual network*

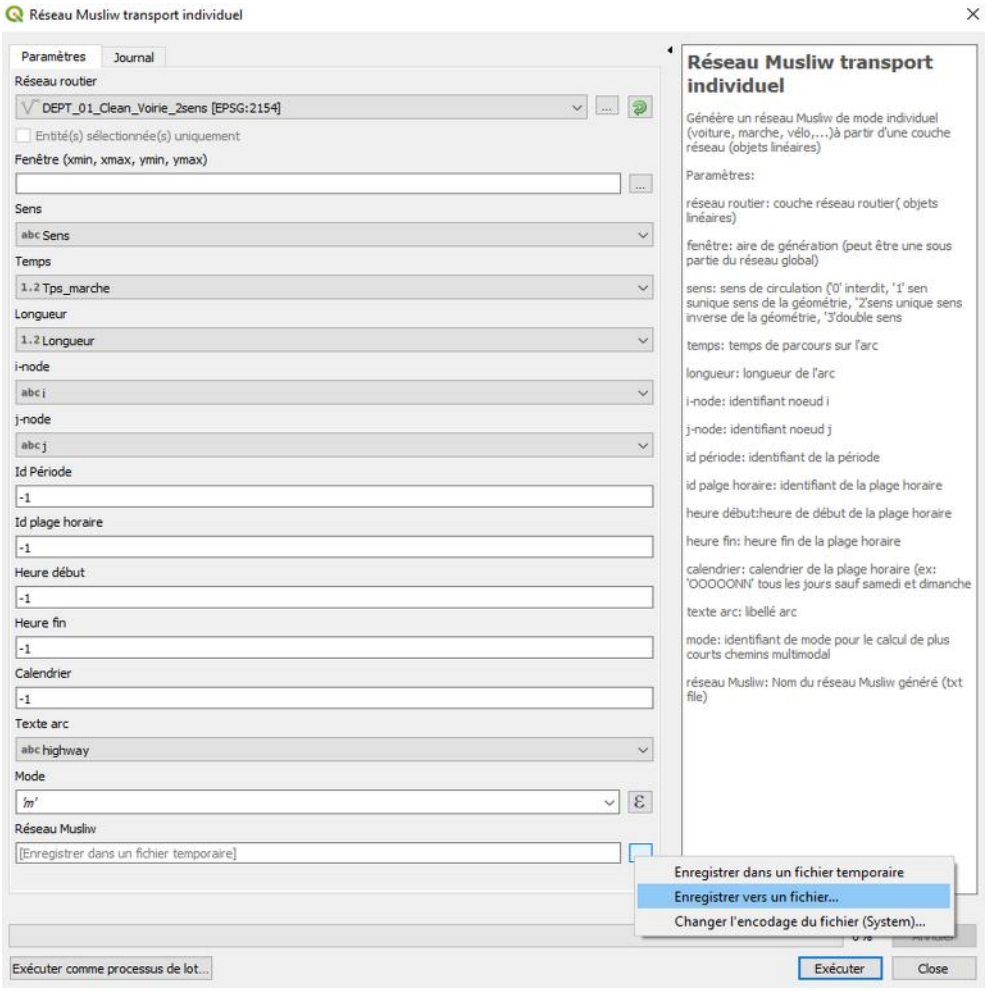
Allow to generate a Musliw individual network from a Qgis road network layer



Load road network layer
Run “Musliw individual network” script

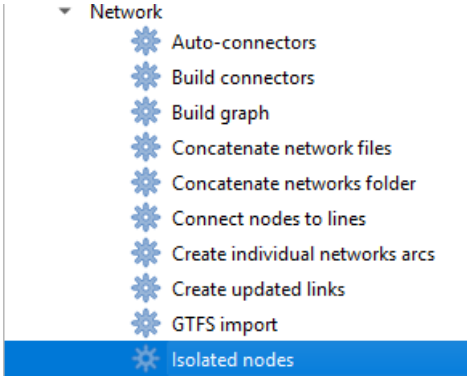
- Road network : **road network layer**
- Sens : “**sens” of direction field**
- Time : **walk-time**
- Length : **longueur**
- i-node : **i**
- j-node : **j**
- Time category id : **keep default value**
- Time period id : **keep default value**
- Start time : **keep default value**
- End time : **keep default value**
- Calendar : **keep default value**
- Arc label : **highway**
- Mode : **m** (si marche)
- Musliw network: **save file (with .txt extension)**
- **Run**

Result

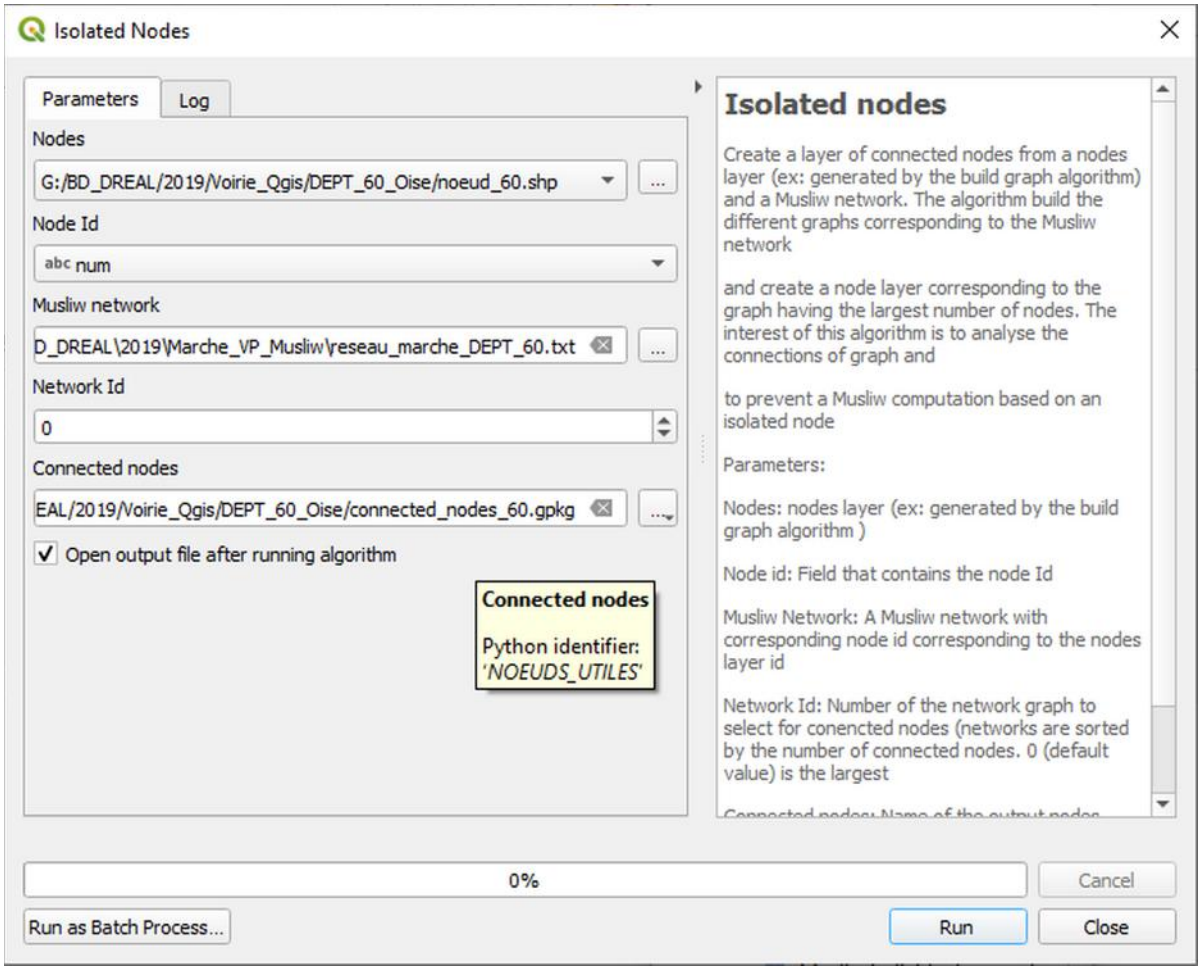


9. Suppress isolated nodes (nodes which are unreachable/ non connected to the main graph))

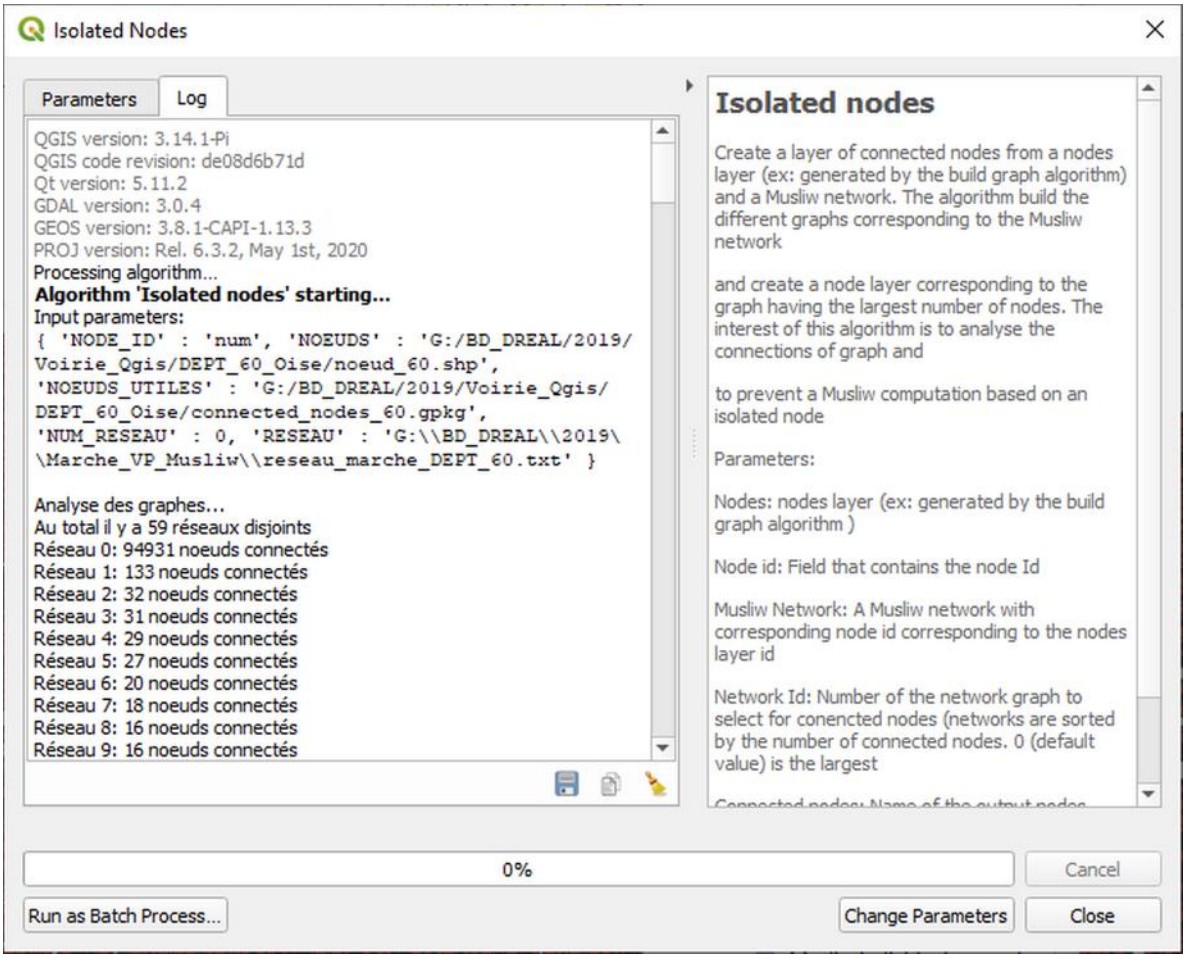
SCRIPT Isolated nodes



Nodes: **open nodes layer**
Node Id : **num**
Musliw network : **open walking network .txt**
Netowrk Id : 0 in a first step (networks are sorted depending on the greater number of nodes, then 1, or even 2, etc. depending on the network analysis and the homogeneous chosen network
Connected nodes : **Save file as...**

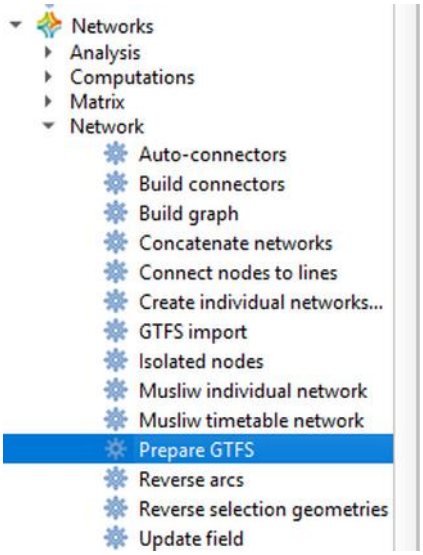


In this example, there are for the network (ID 0),94 931 connected nodes and 59 non-intersecting networks.



10. Prepare GTFS – From a raw GTFS (not prepared for Musliw)

SCRIPT Prepare GTFS
Right click on the script/ **Execute as batch process**



GTFS source folder: Select the **source directory** (where are located the GTFS files)

Network Id : **AE** (AE example for Lille Airport bus line)

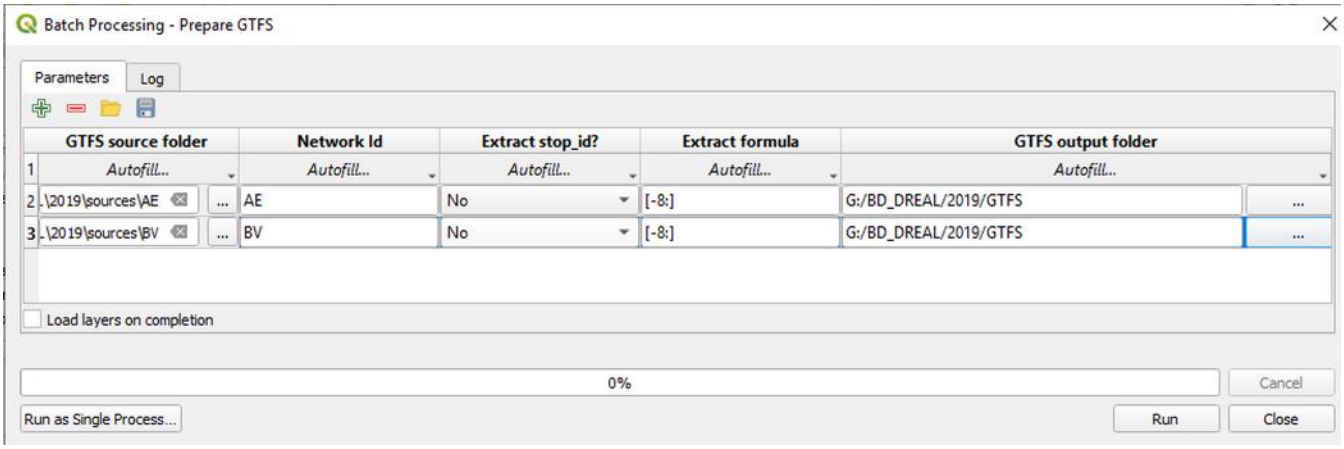
Extract stop_id ? : **Non (select Yes if you want to modify the stop_id string with a formula)**

Exctract formula : **write the formula: Ignored when "Extract stop_id?" is not checked**

GTFS output folder : Save result in the **output folder**. Example « **GTFS** » folder

Run

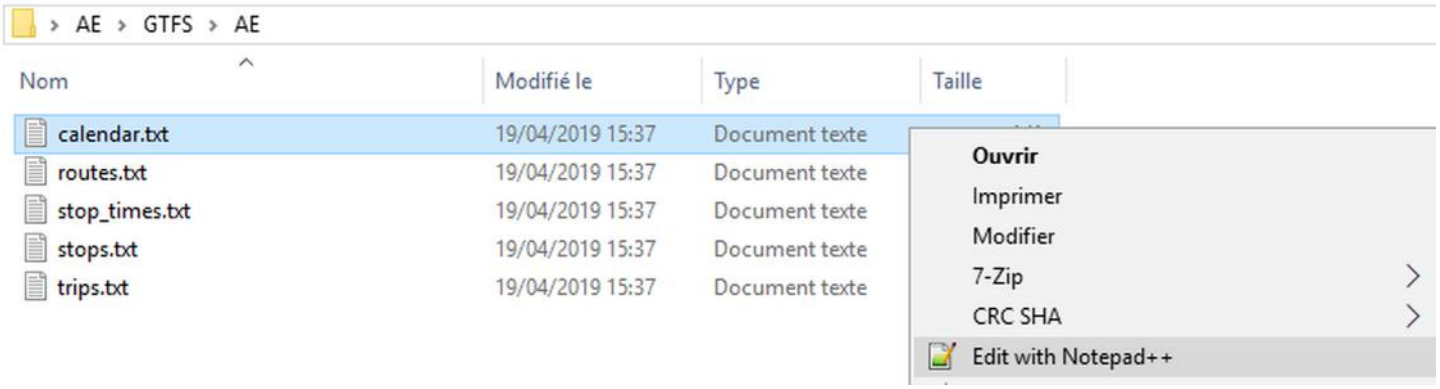
Result:

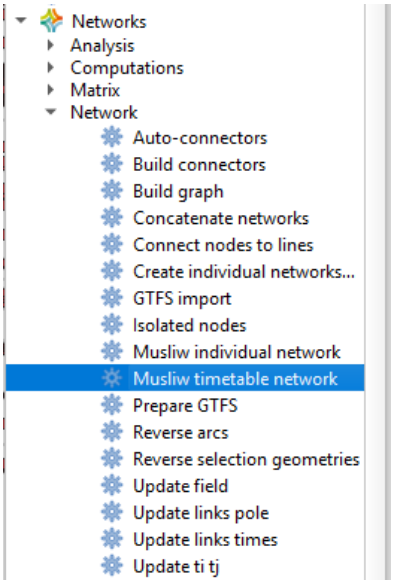


11. Generation of a PT network

As the GTFS extractions od the different PT networks have not been done at the same time, it is very important to select for each GTFS a period where timetables are available.

In the result folder (saved GTFS files), edit (for example with Notepad++) the **calendar.txt ou calendar_dates file from each network** in order to choose a typical week (not during school holidays for example). Example : from 10/06/2019 to 16/06/2019





SCRIPT Musliw timetable network

GTFS Folder : open the directory where the GTFS files are located

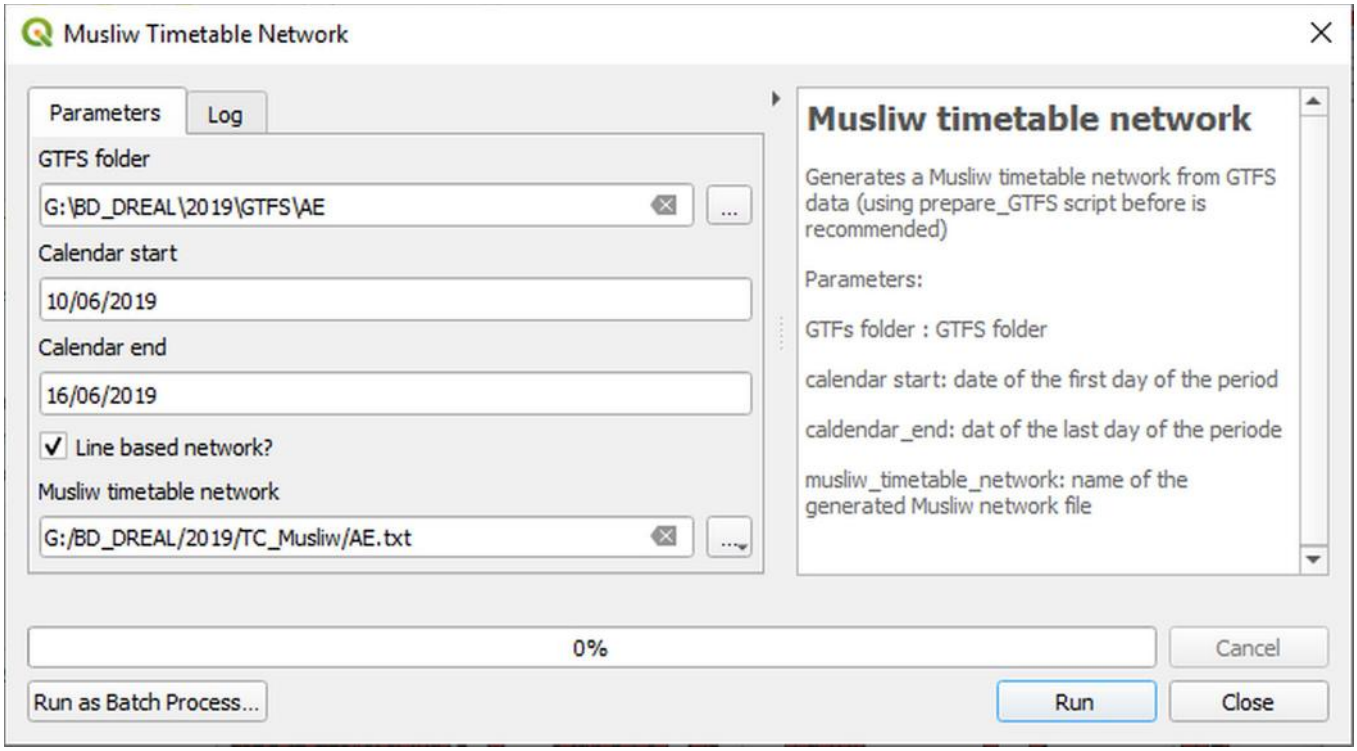
Calendar start: enter the first day of the period **(often a monday for a typical week period, but you can choose any day and any period duration)**

Calendar end: enter the last day of the period **(sunday for a typical week period)**

Lines based network: checked by default for most uses

Musliw timetable network : Save Muslwiwnetwork name as **(with .txt extension) – Example AE.txt**

Run

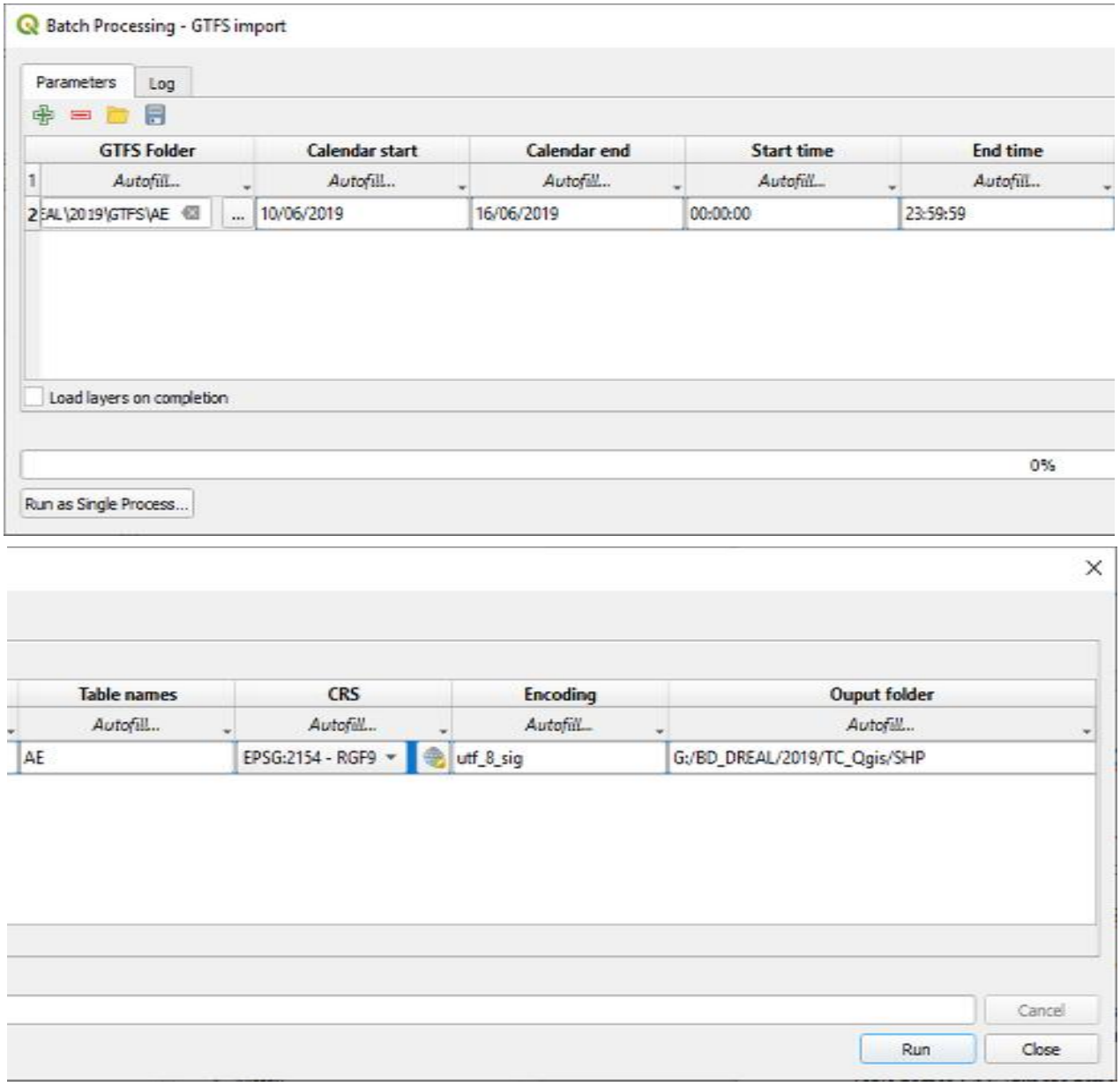


12. Import PT stops in QGIS

SCRIPT GTFS Import
Right click / Execute as batch process (To save time in case of errors or in order to modify the batch process, don't forget to save the batch process script in Json format)

GTFS Folder : **Select the GTFS directory**
Calendar start : ex: **10/06/2019**
Calendar end : **ex: 16/06/2019** (corresponding to the week from Monday, June 10th 2019 to Sunday, June 16th 2019)
Start time : **keep default values (default values corresponds to the whole day. You can change them if you want to reduce the analyses period ex: peak hours)**
End time : **keep default values**
Table names : **AE** (put the network name, for example AE pour Lille Airport shuttle
CRS : **let default values (EPSG 2154 for France)**
Encoding : **« utf8_8_sig » Keep the default value. If this doesn't work, try another encoding« utf8 » or « cp1252 » for example**
Output folder: **Select an output directory** (example "shp" (as output files are.shapefiles))
Run

Results :



13. Generation of the connectors

Connectors are virtual links designed to enable multimodality between two modal networks. For example, you need to create connectors between the walking network and the railway network to enable people to combine tran and walking.

Load the previously generated nodes layer
Load the walking network nodes layer

SCRIPT Build connectors

Generate from two points layers (PT stops and road layers) a Musliw network file for connectors and a connectors layers. **Connectors are visible through the connectors layer**

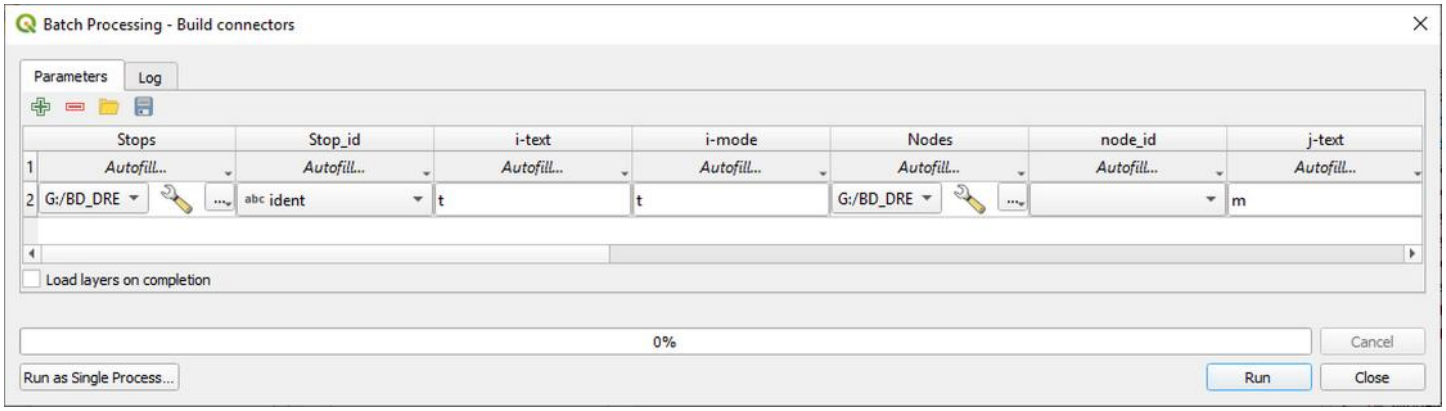
Connect PT stops to the nearest pedestrian nodes

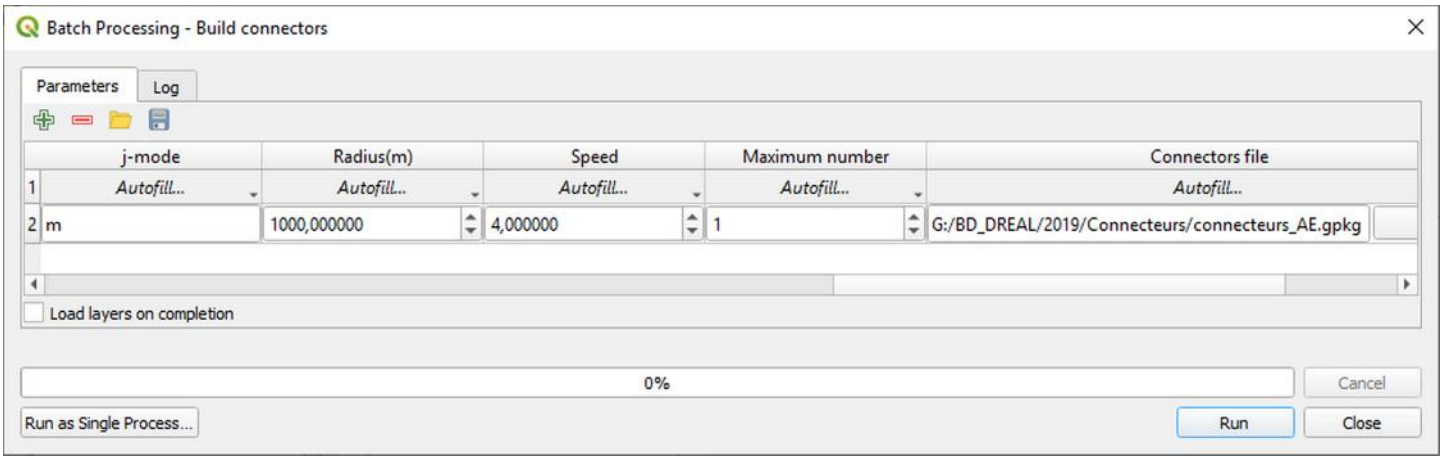
Right Click/ Execute or Execute as batch

stops : **PT stops layer (stops.shp)**

stop id : **ident**
i-text : **t** (transit)
i-mode: **t** (TC) - mode id at i node
Nodes : **pedestrian nodes layer**
node id : **num**
j-texte j : **m** (walking)
j-mode : **m** (walking) – mode id at j node
search radius : **1000**
speed : **0 – if 0 connector wille be with no travel time**
maximum number : **1 – only 1 connector per PT stop**

Connectors file : connecteur_reseau_tc_map (name of the connectors layer)





14. Generation of the multi-modal network

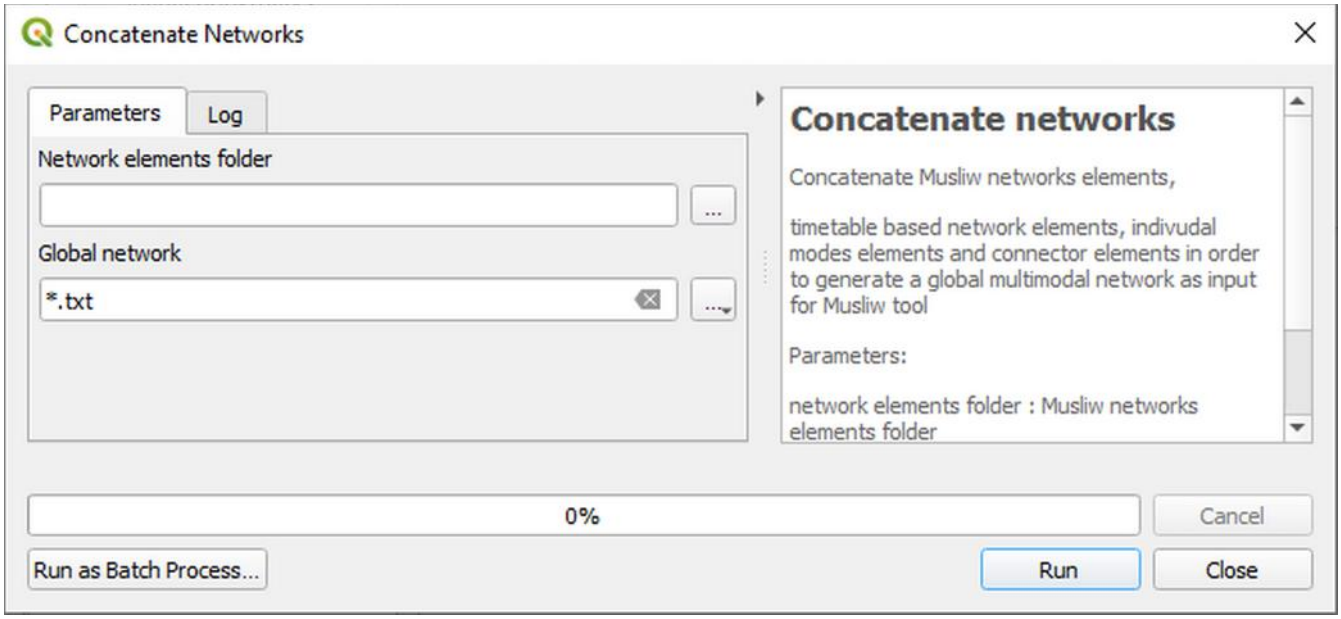
SCRIPT Concatenate networks

Concatenate different networks files in Musliw format (.txt extension)

Walking network
Connectors network
PT timetables networks
...

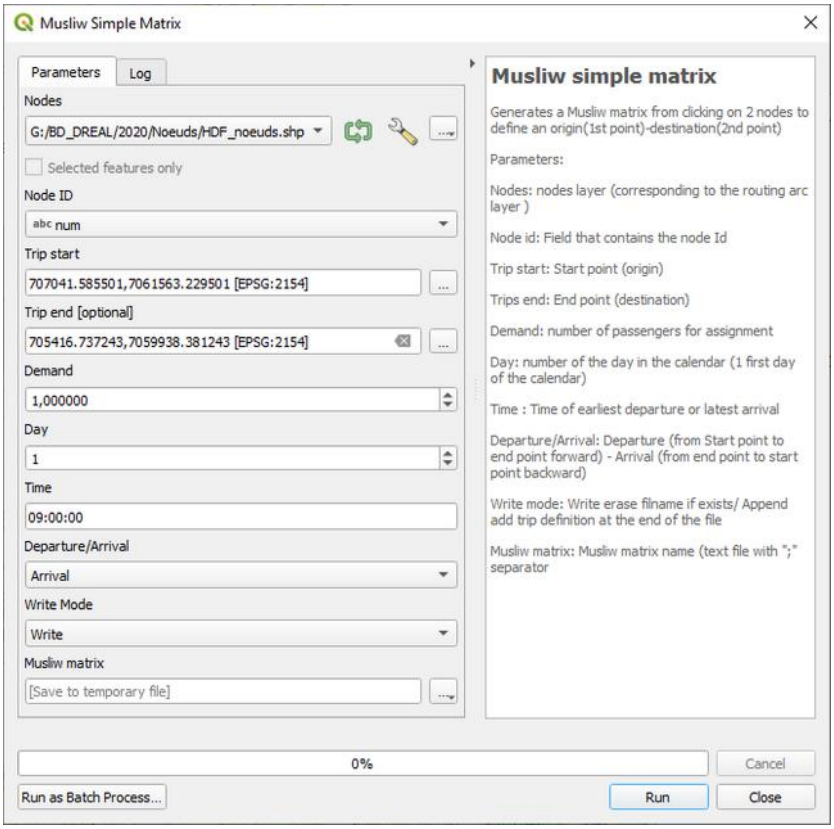
Put each file that need to be used in a specific directory **(example : folder « concatenate »**
network elements foldere : **select the folder where .txt files to be concatenated are located**
global network : **save multimodal network as** (**with txt extension**)
Run

Résultat :


Concatenate Networks
Parameters
Log
Network elements folder
Global network
*.txt
Concatenate networks
Concatenate Musliw networks elements,
timetable based network elements, indivudal modes elements and connector elements in order to generate a global multimodal network as input for Musliw tool
Parameters:
network elements folder : Musliw networks elements folder
0%
Cancel
Run as Batch Process...
Run
Close

Part 2 : generation of the territorial accessibility map

15. Matrix generation for Musliw



SCRIPT **Musliw simple matrix**

Nodes : **nodes layer**

Node Id : **num**

Trip start : **click directly in Qgis the start point**

Trip end : **select directly in Qgis the end point**

Demand : **number of passengers**

Day : **1 (0 is the first day from the period, 1 for a typical week is corresponding to a Tuesday))**

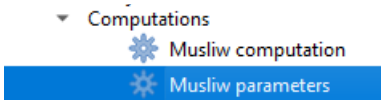
Time : time **format 00:00:00**

Departure/Arrival : **Choose**

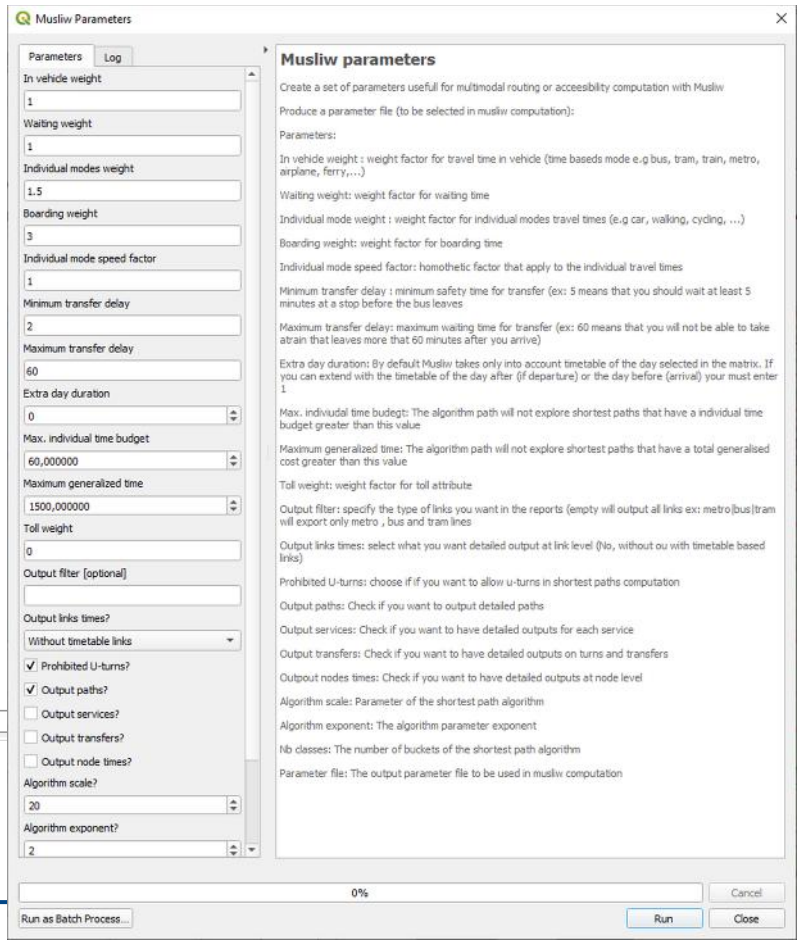
Write mode : **write or append**

Musliw matrix : **save matrix as...**

16. Musliw parameters file



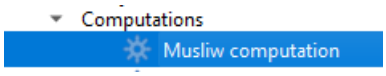
SCRIPT Musliw parameters



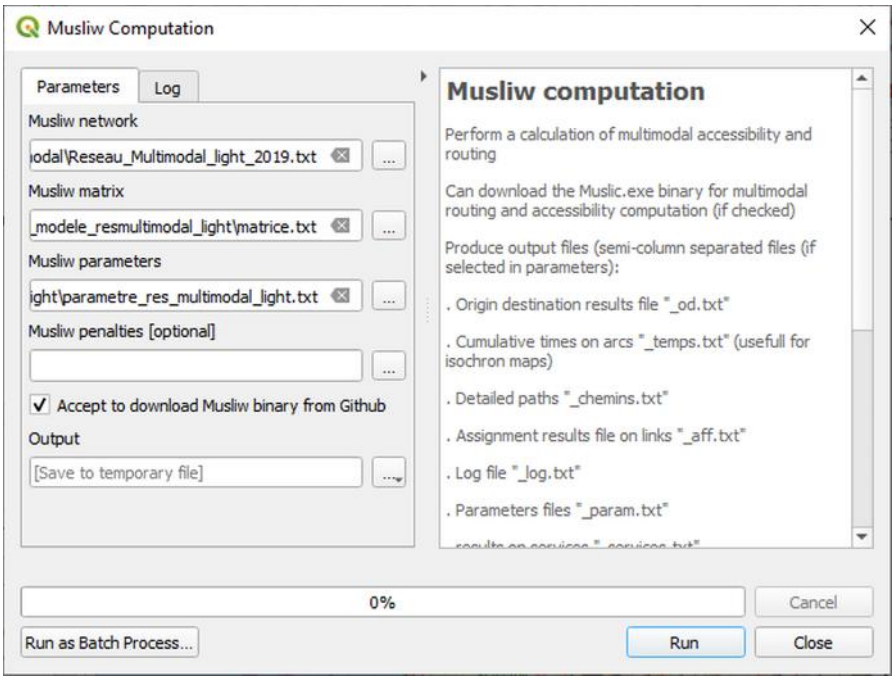
Individual mode speed factor : **0,25 for cycling**

Output links times : **Without timetables links**(travel times file)

17. Musliw computation



Musliw network : **reseau_multimodal.txt**
Matrice Musliw : **matrix.txt**
Penalties Musliw (optional) : **penalties and transfers file**
Accept to download Muslic binary from Github : M**ust be if ckecked if Musliw has not ever being loaded.**

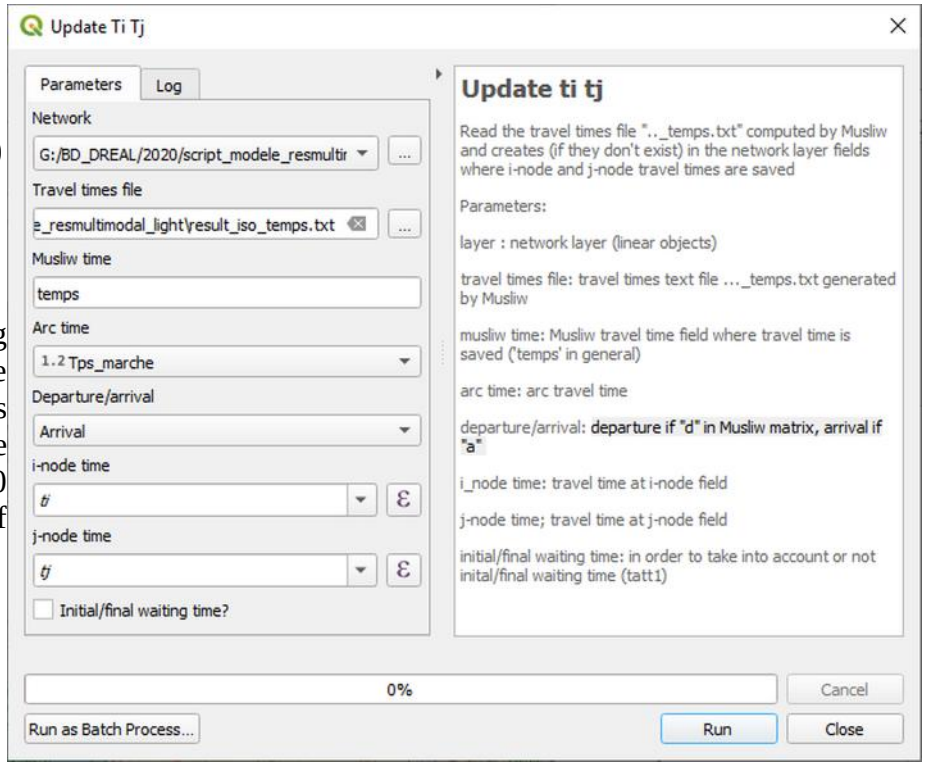


18. Update ti_tj

SCRIPT Update ti tj

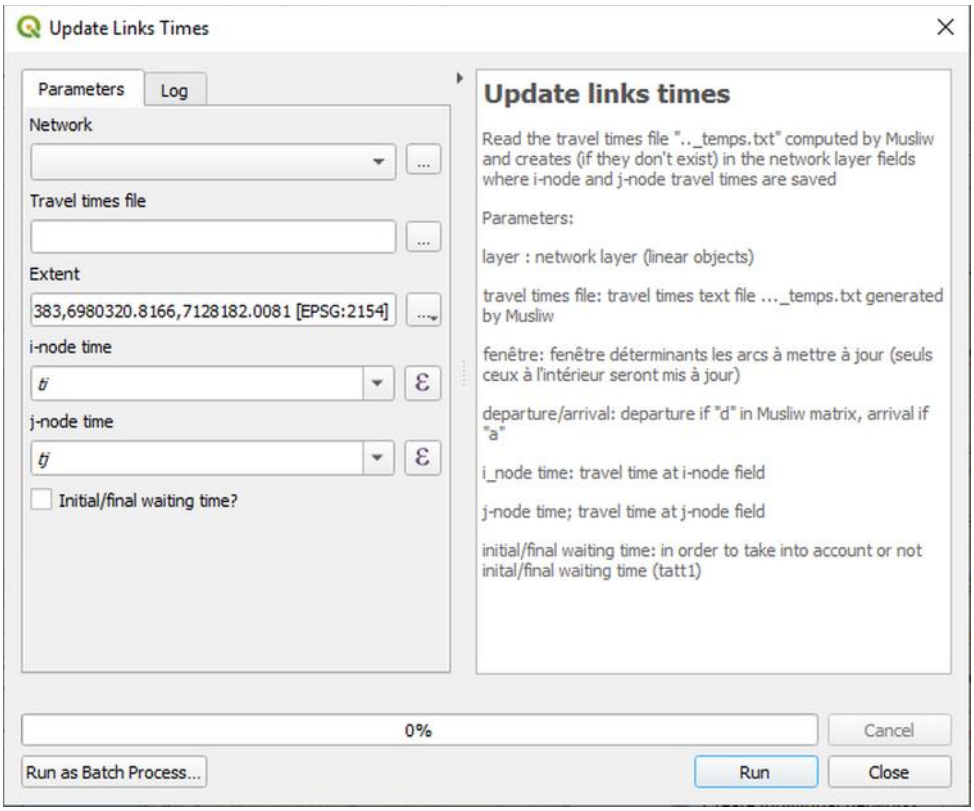
Network: **reseau_voirie.shp**
Travel times file : **Choose the travel times output file generated by Musliw (_temp.txt)**
Musliw time : **temps**
Arc time : **Tps_marche**
Departure/arrival : **departure if "d" in Musliw matrix, arrival if "a"**
i-node time : **ti if this field exists (update) or possibility to create a new field, in this case write « ti_bike » for example.**
J-node time : **tj (as i-node time but in j)**

Check Initial/final waiting time for taking into account the remaining time from the stop (for example if the arrival time is 8h20 and you asked in the matrix to arrive at 8h30 at latest, then the remaining 10 minutes are not taken into account if checked)



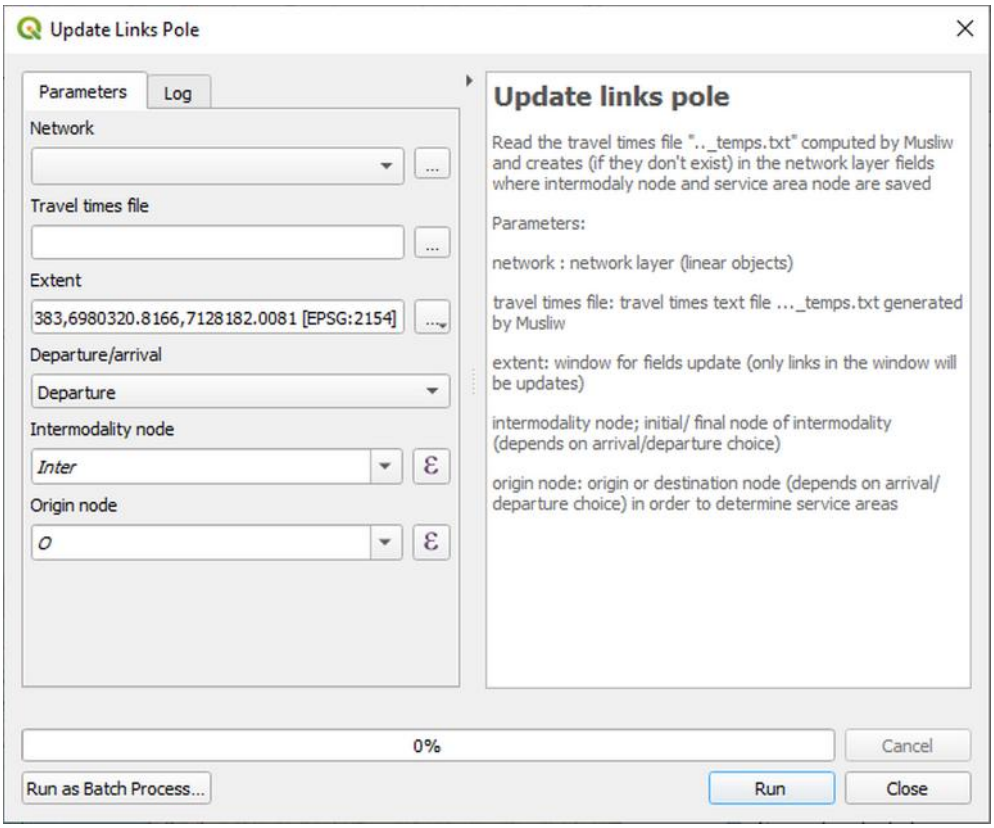
Option 1: SCRIPT update links times

If you just need to make an territorial accessibility map, you case use the script “update links time” instead of “update ti tj”. It uses the Musliw time “temps” by default and get time values directly into travel output times file.



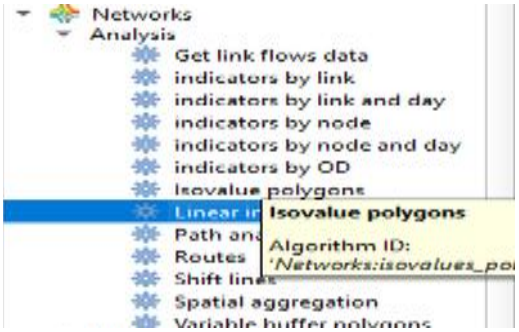
Option 2: SCRIPT update links pole

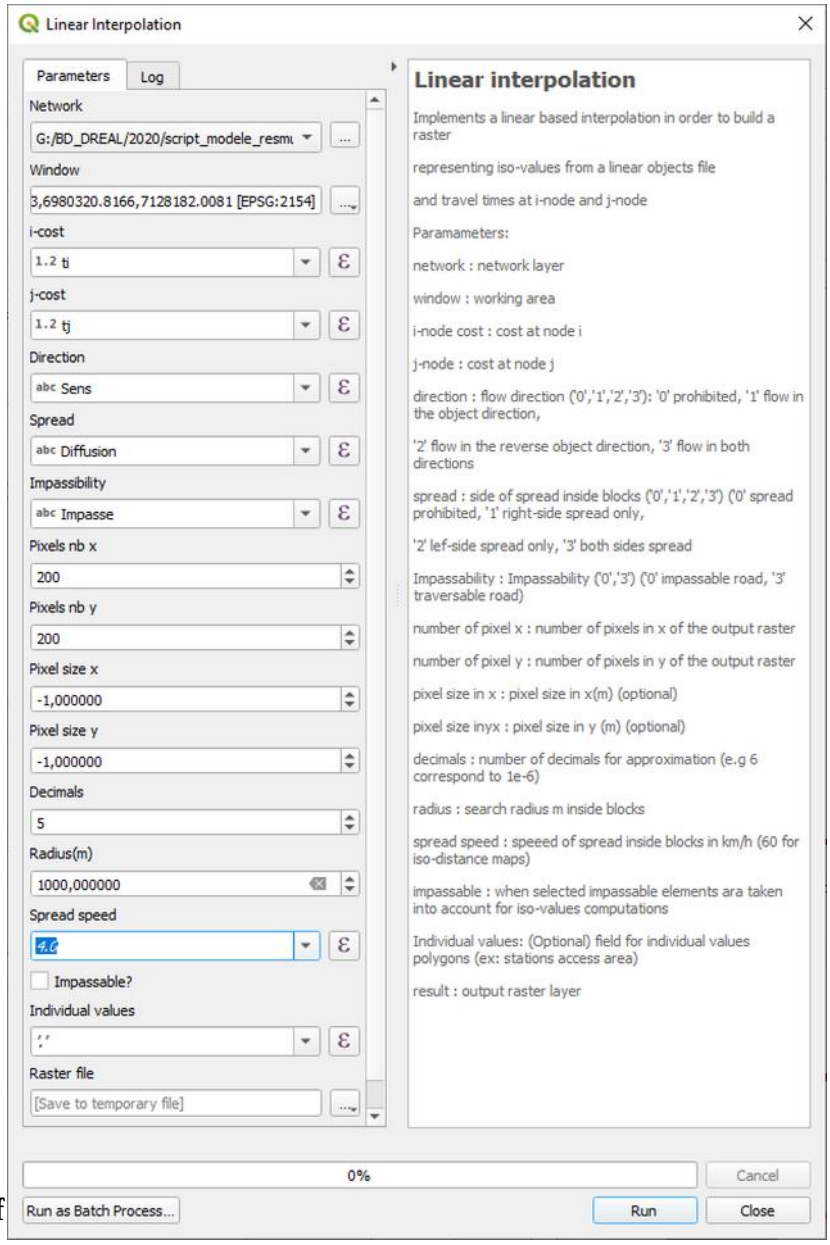
Use this option if you want to make an discrete territorial analyze map (for example, if you compute a territorial accessibility from several points, this option allow you to generate the catchment area of each point.



19. Linear Interpolation

SCRIPT Linear interpolation



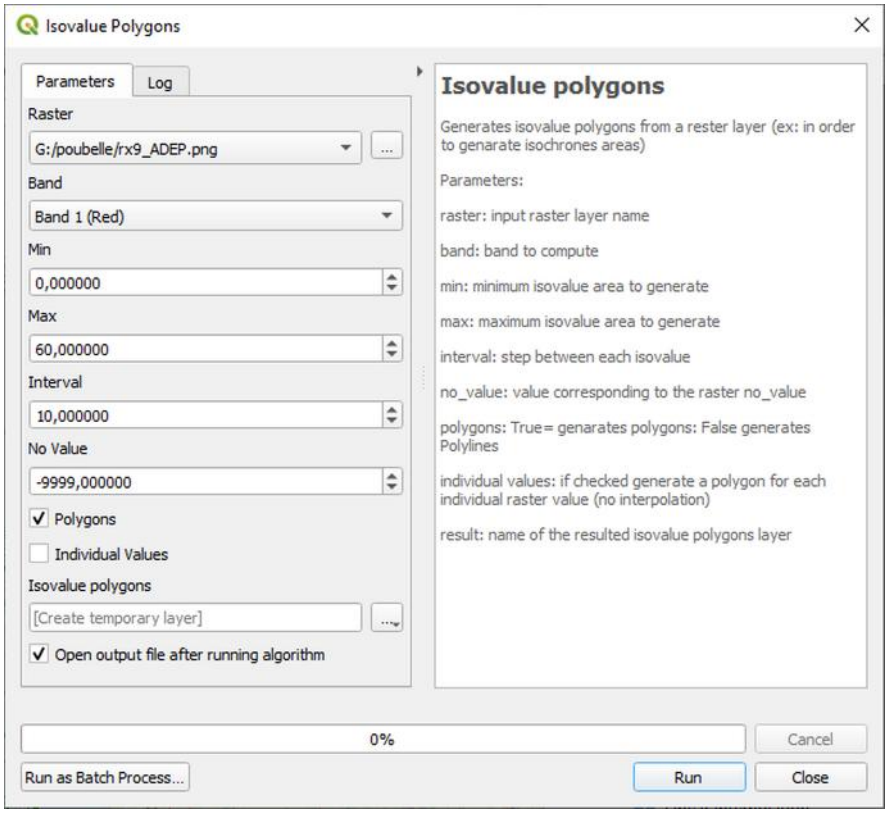


Network : **réseau_voirie_shp**
Window **: Allow to define the analysis area**
i-cost: **ti**
j-cost : **tj**
Direction : **Sens**
Spread : **Diffusion**
Impassability : **Impasse**
Pixels nb x :**200**
Pixels nb y : **200**
Pixel size x : **let default values**
Pixel size y : **let default values**
Decimals : **5**
Radius(m) : **1000**
Spread speed : **4** (4 km/h for walking) speed to be adapted depending on mode (15 km/h cycling)
or for a distance map : **60**
Impassable? : **do not check**
individual values**: choose a field for in case of an individual value analysis (ex: pole) or keep '.' as default**
Raster file : save raster as
Run

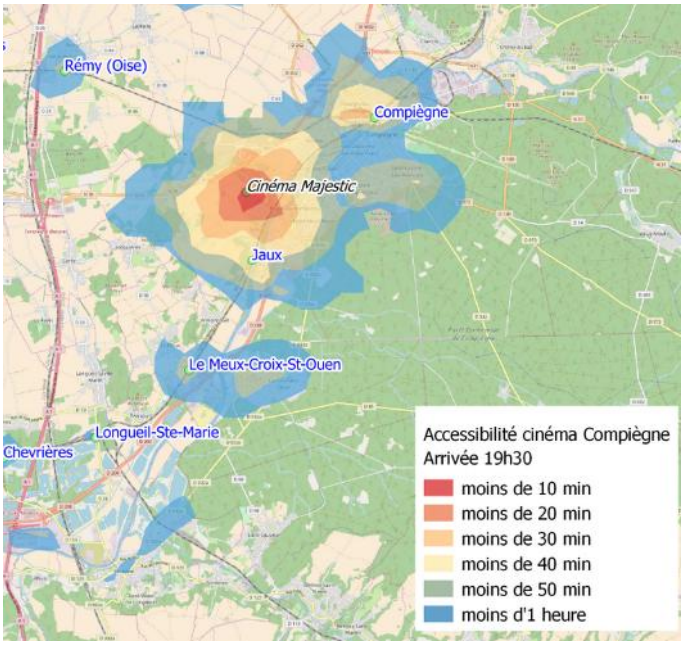
20. Isovalue polygons

SCRIPT Isovalue polygons

Raster : **raster layer**
Band : **1**
Min : **0**
Max : **60** (depends on the duration)
Interval : **10** (iso-value interval)
No value : **let default value**
Polygons : **check it**
Individual values: **check only for individual value analysis**
Contours iso-valeurs : **save layer as..**



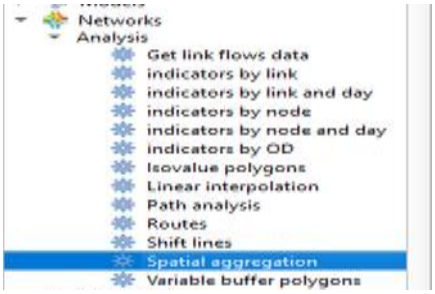
Isochrone example



Partie 3 : To go further...

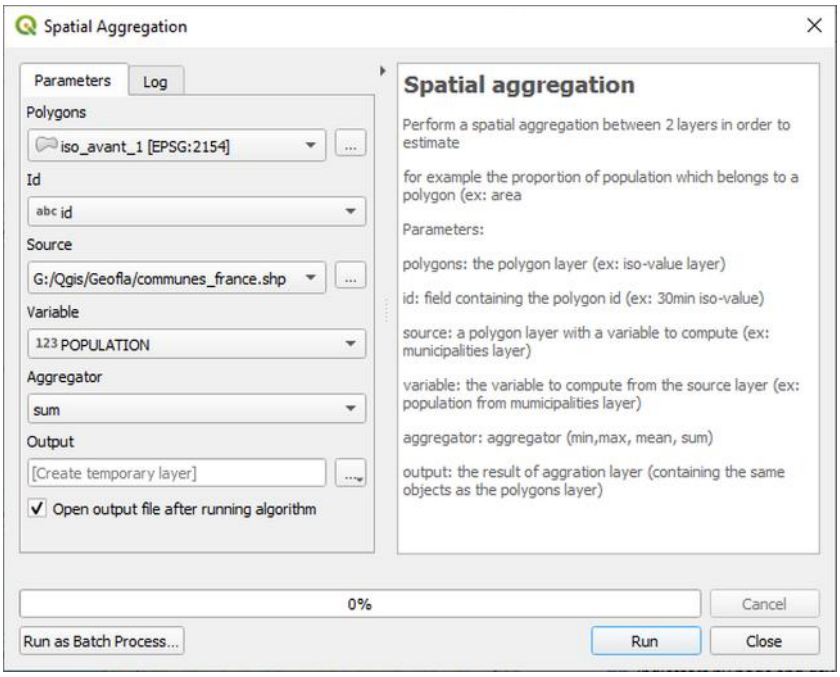
21. Compute population inside iso-value polygons

SCRIPT Spatial aggregation

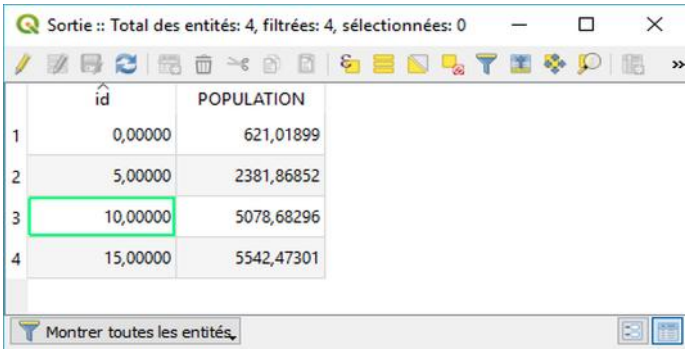


Open the iso-value polygons layer
Open an administrative layer with a population field

Polygons : **select the iso-value polygons layer**
Id : **polygon id** (corresponding to the threshold defined in the polygons layer
Source : **administrative layer**
Variable : **field name containing the population data**
Aggregator : **sum** (to sum). Choose between sum, mean, minimum, maximum
Output : **save result as (polygons layer)**



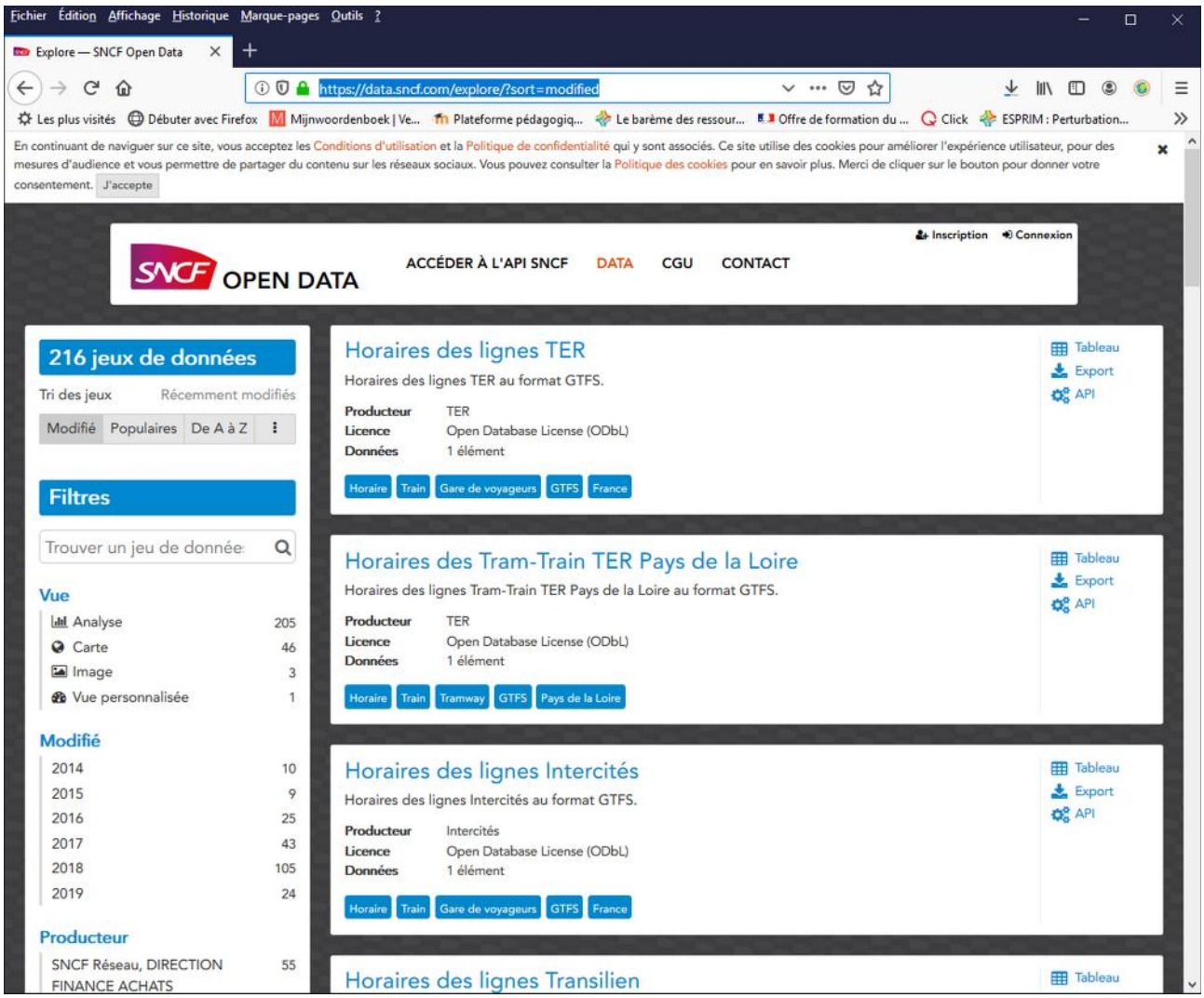
Results



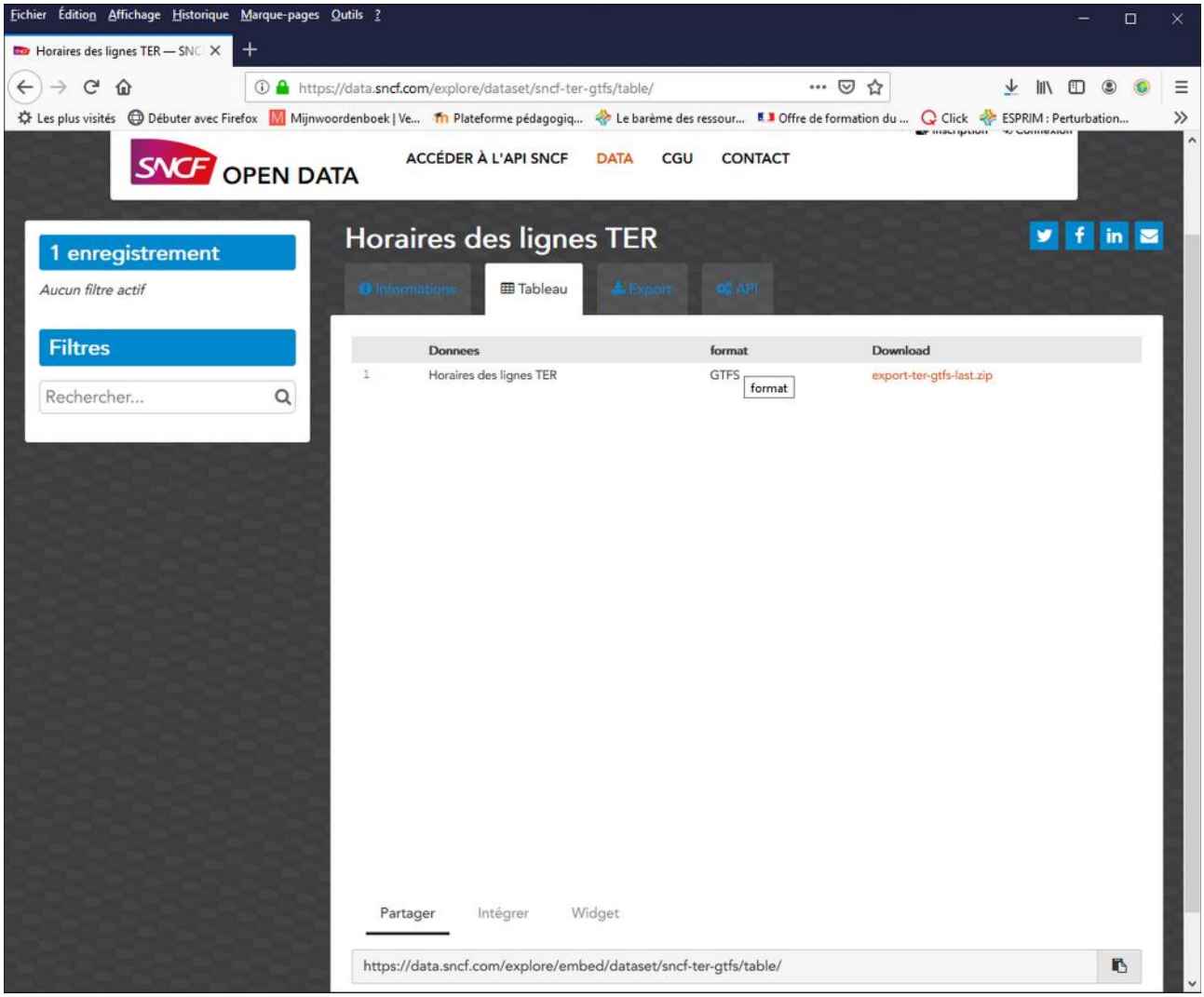
B- Production of a PT services offer (example : french regional railways offer)

1. Get the theoretical offer on the SNCF open data website

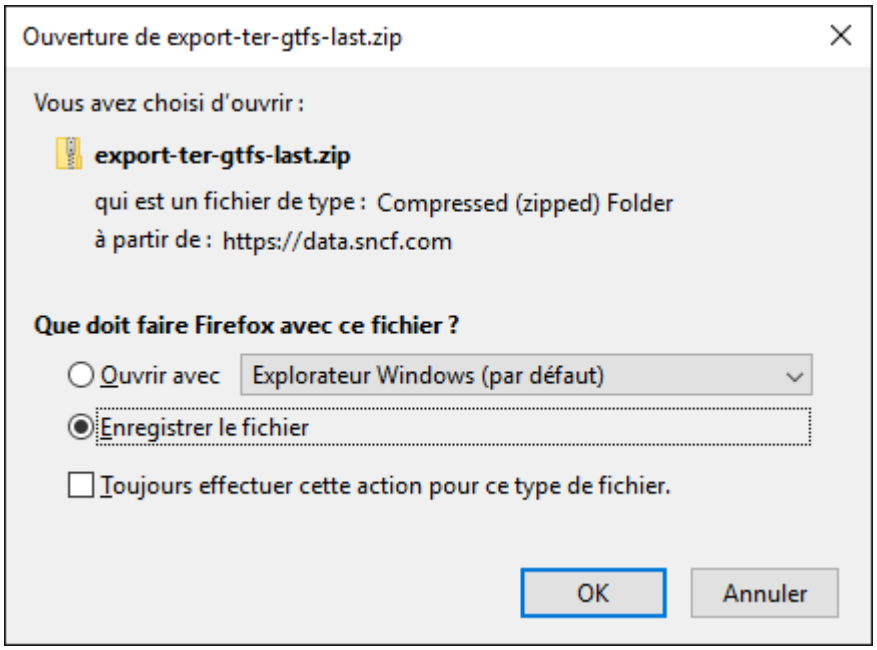
(https://data.sncf.com/explore/?sort=modified)



Choose the services that we want to represent (In our example “Horaire des lignes TER”).
Click on **« export-ter-gtfs-line.zip »** to download and save the file



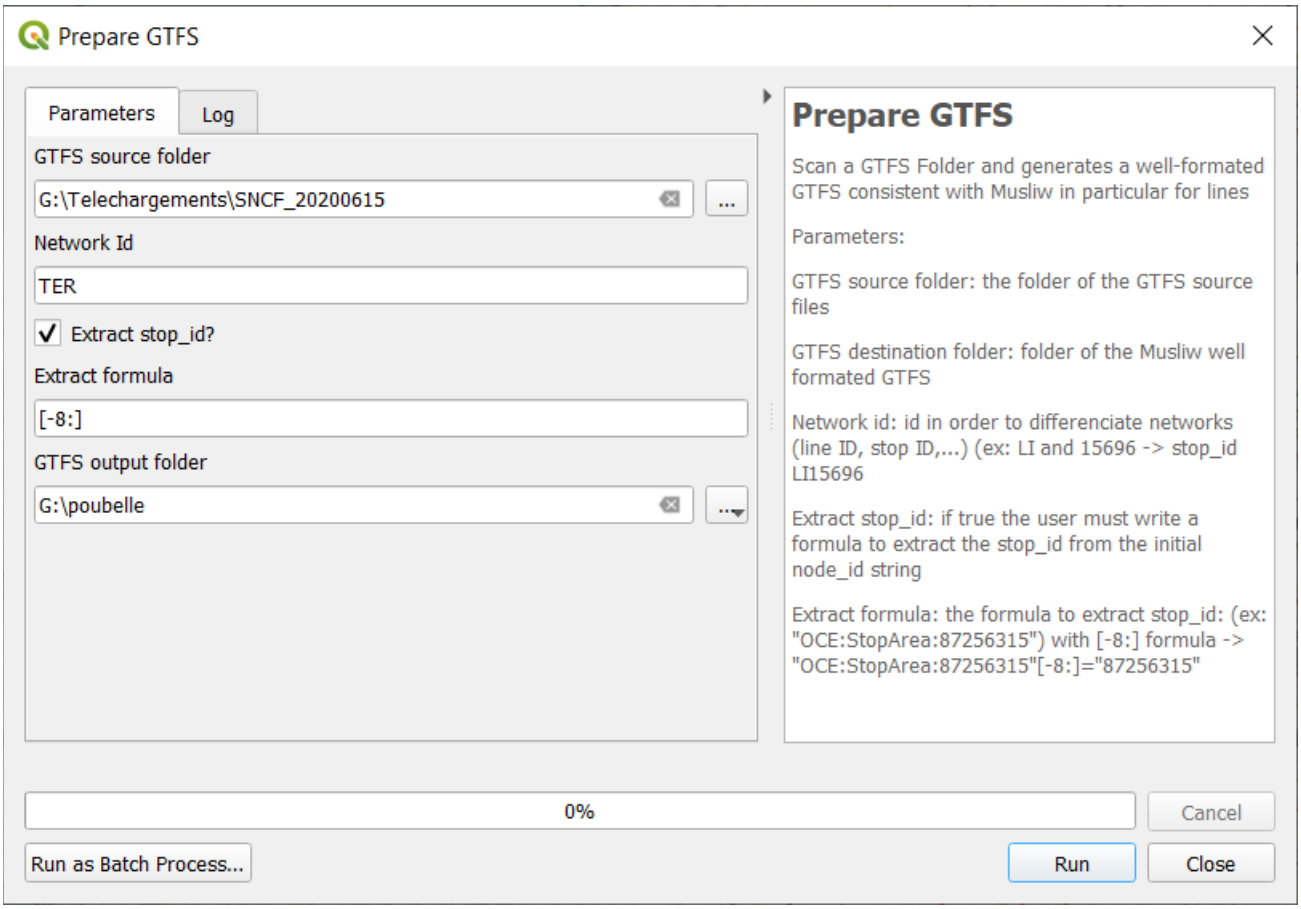
Accept website policy



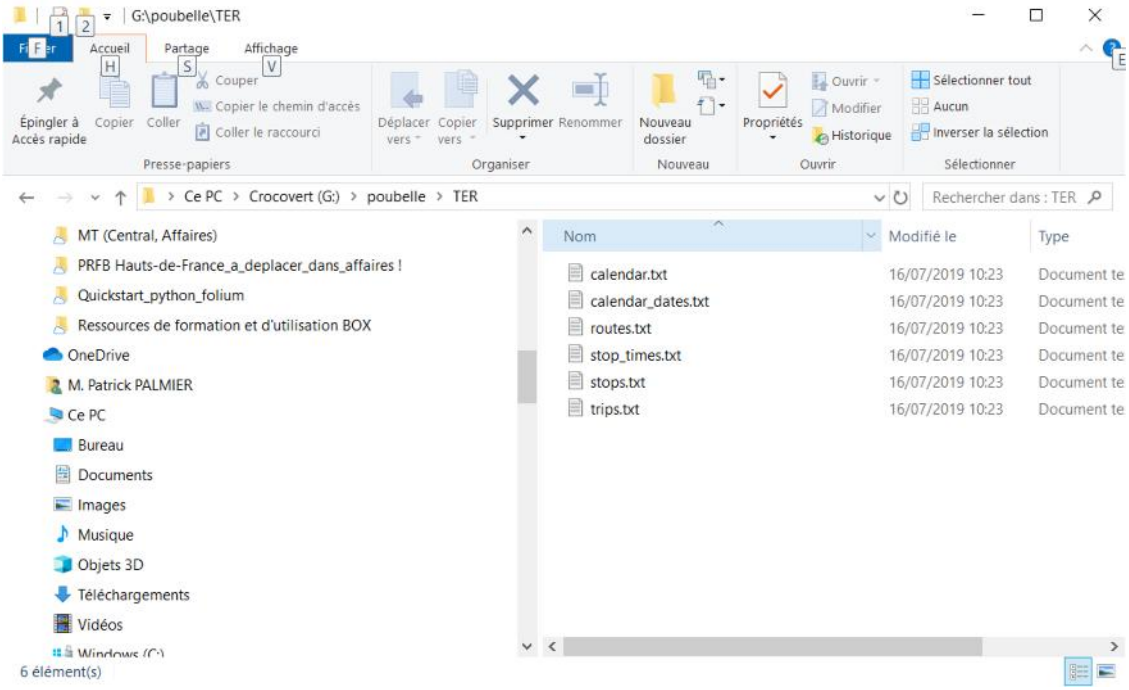
Save and unpack the archive in a directory

2. GTFS pre-processing

A pre-processing has first to be done on the raw GTFS with the **« prepare GTFS »** algorithm in order to be correctly formatted for “networks” plugins scripts and in particular with the mutli-modal calculator Musliw.

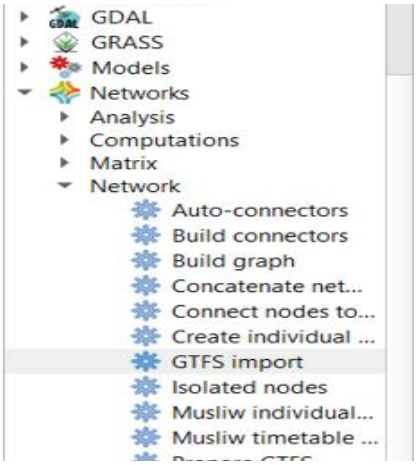


The script generates a new GTFS dataset compliant with networks algorithms

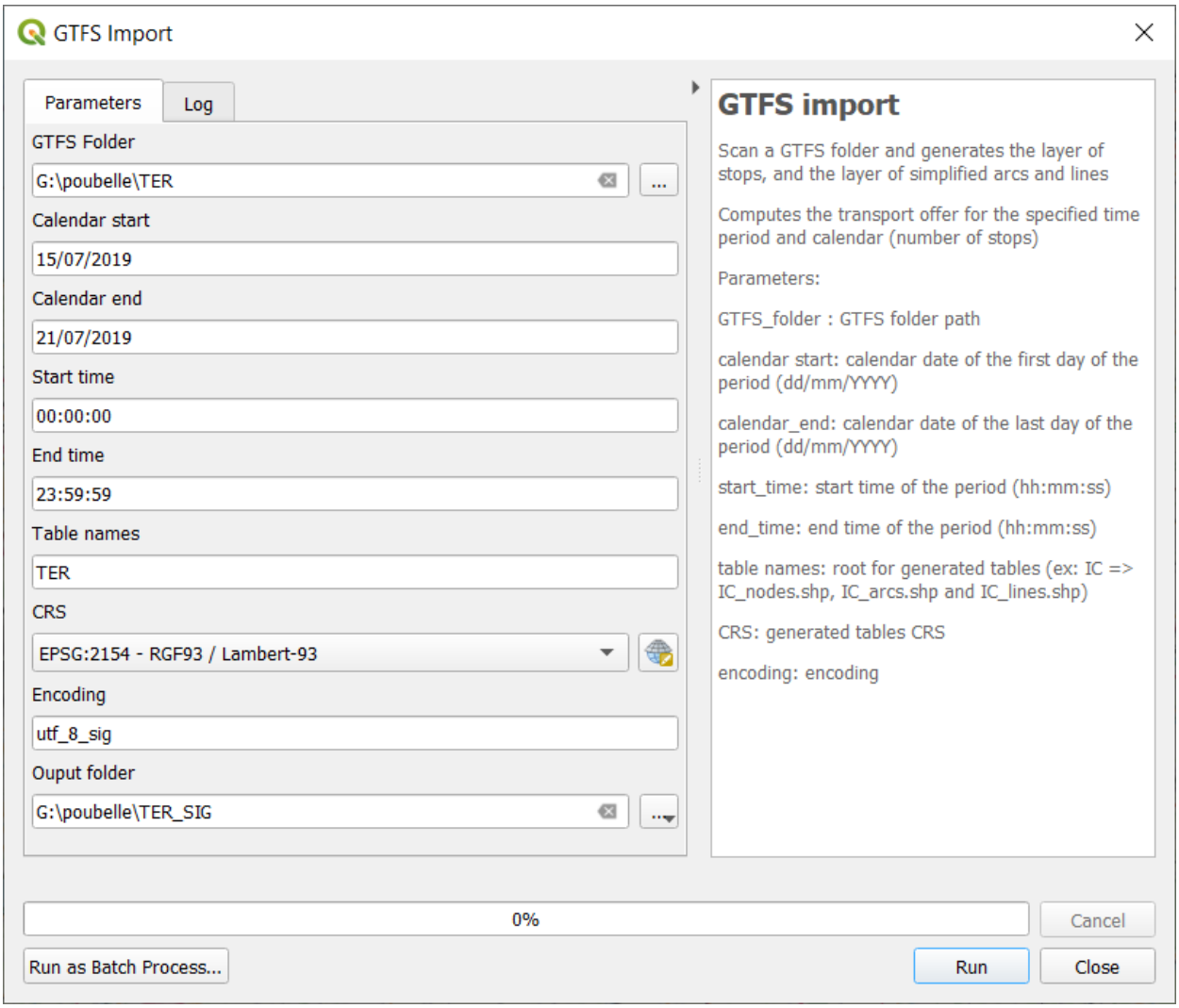


3. GTFS import in Qgis

Run the **« import GTFS »** algortihm to be able to view the corresponding offer in Qgis

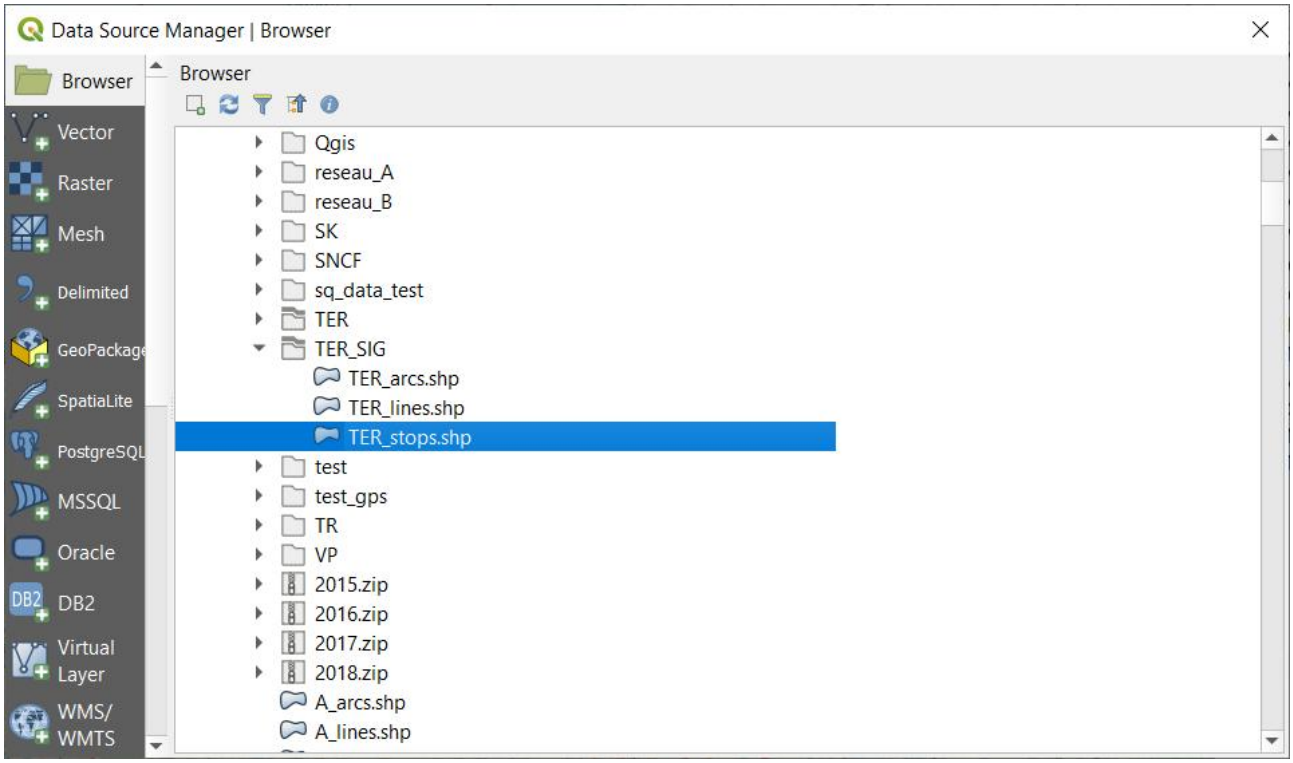


the start and end dates of the calendar must be dates where corresponding data (timetables) are available in the GTFS file.
For that point, it is recommended to edit **calendar.tx**t and/or **calendar_dates.txt** to choose a week or a period where timetables are available

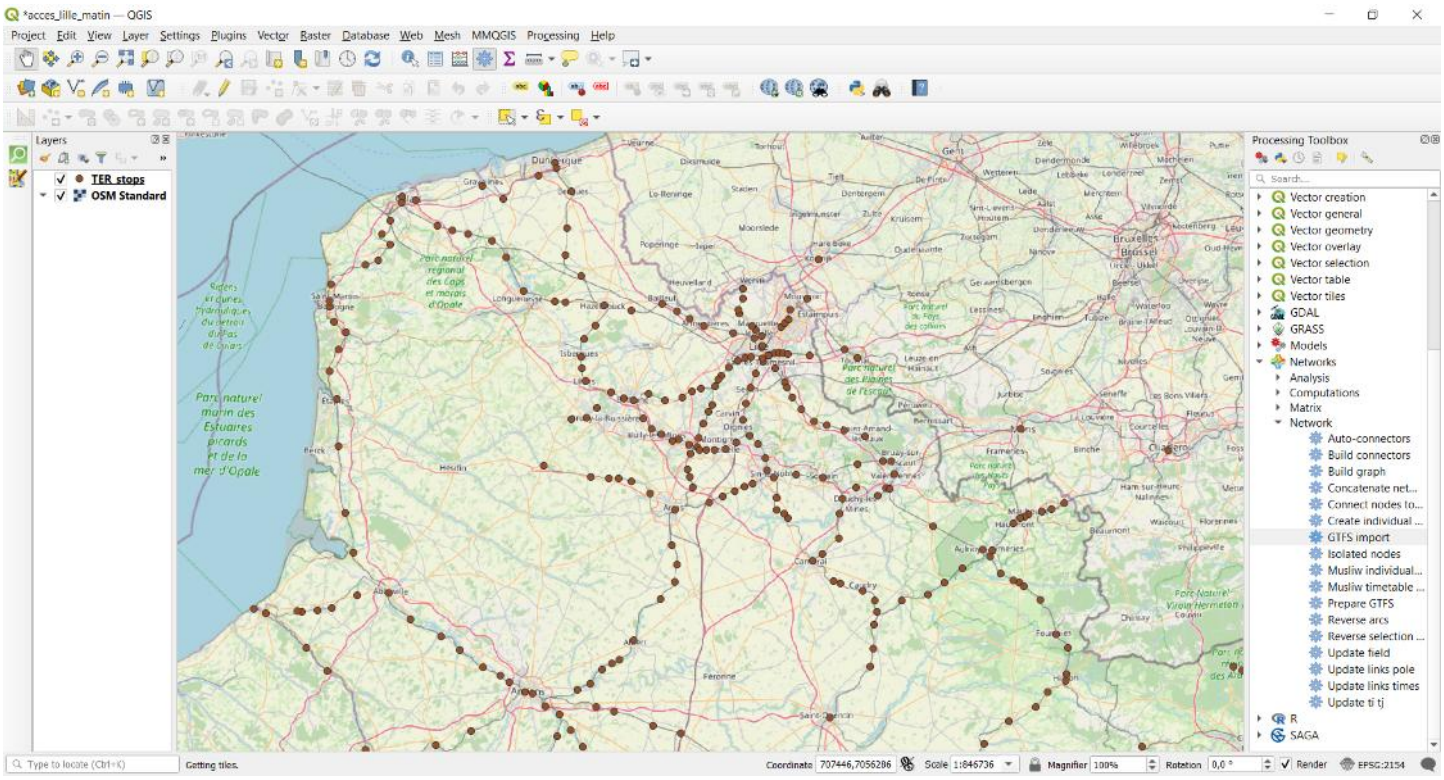


4. View stops

Load in Qgis the **« _stops.shp »** layer which contains regional railways services stops

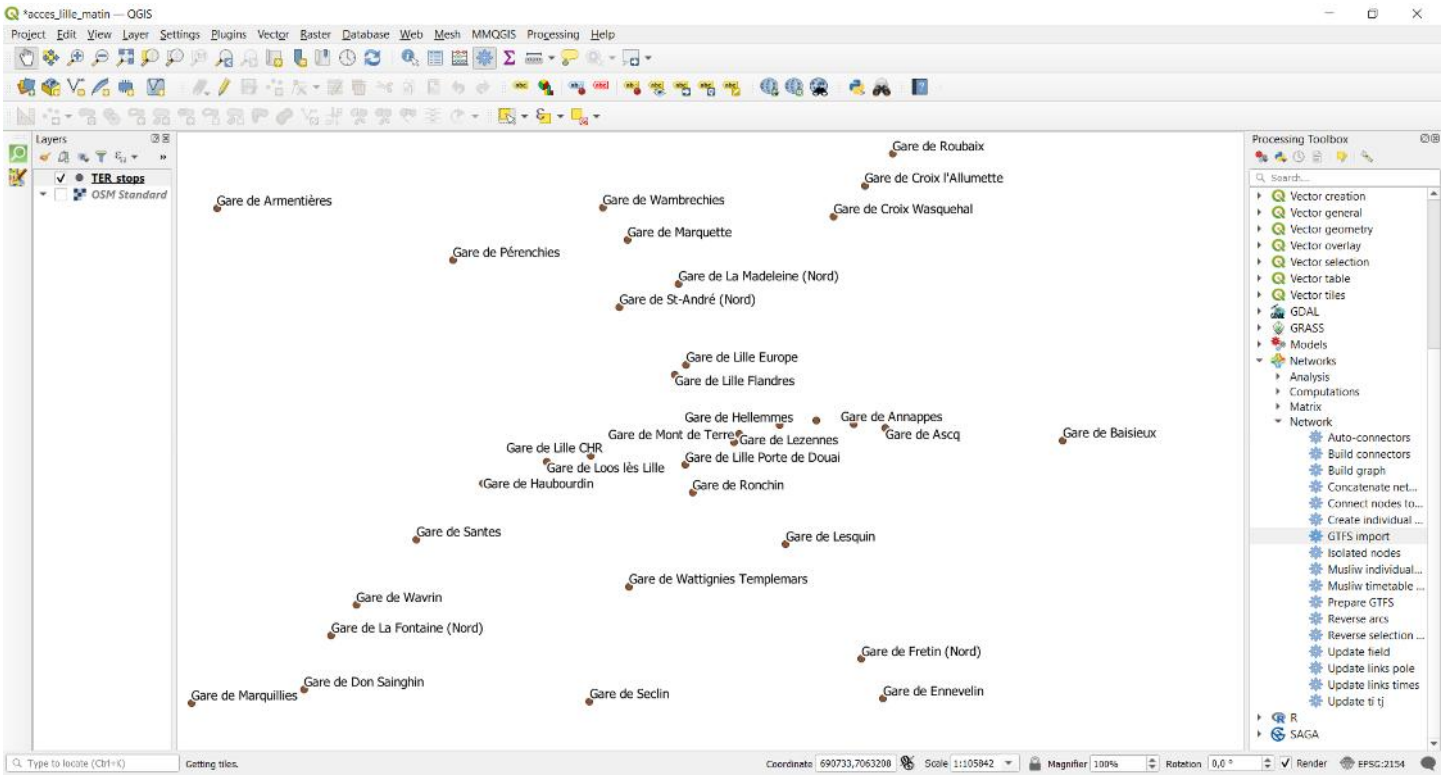


Results (zoom in Lille regional area)

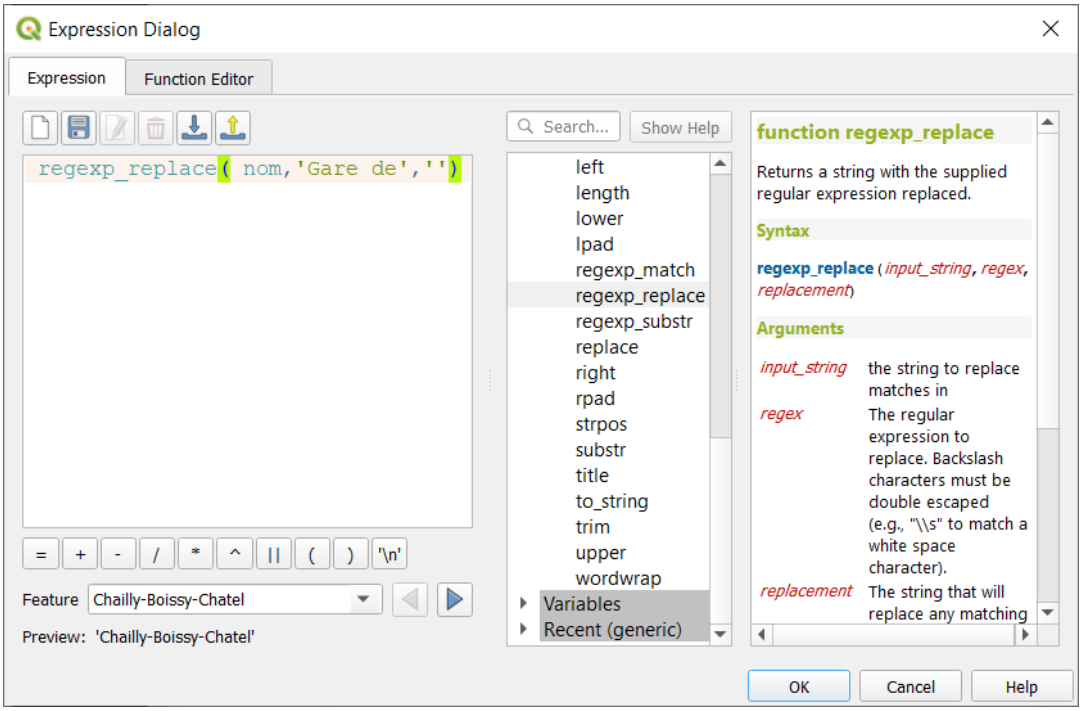


5. Show stations names

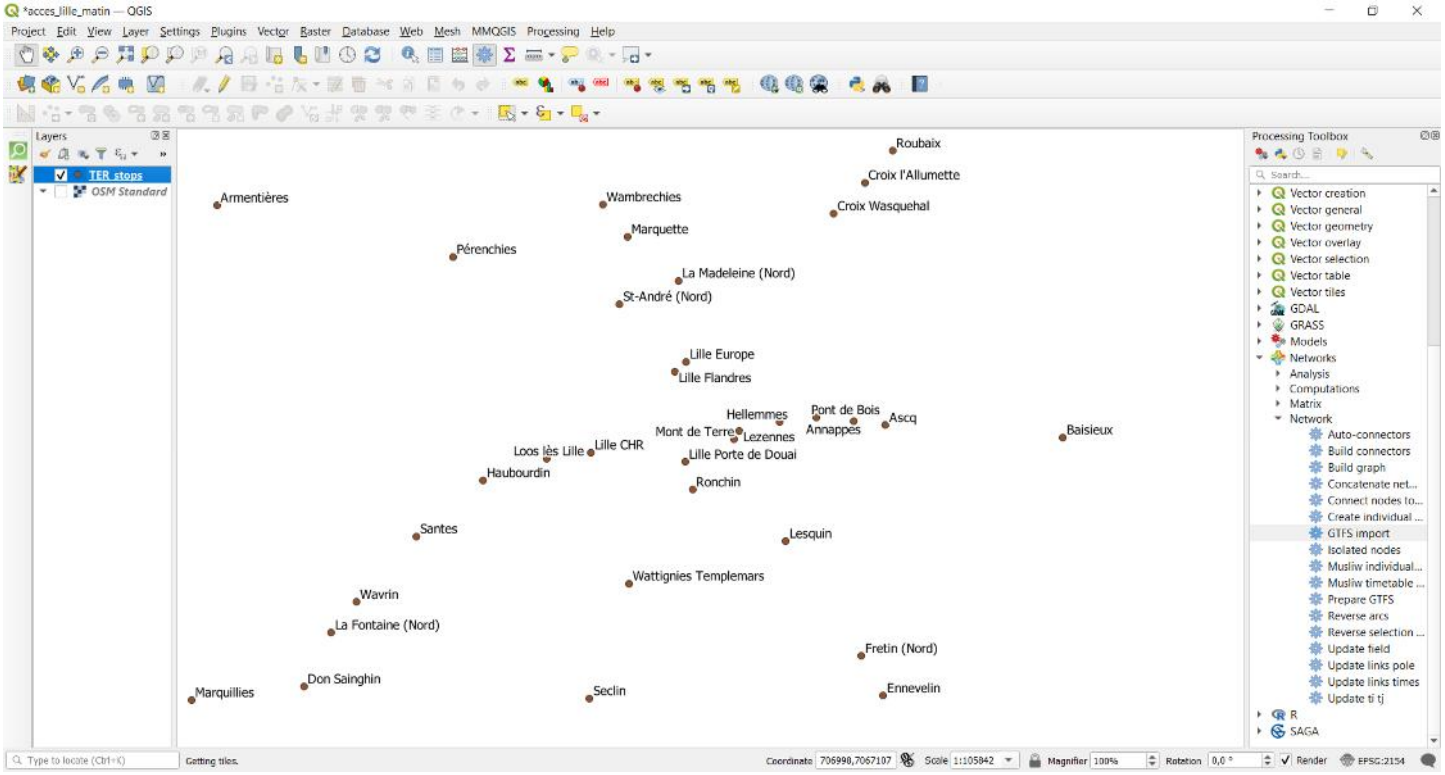
To show the stations names, you have to show labels with the field "**nom**"
On the other hand, we notice that the station name is defined with "**Gare de"** before the names of the station



To not show "Gare de", it is possible to use **a regular expression** which will replace **"Gare de"** by an empty string.



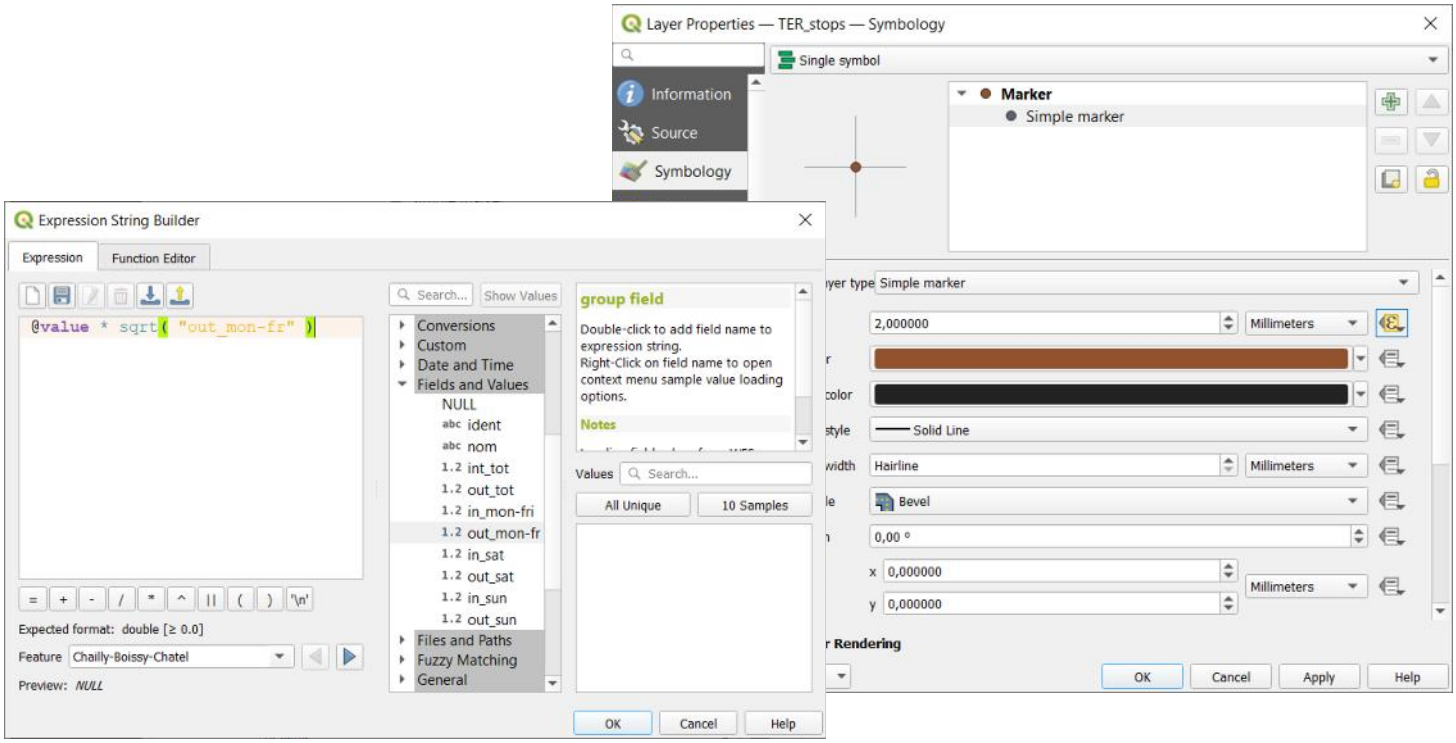
Result


*acces_lille_matin — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh MMQGIS Processing Help
Layers
TER_stops
OSM Standard
Roubaix
Croix l'Allumette
Armentières
Wambrechies
Croix Wasquehal
Marquette
Pérenchies
La Madeleine (Nord)
St-André (Nord)
Lille Europe
Lille Flandres
Hellemmes
Pont de Bois
Ascq
Annappes
Mont de Terre
Lezennes
Baisieux
Loos lès Lille
Lille CHR
Lille Porte de Douai
Haubourdin
Ronchin
Santes
Lesquin
Wattignies Templemars
Wavrin
La Fontaine (Nord)
Fretin (Nord)
Don Sainghin
Marquillies
Seclin
Ennevelin
Processing Toolbox
Search...
Vector creation
Vector general
Vector geometry
Vector overlay
Vector selection
Vector table
Vector tiles
GDAL
GRASS
Models
Networks
Analysis
Computations
Matrix
Network
Auto-connectors
Build connectors
Build graph
Concatenate net...
Connect nodes to...
Create individual ...
GTFS import
Isolated nodes
Musliw individual...
Musliw timetable ...
Prepare GTFS
Reverse arcs
Reverse selection ...
Update field
Update links pole
Update links times
Update ti tj
R
SAGA
Type to locate (Ctrl+K)
Getting tiles.
Coordinate 706998,7067107
Scale 1:105842
Magnifier 100%
Rotation 0,0 °
Render
EPSG:2154

6. Produce a proportional symbol size map

To perform a **map with proportional symbol size**, you have to **customize the symbol size** with an expression containing the variable to represent

“out_mon-fri” field represents the average daily number of departures during the time interval you’ve chosen in script “GTFS import”. The keyword **@value** represents the width attribute value. To keep it in the expression allow to adjust the size directly by modifying the value in the size width, without having to modify the expression.

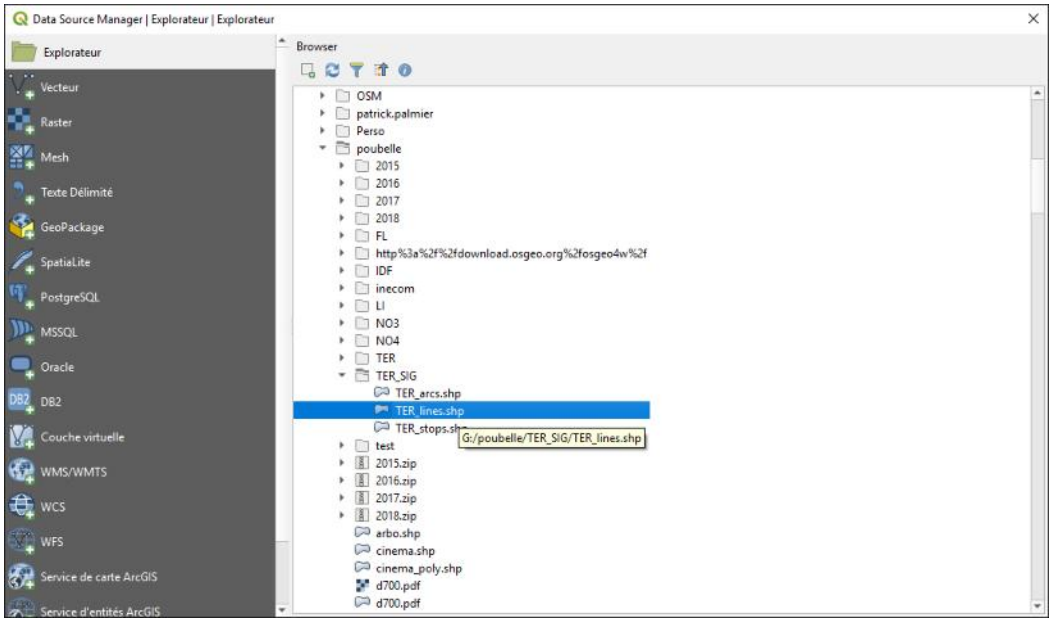


Result

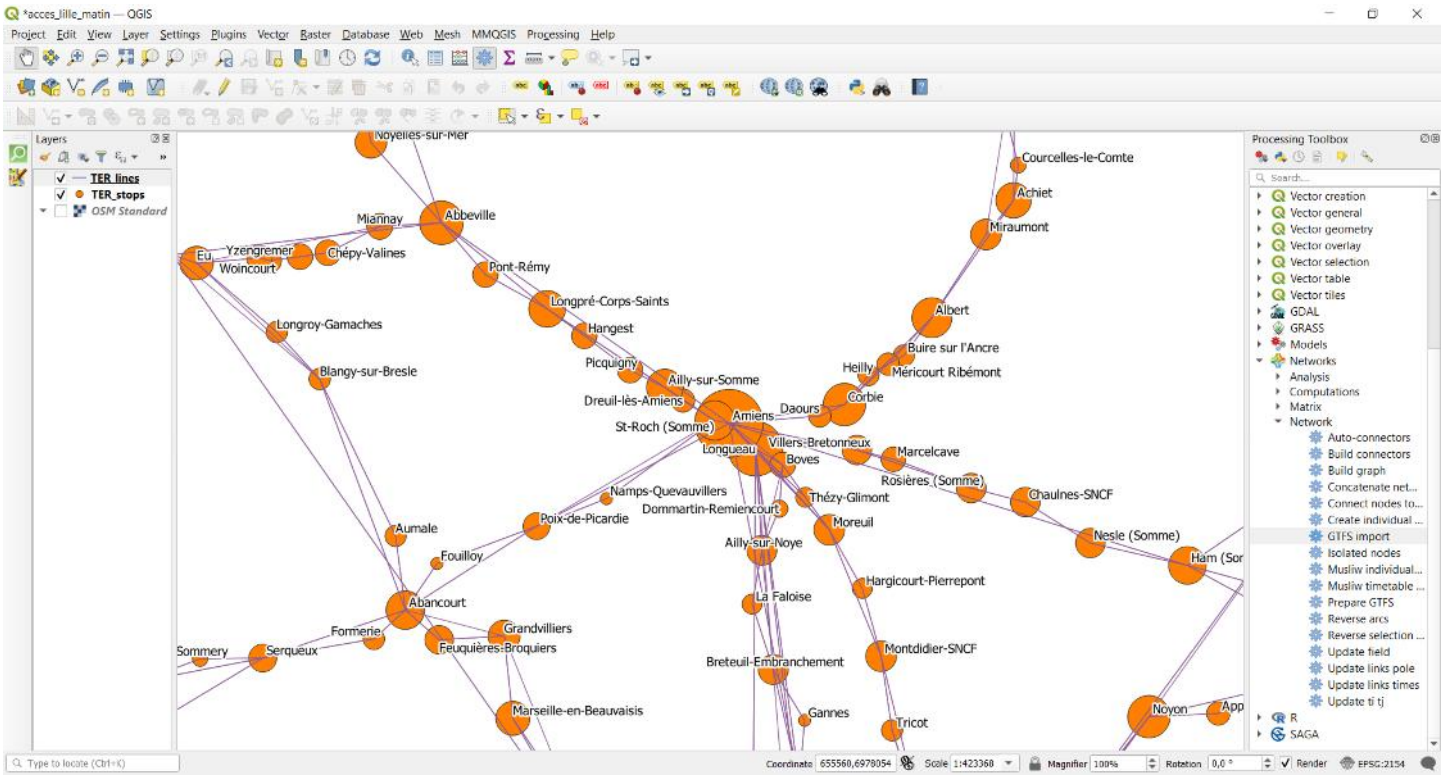
7. View the number of services by link

To view the **number of services by link**, you have to load the **“_lines”** layer**.**
On the other hand, the line shape is simplified. it consists of a straight line between two successive stops or stations. That could generate an unclean representation due to the overlay of direct and omnibus trains

Load the « lines » layer


Data Source Manager | Explorateur | Explorateur
Explorateur
Vecteur
Raster
Mesh
Texte Délimité
GeoPackage
SpatiaLite
PostgreSQL
MSSQL
Oracle
DB2
Couche virtuelle
WMS/WMTS
WCS
WFS
Service de carte ArcGIS
Service d'entités ArcGIS
Browser
OSM
patrick.palmier
Perso
poubelle
2015
2016
2017
2018
FL
http%3a%2f%2fdownload.osgeo.org%2fosgeo4w%2f
IDF
inecom
LI
NO3
NO4
TER
TER_SIG
TER_arcs.shp
TER_lines.shp
TER_stops.shp
G:/poubelle/TER_SIG/TER_lines.shp
test
2015.zip
2016.zip
2017.zip
2018.zip
arbo.shp
cinema.shp
cinema_poly.shp
d700.pdf
d700.pdf

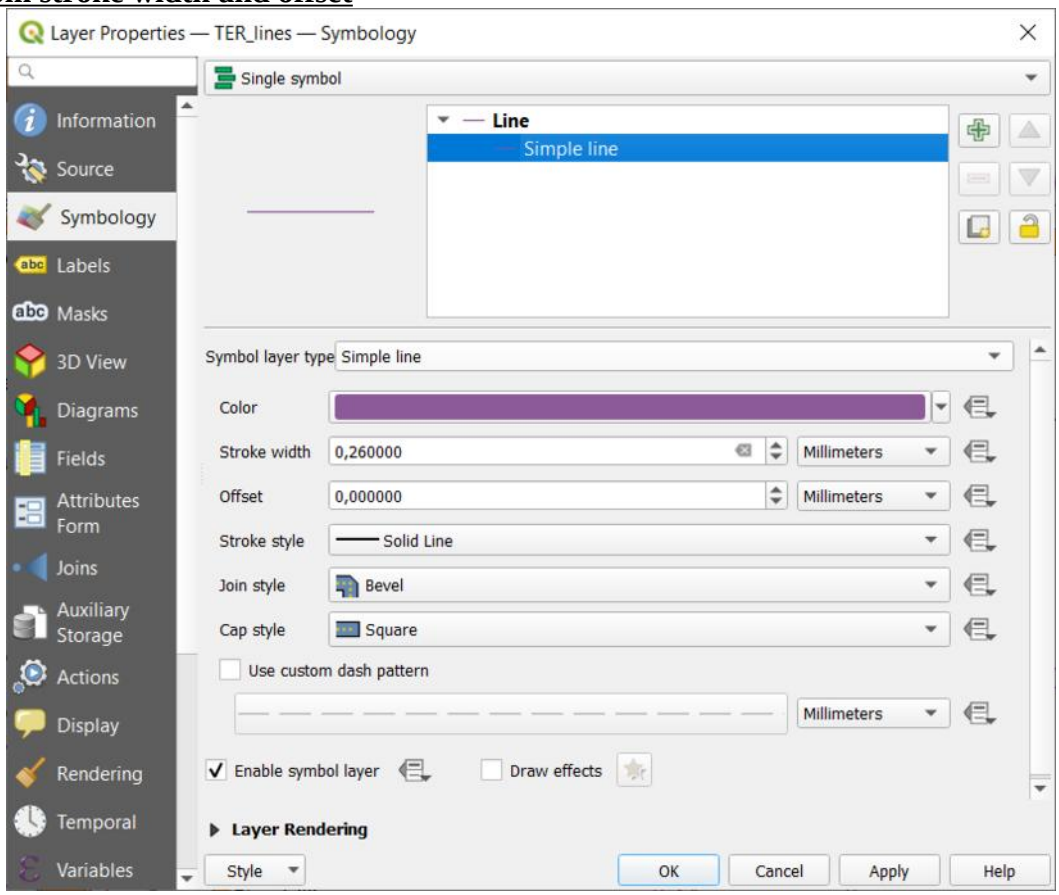
Result


*acces_lille_matin — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh MMQGIS Processing Help
Layers
TER_lines
TER_stops
OSM Standard
Processing Toolbox
Vector creation
Vector general
Vector geometry
Vector overlay
Vector selection
Vector table
Vector tiles
GDAL
GRASS
Models
Networks
Analysis
Computations
Matrix
Network
Auto-connectors
Build connectors
Build graph
Concatenate net...
Connect nodes to...
Create individual ...
GTFS import
Isolated nodes
Musliw individual...
Musliw timetable ...
Prepare GTFS
Reverse arcs
Reverse selection ...
Update field
Update links pole
Update links times
Update ti tj
R
SAGA
Noyelles-sur-Mer
Courcelles-le-Comte
Achiet
Miraumont
Albert
Buire sur l'Ancre
Méricourt Ribémont
Heilly
Corbie
Daours
Amiens
Dreuil-lès-Amiens
St-Roch (Somme)
Longueau
Boves
Villers-Bretonneux
Marcelcave
Rosières (Somme)
Chaulnes-SNCF
Thézy-Glimont
Moreuil
Ailly-sur-Noye
Nesle (Somme)
Ham (Sor
Hargicourt-Pierrepont
La Faloise
Montdidier-SNCF
Breteuil-Embranchement
Gannes
Tricot
Noyon
Marseille-en-Beauvaisis
Abancourt
Fouilloy
Aumale
Poix-de-Picardie
Namps-Quevauvillers
Dommartin-Remiencourt
Ailly-sur-Somme
Picquigny
Hangest
Longpré-Corps-Saints
Pont-Rémy
Abbeville
Miannay
Yzengremer
Woincourt
Eu
Chépy-Valines
Longroy-Gamaches
Blangy-sur-Bresle
Formerie
Grandvilliers
Feuquières-Broquiers
Sommery
Serqueux
Type to locate (Ctrl+K)
Coordinate 655560,6978054
Scale 1:423360
Magnifier 100%
Rotation 0,0 °
Render
EPSG:2154

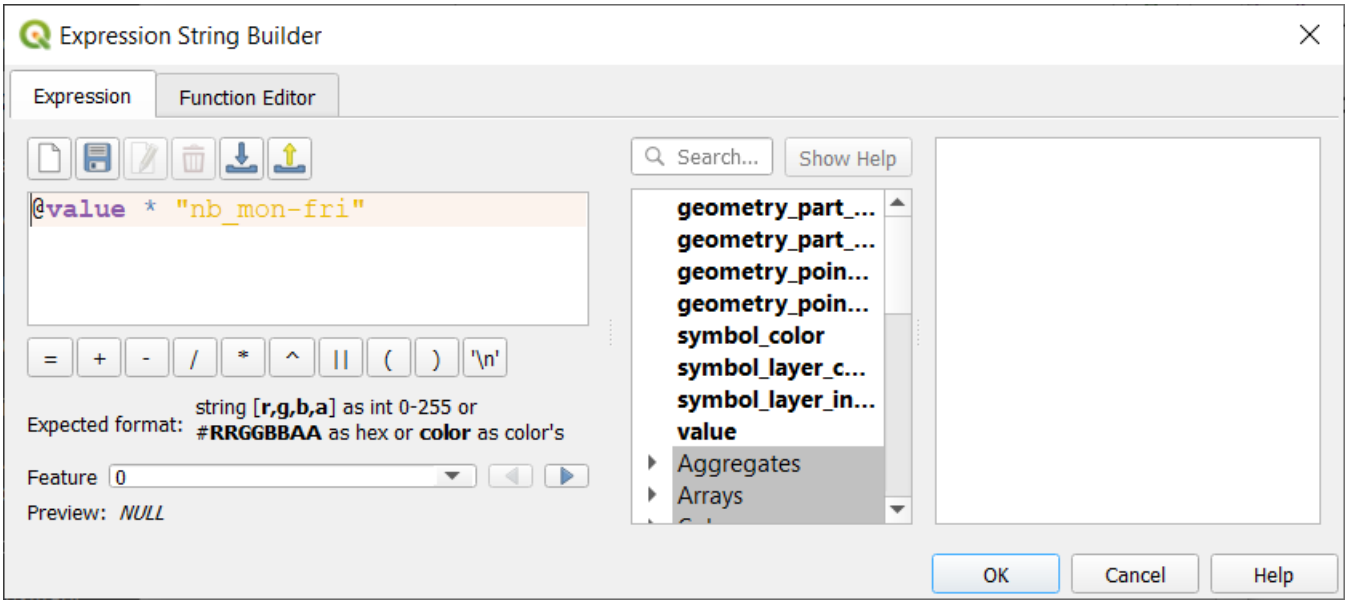
8. Widths proportional to the number of circulations

Warning: If a network contains a common core where several lines pass in the same road (two successive stops), the settings below is no more adapted and need more complex expression not presented in this document that allow to view the different lines juxtaposed instead of overlapped

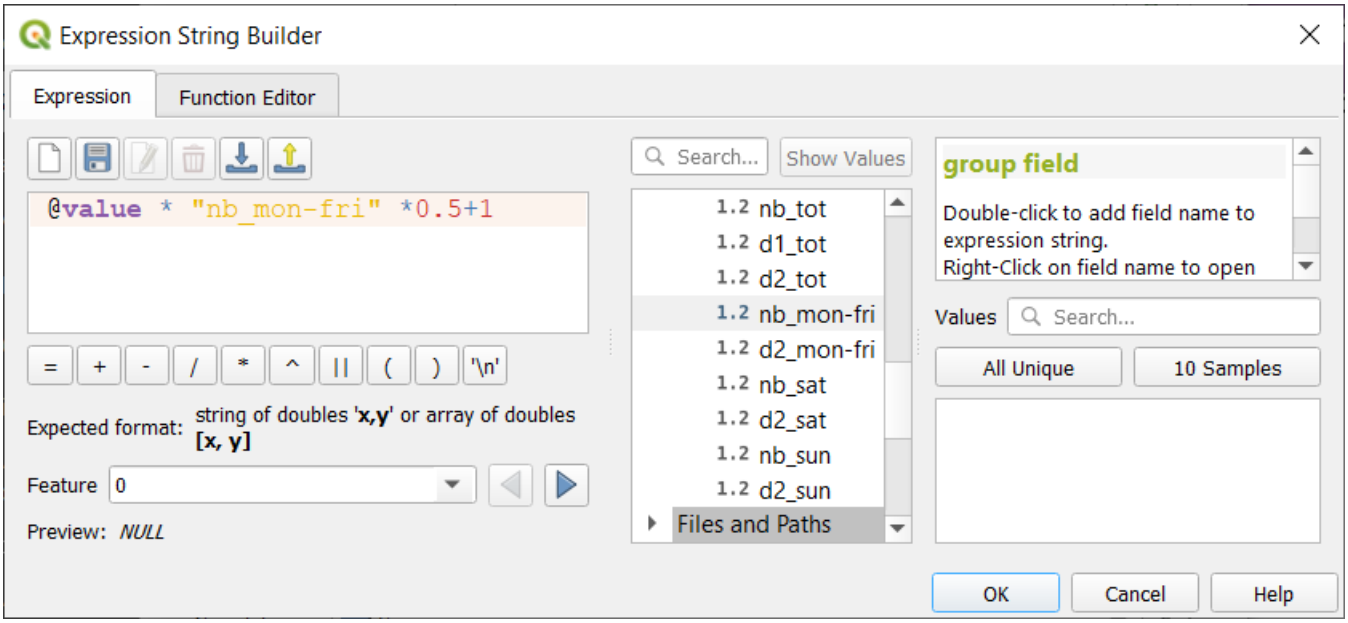
Define a custom stroke width and offset



Stroke width

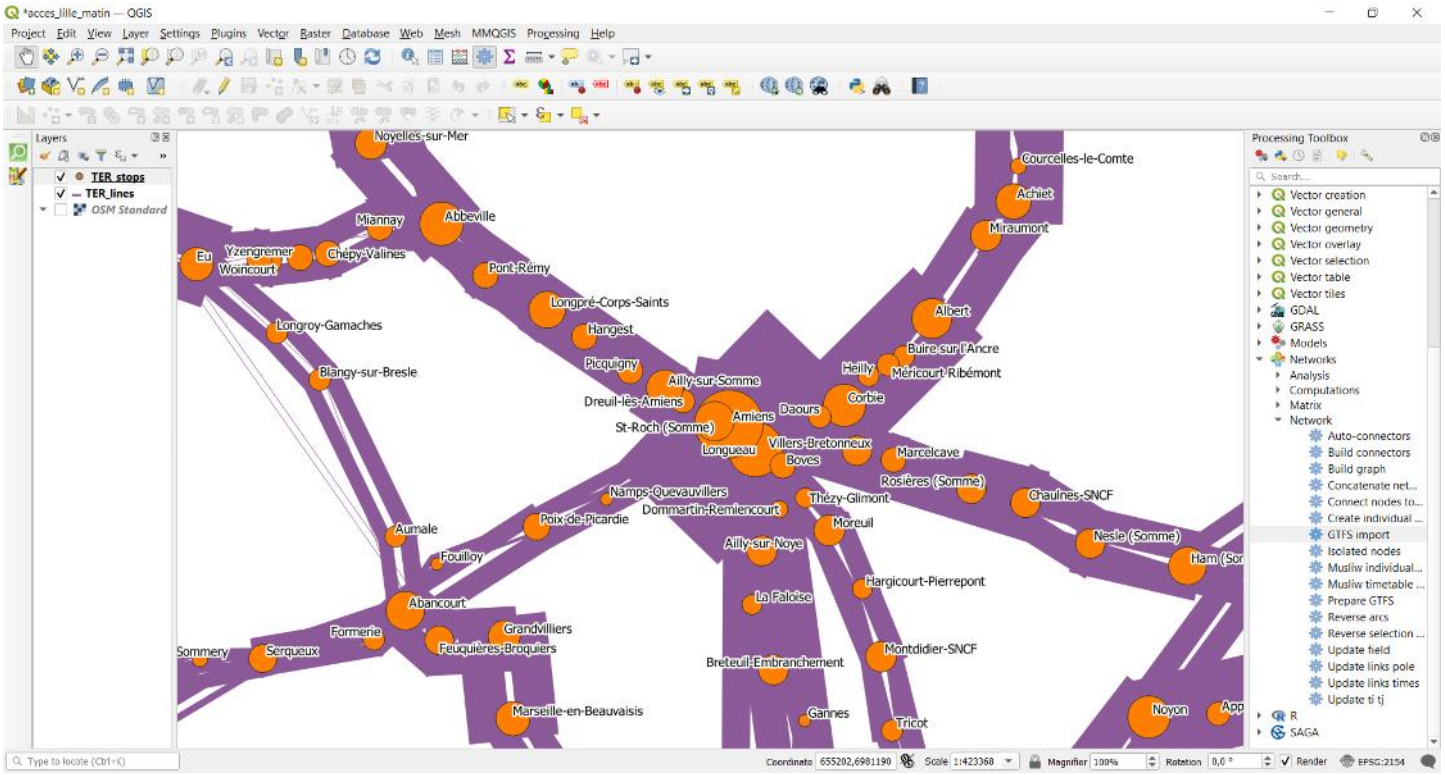


Offset



NB : Numeric values in stroke width form and offset from have to be identical with the expressions defined above.

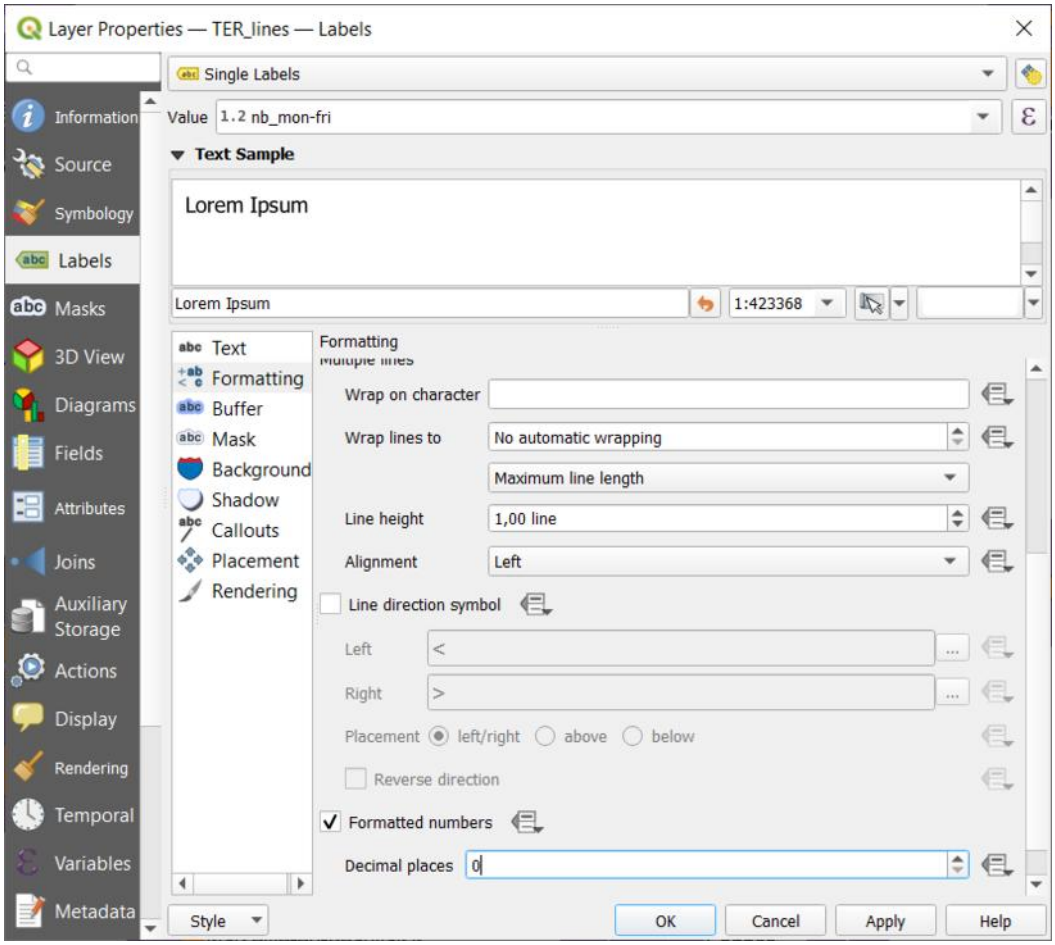
Result:



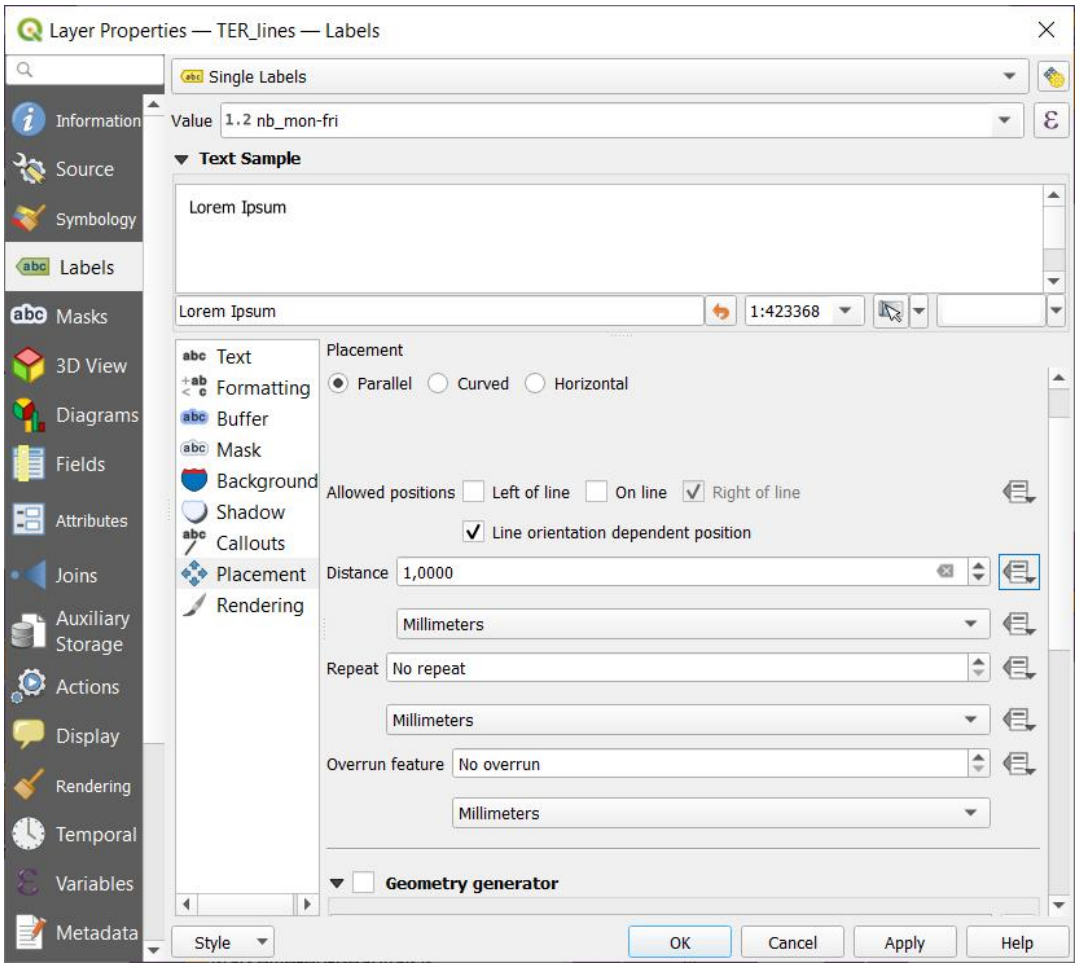
9. Label the number of services

Choose **nb_mon_fri** (average number of train courses the selected week from Monday to Friday included). As it is an average, specify formatted numbers with **“0”** in place of decimals

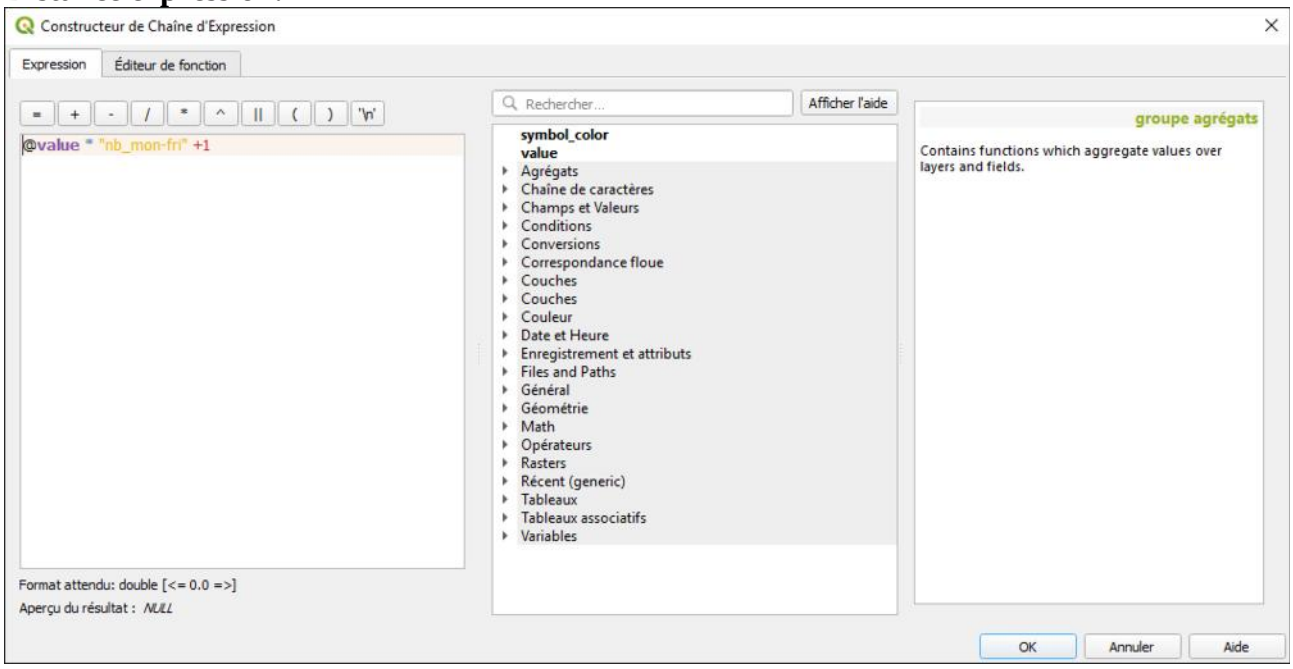
PS: In general, rather than simple labels, choose "rule-based labels". This will allow you to filter the labels more easily, because if you have set up your simple labels in a complex way, changing the type of labels will make you lose this settings which will have to be renewed with the rule-based labels.



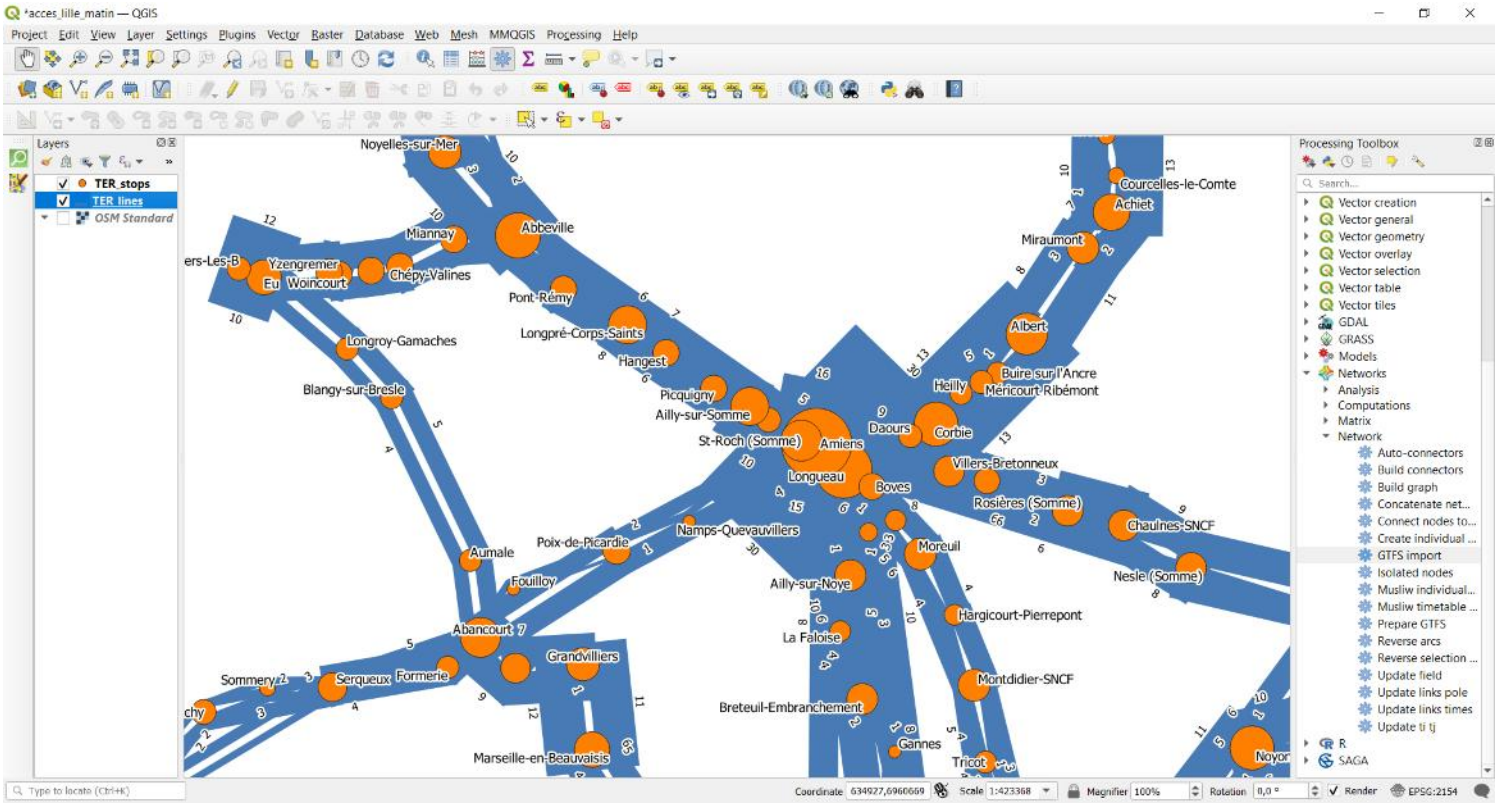
Set the same distance as the strike width and offset value on the lines layer. Check **“line orientation dependent position”** and **“right of line”** a traffic map rendering.



Distance expression:



As railways services aren't all identical, the succession of stops isn't identical too. Then, the map could be quite unreadable. A method exists to face this inconvenience and produce a clean and pretty map. This method is longer and more complex as it needs the generation of a network based on the real shape of lines. Then you have to compute shortest paths for each pair successive stations. This method is not described at the moment in this document.

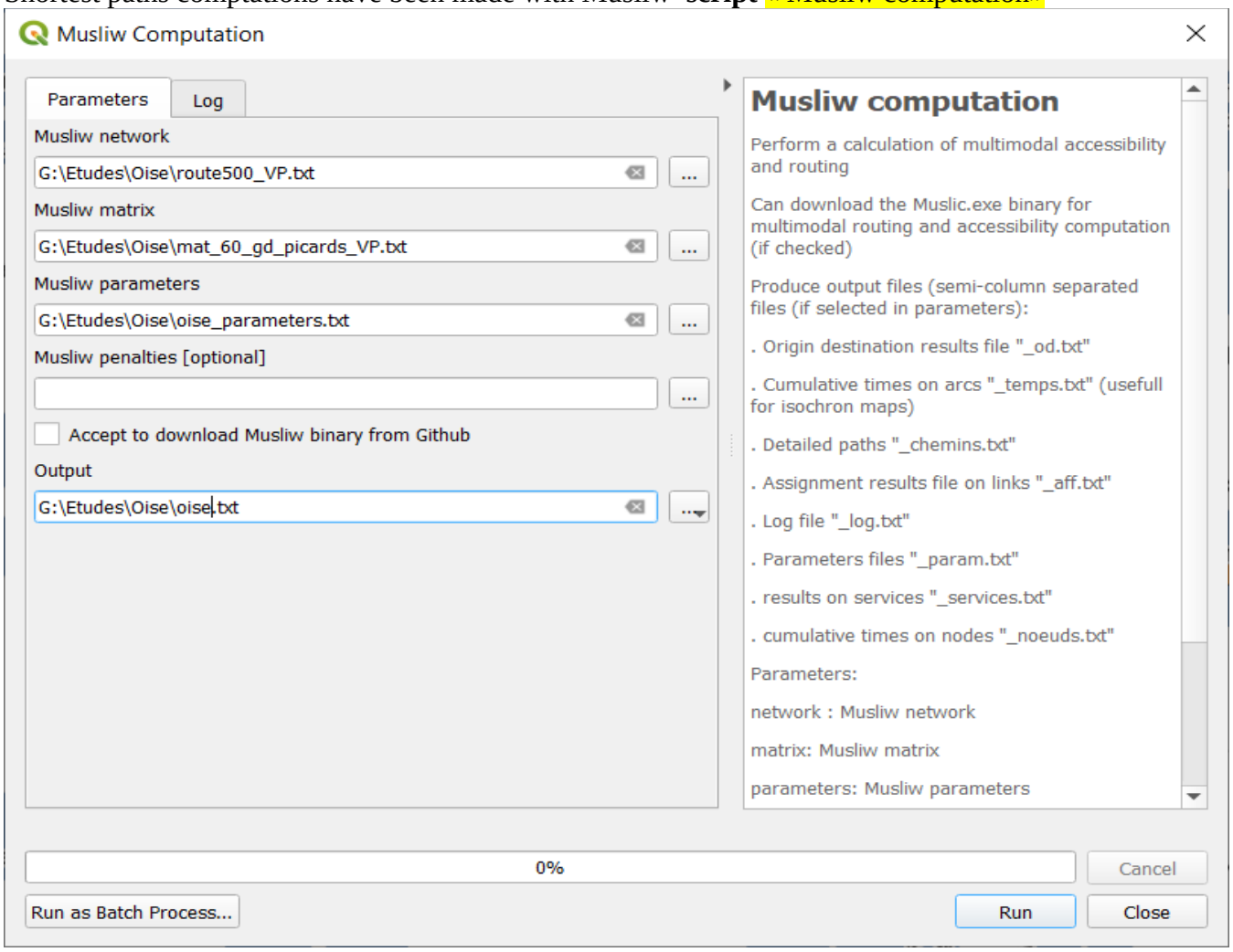


C- Production of a traffic map

The aim of this tutorial is to produce a traffic map

In a first step a Musliw matrix has been generated from the trip file of the “Grands Mobiles Picards” household travel survey. The matrix was built with car as driver trips from people living in the Oise Territory.
The objective is to view desire lines of drivers who live in Oise territory.

Shortest paths comptations have been made with Musliw **script** « Musliw computation»



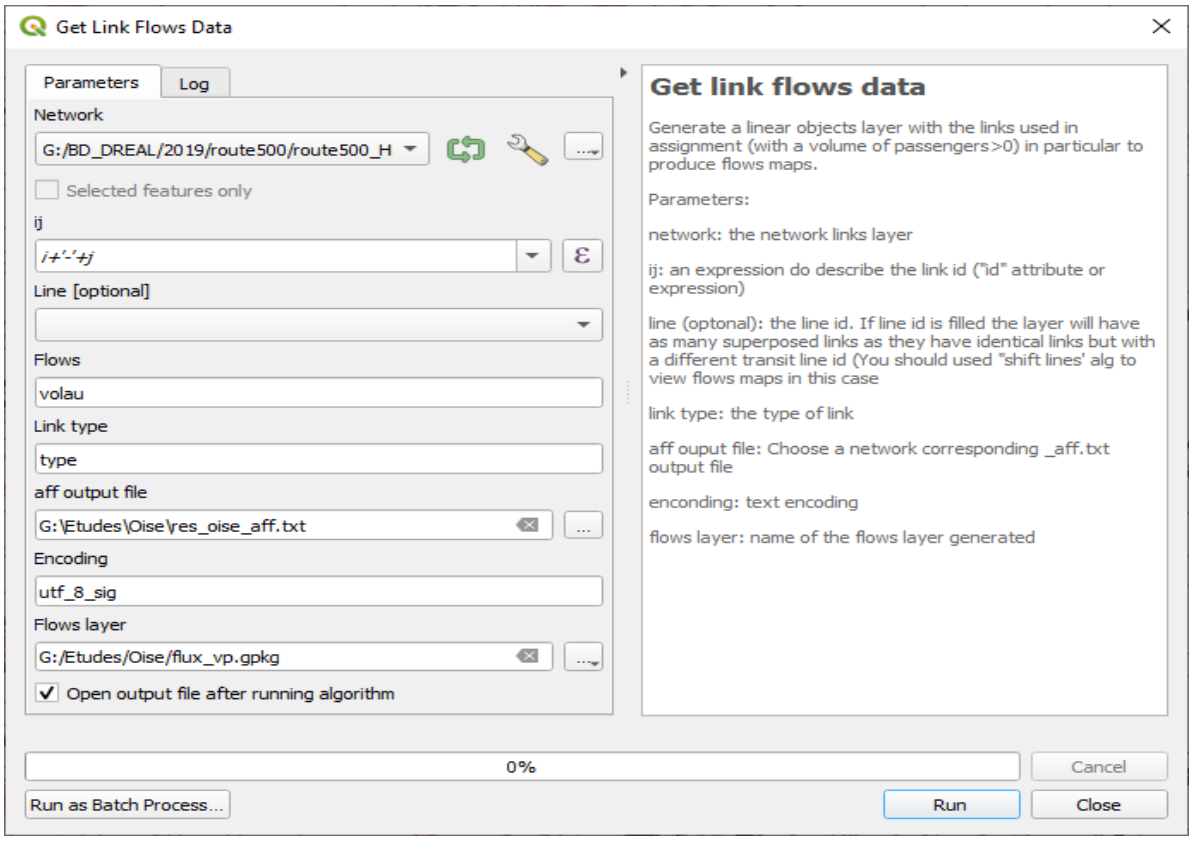
The “_aff” file contains the list of arcs (here route500) with auto-volumes on shortest paths .

1. Get traffic volumes

The road network used here is the Route500 network

As route500 is the used network, it has to be the reference in order to get Musliw results calculations data

The traffic volumes fields is often called **volau** (auto volumes), and you have to get data from t**he ".aff" file** which contains informations about the number of cars on each "route500" arc.



Le **script** génère une **table flux_vp.shp** qui contient les arcs route500 ayant été empruntés par au moins une OD de la matrice.

flux_vp — Features Total: 20958, Filtered: 20958, Selected: 0

	i	j	ij	volume	type
1	v182533320228...	v182534285228...	v182533320228...	271,0000000000...	v
2	v182319291228...	v182326723228...	v182319291228...	56,00000000000...	v
3	v182074827229...	v182084313229...	v182074827229...	5644,000000000...	v
4	v182475884228...	v182480714228...	v182475884228...	4444,000000000...	v
5	v181562335229...	v181573797229...	v181562335229...	66,00000000000...	v
6	v182360066229...	v182396484229...	v182360066229...	670,0000000000...	v
7	v181663081228...	v181653220228...	v181663081228...	34,00000000000...	v
8	v181814382229...	v181819873229...	v181814382229...	1572,000000000...	v
9	v181431278229...	v181414171229...	v181431278229...	83,00000000000...	v
10	v182685000229...	v182693442229...	v182685000229...	113,0000000000...	v
11	v180499802226...	v180506992226...	v180499802226...	23,00000000000...	v
12	v181873015230...	v181866836230...	v181873015230...	128,0000000000...	v
13	v182428572229...	v182430210229...	v182428572229...	587,0000000000...	v
14	v182481685229...	v182462523229...	v182481685229...	102,0000000000...	v
15	v184065303229...	v184057513229...	v184065303229...	36,00000000000...	v
16	v182620965228...	v182631450228...	v182620965228...	168,0000000000...	v

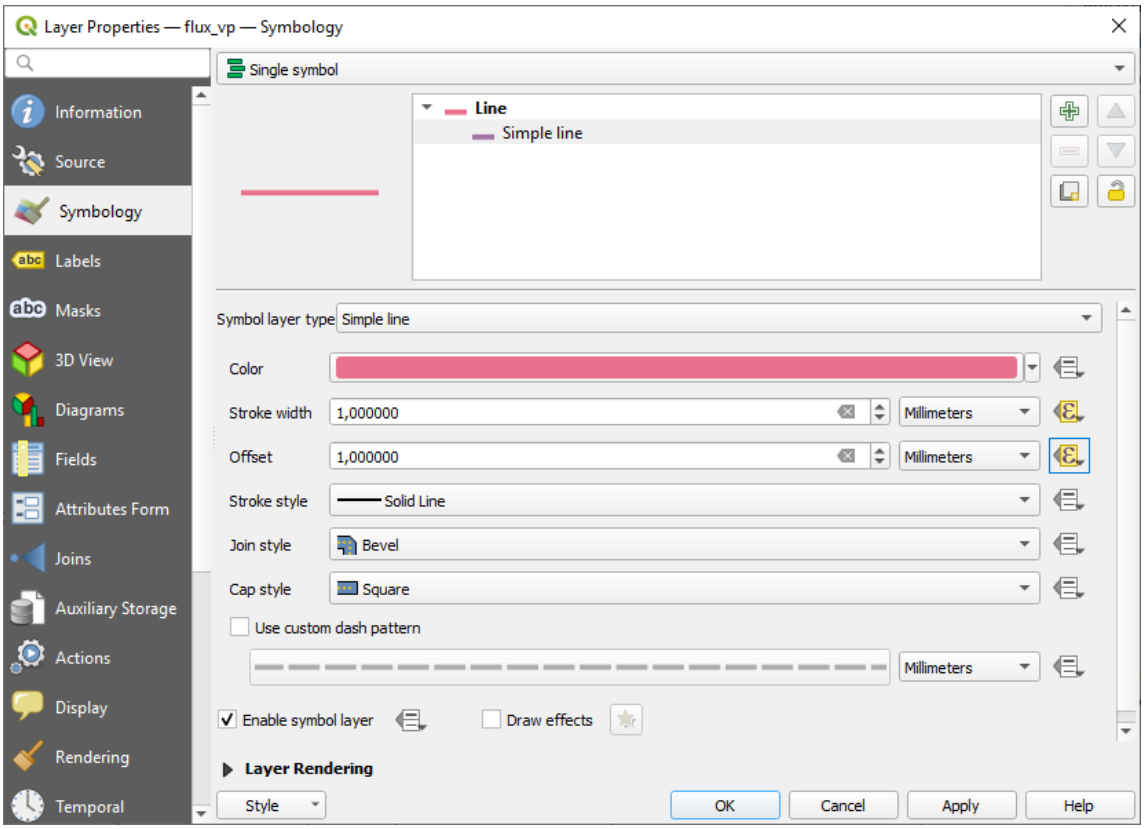
type
String(35) NULL

Show All Features

flux-vp data table

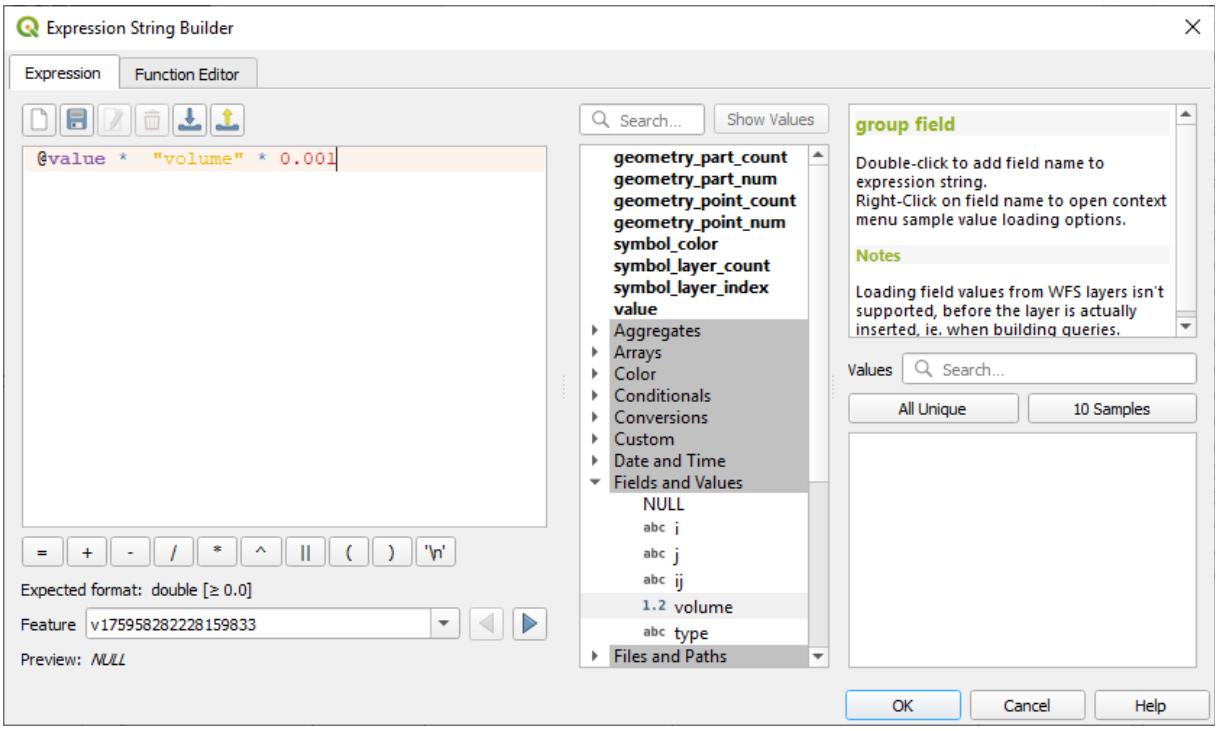
2. Define proportional widths

In layer properties choose symbology to **define the line style**. Numeric values of **« stroke width»** and **«oOffset »** must be identical

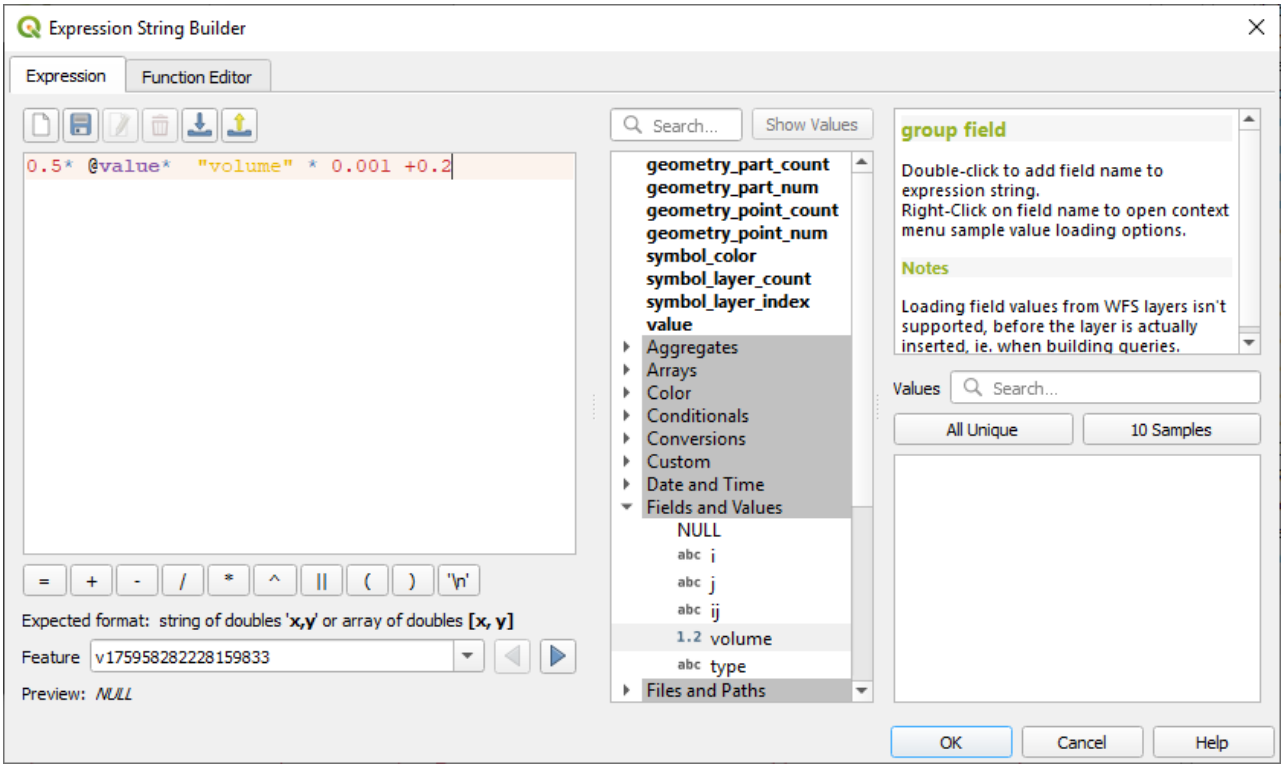


3. Set stroke width with an expression

Stroke width

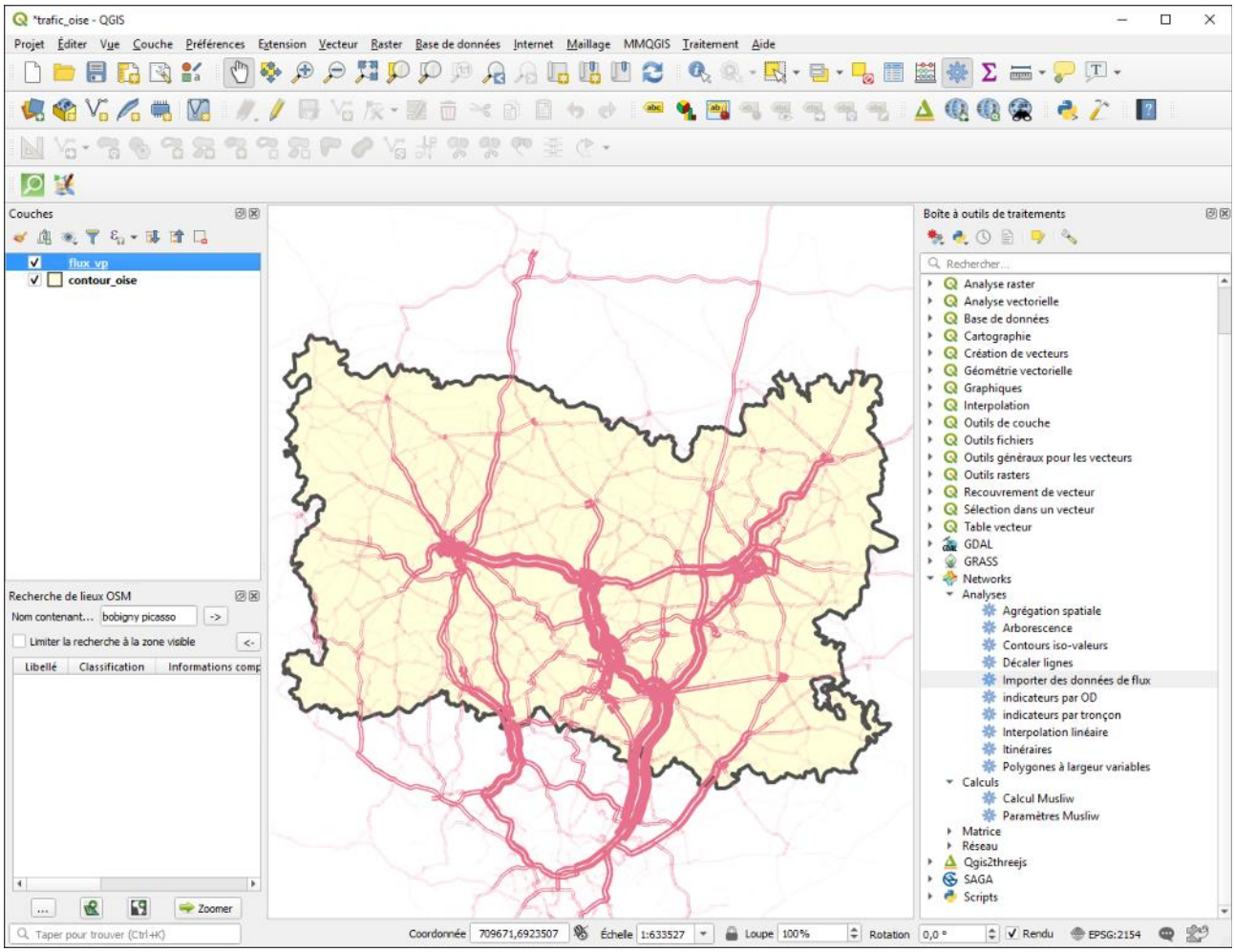


Then set offset as volumes are defined by traffic directions. Then set **offset** expression as traffic is depending of directions. The **+0.2 value** is useful in order to visually differentiate both directions.



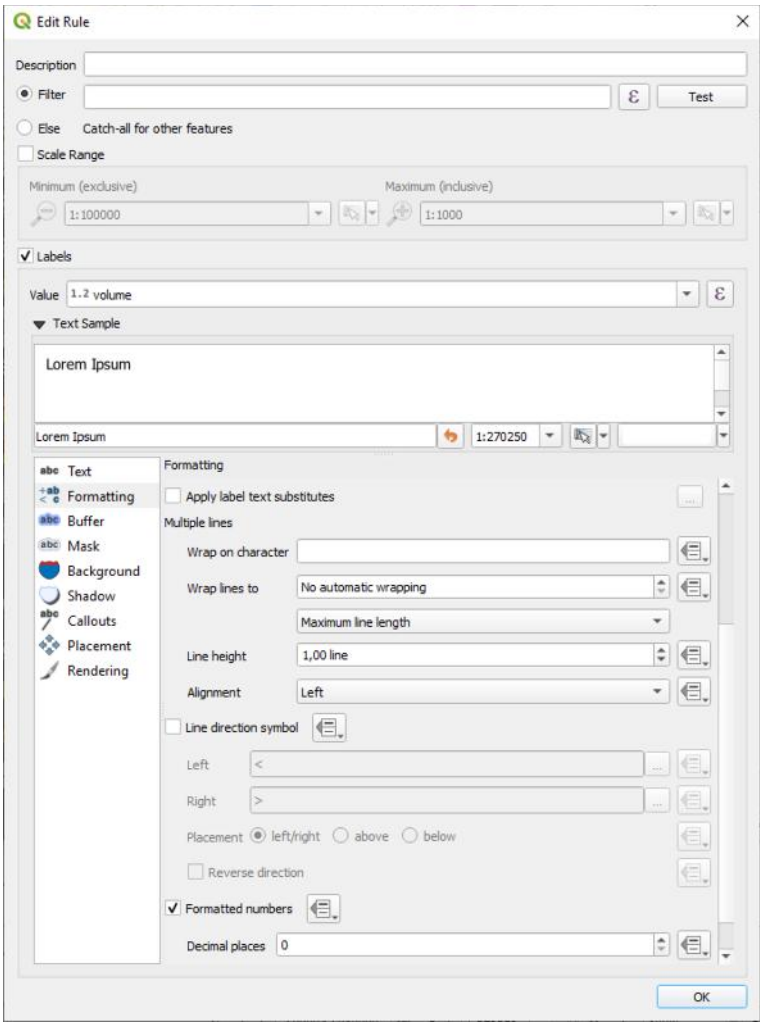
@value corresponds to the numeric value defined in stroke width and offset parameters. It is important to introduce @value in expression formula because it allows to customize the width by adjusting the parameters value without having to modify the formula.

Result



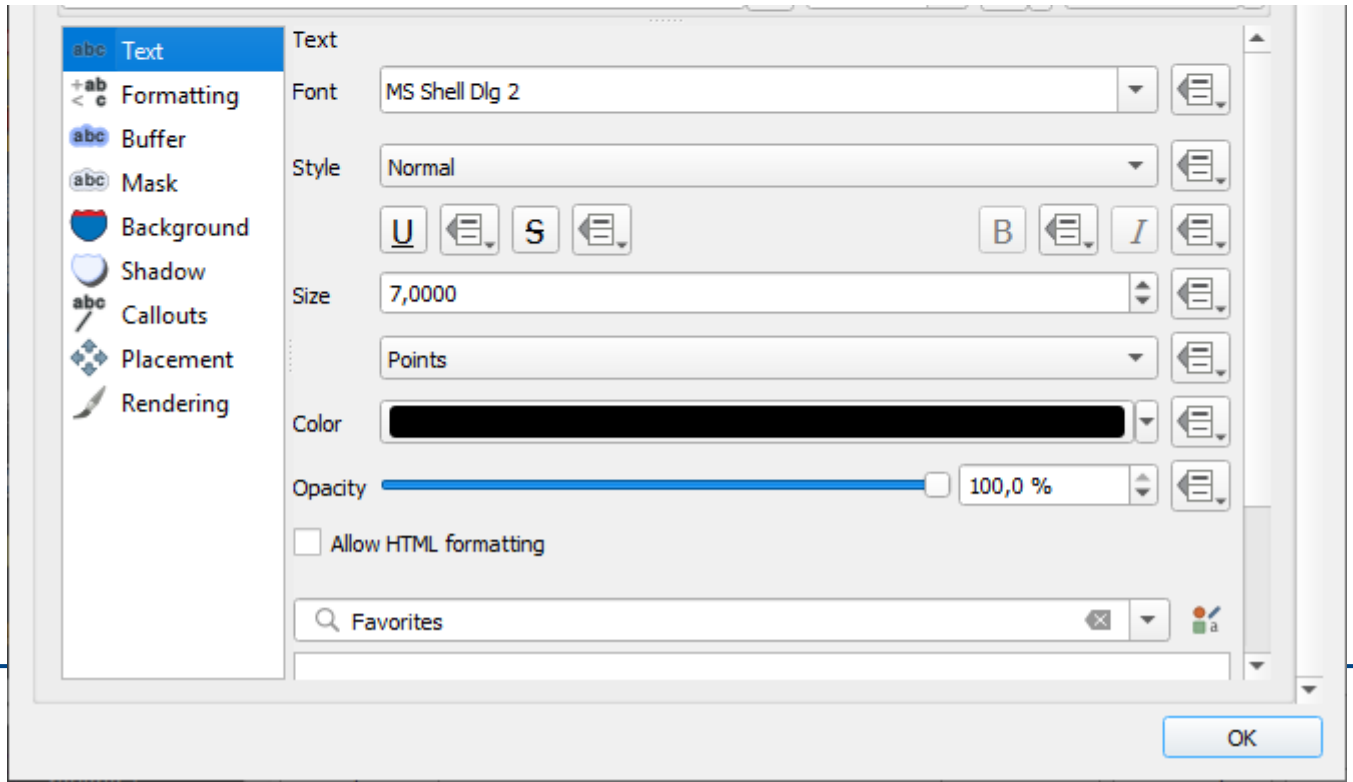
4. Show traffic volumes flows

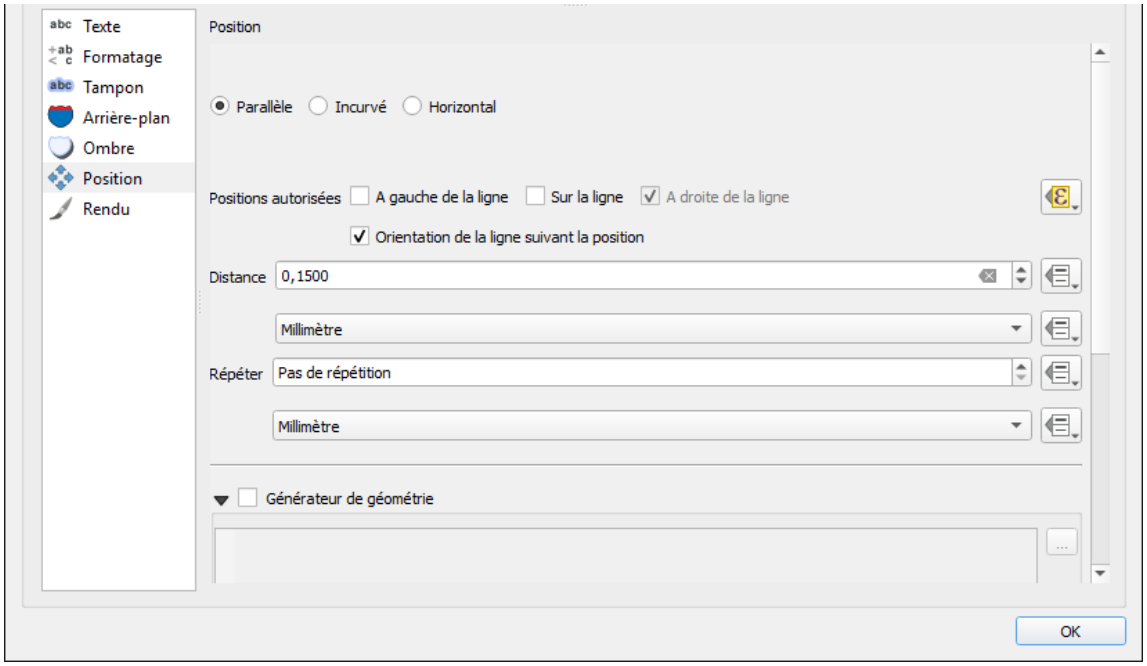
Create **rule based labels** because you will filter small figures later. It is important to click on the  in order to add a rule, even if there is no filter in a first step.



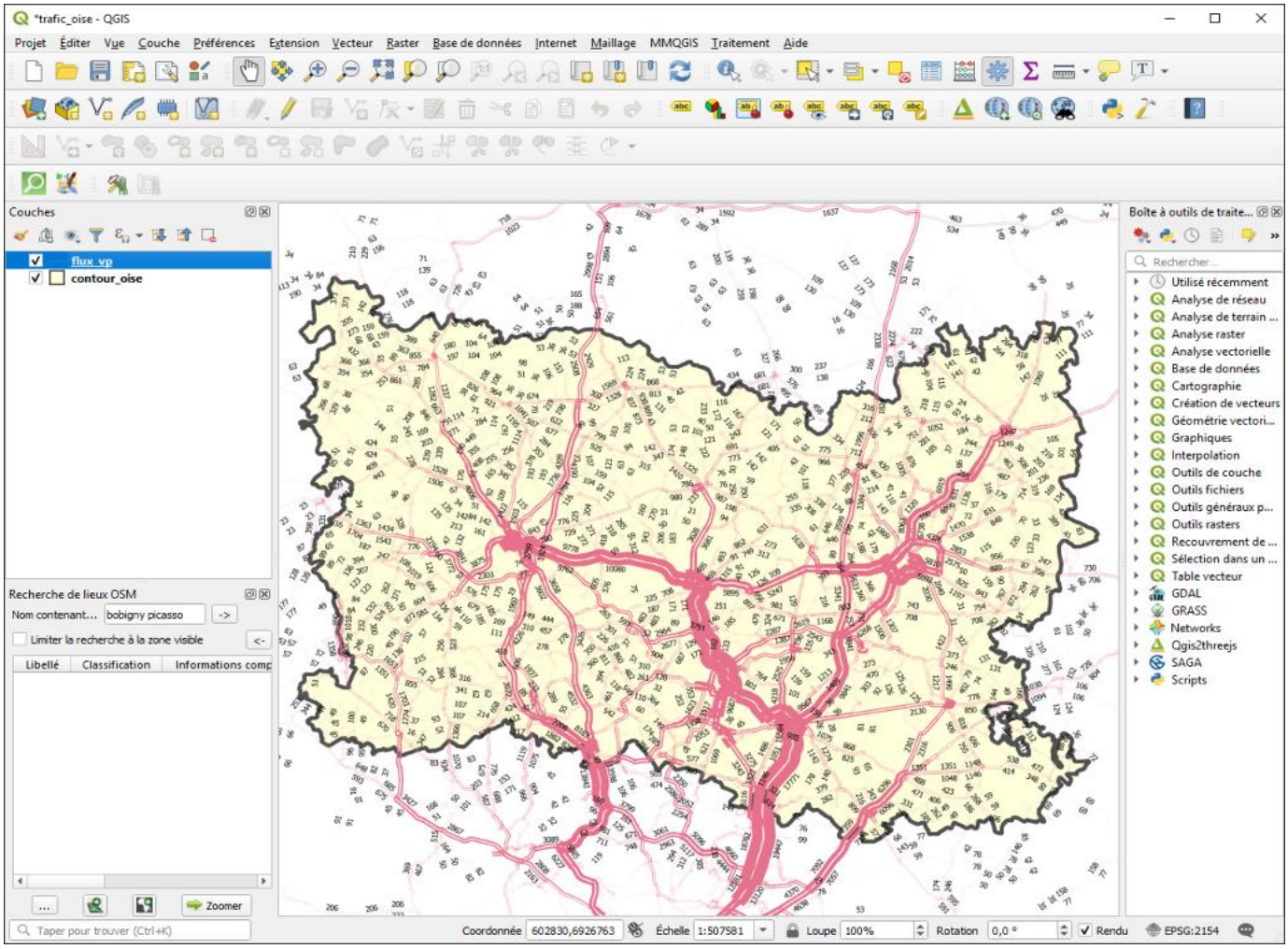
Formatting text

Possibilité de définir la position de l'étiquette



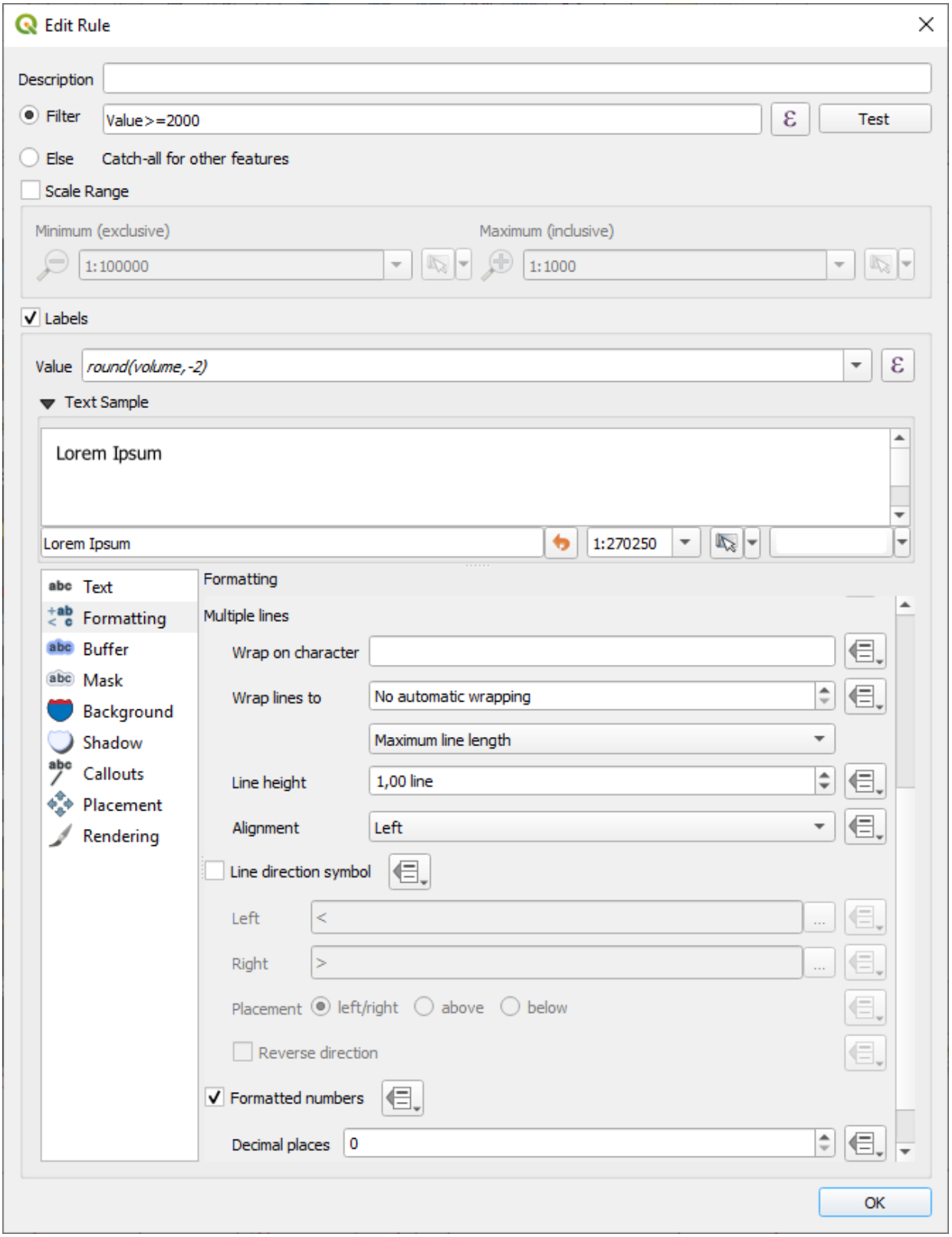


Result



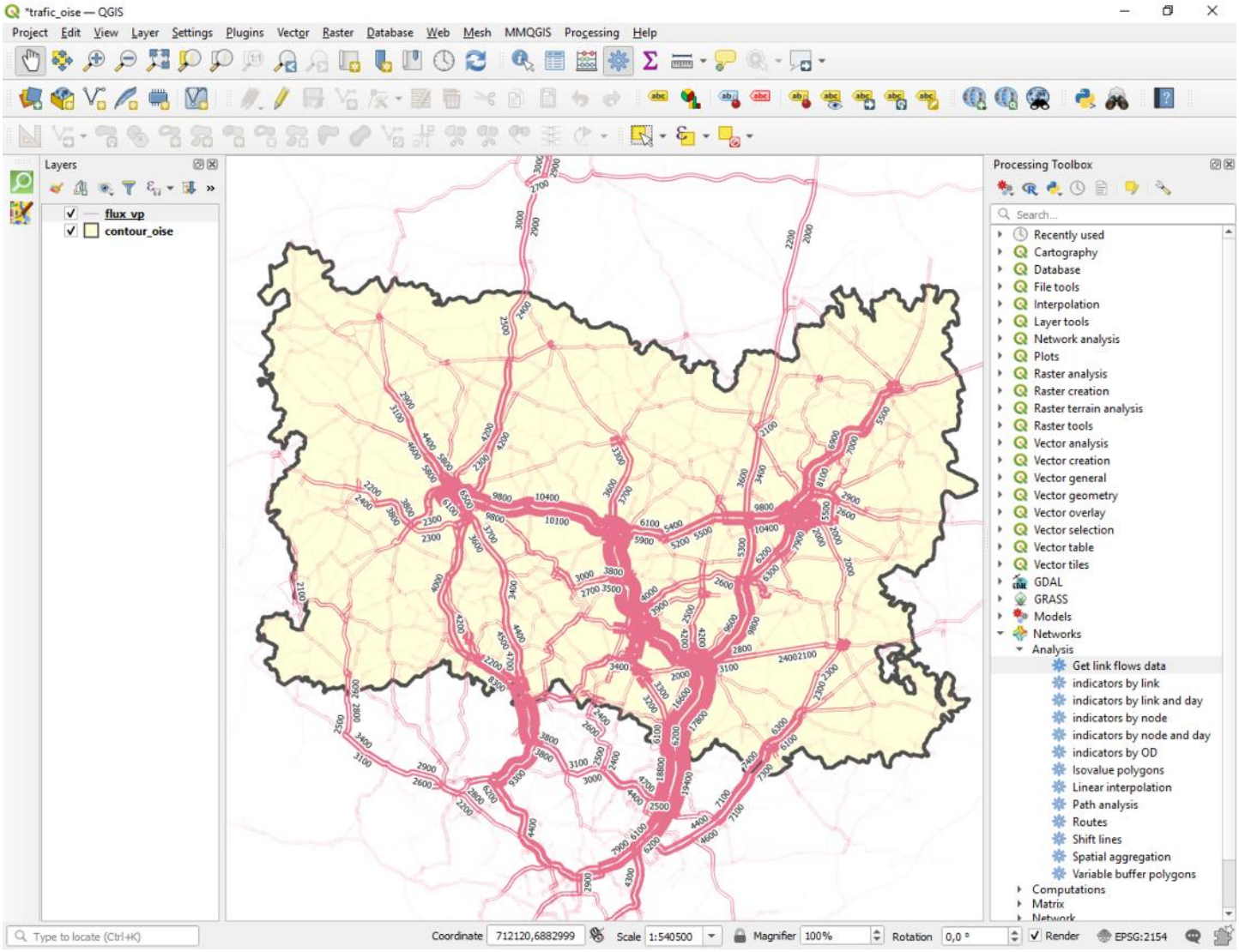
The map has too many labels and is becoming unreadable. Moreover, the precision to the nearest unit is illusory considering the survey sample rate.

We will filter labels for volumes<2000, **merge labels** for adjacents links sharing the same value and round volumes to the nearest hundred.



Define the s**ame distance as the stroke width and offset values.** Check **“Line orientation dependent position”** and **“Right of line”** for a traffic map-like rendering

Result


*trafic_oise — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh MMQGIS Processing Help
Layers
flux_vp
contour_oise
Processing Toolbox
Search...
Recently used
Cartography
Database
File tools
Interpolation
Layer tools
Network analysis
Plots
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector creation
Vector general
Vector geometry
Vector overlay
Vector selection
Vector table
Vector tiles
GDAL
GRASS
Models
Networks
Analysis
Get link flows data
indicators by link
indicators by link and day
indicators by node
indicators by node and day
indicators by OD
Isovalue polygons
Linear interpolation
Path analysis
Routes
Shift lines
Spatial aggregation
Variable buffer polygons
Computations
Matrix
Network
Type to locate (Ctrl+K)
Coordinate 712120,6882999
Scale 1:540500
Magnifier 100%
Rotation 0,0 °
Render
EPSG:2154

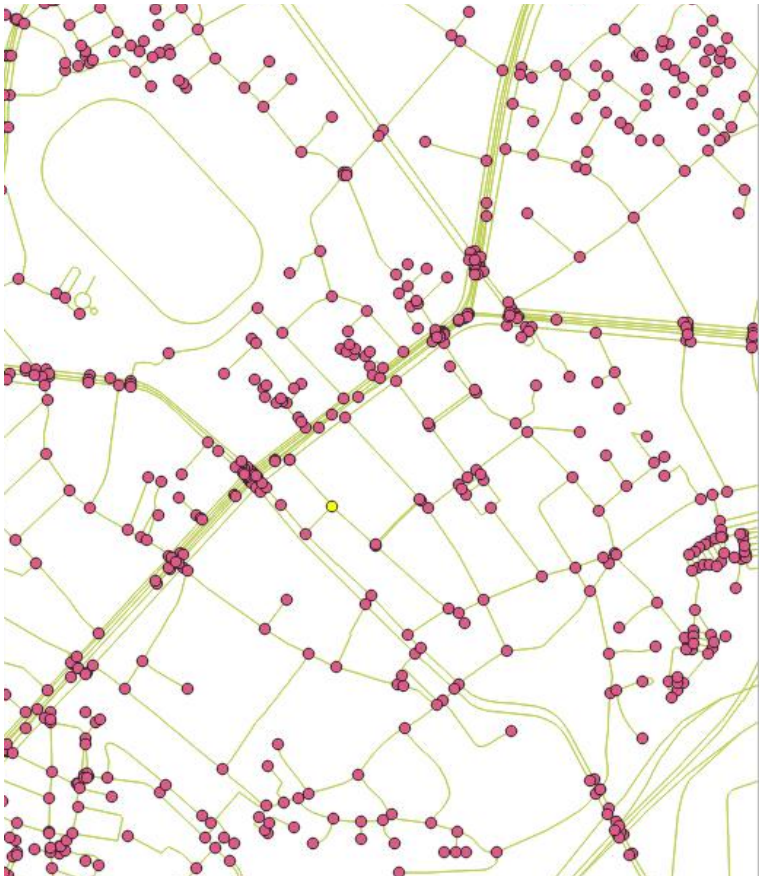
D- Create a territorial accessibility map on a time period and/or from/to several points

1. Create the accessibility map and the area of influence map

1.1. Matrix creation

Example with a point during a period (several arrival or departure times)

Select a point (in yellow on the map)



SCRIPT Musliw matrix simple list

nodes : **select the nodes layer** (the layer where the node is selected)

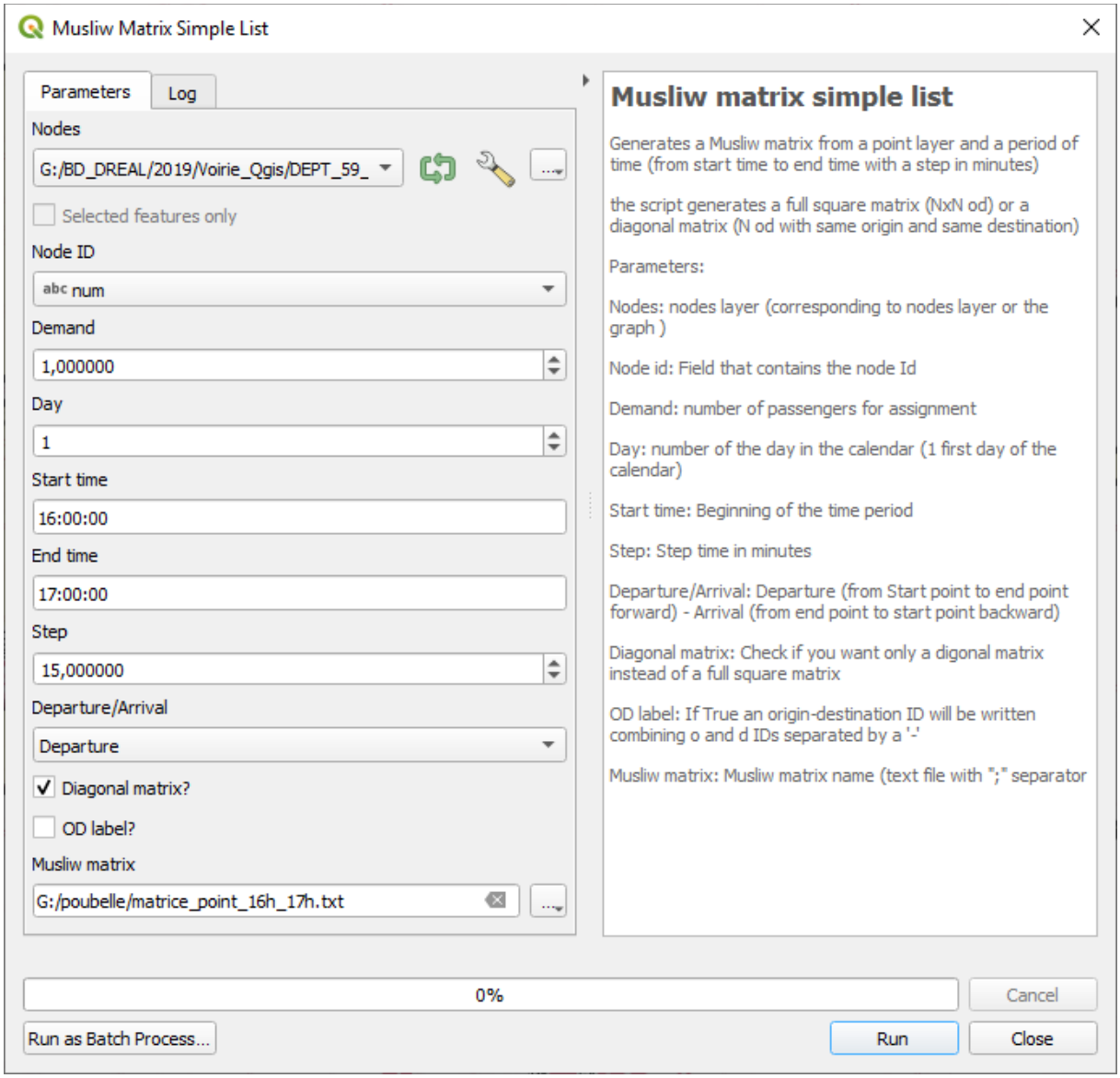
Check **Selected features only**

Fill **start_time** and **End time fields** of the period : example 16:00:00 – 17:00:00

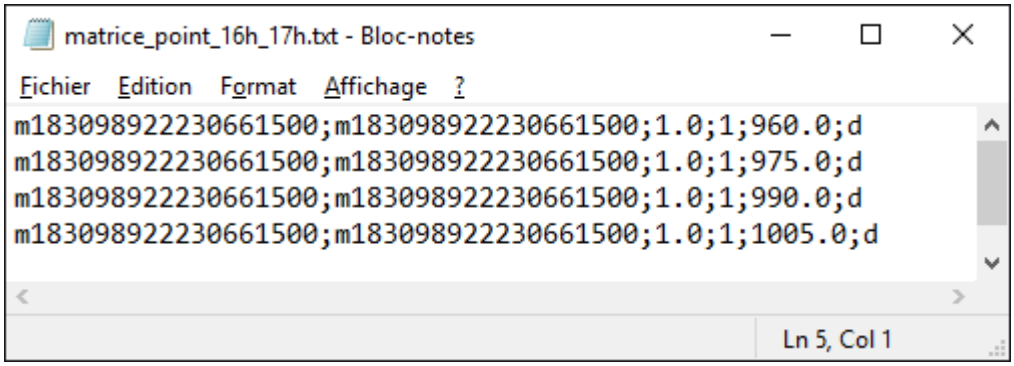
Fill **time interval in minutes** : example 15,0000

Check **diagonal matrix** (same origin, same destination.

Save matrix as.

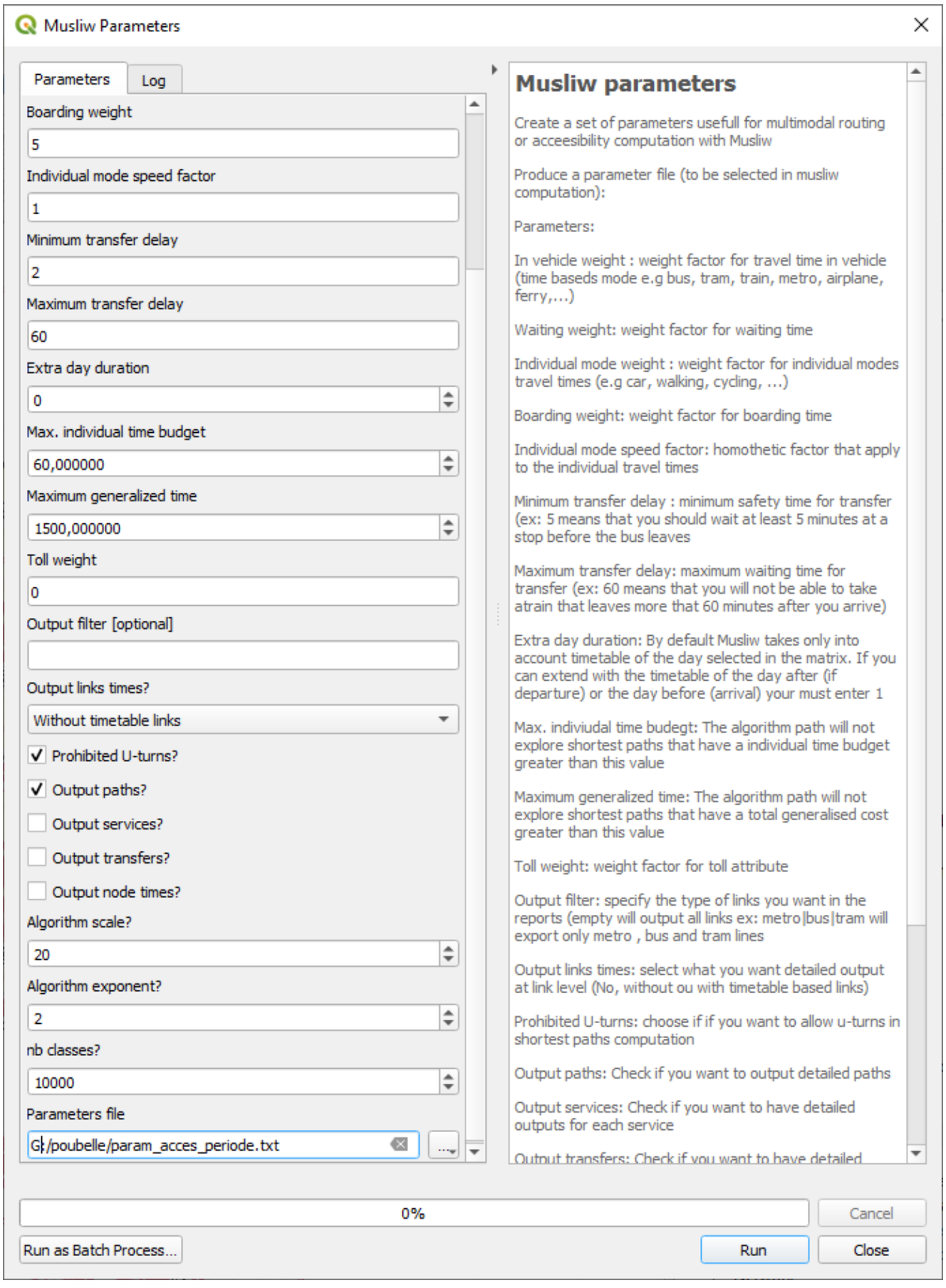


Result: The generated matrix



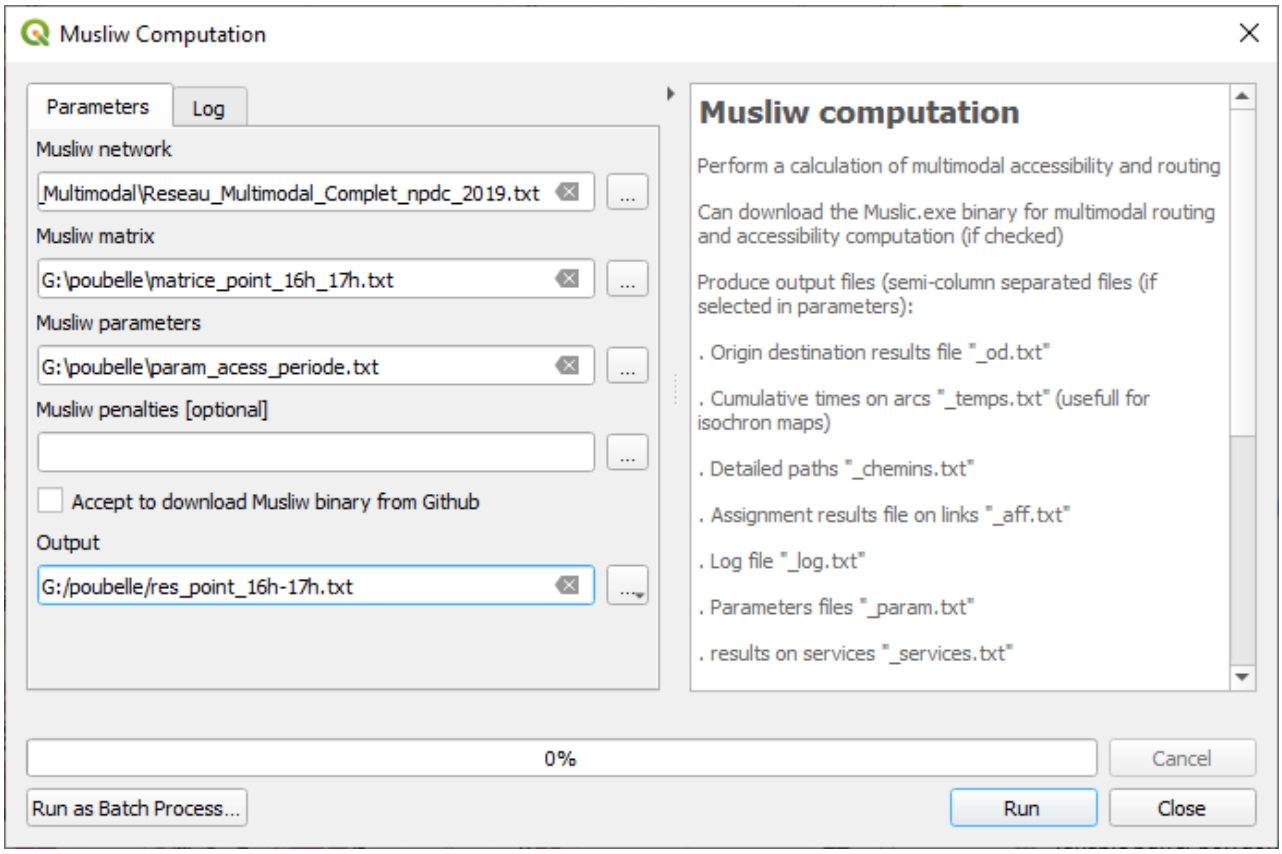
1.2. Customize Musliw computation parameters

In the field “output link times” choose **« Without timetable links »**



1.3. Musliw computation

SCRIPT Musliw computation



1.4. Compute travel times aggregation

From Musliw computation out files, we are going to use the **travel times output file**.

In this file, there are several travel times for each link, because there are 4 simulations during the period.

Here, we have at most 4 travel times per link, as there are 4 simulation during the period (one simulation every 15 minutes) (cf matrix)

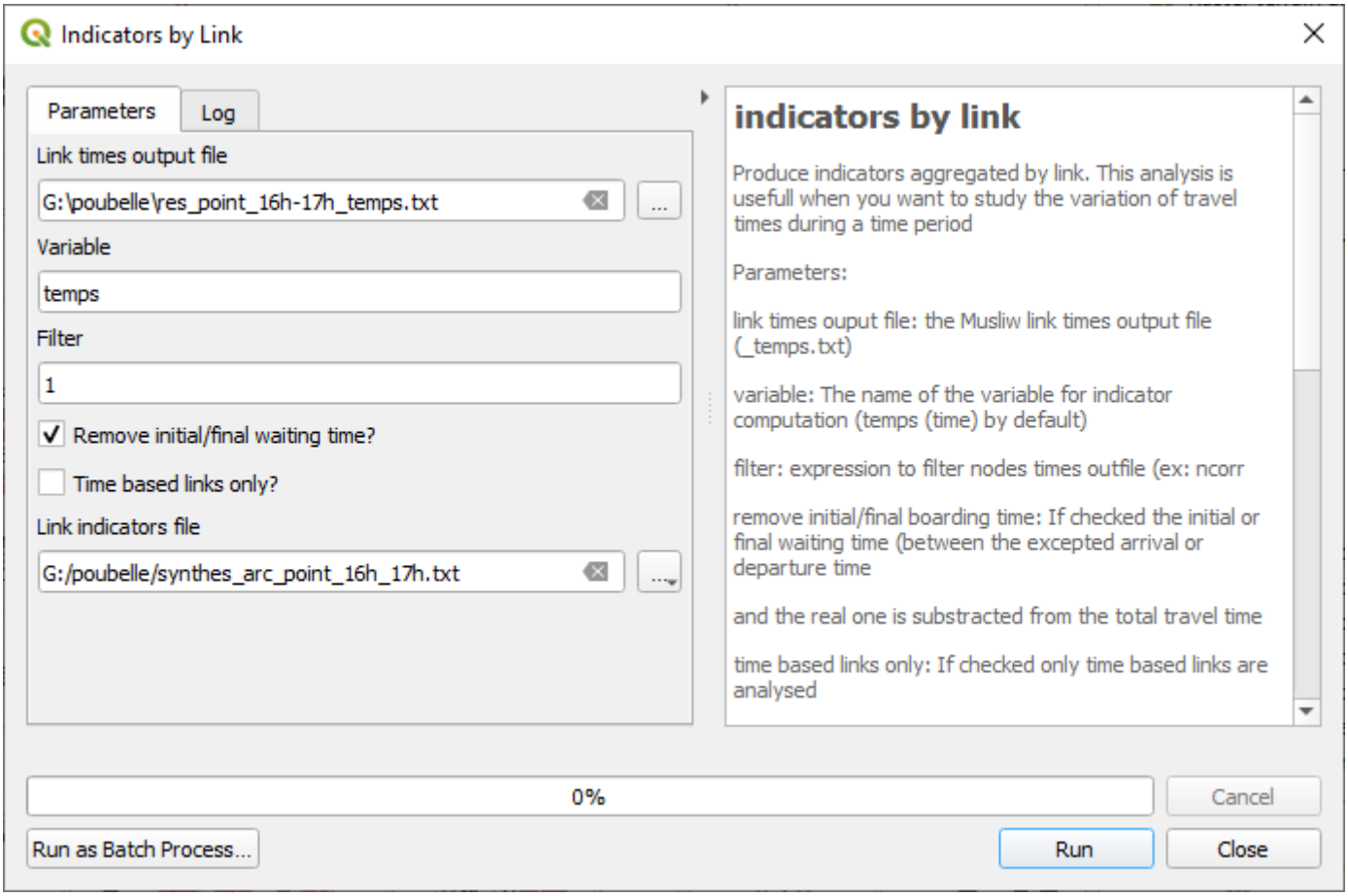
Then, we can't represent directly the isochrones. It is necessary to perform an statistic analysis (min, mean, max, …)

To do this we will use the **SCRIPT indicators by link** (You can also use if you want "indicator by node" if you want a representation based on nodes

Link times output files : **fichier temps** (computation result)

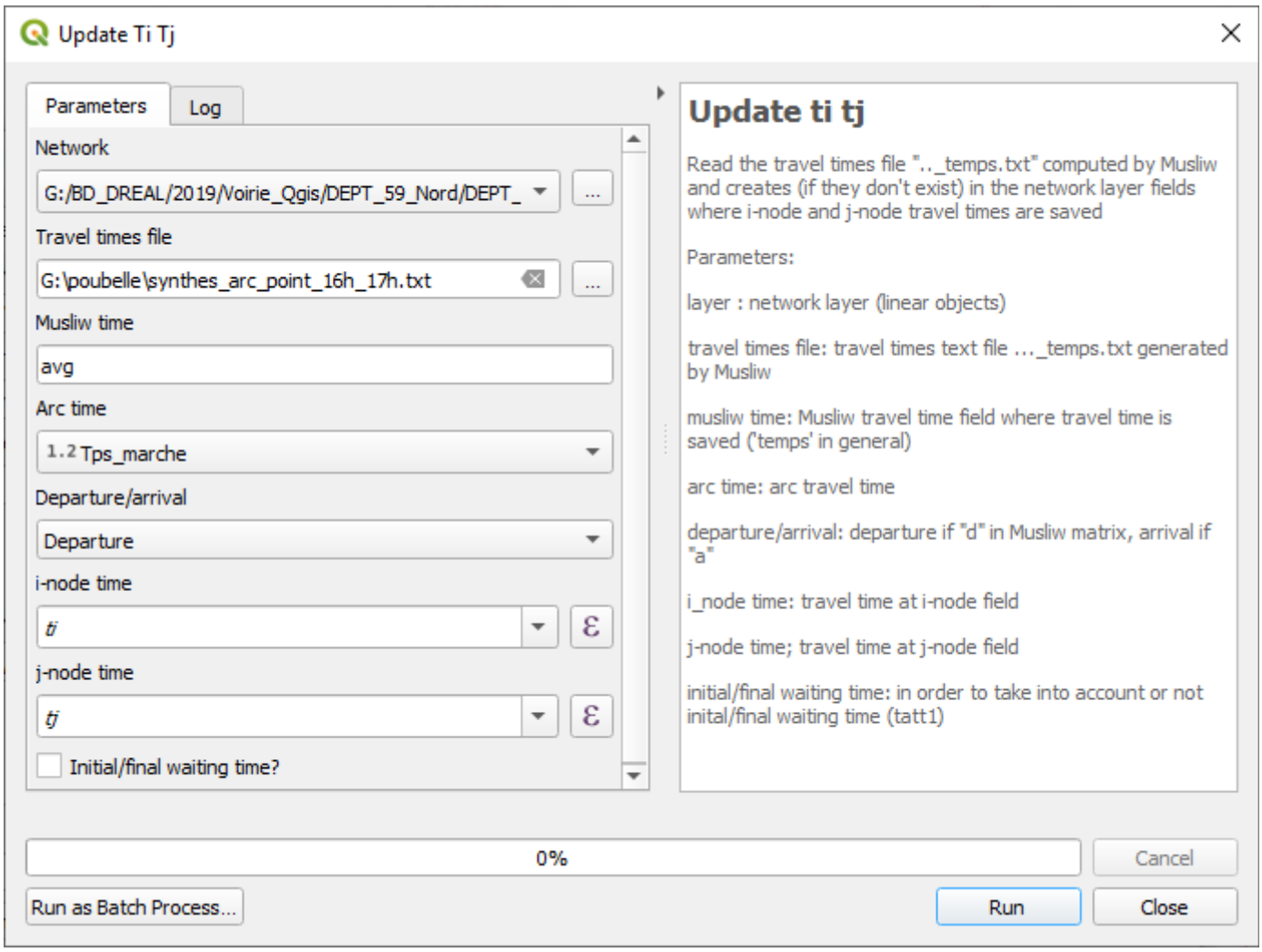
Variable : choose **temps**

Check Remove initial/final waiting time

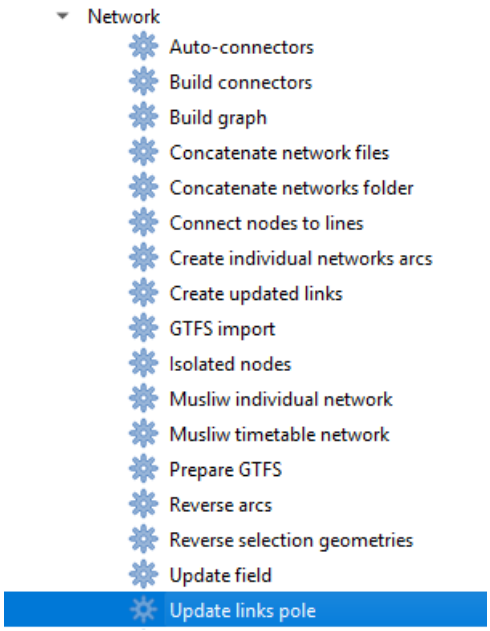

Indicators by Link
Parameters
Log
Link times output file
G:\poubelle\res_point_16h-17h_temps.txt
...
Variable
temps
Filter
1
Remove initial/final waiting time?
Time based links only?
Link indicators file
G:/poubelle/synthes_arc_point_16h_17h.txt
...
indicators by link
Produce indicators aggregated by link. This analysis is usefull when you want to study the variation of travel times during a time period
Parameters:
link times ouput file: the Musliw link times output file (_temps.txt)
variable: The name of the variable for indicator computation (temps (time) by default)
filter: expression to filter nodes times outfile (ex: ncorr
remove initial/final boarding time: If checked the initial or final waiting time (between the excepted arrival or departure time
and the real one is substracted from the total travel time
time based links only: If checked only time based links are analysed
0%
Cancel
Run as Batch Process...
Run
Close

1.5. Update aggregated ti tj

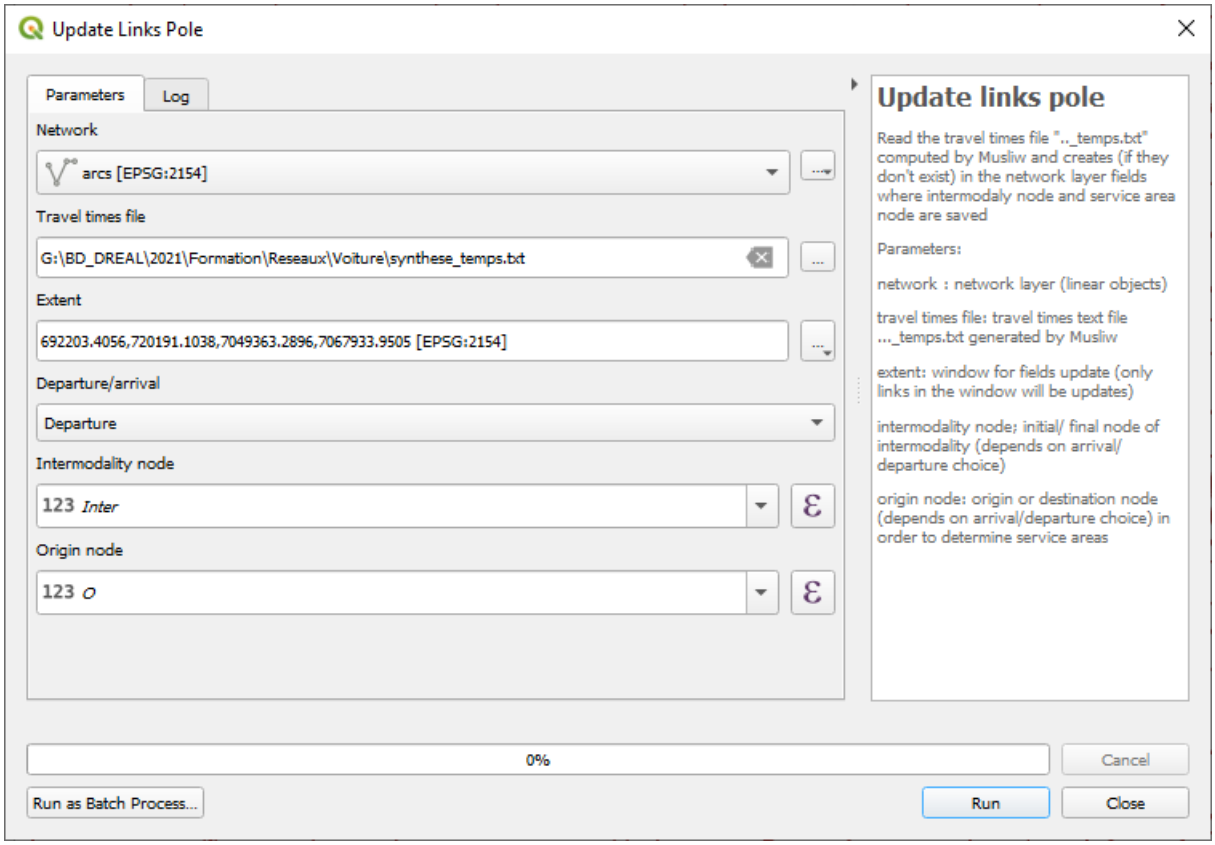
The following process is similar as a standard creation of a territorial accessibility map, except that for the **"update of ti tj"** script, you must choose which time we want to represent in the variable **«Musliw time »** (the minimum travel time : min, the average travel time : avg, the maximum travel time : max).



1.6. Update of the links pole

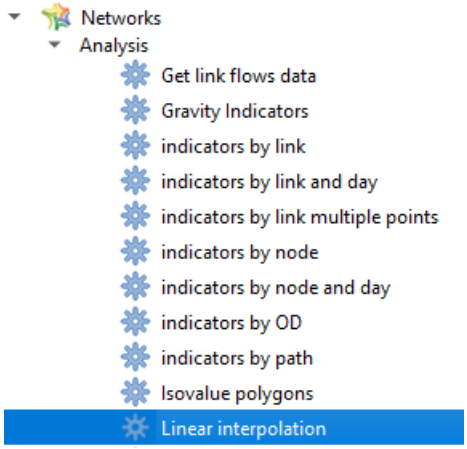


If you also want to create a map of the areas of influence or catchment areas (if there are several points), you must then update for each arc the pole to which it is attached, i.e. the node identifier from which the travel time is the smallest



To make the map, start again from A- Making an accessibility map, part 2 (making the accessibility map), level 5 (starting the linear interpolation).

1.7. Linear interpolation



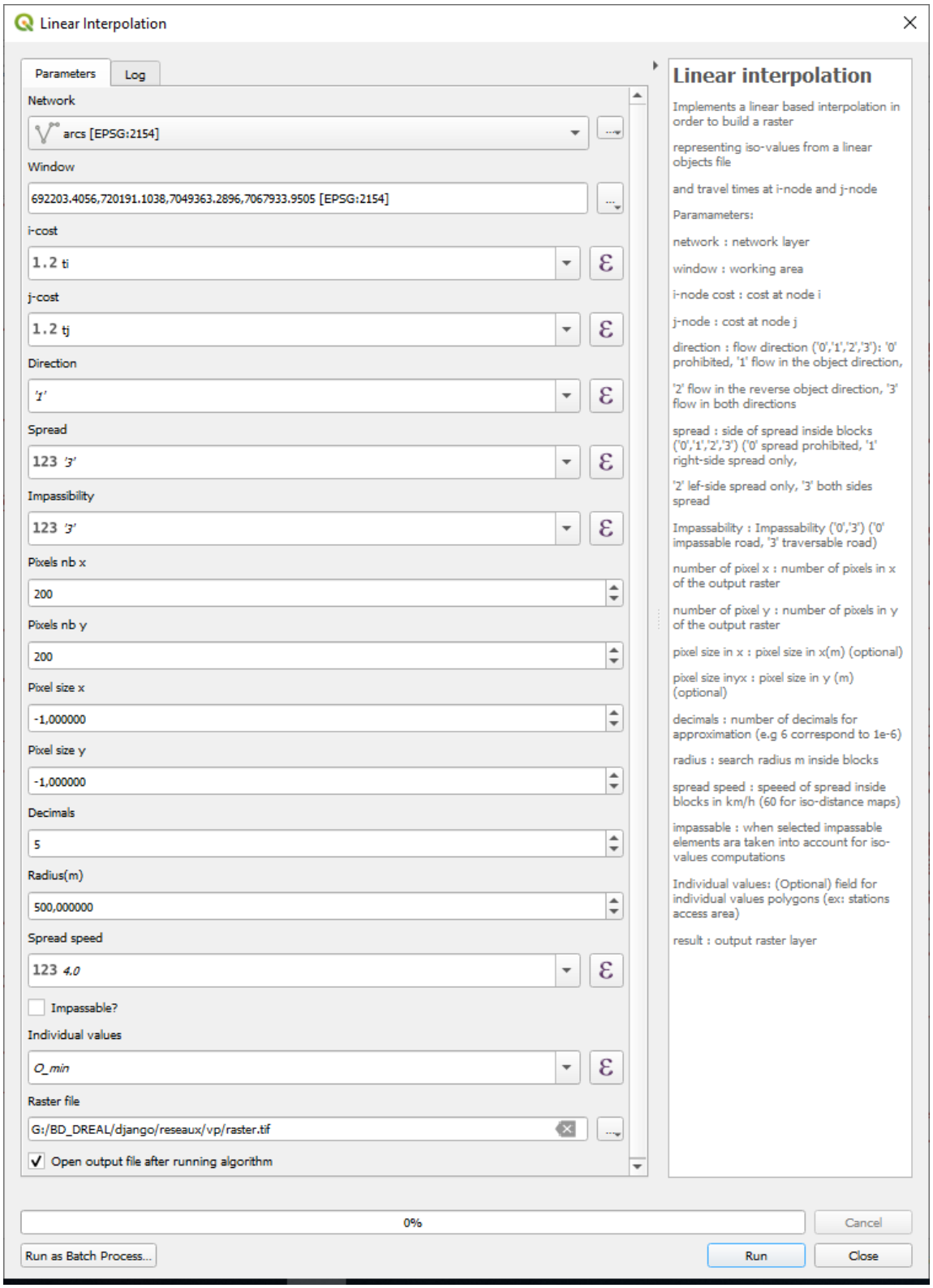
To be able to create the map of areas of influence or catchment areas, you must, in addition to the identical settings made for the single-point map, fill in the individual values field.

You must indicate the identifier that defines your areas of influence.

It is a text variable:

For example:
O, O_min,... For the closest starting or ending point

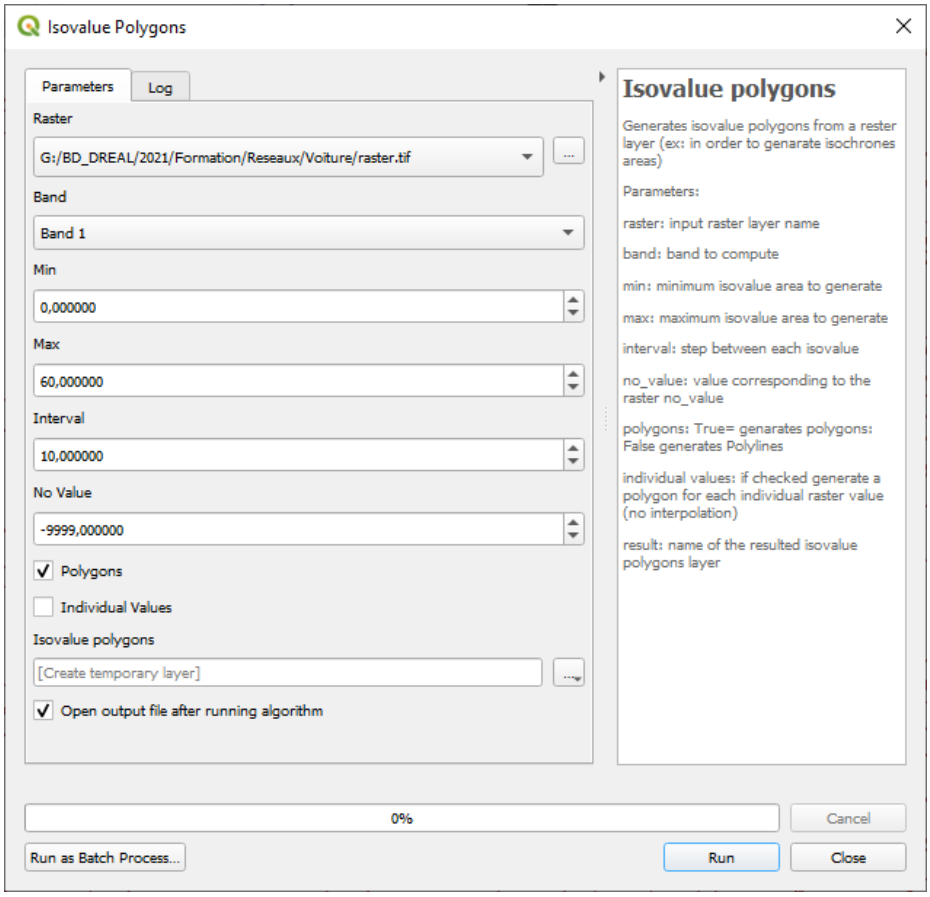
Pole, pole_min, ... : For the intermodality point



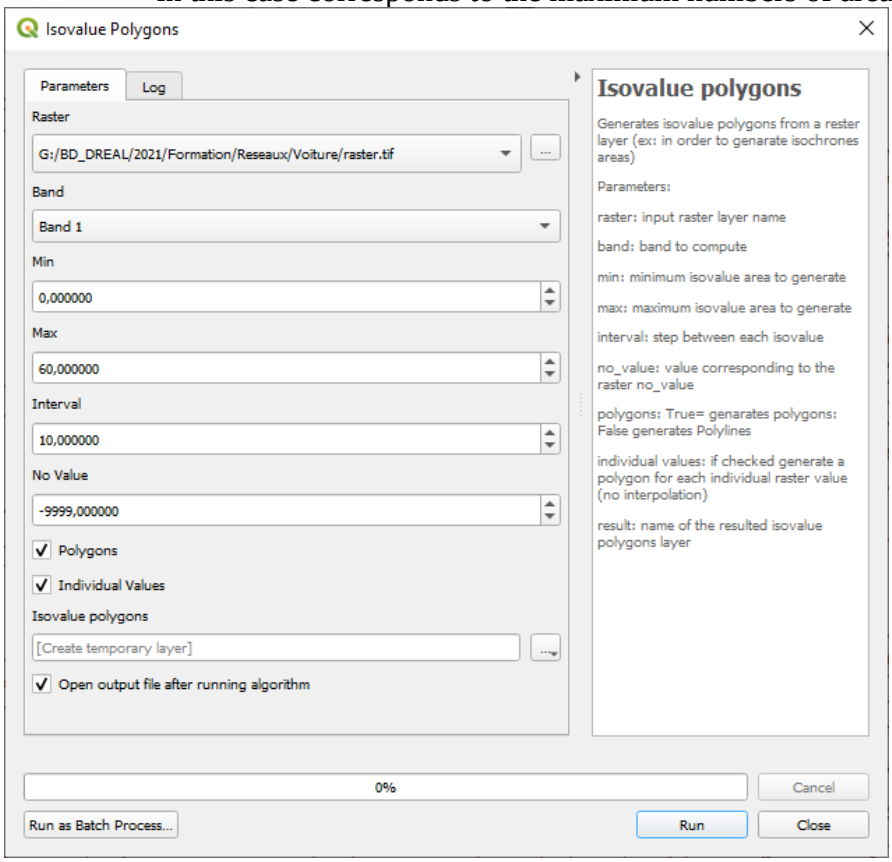
1.8. Drawing of isovalue polygons and areas of influence.

For the drawing of isovalue polygons, the settings are identical to those of the single point map.
For the drawing of the areas of influence or catchment areas, the "individual values" option must also be checked.
So, to have both maps, you have to launch the "iso value polygons" algorithm twice

- Once with the box "individual values" unchecked for the drawing of iso-value polygons



- Once with the box "individual values" checked for drawing areas of influence
  - For areas of influence if their number is large, it is necessary to increase the max parameter, which in this case corresponds to the maximum numbers of areas of influence to be represented



E- Simplified accessibility map

The interest of developing plugins in the form of processing algorithm managers is to :

- Perform batch processing: Run the algorithm in batch several times
- Build models by chaining the algorithms together (can also be run in batch)

The realization of accessibility maps requires to execute successively several scripts or algorithms.

If the understanding of the process of realization of these maps is important for the consultant so that he can understand the process, the parameters that he can modify, it seems useful to have a tool that allows to realize the maps in an automatic way with predetermined default parameters.

For this purpose, Cerema has developed, thanks to the model tool of the processing box, 3 models (1 per network) to generate accessibility maps very simply.

- isochrone_multimodal (multimodal network with detailed road network, large scale)
- isochrone_multimodal_light (multimodal network with simplified walking road network, small scale)
- isochrone_route500 (car road network, medium scale)

1. Installation

Each model is delivered in the form of a directory containing :

- Musliw network
- GIS network (arcs, nodes)
- Musliw parameters file
- Qgis style file
- model file (.model3)

Copy each directory to your disk:

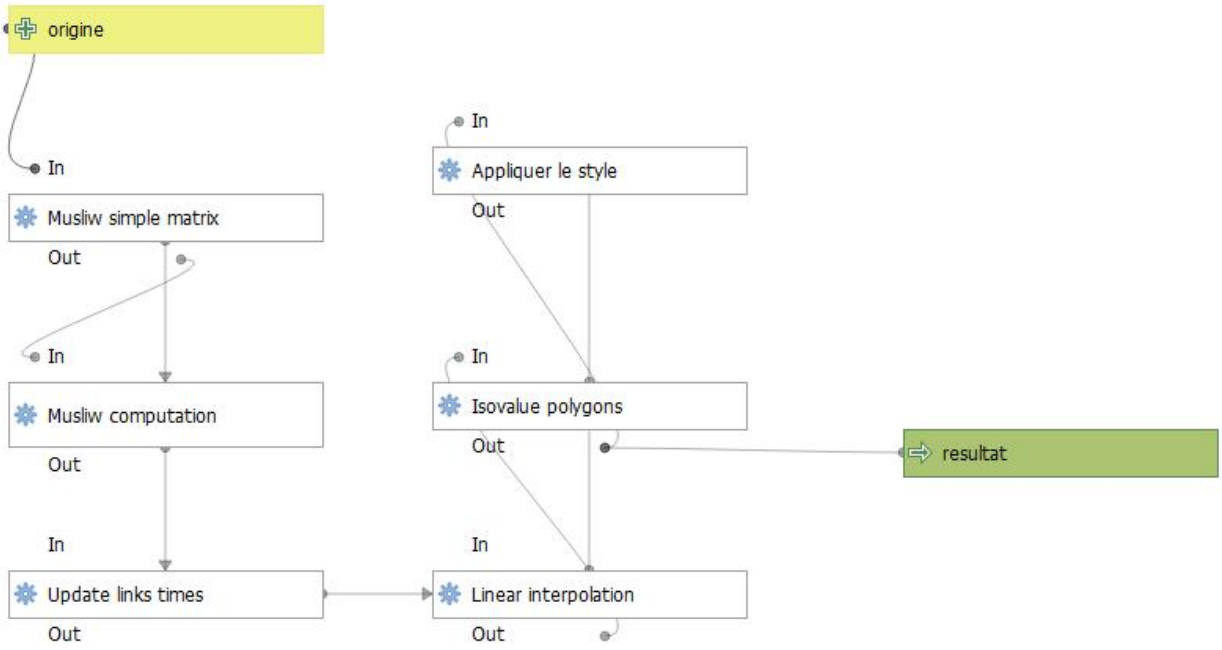
Copy the model files (.model3) to your processingmodels directory: C:\Users\<username>AppData\Roaming\QGIS\QGIS3\profiles\default\processing\models

2. Settings

The principle of models in Qgis is to be able to chain several algorithm scripts.

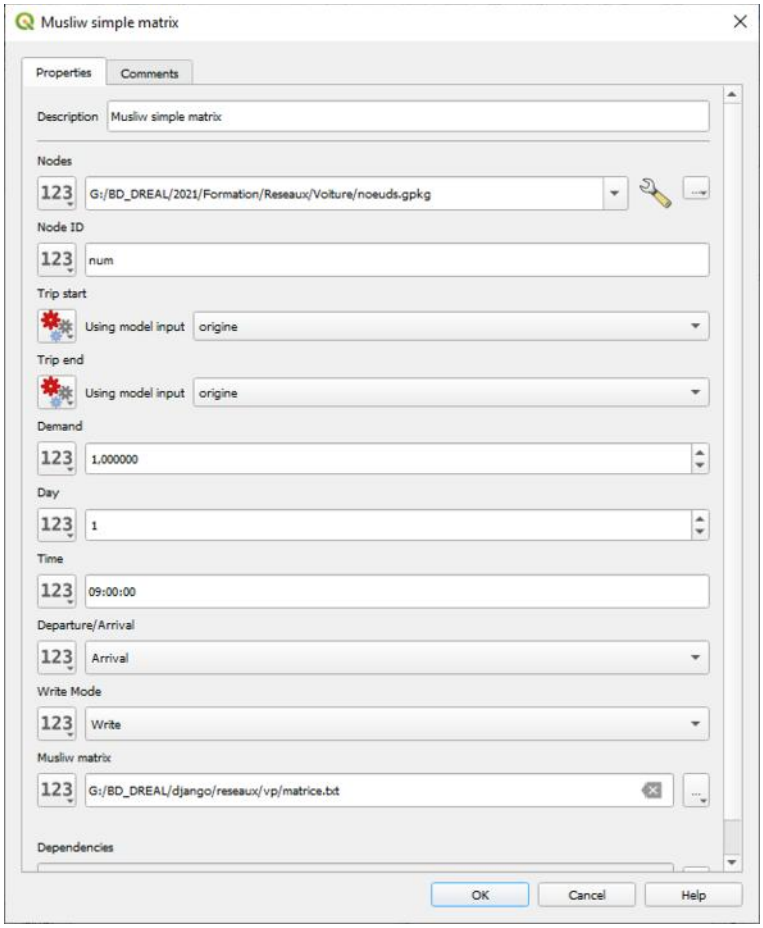
A model is made of :

- algorithms (scripts)
- inputs (parameters to be entered by the user)
- links between algorithms



In the example of the multimodal light network model, we can see that it is composed of 6 successive algorithms and 2 inputs (point of origin and result file)

All the other parameters of the different scripts are entered as parameters in the scripts (e.g. simple musliw matrix script).



Each parameter of each script can be defined by either :

- a value: constant defined by the user for the parameterization (e.g. "num" for the name of the node number in the node table)

- an input (here origin for the start and end points)

- an expression

- the result of another algorithm

The user must initially define all the parameters of each algorithm to adapt them to his needs and context (file paths, day, time, number of pixels, etc.)

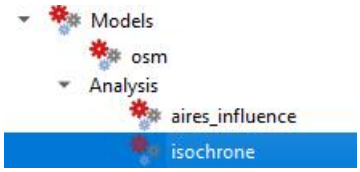
He can also add inputs if he wants the user to enter a parameter at each use. For example, here the time has been set to 9am. It could be useful to add the time input so that the user has to specify the desired arrival or departure time each time.

Once the template has been modified and customized, the changes must be saved. However, the models are only loaded when Qgis is started. In order to use the modified version, it must also be loaded into the project (green icon to the right of "save as"), otherwise the old version is used.

3. Making the map

To make the map, you need to run the template in question as a script. They are located in "Models".

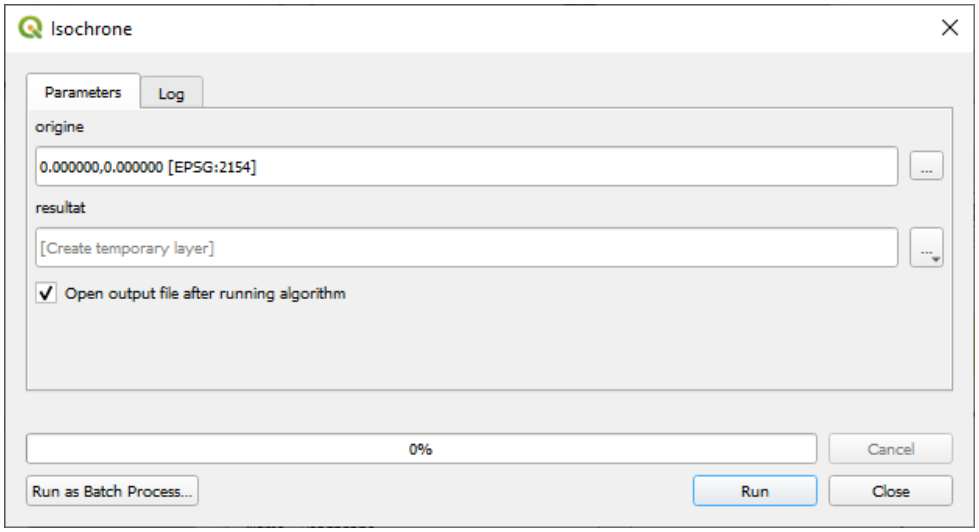
First of all, you need to define in the map window the area of study of the map. The map will be defined on the visible part of the map only



We can see that now only two parameters are required for the accessibility card

multimodal accessibility map :

- the origin: point to click on the map
- the result : vector file in which the isochrones will be generated



All other parameters are defined by the user beforehand by default. The user can always change them if he wishes but he will have to save the model again, or add an input if this parameter is to be changed often.

4. Tips for setting parameters

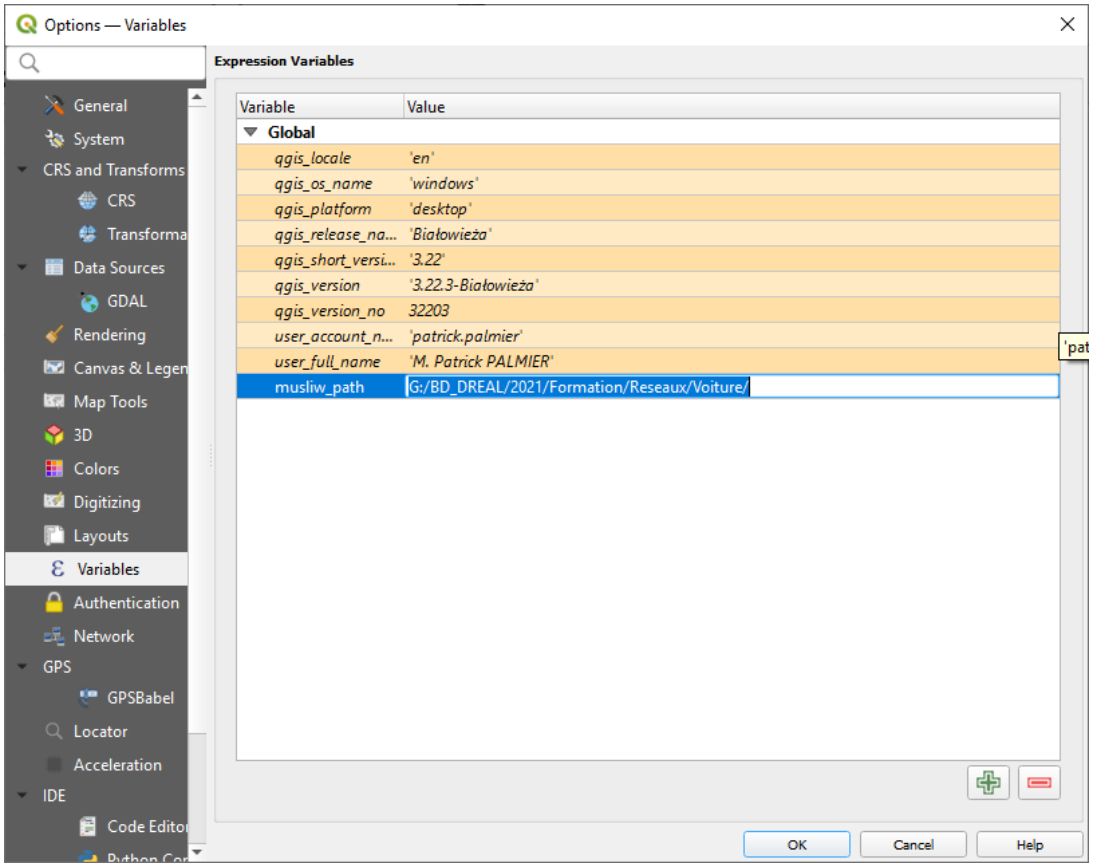
When building models composed of a succession of several scripts, it may be interesting to set the paths of frequently used tables or files dynamically.

In the creation of accessibility maps in particular, the user may need to create maps for different modes (walking, cycling, public transport, car, ... and on several study areas).

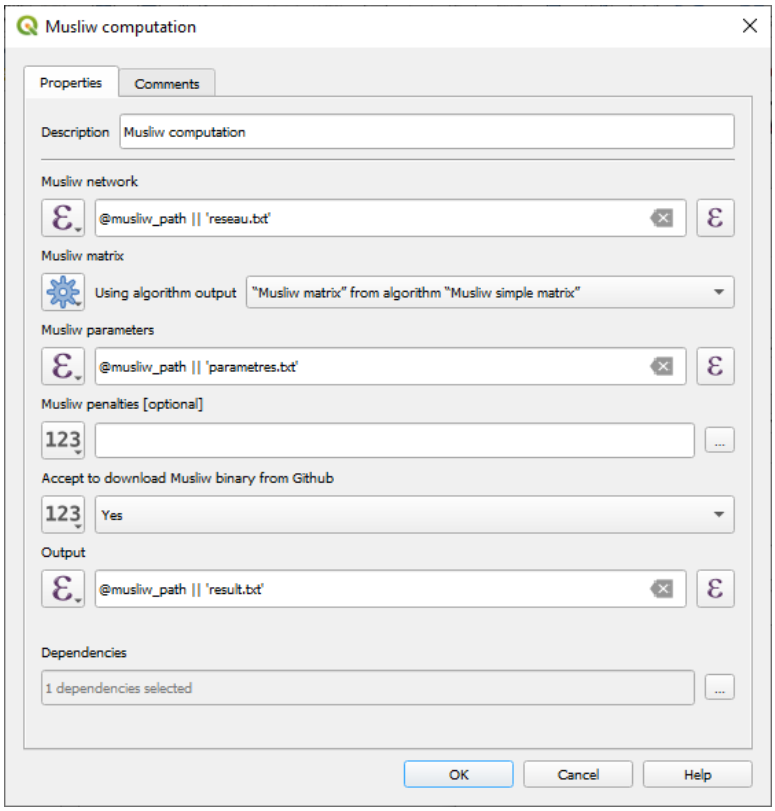
For this purpose, an interesting method consists in naming the files used for each mode in the same way and placing them in a different directory with the name of the mode, for example

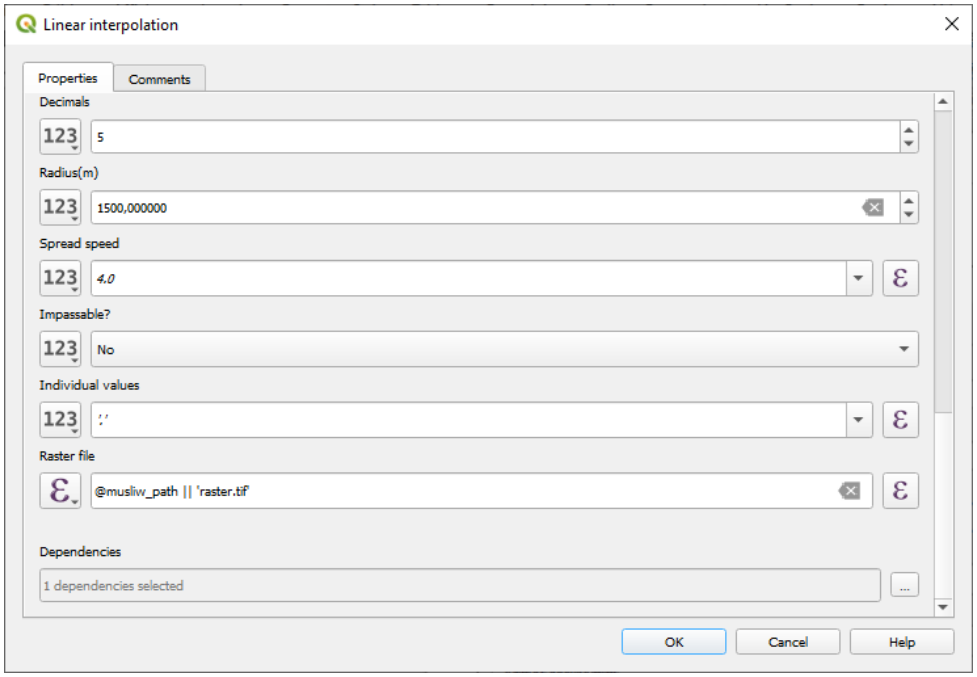
- reseau.txt : Multimodal network in Musliw format (txt)
- arcs.gpkg : The table of the road network used by the mode we want to study (walking, car, bicycle, ...)
- nodes.gpkg : The table of nodes associated with the arcs.gpkg table
- parametres.txt : The calculation parameters of the mode in question
- result.txt : The name of the result file

Then the trick is to create a variable in Qgis that will point to the directory of the mode you want to study. For example, here, we must add the variable "musliw_path".



In the following two images you have an example of how to parameterize directory names using the @musliw_path variable. You have to define the parameter as an expression and concatenate the name of the variable with the name of the file (same as above for each directory.




Linear interpolation
Properties
Comments
Decimals
5
Radius(m)
1500,000000
Spread speed
4.0
Impassable?
No
Individual values
','
Raster file
@musliw_path || 'raster.tif'
Dependencies
1 dependencies selected
OK
Cancel
Help

F- Create an accessibility map based on a grid

1. Advantages and drawbacks

The main advantage of this method is that the realization of the map is much faster since it only requires the steps of creating the matrix and the calculation and then a thematic analysis.

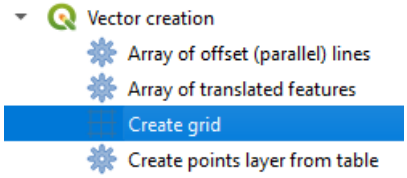
Moreover, this method is particularly interesting when there is a need to cross-reference accessibility results with other data (e.g. gravity accessibility) since all the necessary data can be disaggregated to the level of each grid cell.

However, as the grid is fixed, it is scale dependent. If we zoom in too much, the meshes become too big and this gives a pixelated effect and a loss of precision. Finally, as the grid is fixed, the initial step of creating the grid and preparing the data is a bit time consuming.

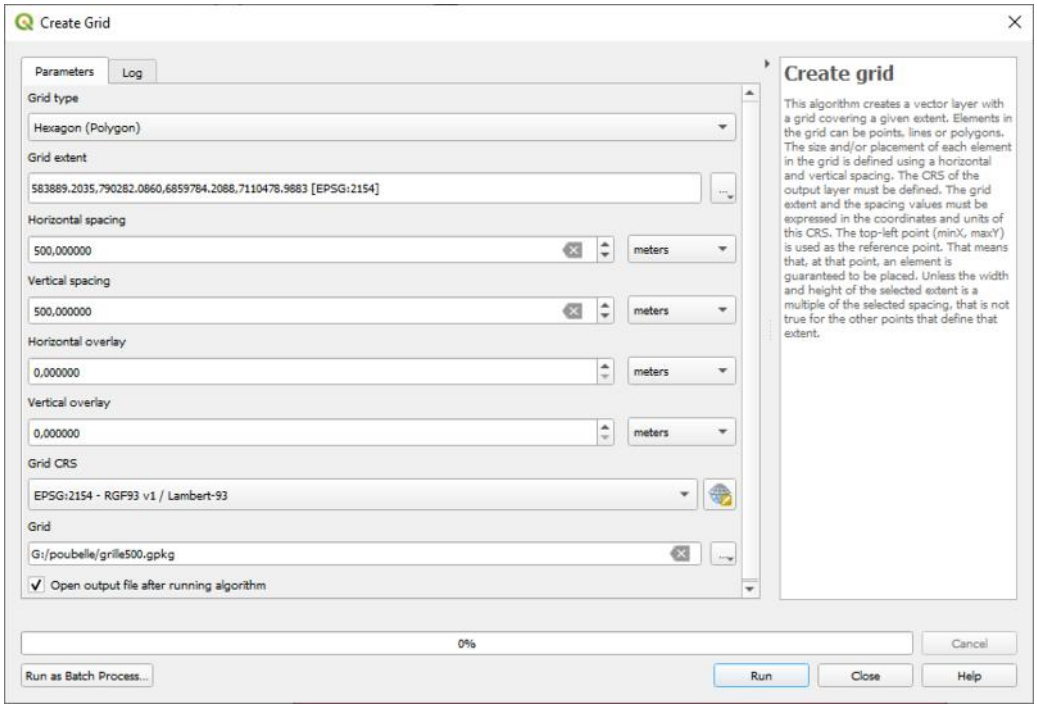
2. Creating the grid

The grid can be created in several ways.

To stay with the providers of processing algorithms, we can use the Qgis script "Create grid" located in the "Vector creation" group

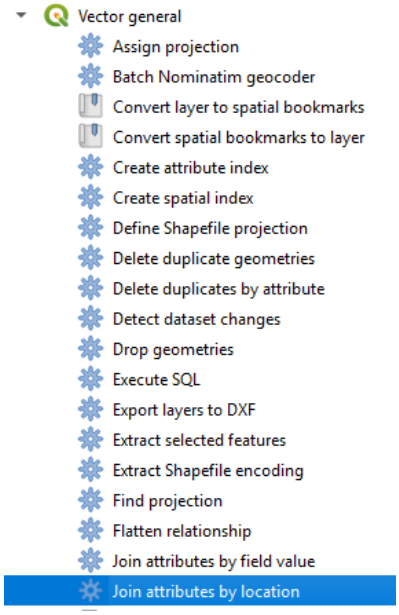


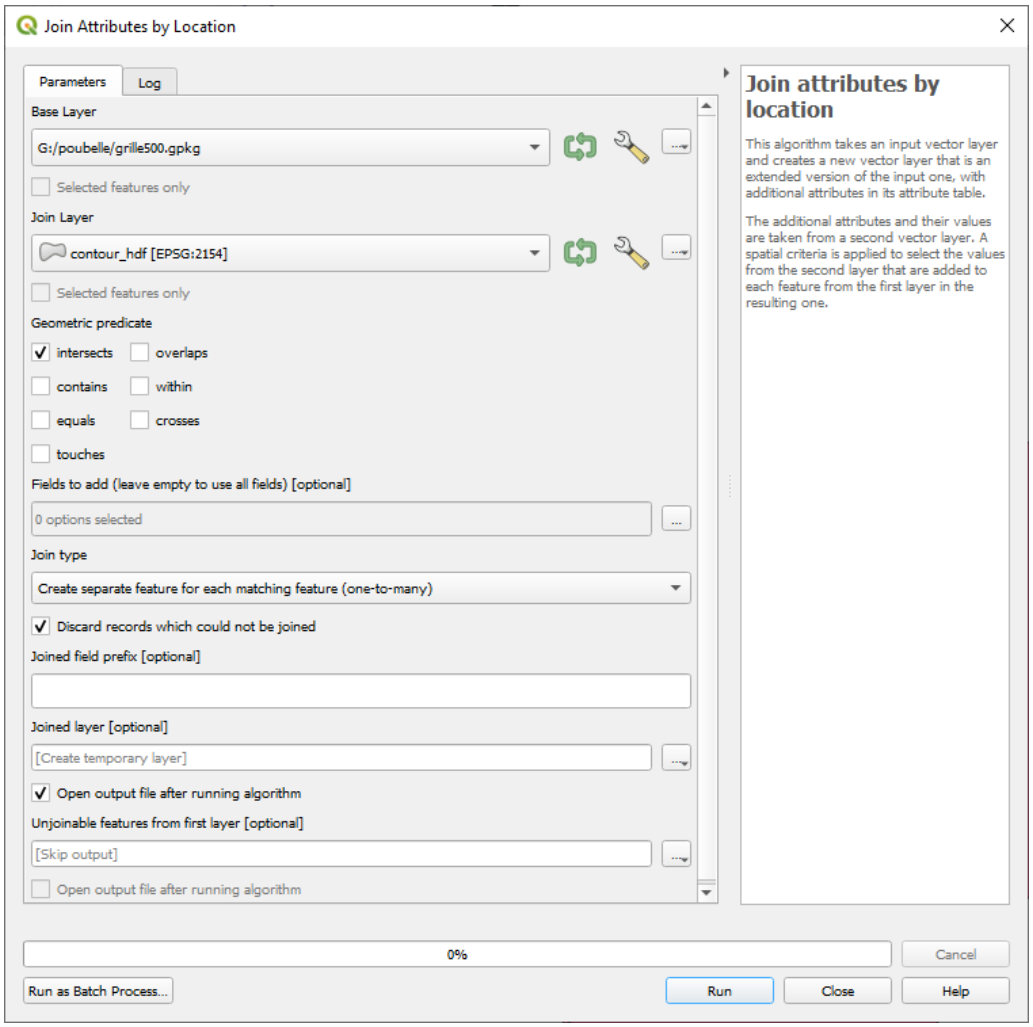
The following script generates a hexagonal grid on a defined area from a loaded layer (Hauts-de-France Region) with a spacing of 500m.



3. Cut the grid according to the outline of the study area

The grid can then be lightened by deleting the cells located outside the study area (here the Hauts-de-France Region) with the script “join attributes by location”



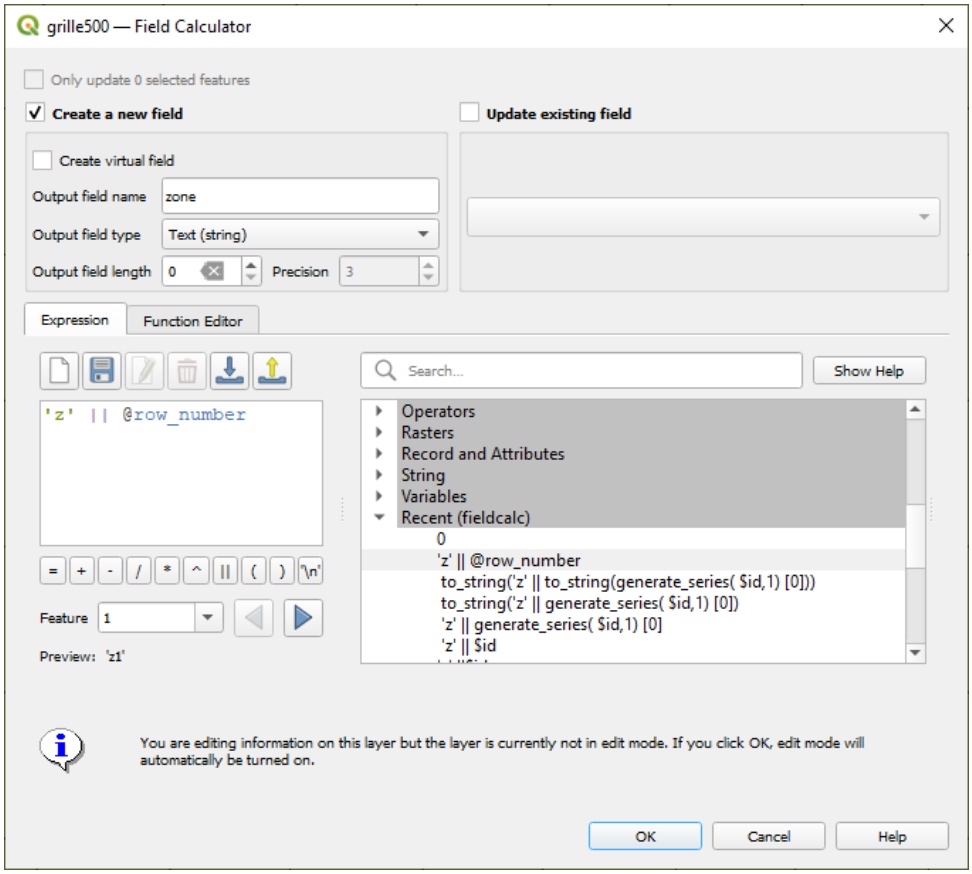


Make sure to check "delete records that cannot be joined".

4. Create a zone identifier

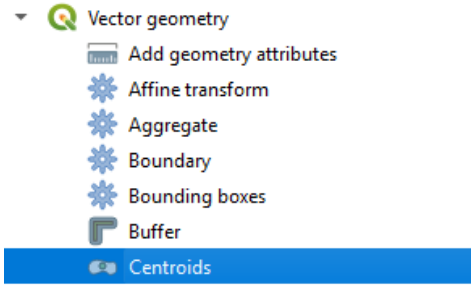
Then you have to add a field in the grid corresponding to a cell identifier (or zone identifier)

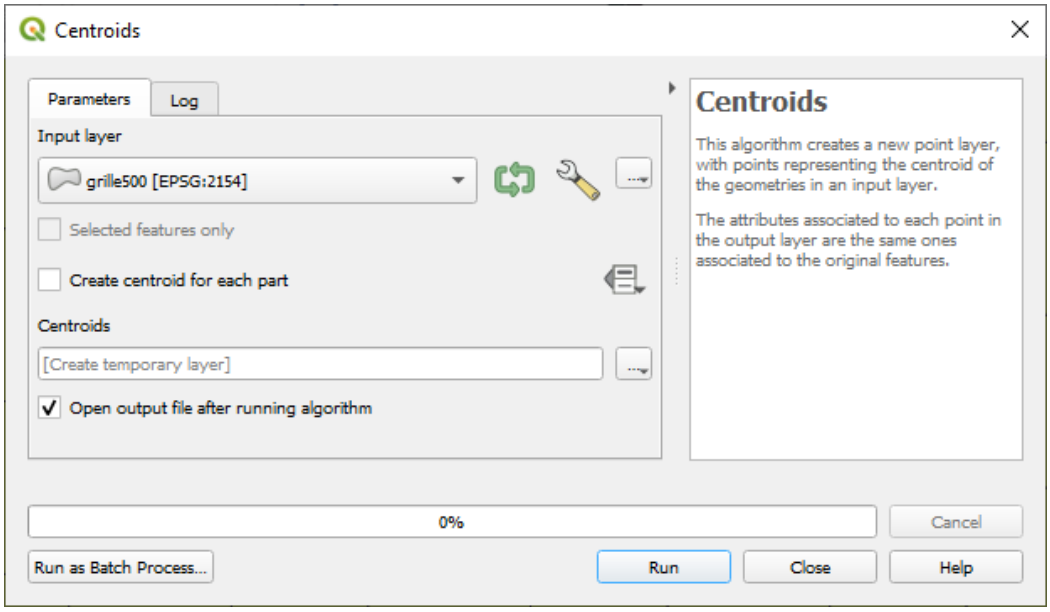
To do this, you must create a text field "zone" for example defined by the expression 'z'+@row_number



5. Make a copy of the grid in centroids

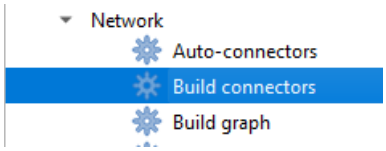
The objective is to have each grid cell connected to the road network. This means treating each cell as a point of interest (POI). To do this, it is necessary to create a layer of points corresponding to the centroids of each cell of the grid. To do this, we must use the "centroids" algorithm.





6. Create the grid - road network connectors

In order to integrate the centroids of the grid cells into the accessibility calculation, it is necessary to generate the connectors between each of the centroids and the nodes of the road network.



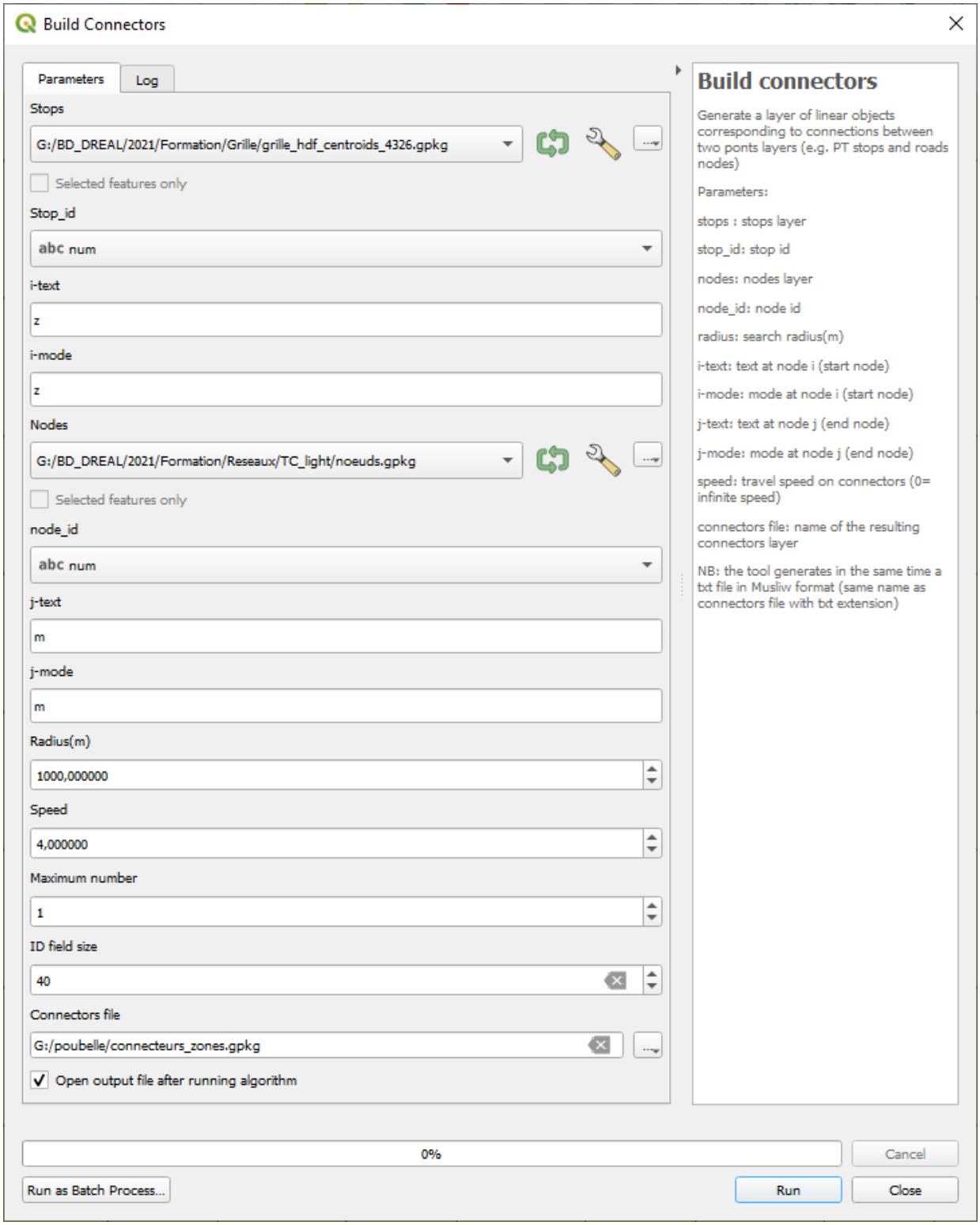
Special attention must be paid to the values entered for "node mode i" and "node mode j".

In the example below, 'z' is entered for 'node mode i' and 'm' for 'node mode j'. This means that the connectors :

- from zone 'z' to road node 'm' will have the type 'zm

- from road node 'm' to zone 'z' will have the type 'mz'.

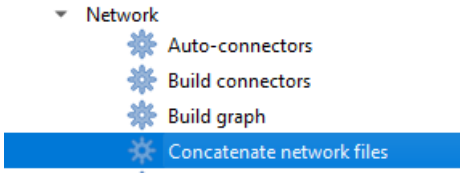
These types will be the ones used in the calculation parameters to filter the results only on the zones and not on the whole network.

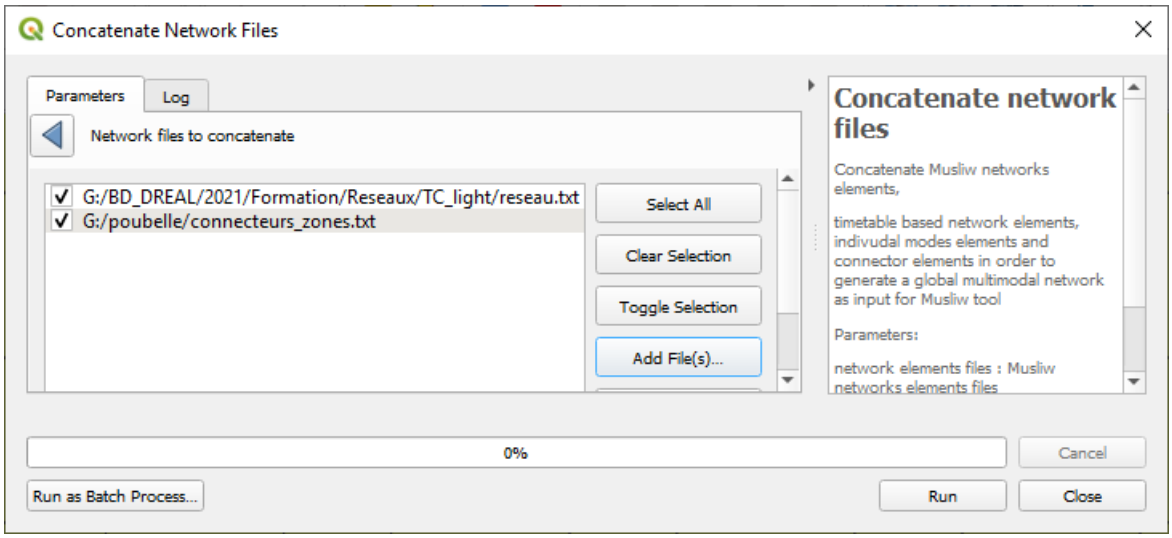

Build Connectors
Parameters
Log
Stops
G:/BD_DREAL/2021/Formation/Grille/grille_hdf_centroids_4326.gpkg
Selected features only
Stop_id
abc num
i-text
z
i-mode
z
Nodes
G:/BD_DREAL/2021/Formation/Reseaux/TC_light/noeuds.gpkg
Selected features only
node_id
abc num
j-text
m
j-mode
m
Radius(m)
1000,000000
Speed
4,000000
Maximum number
1
ID field size
40
Connectors file
G:/poubelle/connecteurs_zones.gpkg
Open output file after running algorithm
Build connectors
Generate a layer of linear objects corresponding to connections between two ponts layers (e.g. PT stops and roads nodes)
Parameters:
stops : stops layer
stop_id: stop id
nodes: nodes layer
node_id: node id
radius: search radius(m)
i-text: text at node i (start node)
i-mode: mode at node i (start node)
j-text: text at node j (end node)
j-mode: mode at node j (end node)
speed: travel speed on connectors (0= infinite speed)
connectors file: name of the resulting connectors layer
NB: the tool generates in the same time a txt file in Musliv format (same name as connectors file with txt extension)
0%
Cancel
Run as Batch Process...
Run
Close

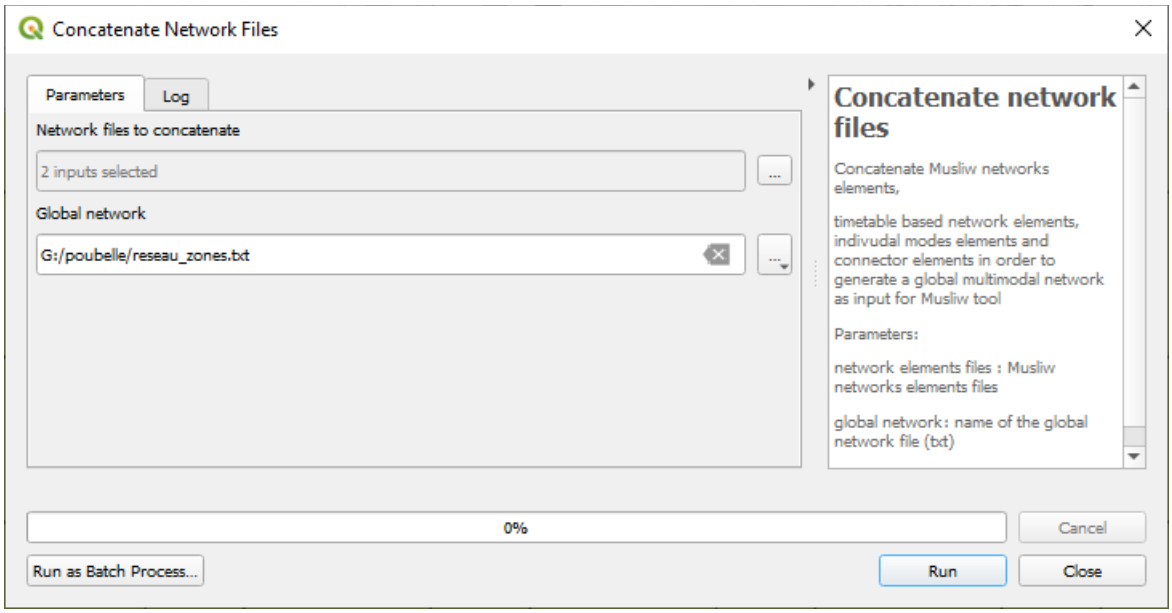
7. Insert connectors into the multi-modal network

The "Build connectors" algorithm generates a table of linear objects corresponding to the connectors (here connectors_zones.gpkg) and the "txt" file of the connectors in Musliw format ready to be integrated into the multimodal network.

To do this, the file connectors_zones.txt (connectors in Musliw format) must be concatenated with the file "reseau.txt" of the multimodal network in Musliw format.

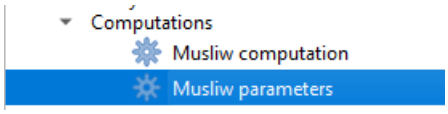






8. Create an adapted parameter set and launch the calculation

To be able to visualize the results, it is necessary to generate a specific parameter set a little bit different from the usual parameter set for the realization of classical isochrones maps.



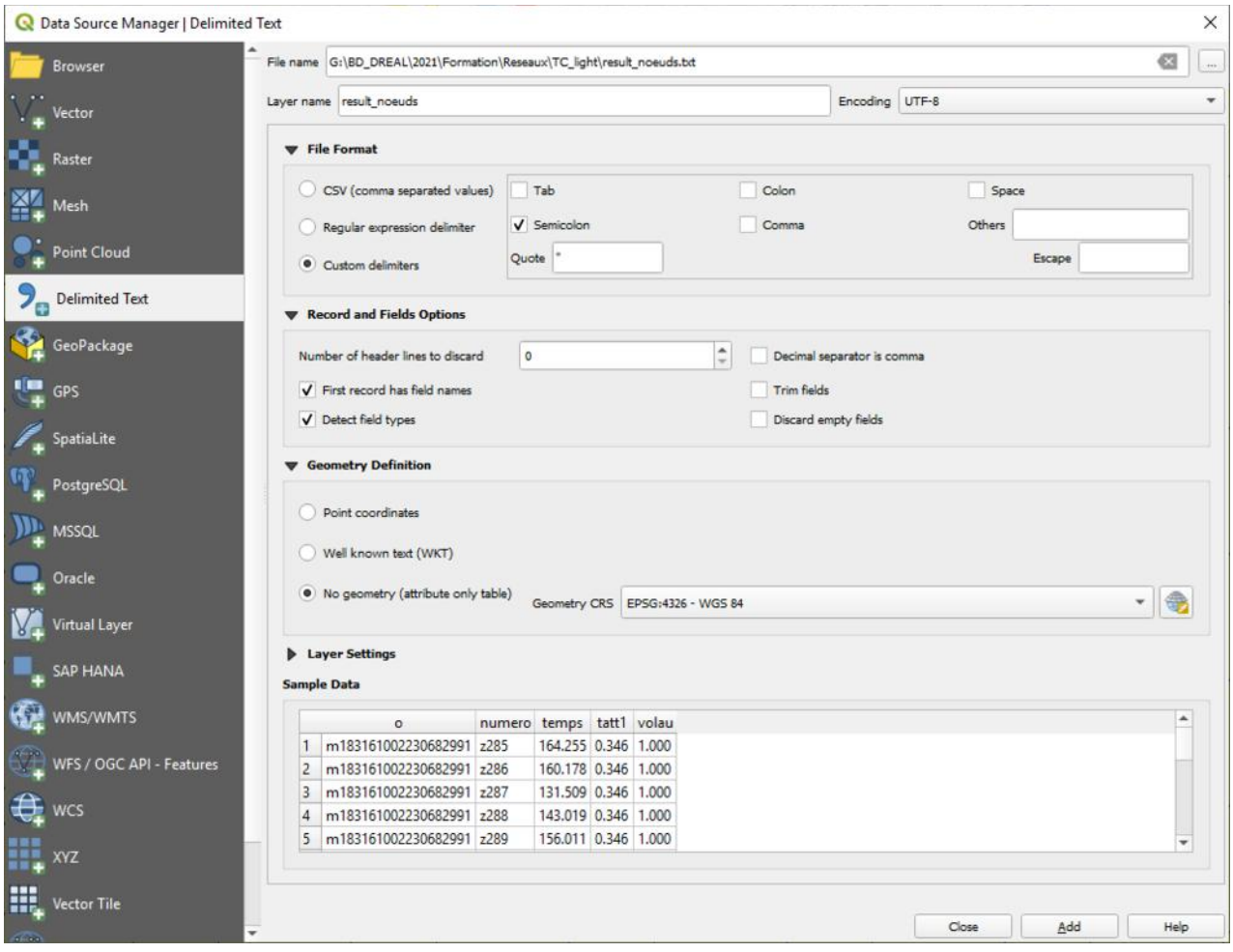
In particular, you should :

- Enable "output nodes?
- Enable "detailed travael times" "compact mode" if you want to reduce the size of the node file
- Set up the "output filter".
  - zm (if the mode identifier is z for zones, and m for the network walk) with an "endpoint" calculation)
  - mz (if the mode identifier is z for the zones, and m for the network on) with a "starting point" calculation)

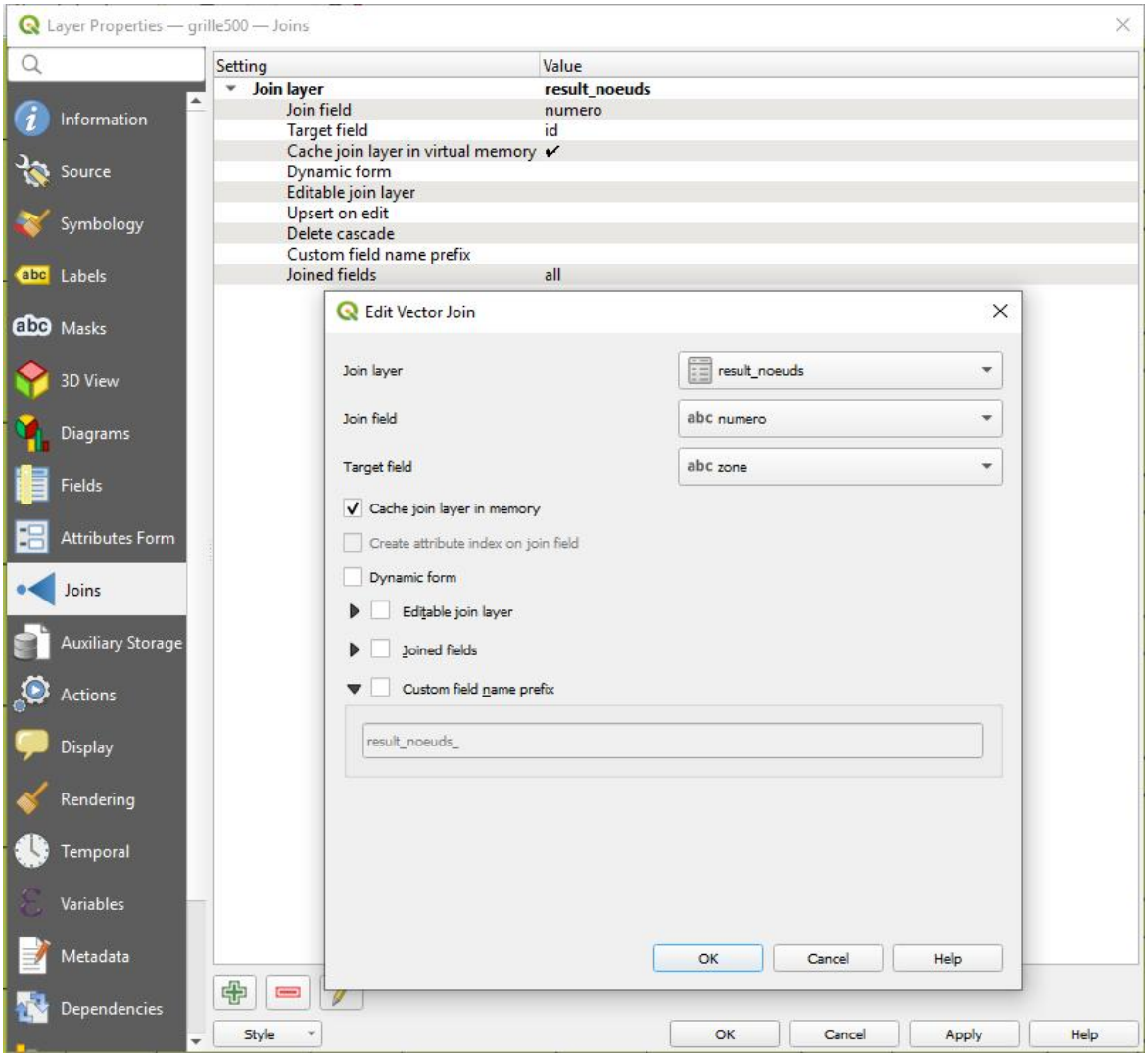
Then, it is necessary to launch the Musliw calculation

9. Visualize the results

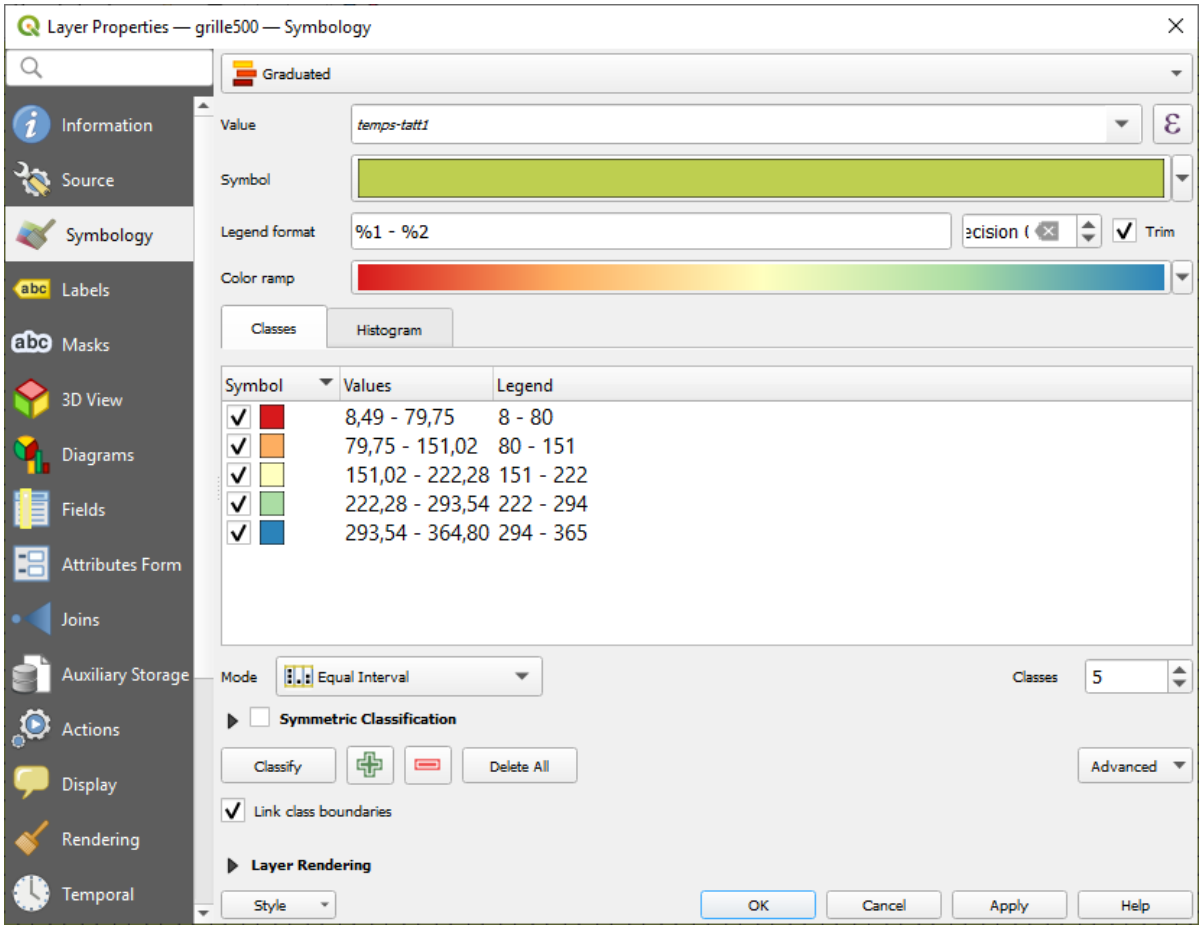
Import the Delimited Text file ..._noeuds.txt



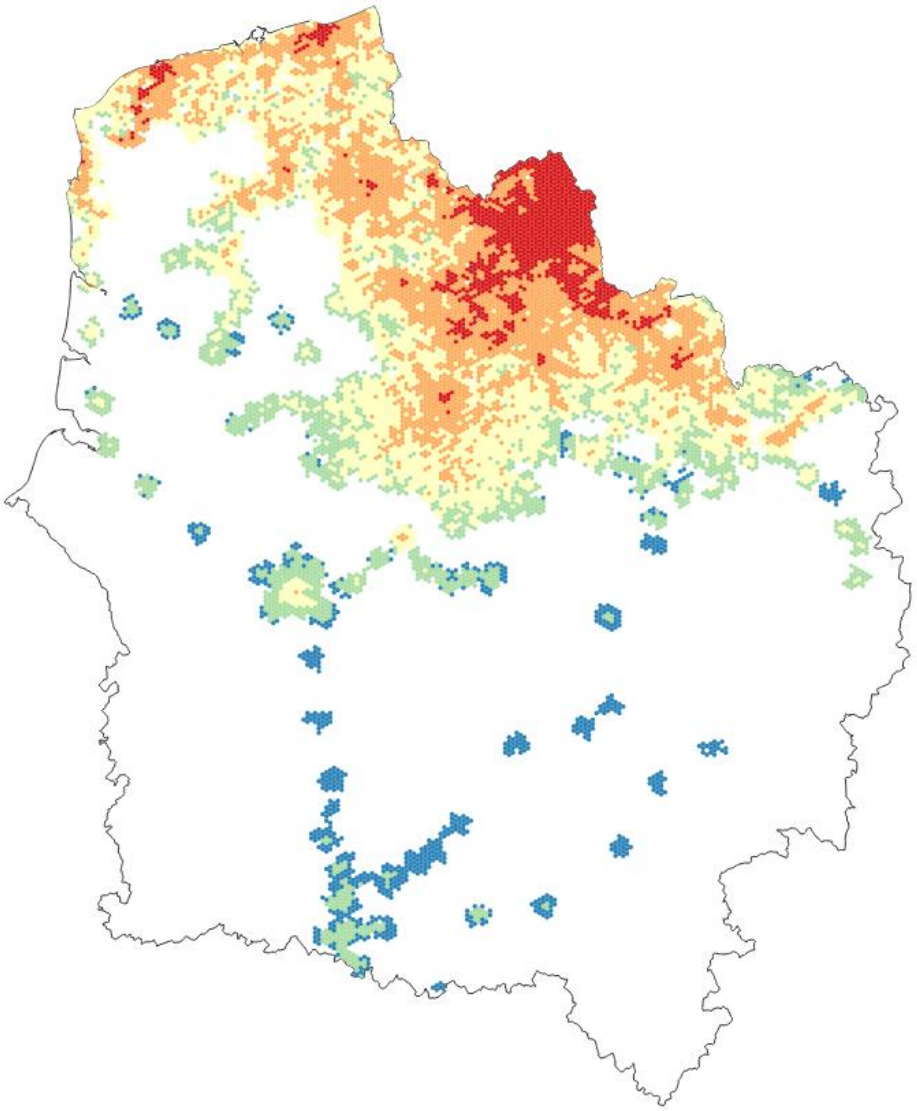
Perform the join with the grid layer



Make a graduated representation on " temps " or " temps -tatt1 ".



We obtain the following map



G- Creation of gravity accessibility maps with or without spatial and/or modal competition

1. Presentation

1.1. Introduction

Unlike accessibility to the nearest service, which provides information on the fastest travel time to access the nearest facility, gravity accessibility makes it possible to construct indicators that take into account the accessibility not of the nearest facility but of all facilities.

Indeed, it is not the same thing to be located 10 minutes from the nearest general practitioner who happens to be the only one within an hour, for example, as it is to be located 10 minutes from the nearest one, but with the presence of one or more others within 15 minutes. The gravity method also makes it possible to determine whether it is more interesting to be located 10 minutes from a single doctor, or 15 minutes from two doctors

1.2. Principle

The principle of gravity accessibility is to give a value or a weight to each equipment (e.g. to each general practitioner) which depends on its accessibility. Gravity accessibility then consists in summing up the weights of each equipment. The value of the weight is defined by a decreasing function of the distance called the conductance or resistance function (distance, access time, generalized access time, cost, etc.).

1.3. Function used for accessibility by gravity

For accessibility to services, the principle chosen is as follows:

- We want a service on the spot (t=0) to count as 1

- A service not accessible or at an infinite time counts as 0

- A service located at t=t0 counts as 0.5

The choice of t0 is a parameter defined by the user. In the example we have taken it equal to the average time observed in the Household Travel Surveys according to the mode of transport:

- Car: t0=15 minutes

- Public transport: t0=30 minutes

Ideally, these parameters should be calibrated on the basis of survey results that make it possible to assess the attractiveness of one facility in relation to another, from the standpoint of accessibility alone.

In this study, the resistance function used is of the form $2^{-\left(\frac{t}{t_0}\right)^n}$

1.4. Required data

To calculate the gravity accessibility for a service in a territory, it requires :

- A table of areas of the territory (polygons or grid) that contains data on the population(s) for which we are looking for territorial accessibility to the service

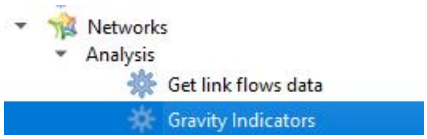
- A table of facilities (amenities representing the attractiveness of the service, e.g. table of doctors, table of schools, jobs). This table contains a column characterizing the attractiveness of the service (e.g.: number of pupils for schools, number of jobs, '1' for doctors if we do not know their average number of consultations)

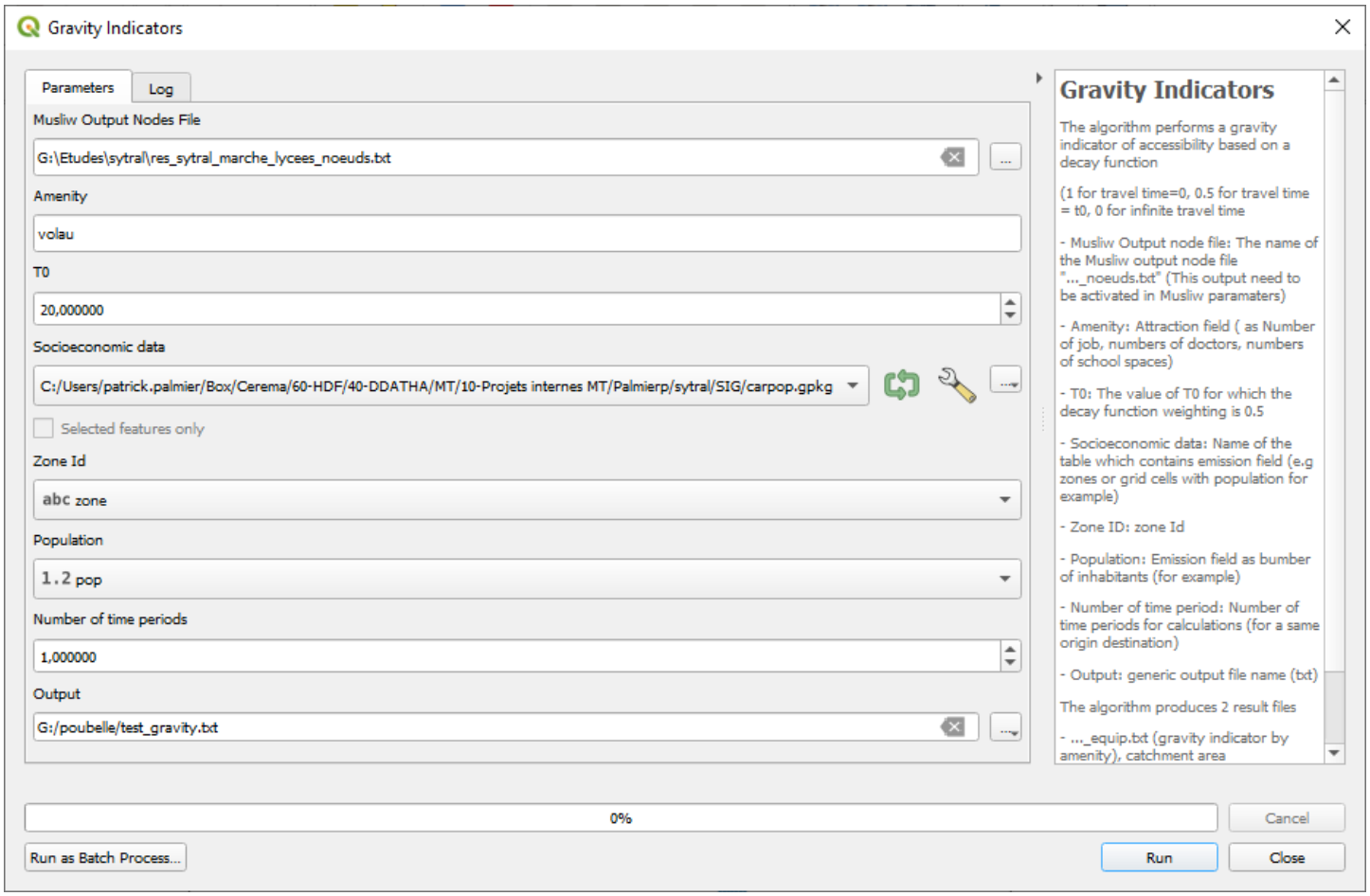
- An accessibility calculation based on a matrix between each generator and each population zone.

2. Gravity indicators with or without spatial competition

For the gravity calculation, it is necessary to check that in the parameterization file, the output 'Nodes file' is activated and that in the 'output filter' only the sections connecting the zones to the road network (outbound or inbound) are activated, depending on whether the calculation is done with a starting point or an end point.

Depending on the size of the matrix (at least nb zones*nb equipments * nb hours per od) the script will, based on the result node file, calculate gravity indicators taking into account the socio-economic data of the concerned population.





The script produces two result files of delimited text type with ";" as separator.

- A file ..._zones.txt: This file contains the gravity accessibility indicators related to each zone with the following fields:
  - zone: zone identifier
  - nb: number of DOs that contributed to the calculation of the indicator
  - n_tot: number of accessible amenities without taking into account spatial competition
  - w_n: gravity accessibility indicator with spatial competition without taking population into account
  - w_pop: gravitational accessibility indicator with spatial competition with population taken into account (nb of amenities per capita). In the example below, this is the number of college seats per capita.

zone	nb	w_n	n_tot	w_pop
Z2	117	0.444	1'909	0.00810
Z3	117	0.303	1'263	0.00562
Z4	117	0.334	1'405	0.00618
Z5	117	0.295	1'228	0.00548
Z6	117	0.318	1'322	0.00594
Z7	117	0.370	1'563	0.00690
Z8	117	0.400	1'701	0.00743
Z9	104	0.035	126	0.00086
Z10	107	0.038	137	0.00092
Z11	117	0.165	677	0.00337
Z12	117	0.183	755	0.00372
Z13	117	0.160	653	0.00326
Z14	117	0.332	1'421	0.00654
Z15	117	0.331	1'417	0.00652
Z16	117	0.323	1'339	0.00590
Z17	117	0.332	1'389	0.00610
Z18	117	0.396	1'688	0.00727
Z19	117	0.342	1'438	0.00631
Z20	117	0.386	1'641	0.00709
Z21	117	0.376	1'595	0.00691
Z22	117	0.392	1'671	0.00718
Z23	117	0.280	1'153	0.00526
Z24	117	0.435	1'866	0.00806

- A file ..._equip.txt : This file contains the gravity accessibility indicators relative to each of the equipments
  - equip: equipment identifier
  - nb: number of DOs that contributed to the calculation of the indicator
  - pop: total population of areas that contributed to the calculation of the indicator
  - w_n: gravity accessibility indicator with spatial competition without taking population into account.
  - w_pop: gravity accessibility indicator with spatial competition with population taken into account (catchment area)
  - w_pop2 : population by amenity (by doctor, by job, by college place, by equipment, ...). . In the example below, this is the number of inhabitants per college place

- vol : volume of amenities of the equipment (here number of college places)

equip	nb	pop	w_n	w_pop	w_pop2	vol
0690636S	45942	1'351'589	1'072	39'900	4.0	81
0690514J	46271	1'353'322	6'753	484'244	25.3	646
0690653K	46839	1'355'298	7'334	487'667	28.0	245
0690032K	46680	1'354'660	8'195	573'483	29.4	858
0691626T	46179	1'352'939	3'072	215'994	12.2	294
0690281F	45780	1'352'187	4'550	300'998	17.7	405
0690698J	46654	1'354'617	6'876	486'064	25.0	562
0690554C	46764	1'354'966	5'695	467'076	23.6	530
0011276G	1673	96'284	114	387	0.8	1261
0690125L	46866	1'355'225	5'137	412'216	19.5	351
0691724Z	46836	1'355'284	7'502	551'326	27.6	166
0690651H	46619	1'354'702	6'728	508'744	25.0	492
0693330V	43923	1'345'727	985	18'615	14.2	1141
0690035N	45895	1'352'216	5'598	419'154	21.4	943
0690563M	46391	1'353'739	4'370	346'552	16.3	608
0694197M	45690	1'350'843	2'411	124'537	9.4	137
0693619J	46733	1'354'408	3'281	255'085	13.6	816
0692968B	46722	1'354'785	4'827	303'898	19.6	303
0690027E	46622	1'354'728	7'496	564'146	27.9	1083
0693045K	46441	1'353'837	3'192	244'540	13.2	361
0693368L	46802	1'354'954	7'971	556'225	29.6	34
0693373S	45846	1'351'953	5'870	490'006	23.1	375
0690712Z	46391	1'353'739	4'370	346'552	16.3	156
0690642Y	46339	1'353'635	5'751	439'282	21.9	118
0693478F	46621	1'354'391	4'413	170'467	15.3	1254
0693200D	43699	1'344'959	927	17'758	13.4	270
0690109U	46379	1'353'997	4'859	418'424	20.6	284
0690103M	46252	1'353'124	5'500	430'451	22.3	1043
0690522T	46165	1'353'006	6'948	491'457	24.6	1405
0690074F	45591	1'351'483	4'387	297'859	16.8	925
0693974V	46572	1'354'004	6'081	462'877	24.2	94
0693446W	45651	1'351'909	4'215	318'111	16.2	835
0691680B	45690	1'350'843	2'411	124'537	9.4	380

3. Gravity indicators with spatial and modal competition

The gravity indicators with spatial and modal competition are a generalization of the above indicators. In addition to spatial competition for access to facilities, the calculation also incorporates modal competition between several subpopulations that differ in their use of transport modes.

In order to calculate this indicator, the area layer must contain columns corresponding to the subpopulations to be studied.

Here in the example, the population is divided into 3 subpopulations in the following columns

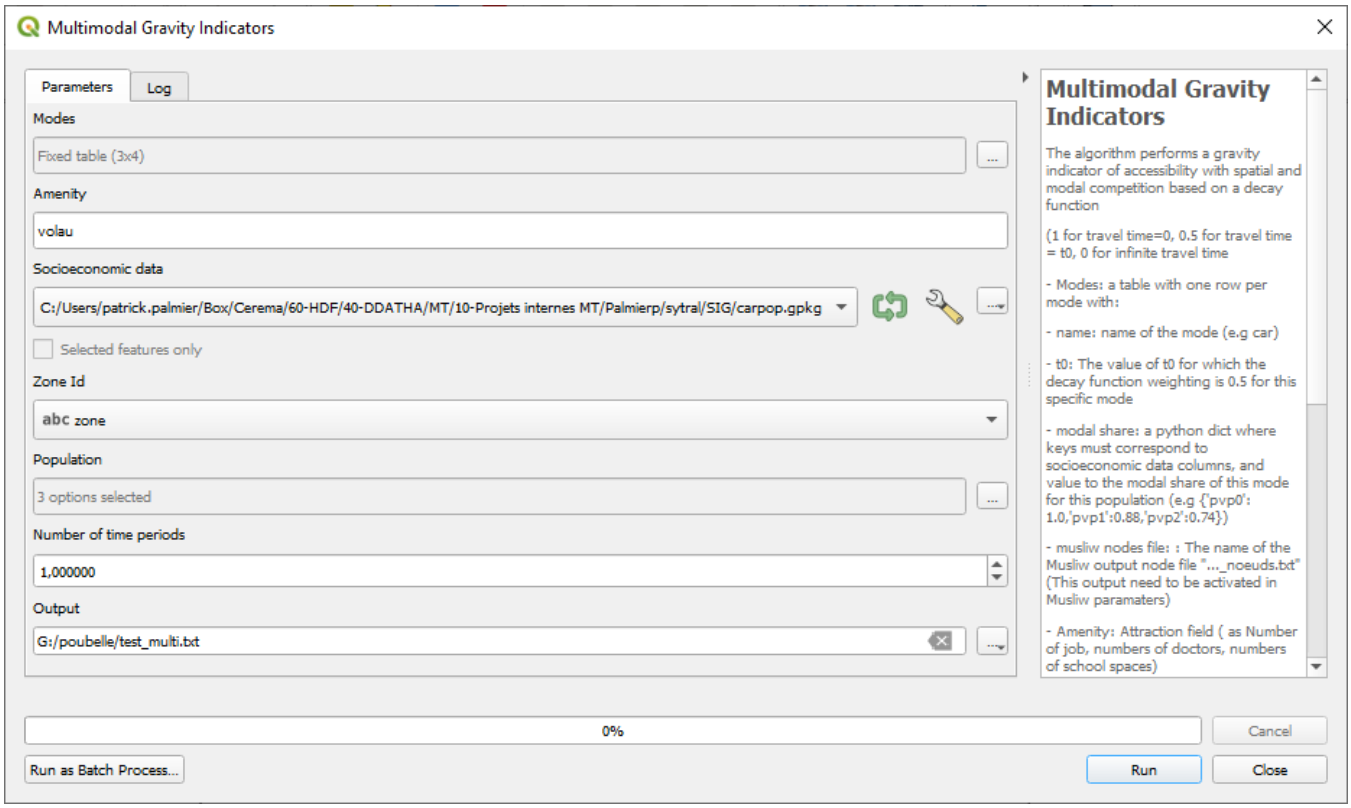
pvp0: the population of households without cars

pvp1 : the population of households with one car

pvp2: the population of households with 2 or more cars

For each of the sub-populations, we need to know the modal split of the trips (here for the home-college pattern) if we are interested in the accessibility to the colleges.

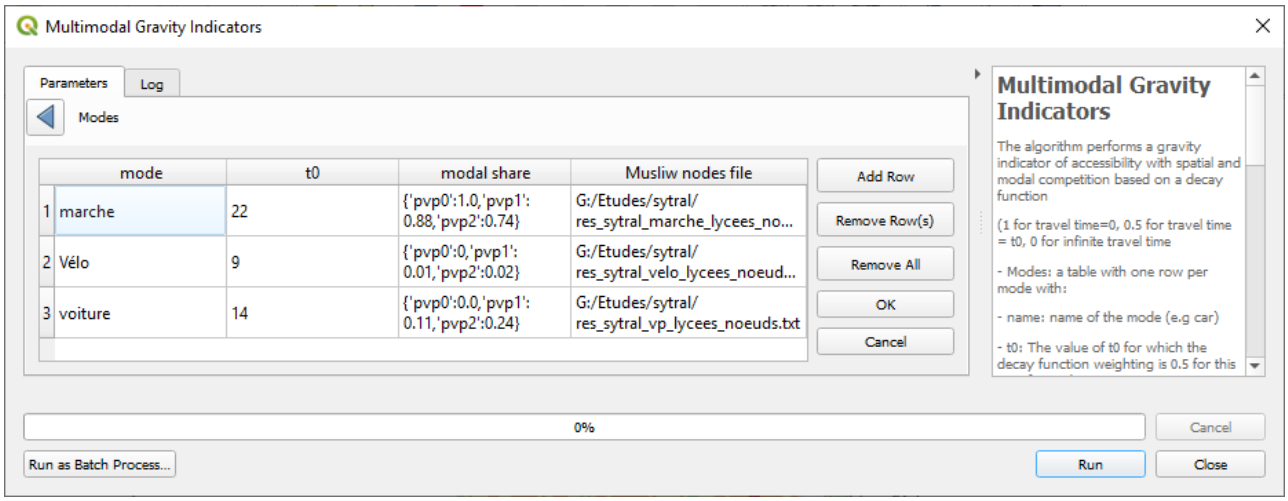
	Vélo	TC+marche	Voiture
pvp0	0%	100%	0%
pvp1	1%	88%	11%
pvp2	2%	74%	24%



The mode table must be filled in as follows:

- mode : The mode label
- t0 : The value of the time in minutes for which the resistance function is 0.5
- modal share: The modal share of the mode for each of the subpopulations (in the form of a Python dictionary where the keys are the column names of the subpopulations)
- Musliw node file: Musliw node file of the gravity indicator of the mode in question (see G 2)

It is therefore necessary to have previously calculated the gravity indicators for each of the modes (see G 2)



The script produces two result files of delimited text type with ";" as separator.

- A file ...multi_zones.txt: This file contains the gravity accessibility indicators related to each zone with the following fields:
  - zone: zone identifier
  - nb: number of DOs that contributed to the calculation of the indicator
  - pop: total population of the area
  - n_tot: gravity accessibility indicator with spatial competition without taking population into account
  - w_pop: gravity accessibility indicator with spatial and modal competition with population taken into account (nb of amenities per capita). In the example below, this is the number of college seats per capita.
  - pvp0,pvp1,pvp2: This is the w_pop indicator but relative to each subpopulation. The name is that of the corresponding column name in the area table.

zone	nb	n_tot	w_pop	pvp0	pvp1	pvp2
Z2	1072	233	0.0178	0.0281	0.0123	0.0196
Z3	1072	143	0.0105	0.0150	0.0072	0.0125
Z4	1072	158	0.0119	0.0177	0.0082	0.0137
Z5	1072	139	0.0101	0.0144	0.0070	0.0122
Z6	1072	151	0.0114	0.0169	0.0079	0.0133
Z7	1072	272	0.0186	0.0224	0.0127	0.0244
Z8	1069	290	0.0201	0.0255	0.0138	0.0258
Z9	1018	51	0.0024	0.0008	0.0016	0.0043
Z10	1027	51	0.0024	0.0008	0.0016	0.0043
Z11	1057	71	0.0045	0.0049	0.0031	0.0061
Z12	1057	76	0.0050	0.0059	0.0034	0.0066
Z13	1057	69	0.0044	0.0046	0.0030	0.0060
Z14	1057	157	0.0114	0.0164	0.0079	0.0133
Z15	1057	156	0.0114	0.0163	0.0079	0.0132
Z16	1072	148	0.0102	0.0149	0.0070	0.0120
Z17	1072	155	0.0111	0.0161	0.0077	0.0132
Z18	1072	205	0.0154	0.0234	0.0106	0.0175
Z19	1072	175	0.0126	0.0181	0.0087	0.0151
Z20	1072	179	0.0139	0.0221	0.0096	0.0152
Z21	1072	174	0.0134	0.0212	0.0093	0.0148
Z22	1072	192	0.0146	0.0227	0.0101	0.0163
Z23	1072	137	0.0099	0.0137	0.0068	0.0122
Z24	1069	313	0.0223	0.0296	0.0153	0.0277
Z25	1072	325	0.0233	0.0316	0.0160	0.0286
Z26	1057	24	0.0023	0.0031	0.0016	0.0027
Z27	1057	61	0.0037	0.0033	0.0025	0.0054
Z28	997	50	0.0023	0.0006	0.0015	0.0042
Z29	997	50	0.0023	0.0006	0.0015	0.0042
Z30	1057	69	0.0043	0.0046	0.0030	0.0060

- A file ..._equip.txt : This file contains the gravity accessibility indicators relative to each of the equipments
  - equip: equipment identifier
  - nb: number of DOs that contributed to the calculation of the indicator
  - pop: total population of areas that contributed to the calculation of the indicator
  - nb: number of DOs that contributed to the calculation of the indicator
  - w_pop: gravity accessibility indicator with spatial competition with population taken into account (catchment area)
  - pvp0,pvp1,pvp2: gravity accessibility indicator with spatial competition with population subpopulations taken into account (catchment area by subpopulation)(column names identical to those of the zones table)
  - w_pop2 : population by amenity (by doctor, by job, by college place, by equipment, ...) . In the example below, this is the number of inhabitants per college place

- w_pop_pvp0, w_pop_pvp1, w_pop_pvp2: sub-population by amenity (per doctor, per job, per college place, per facility, ...). . In the example below, this is the number of inhabitants per college place for each population category
- vol : volume of amenities of the equipment (here number of college places)
-

equip	pvp0	pop_pvp0	w_pop_pvp0	pvp1	pop_pvp1	w_pop_pvp1	pvp2	pop_pvp2	w_pop_pvp2	nb	pop	w_pop	w_pop2	vo
0691614E	4'588	257'118	0.29	15'276	625'589	2.80	14'934	422'024	5.40	409'125	1'304'731	34'799	8.48	
0690546U	8'721	257'099	0.34	16'828	625'541	2.33	11'614	422'064	5.23	418'554	1'304'704	37'163	7.90	
0691675W	49'555	257'114	3.03	92'009	625'671	7.56	32'019	422'310	3.57	422'928	1'305'095	173'583	14.16	
0692337R	29'179	257'109	1.77	62'013	625'679	6.33	30'949	422'281	5.63	417'258	1'305'070	122'141	13.73	
0692898A	5'411	257'117	0.32	15'136	625'562	2.66	12'983	421'911	4.61	401'733	1'304'591	33'530	7.59	
0011338Z	0	7	0.00	406	73'290	0.52	1'252	106'406	1.81	220'266	179'702	1'659	2.32	
0691484N	8'574	257'038	0.62	21'799	625'414	3.58	14'641	422'222	4.59	423'201	1'304'674	45'015	8.79	
0691497C	6'391	257'112	1.12	23'487	625'590	6.08	20'623	422'041	6.89	410'034	1'304'742	50'500	14.09	
0692159X	9'800	257'104	1.80	24'766	625'562	6.31	16'182	422'104	6.18	418'653	1'304'770	50'749	14.29	
0691650U	1'853	257'028	0.07	7'331	625'058	1.56	8'540	421'349	3.90	396'444	1'303'436	17'724	5.53	
0690621A	8	251'153	0.00	182	601'588	0.25	495	393'411	0.82	345'693	1'246'152	685	1.07	
0691479H	33'639	257'110	2.12	71'862	625'689	7.79	36'502	422'303	7.07	417'654	1'305'102	142'003	16.98	
0693365H	3'492	257'030	0.18	9'019	625'356	1.29	6'766	422'085	2.43	421'092	1'304'471	19'276	3.90	
0011151W	15	189'537	0.00	1'141	452'564	0.71	1'864	294'995	1.57	323'019	937'096	3'020	2.28	
0694188C	73'206	257'120	5.32	126'643	625'708	11.48	46'766	422'378	6.96	425'793	1'305'206	246'615	23.76	
0691481K	9'744	256'960	0.60	21'005	625'023	3.01	14'286	421'582	4.79	418'869	1'303'565	45'035	8.40	
0693890D	2'259	257'109	0.21	7'221	625'642	1.57	6'029	422'169	2.58	412'635	1'304'920	15'509	4.36	
0690002C	0	3	0.00	512	71'738	0.61	1'534	105'251	2.20	209'235	176'991	2'047	2.80	
0691664J	46'302	257'118	3.05	92'550	625'700	8.59	36'308	422'340	4.82	422'841	1'305'159	175'159	16.46	
0690624D	112	256'532	0.00	2'837	622'772	1.46	5'429	418'455	3.79	380'898	1'297'759	8'379	5.26	
0692942Y	21'920	257'037	1.22	42'817	625'435	4.71	20'785	422'278	4.52	423'858	1'304'750	85'523	10.45	
0691663H	95'294	257'119	7.61	165'245	625'712	14.90	57'114	422'360	6.47	423'735	1'305'191	317'653	28.98	
0691669P	35'214	257'023	2.16	68'566	625'392	7.48	32'150	422'186	6.72	419'313	1'304'601	135'930	16.35	
0694185Z	23'980	257'041	1.73	57'228	625'294	8.56	36'329	421'949	10.74	412'701	1'304'284	117'537	21.03	
0693286X	2	237'542	0.00	401	564'247	0.75	1'481	362'984	3.11	316'308	1'164'772	1'885	3.86	
0694317T	856	256'880	0.11	3'339	624'740	0.98	3'229	420'911	1.87	405'396	1'302'532	7'423	2.96	
0692583H	1'262	256'990	0.65	4'136	622'266	2.48	3'660	416'526	3.09	337'638	1'295'782	9'058	6.21	
0692390Y	64'499	257'117	4.46	116'605	625'680	10.22	42'470	422'318	5.12	421'842	1'305'114	223'574	19.80	
0693479G	34'473	257'103	1.89	59'220	625'643	4.98	22'780	422'276	3.71	422'121	1'305'023	116'472	10.58	
0693995T	38'395	257'113	2.13	72'799	625'657	7.26	34'918	422'292	7.12	424'773	1'305'062	146'112	16.51	
0692520P	5'208	256'978	0.28	11'805	624'972	1.53	7'743	421'443	2.27	407'508	1'303'393	24'756	4.08	
0690599B	74'858	257'111	5.18	130'499	625'654	10.72	44'108	422'278	4.64	421'827	1'305'044	249'465	20.54	
0692343X	17'632	257'108	2.18	35'872	625'638	5.95	17'622	422'223	4.75	417'636	1'304'968	71'126	12.89	
0692340U	41'461	257'113	2.64	76'038	625'669	7.04	29'383	422'296	4.87	421'575	1'305'078	146'882	14.55	
0690551Z	39'582	257'116	2.90	72'098	625'705	7.50	28'355	422'350	4.78	424'515	1'305'171	140'035	15.18	
0692703N	65'439	257'108	4.59	114'619	625'655	10.54	47'080	422'294	7.49	422'685	1'305'057	227'138	22.62	
0693287Y	4'260	257'115	0.46	16'067	625'636	4.08	16'042	422'148	6.63	414'093	1'304'898	36'369	11.17	
0692419E	1'743	256'967	0.85	6'674	624'207	3.85	7'546	419'534	5.39	368'418	1'300'708	15'964	10.08	
0694312M	3'701	257'118	0.17	10'889	625'540	2.33	10'197	421'785	4.54	398'109	1'304'443	24'788	7.04	
0692335N	2'750	256'995	0.32	12'613	625'141	3.40	13'824	421'686	6.21	408'519	1'303'821	29'187	9.93	
0692920Z	87'991	257'120	6.92	152'592	625'726	14.14	58'983	422'391	8.25	425'958	1'305'238	299'566	29.31	

H- Using processing models

1. Introduction

The advantage of developing the "networks" extension in the form of an algorithm provider is the ability to create complex processing models by combining different algorithms and also to carry out batch processing. To make things easier for users, processing models have been developed and made available to perform the most frequent complex tasks.

These models are as follows (sorry in french):

- Isochrone: Drawing a monosite isochrone
- Isochrone multiples: Drawing of a multi-site isochrone
- Réseau individuel choix référence comple: Drawing of catchment areas
- Individual network full reference choice: Generation of a road network from OSM over the study area (exhaustive network)
- Réseau individuel choix référence light: Generation of a road network from OSM on study area (simplified network)
- Réseau individuel fenêtre complet: Generation of a road network from OSM on the study area defined by the visible canvas (exhaustive network)
- Réseau individuel fenêtre light: Generation of a road network from OSM on the study area defined by the visible canvas (simplified network).
- Réseau Transport en commun: Generate a public transport network from GTFS data and connectors to the road network.

2. Installation

- Go to Preferences/User profile/Open active profile folder to open the corresponding folder
- Then go to python/plugins/networks and copy the .model3 files into the processing/models directory from the active profile folder.
- Deactivate and reactivate the processing extension so that models are taken into account

3. Running the model



Go to the edit box in the Templates/Cerema menu and double-click on the template you want to run.

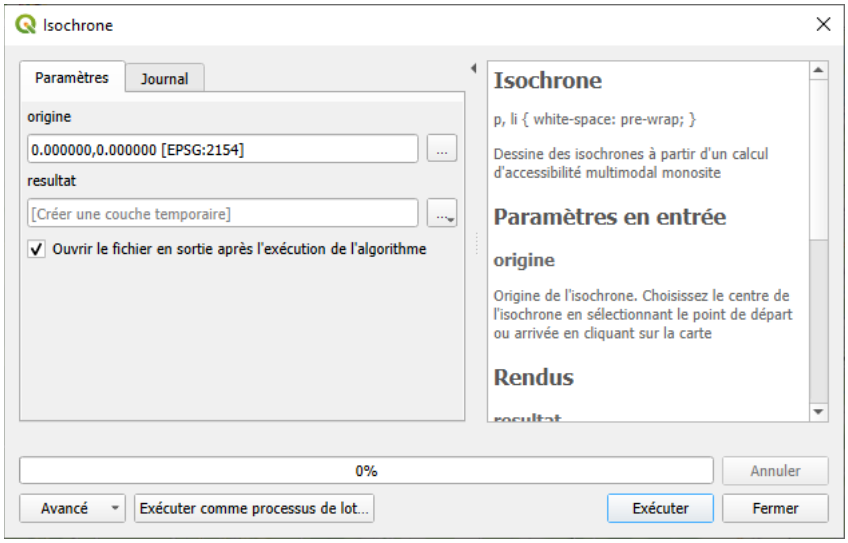
Enter the required fields and run the procedure.

Only the essential parameters are requested.

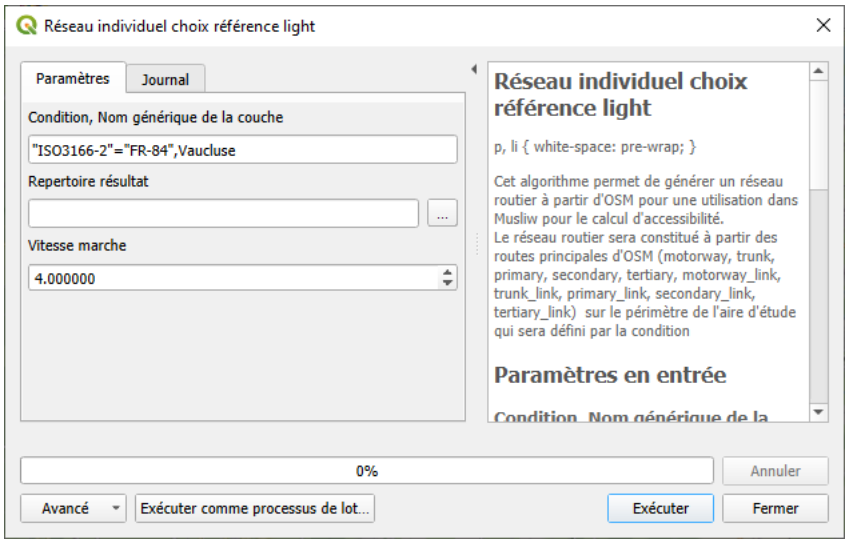
However, you can customize the template by editing it. This allows you to change the value of the various algorithm parameters.

It is also possible to add entries to the dialog box to ask the user to specify one or more other parameters.

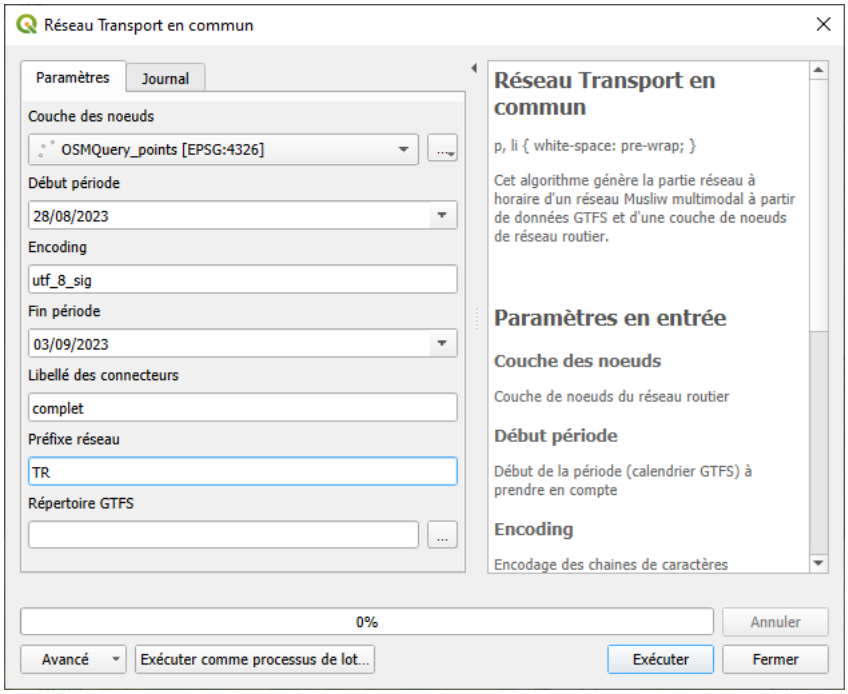
Ex: Isochrone dialog box



Ex: individual network by reference light dialog box



Ex: Public transport network dialog box



4. Description of model outputs

- Isochrons and multiple isochrons:
  The model produces a layer of iso-value polygons representing the isochrons by time step (default 0 to 120 minutes in 10-minute steps). These values can be customized by modifying the parameters of the "iso-value contours" algorithm.

- Catchment area :
  The model produces a layer of polygons representing the catchment area of the nearest point in terms of multimodal accessibility (network indicated in the Musliw calculation) of each hub indicated in the matrix (departure or arrival point).

- Individual networks :
  These models generate the following files:

- A table of road arcs with the addition of the following columns (direction, diffusion, dead end, length, time_m, i, j , ij)
- The corresponding road network file in Musliw format (.txt)

- Public transport network :
  - a GIS file of nodes: a layer of point objects made up of stop points with information on the name of the stop and the frequency of use
  - an GIS file of arcs: a linear object file representing a line linking two consecutive stops, with indications of the initial node number and final node number.
  - a GIS file of lines: a file of linear objects made up of lines linking two consecutive stops on the same public transport line, with the initial node number, the final node number and the line number indicated. The layer also contains frequency information.
  - a public transport network file in Musliw format (.txt)
  - a GIS file of connectors between public transport stops and road network nodes
  - a file of connectors in Musliw format (.txt)

I- Calculating nodal potential and intensity

1. Introduction

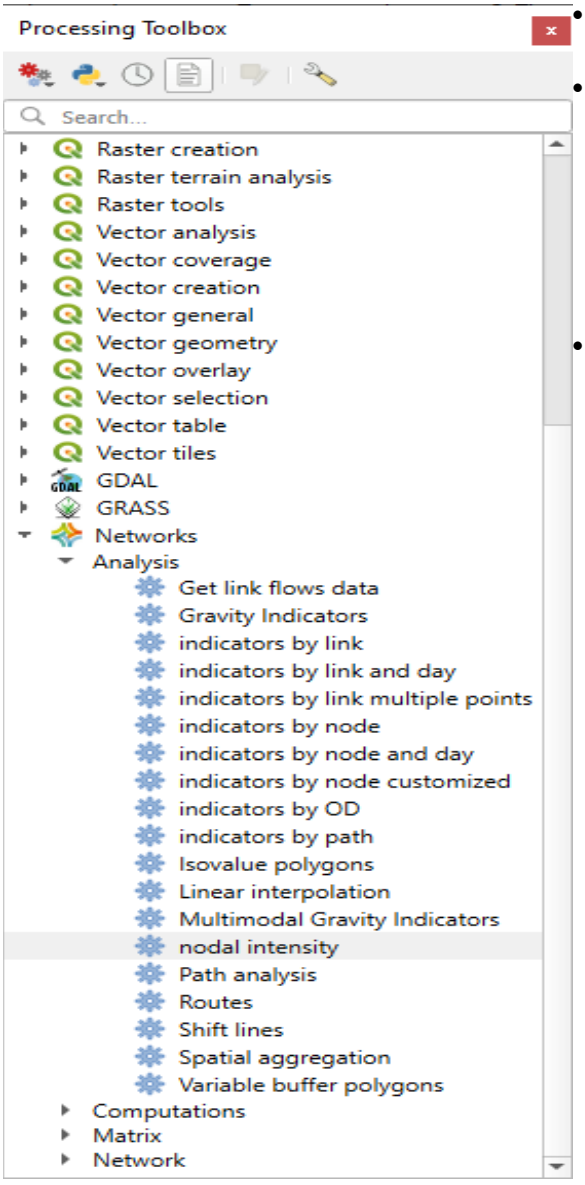
The "Networks" tool can be used to determine the intermodal performance of hubs by calculating nodal potential and nodal intensity indicators. These indicators were defined by Beaucire in 2004 and then developed by Richer and Vuidel in 2013.

The nodal potential of an interchange is the theoretical number of connections that can be made in a day. It expresses the maximum potential number of connections offered by the interchange.

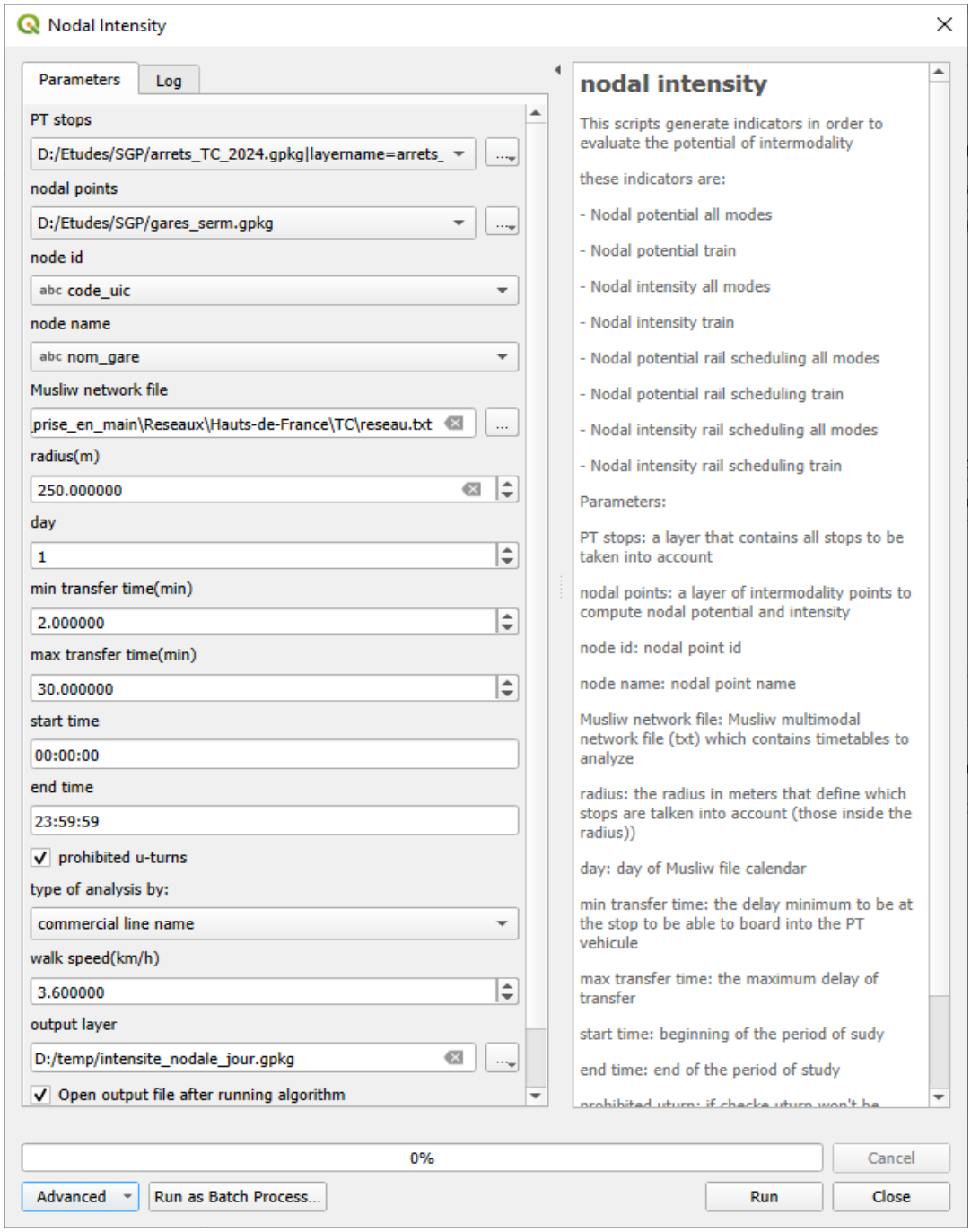
Nodal intensity is the proportion of nodal potential that can be achieved within a given waiting time. This index reflects the rate of possible connections within a given timeframe.

2. Procédure

To calculate these indicators, simply run the "Nodal intensity" script in the analyses.0 group.
You will need the following data:



- A table of point objects corresponding to the intermodal nodes you want to study
- A table of public transport stops. This table must contain all the stops whose timetables you want to take into account for your analysis. This stop table must contain a stop identifier called "i". You can directly use the stop tables generated by the tool's "Import GTFS V2" script and combine the stops from the different networks you want to take into account in a single layer.
- A network file in Musliw format. This file must be a network file in Musliw (txt) format and must contain arcs with timetables. The numbering of the stops in the Musliw network file must be consistent with that in the TC stop file.



It is up to you to define :

- The analysis time period
- The minimum connection time. This determines how many minutes the user must be at the stop before the vehicle leaves in order to be able to use it.
- Maximum transfer time. This determines the maximum acceptable time for making a connection.
- Walking speed as the crow flies.
- Intermodality radius. This determines which stops and their associated services to take into account when calculating the indicators. The centre is determined by the coordinates of the nodal point to be studied.
- The possibility of calculating by commercial line name or by mission, as well as authorising or not authorising U-turns on the same line.

3. Results

The script generates a table of points representing the nodal poles with the various nodal potential and intensity indicators

- potential_tot: The total nodal potential combining all the possible transports described in the data

- intensite_tot : The nodal intensity corresponding to the total potential
- train_potential: The total nodal potential, but only for connections that include at least part by train
- intensite_train: The nodal intensity corresponding to the train potential

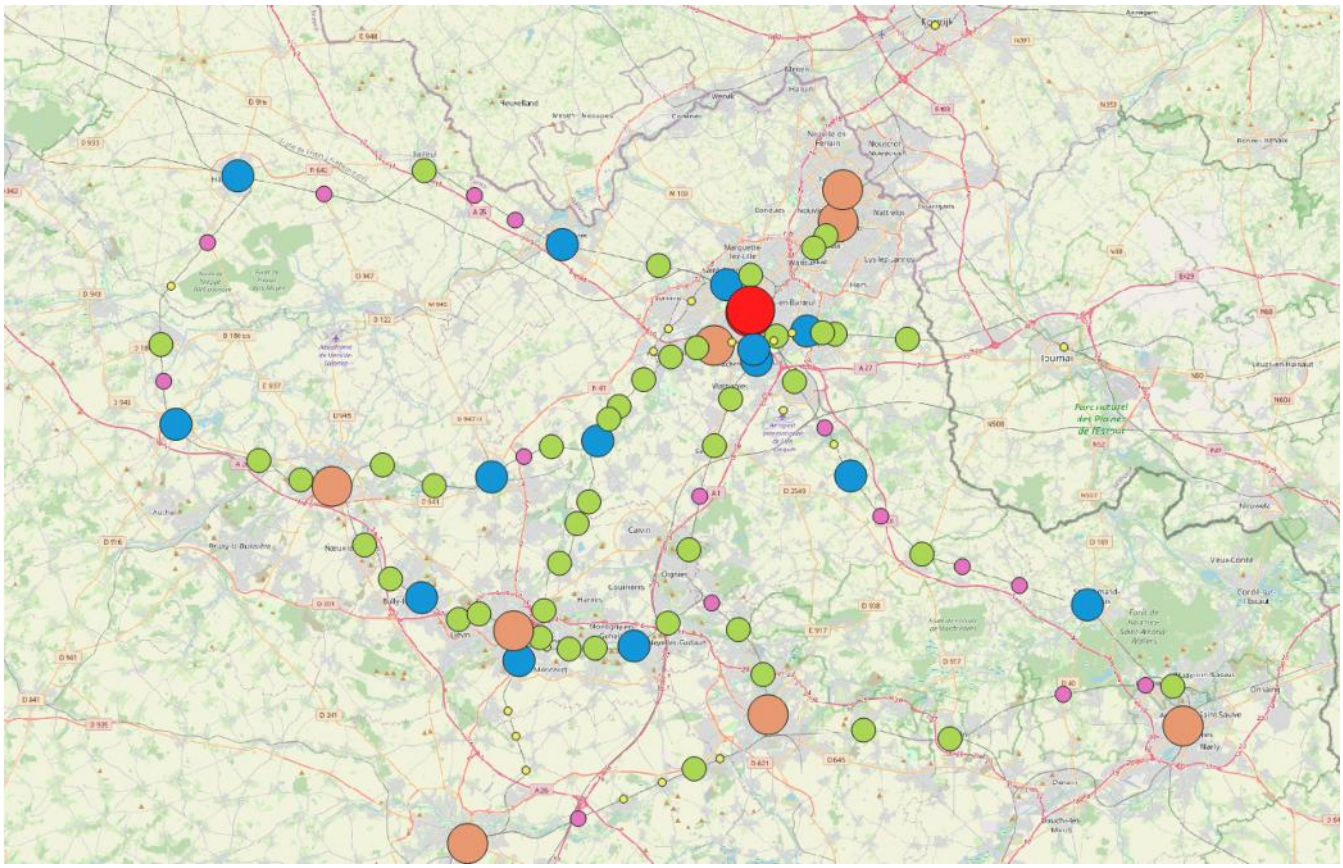
The "cadenced" indicators correspond to different pairs of incoming and outgoing transport. For example, for a metro+train connection, it is often the case that for 2 consecutive metros the connecting train is the same. In this case, only one of the two connections is counted, as the outgoing transport is identical.

intensite_nodale_jour — Total des entités: 100, Filtrées: 99, Sélectionnées: 0

	fid	i	name	potentiel_tot	potentiel_train	intensite_tot	intensite_train	potentiel_cadence_tot	potentiel_cadence_train	intensite_cadence	intensite_cadence_train
1	100	NG0100	Lille Flandres (souterrain)	162709	93808	78548	26692	162709	4400	77984	3488
2	70	87286005	Lille-Flandres	148869	89856	66918	25656	148470	3575	66194	2842
3	15	87342014	Arras	53366	28056	15561	3032	53366	2596	15340	1126
4	99	87343004	Valenciennes	20962	5117	10203	1955	20962	2040	10111	903
5	8	87286831	Pont-de-Bois	20652	1689	11964	1275	17270	288	11722	188
6	43	87345025	Lens	18232	5730	7044	1433	18232	1734	7042	731
7	4	87342006	Béthune	13705	3736	4940	923	13636	1394	4940	569
8	86	87286146	Armentières	9620	1041	3600	636	9620	1041	3600	560
9	69	87286724	Lille-CHR	9548	5044	5382	2366	9548	309	5382	210
10	31	87286732	Roubaix	9538	4080	5230	954	9511	200	5230	163
11	55	87345009	Douai	8660	5010	3284	1257	8654	500	3284	423
12	40	87286542	Tourcoing	8405	3645	3773	533	8383	222	3773	152
13	62	87342436	La Bassée-Violaines	5591	504	1608	285	5591	504	1608	247
14	59	87286302	Hazebrouck	4719	890	1133	280	4548	890	1095	235
15	84	87287250	Lille-Porte-de-Douai	3690	1144	2411	538	3690	172	2411	144
16	54	87286435	Don-Sainghin	3289	744	1150	448	3289	612	1117	169

Ne montrer que les entités visibles sur la carte

The map below illustrates the use of nodal potential to automatically classify the intermodal performance of stations around Lille Metropole.



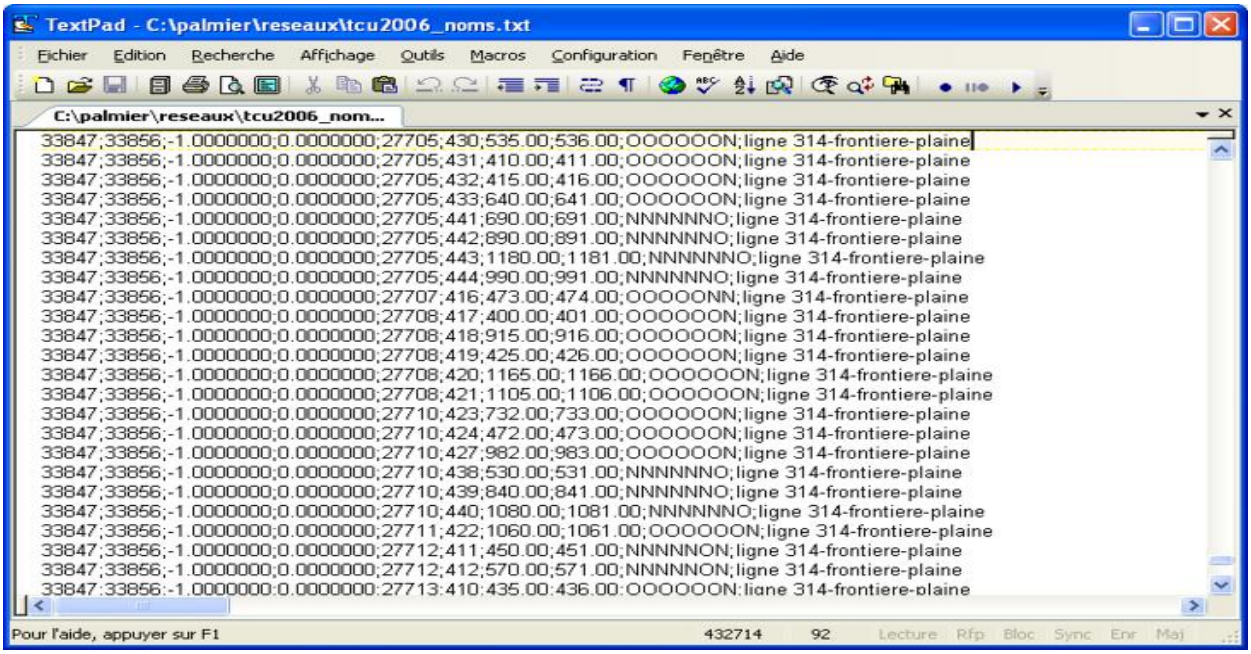
J- Musliw documentation

For the computation of multimodal accessibility based on timetables and time on links, it is necessary to import a network and a demand matrix. It is also necessary to set up parameters for calculating the TC generalized time. The following paragraphs describe the process.

This calculation also works if there are no links based on timetables (ex : walking, cycling or car network only)

1. The « network » file

To define a time based network in MUSLIW, you need a file of the following type .



The network file is a "Delimited Text File" with "; » As separator.
NB: MUSLIW automatically handles the problems of decimal separator '.' or ','. It converts them automatically according to the settings defined by your operating system.

•The different fields in order are:
- origin node number of the link; maybe also an alphanumeric string;
- destination node number of the link; maybe also an alphanumeric string;
- travel time of the section:
  - number of minutes, if the mode of transport on the link is individual: car, walking, cycling;
  - -1 or , if the mode of transport on the section is a public transport with schedules.
- length of the section :
  - length ;
  - «0» if length is unavailable.

•PT line number : (a number)
- if positive (> 0) :

- –line number (integer), for links of public transport with schedules ;

- •si negative (< 0) :
  - ◆◦ period id during which the travel times will be identical The link is then considered to be of individual transport type variable in time. You need to describe the time periods during which time is applied. The identifier must be different for each period with a different travel time. An example of coding can be found below:
  - ◆ • -1 for the morning peak period;
  - ◆ • -2 for the evening peak period;
  - ◆ • -3 for the off-peak period;
  - ◆ • -4 for the night;
  - ◆ • -5 for Sunday and holidays.

- •Service id :
  - –service id (> 0) id (integer), for public transport sections at times;
  - – service id (> 0) the range number (integer) for the individual transport links taking into account schedules and the calendar; ;
  - – -1, for individual transport links (VP, walk, bike) without taking into account schedules or calendar. The section will be accessible 24/7, and will invariably present the same travel time.

- •Departure time at start node :
  - –departure time, in minutes (past midnight), for public transport sections at times (eg 6h00 = 6 * 60 = 360minutes);
  - –start time of the usable period, in minutes, for individual transport sections taking into account timetables and schedules;
  - – -1, if the mode of transport is individual without taking into account schedules and calendar: VP, walking, cycling

- •arrival time at end node:
  - –arrival time, in minutes (past midnight), for public transport sections at times (eg 6:00 = 6 * 60 = 360minutes);
  - –end time of the usable period, in minutes, for the individual transport sections taking into account schedules and calendar;
  - – -1, if the mode of transport is individual: car, walking, cycling.

- •service circulation schedule :
  - –◦ string of "n" characters: n = length in days of the period. The circulation of the service for each day of the period is determined by "O" for "circulates" or "N" for "does not circulate". For example, for a service that only runs on the 10th day of the period, the corresponding string of characters will consist of 9 "N" and then an "O" in 10th position. The chain of a service that runs every day will be composed of n "O". This chain determines in the same way, the days of circulation the links of the type individual transport and those defined with schedules and calendar;
    - ◦ -1, if the mode of transport is individual: car, walking, cycling.
  - – -1, if the mode of transport is individual: car, walking, cycling.

- •label :
  - –the label is useful in the result files for identifying sections and routes by name rather than just node, line and service numbers. The part of the label before the first vertical separator «| » Generally determines the line identifier. The user is not limited in number of characters.

- •Link type :
  - –the type of link makes it possible to define time and cost weighting parameters differently according to its considered type. Thus, a default connection time of 5 minutes for the network type 0 and 35 minutes for the Eurostar or aerial sections can be set to take account of the registration. Link type

could be a text string (ex : Tram)

•Toll:

–The value to enter is the monetary cost on the section. Musliw will then calculate the optimal path taking into account the toll that is introduced in the generalized time thanks to a new weighting parameter that can depend on the type of link. In output Musliw will also provide a "toll" element that indicates the cumulative toll on the origin-destination.

The limit of the number of links and services is not defined. It is the memory of the computer that will set the limit of the allowable size..

For 32-bit operating systems, the maximum size of an object is 2 GB. Thus, MUSLIW will not be able to handle larger networks, even if the RAM is higher (4 GB is the maximum manageable size).

Conversely, this limit is much higher for 64-bit systems such as Windows 7, Linux 64, etc.

The major advantage of having a network integrating the service circulation calendar is to be able to vary the periods of the accessibility study without having to recode the network accordingly (especially with regard to Saturdays, Sundays , holidays, etc.).

The example below shows the coding of node numbers by strings and the type of network coding (0 for all sections except "station; bruxelles_E *" which is type 1).

```
depart;gare;5;-1;-1;-1;-1;-1;-1;MARCHE,0
gare;bruxelles_tgv;-1;-1;1;1;480;530;OOOOOOO;TGV|LILLE-BRUX_TGV;0
gare;bruxelles_E*;-1;-1;2;2;500;532;OOOOOOO;E*|LILLE-BRUX_E*;1
bruxelles_tgv;bruxelles_centre;5;-1;-1;-1;-1;-1;-1;MARCHE;0
bruxelles_E*;bruxelles_centre;5;-1;-1;-1;-1;-1;-1;MARCHE;0
```

1.1.1 Possible use of the type of link

•The type of link has two essential uses:

- differentiate time and cost weighting parameters by type;
- perform calculations of shorter paths on a part of the network;
- to filter the individual sections of a particular type (must enter a corresponding negative"cmap" negative;)
- To filter the sections of a particular type of time, you have to enter a negative "cveh".
- allow statistical operations by type of network

1.1.2 Example of coding of individual LINKS taking into account time periods and calendar

The following excerpt provides an example of a stretch definition whose travel time depends on the time period and calendar :

```
11;151;5.3;0;-1; 1;420;540;OOOOOON
```

```
11;151;4.1;0;-3; 1;540;960;OOOOOON
11;151;4.1;0;-3; 2;1140;1200;OOOOOON
11;151;4.6;0;-2; 1;960;1140;OOOOOON
11;151;3.5;0;-4; 1;1200;1440;OOOOOON
11;151;3.5;0;-4; 2;0;420;OOOOOON
11;151;3.6;0;-5; 1;360;1140;NNNNNNO
11;151;3.1;0;-6; 2;0;360;NNNNNNO
11;151;3.1;0;-6; 3;1140;1440;NNNNNNO
```

The travel times are:

- 3.5 minutes at night (20h-24h, 0h-6h);
- 4.1 minutes a day (9am to 4pm, 7pm to 8pm);
- 4.6 minutes in the evening (16h-19h);
- 5.3 minutes in the morning (7am to 9am) from Monday to Saturday;
- 3.1 minutes Sunday at night (0h-6h, 20h-24h);
- 3.6 minutes on Sunday during the day (6h-20h).

NB: Do not forget that if you enter the time periods and a calendar for individual sections and that there are periods of the day or days in the calendar that are not defined, the link will be inaccessible for these periods (which would occur in the case of a closed road or street). It is therefore important to cover the time and calendar spectrum in the description

2. The penalties and transfers definition file

This file contains all the necessary information for the introduction of penalties and prohibitions of turning movements and transfers.
The penalties and transfers definition file is a "Delimited Text" file with "; » as delimiter

```
35046;35482;3802;35047;302;1.5
28325;28264;-1;28347;-1;-1
28347;28325;-1;28442;-1;0.25
```

•The different fields are in order:

- nj: node number of the intersection or stop considered;
- ni: origin node number;
- line number of the incoming link (ni-> nj);
- nk: final node number;
- line number of the outgoing link (nj-> nk);
- penalty time:
- 0: no penalty;
- -1: movement or transfer prohibited;
- otherwise: value of the penalty time in minutes.

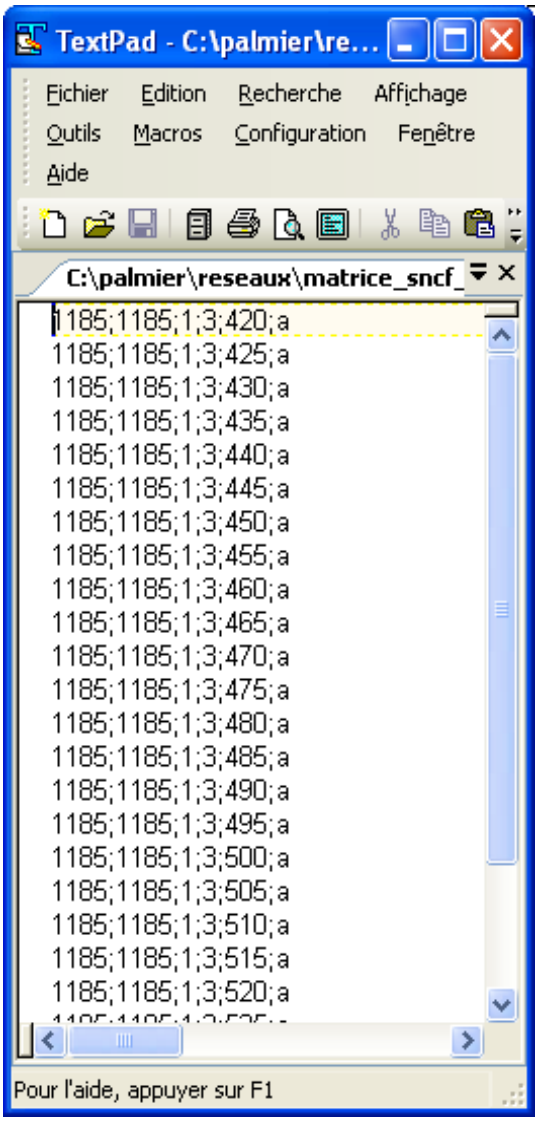
If the penalty time is relative to a transfer, ie one of the two line numbers is not "-1" the penalty time will be used as the transfer time instead of the default transfer time set in the parameters that will be used by default for all other network transfers

The fact of having distinct line numbers for different periods describing the travel time links sections of the individual transport type makes it possible to describe penalties for turning movements or for different transfers depending on the periods.

3. The «Matrix» file

3.1.1 Standard specifications

The matrix file is a "Delimited Text" file with "; » as separator.



The different fields in order are:

- origin node: number of the origin node: this can be any node of the network;
- destination node: destination node number: this can be any network node;
- demand: the volume of the demand that you want to assign between the origin node and the destination node;
- day: indicates the desired day of departure or arrival. This number refers to the period defined for each service in the "Network" file. The first day of the period is day 1. Thus, day 3 is the 3th day of the period. Be careful the number of the day must always be strictly smaller than the number of days of the period;
- time: indicates the desired departure or arrival time in minutes. The time in a day varies between 0 and 1439 minutes. The software tolerates negative hours or more than 1439. It will increment or decrement for the calculation of the number of days * 1440 for the time to be between 0 and 1439;

- type of time desired:
  - "d" for departure: the calculation will be done from the origin node starting at the indicated day and time by iteratively searching for the different successors to the destination node;
  - "a" for arrival: the calculation will be done from the destination node at the indicated day and time by iteratively searching the different predecessors to the origin node.

There is no size limit for the number of rows in the matrix file.

When looking for the shortest path, a link is taken into account in the route if it allows to reach a successor section with a cost strictly lower than the path chosen as the shortest in the current iteration . Thus, if several itineraries are equivalent, the first found will be the one selected; the entire demand of the row of the corresponding matrix file will be assigned to it.

3.1.2 Advanced specifications

MUSLIW offers the possibility to detail for each line of the matrix file the parameters of the calculation (in blue) and even possibly the parameters of the algorithm and outputs (in green).

However, if the calculation parameters (blue part) can be indicated without being those of the algorithm and outputs (green part), the reciprocal is not true. In order to be able to indicate the algorithm and output parameters in batch mode, it is also necessary to indicate the calculation parameters.

As shown in the example below, the fineness of the setting can vary depending on the lines of the matrix file. On the other hand, to be active, each color block must be complete.

1546;1368;1;1;420;a;od1_420 ;1;1.5;3;5;1;2;120,0,30,2,120

1546;1368;1;1;435;a;od1_435 ; 1;1.5;3;5;1;2;120,0;30,2,120, true;0;0;50;10000;2
1546;1368;1;1;450;a;od1_450 ; 1;1.5;3;5;1;2;120,0;30,2,120, true;0;0;50;10000;2
1546;1368;1;1;465;a
1546;1368;1;1;480;a ;od1_480
1546;1368;1;1;495;a;od1_495;1;1.5;3;5;1;2;120,0;30,2,120, true;1;0;50;10000;2
1546;1368;1;1;510;a;od1_510 ;1;1.5;3;5;1;2;120,0,30,2,120
1546;1368;1;1;525;a;od1_525 ; 1;1.5;3;5;1;2;120,0;30,2,120, true;0;0;50;10000;2
1546;1368;1;1;540;a;od1_540 ; 1;1.5;3;5;1;2;120,0;30,2,120, true;0;0;50;10000;2

The field between the red and blue blocks is a text field to identify the row of the matrix. If this field is empty, Musliw will take by default the line number of the file.

The usefulness of this field lies especially in the case of important matrix files for which the calculation of one or more lines did not give the expected results. This identifier makes it possible to restart the calculation only on Ods you need to recalculate (by the constitution of a specific file matrix) and to easily replace the results of the first calculation by those of the new one.

Blue block in order :	Green block in order :
•In vehicle weight ;	•Detailed paths (true/false) ;
•Waiting weight;	•Detailed travel times(0 None, 1 without tum based links, 2 with time based links) ;
•Individual mode time weight ;	•Algorithm (0 GGA with buckets, 1 Dijsktra with

	buckets) ;
•Transfer weight ;	•Algorithm scale parameter ;
•Individual mode time scale;	•Max of buckets ;
•Minimum transfer time;	•Algorithm power parameter.
•Maximum transfer time ;	•Output filter
•Number of days.	
•Maximum individual mode cumulative time	
•Toll weight	
•Maximum generalized cumulative time	

Since the calculation weighting parameters can be differentiated according to the type of links, they can be entered separated by a "| ". Ex 1,1,5; 3; 2; 1; 2 | 5 | 35; 0 specifies a different connection time per link type (2 minutes for type "0", 5 minutes for type "1", and 35 minutes for type "2")..

4. Computation time optimization

To minimize the calculation time, it is necessary to sort the matrix file according to the rules below:

For calculations from a starting point, the file must be sorted by:

- Origin;
- Day;
- Hour.

For calculations from an end point, the file must be sorted by:

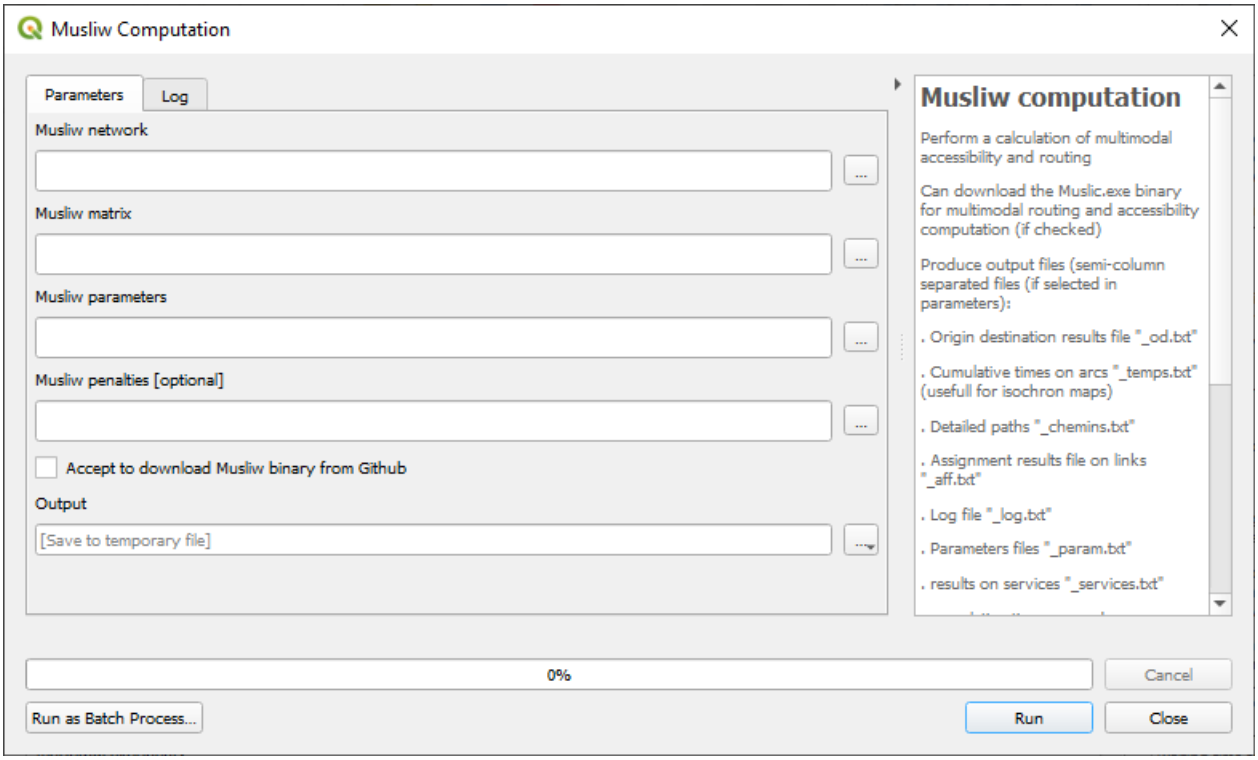
- Destination ;
- Day;
- Hour.

Indeed, MUSLIW calculates the shortest paths of one point to all the other points, so if the origin, the day and the hour of departure or arrival wished are identical between two consecutive lines of the matrix file, it is not necessary to recalculate all the shortest paths, resulting in significant computing time savings for large networks and large matrix files.

5. Computation procedure

5.1. Basic case

The procedure is launched via the menu bar "Procedures> Time based assignment". You must then fill in the dialog box that appears, as shown below.



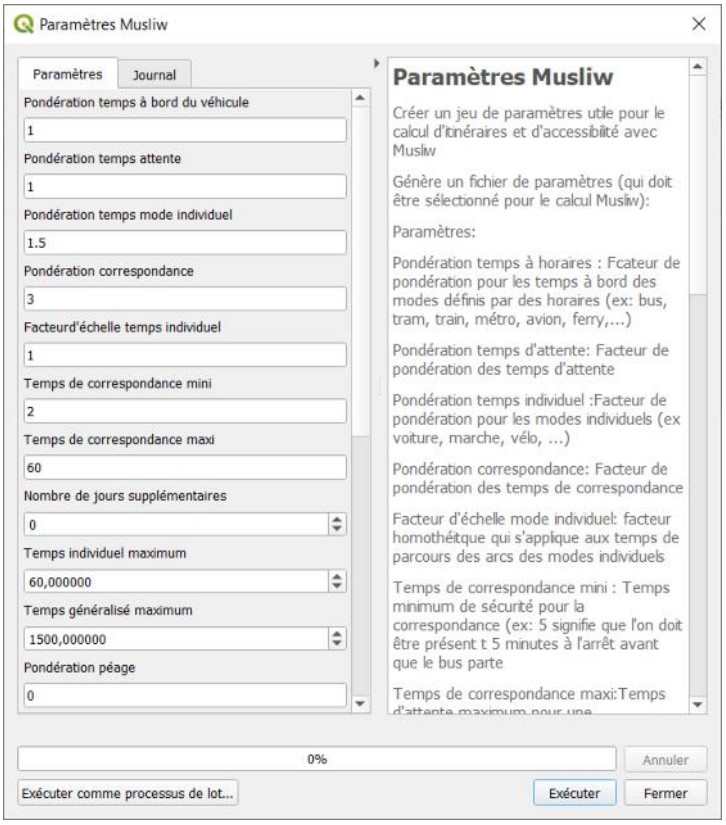
The user must enter :

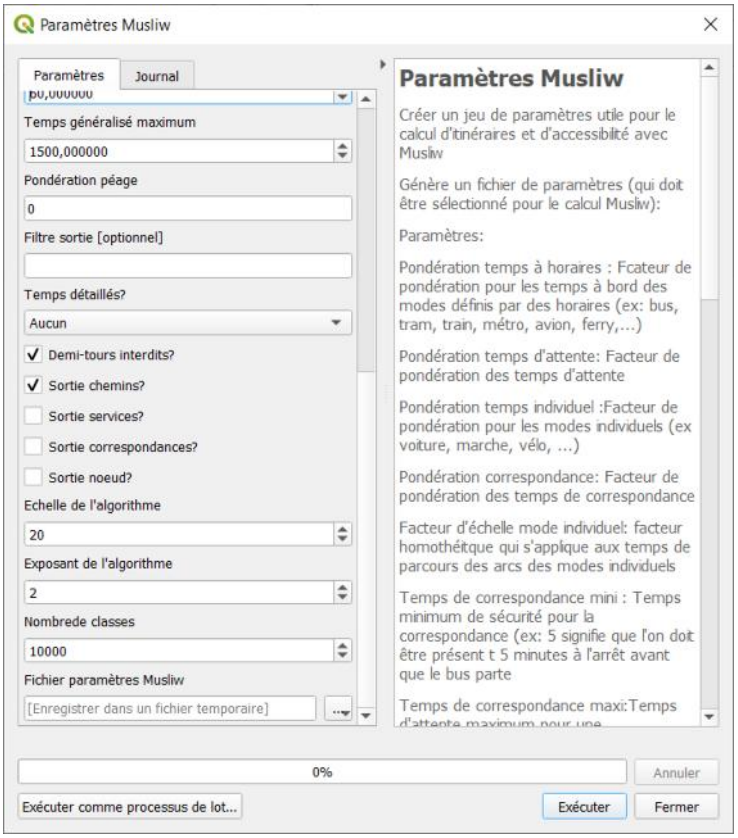
- The network on which he wants to calculate the accessibility or the multimodal routes
- The matrix (origin-destination matrix, also defining the days and times of departure or arrival and the number of passengers)
- The penalties for turning movements or connections
- Calculation parameters and result data (from a parameter file)

The parameter file can be generated in two ways:

- From the "Musliw parameters" script
- By reusing an existing parameter file or one produced by a previous calculation.

Each Musliw calculation generates a "_param.txt" file that corresponds to the parameter file used for the calculation. You have to fill the following settings





• In vehicle weight : weight for time based travel times (on board the vehicle);

• Waiting weight : weight for waiting times;

• Individual mode weight: weighting for individual transport mode travel times (Walking, MAP, Bike);

• Maximum generalized time : Maximum cumulative generalized cost

• Maximum individual mode time: Maximum cumulative individual time

• Boarding weight: weight for transfer times;

• Individual mode factor speed: uniform multiplier coefficient of all individual transport travel times. Performed once before launching all calculations. In the case where the individual transport times have been defined from a constant speed, this coefficient makes it possible to test a different walking speed without

having to modify the travel times in the description of the network. For example, for a network whose walking times have been defined with a speed of 4km / h, if we want to perform calculations for people walking at 5km / h, we will take a TMAP coefficient equal to 1, 25 without having to change the travel times in the description of the network;

• Minimum transfer delay : minimum connection time. This is the minimum time possible between arrival at a stop and the possibility of getting into a vehicle. This time will be weighted by "COR weight" and taken into account in the generalized time;

• Maximum transfer delay : maximum match time. This is the maximum permissible time between arrival at a stop and the possibility of getting into the next vehicle. Beyond this value the correspondence is considered unattractive and will not be taken into account in the search for a shorter path;

• Extra day duration: determines the number of days during which the algorithm will search for the shortest paths. '0' indicates that only services on the day indicated in the origin or destination concerned will be examined. Without this possibility, the search for itineraries lasting several days or very infrequent, such as international regular bus lines or even questions of the "accessibility by staying N days at destination" type, can not be carried out

- Toll weight: toll weighting. This weighting may be different depending on the type of links.
- Algorithm: Graph Growth Algorithm at intervals; Dijkstra at intervals;
- Algorithm scale: 200 (see algorithm parameters);
- nb classes: 10000 (see algorithm parameters);
- Algorithm exponent: 2 (see algorithm parameters);
- Output paths: by activating this box, MUSLIW will write, for each origin-destination, the route, that is to say the succession of links and services with the different components of intermediate time reconstituting the route. In case of many lines in the matrix file, this file can quickly be very huge;
- Output links times?: in case of many lines in the matrix file, the output file with the detailed time can quickly be very large. It is therefore possible to choose the following options:
  - no outpu: no output;
  - without timetable links : write for each origin-destination, the travel times of all accessible individual transport-type links, but no time based links.
  - with timetable links: writing for each origin-destination the travel times of all accessible links of the individual or time based transport type.
  - compact mode: writing in the file _nœuds.txt only the fields strictly necessary to generate the isochrones, (o, numero, temps, tatt1, volau), useful in particular for the gravitational calculation which generates very voluminous files

•Output filter : This field allows you to specify the types of links that you want to include in the detailed output file. If nothing is specified, all accessible links will be printed in the result file. This filter is cumulative with the detailed time option selected. To filter several types of sections it is necessary to enter a list separated by a "| ". Ex: entering "1 | 2 | 4" as a filter will lead to include only accessible sections of type 1,2 or 4 in the detailed time output file

•Prohibited U-turns: If checked, it prohibits by default the ability to perform U-turns (same origin node and same end node in a rotating movement or a transfer). These prohibitions will also be effective for links of transport schedules;

•Output services : this option allows you to generate a results file giving details of the flow volumes per service, for non-zero ones;

• Output transfers: this option makes it possible to generate a file result specifying for each turning movement or transfer node, line by line.

•Output nodes: This option allows to generate a results file for nodes (from the links connected to the node with the minimum cost)

5.2. Advanced settings

The advanced settings allow you to apply weights and values depending on the type of network.

The parameters that can be set according to the type of network are:

- On-board weighting;
- Waiting time weighting ;
- Individual time weighting;
- Correspondence weighting;
- Individual time scaling factor;
- Minimum transfer time;
- Toll weighting

The parameterization is done by a key/value system per type of section, separating the different types by a "|" and the key and the value by a ":". Thus, in the example above, "5|2:35" means that the connection time is 5 minutes by default and 35 minutes for network type "2". When there is no particular value defined for a type in question, MUSLIW defaults to the value of the modality for which the key is absent or is equal to "0" (to be expected).

If no type is entered, the type of all the sections is set to "0" by default.

5.3. The “results” files

MUSLIW outputs four result files that provide different information:

- time and volume by origin-destination, for those requested in the input matrix;
- all accessible links within a time limit fixed by origin or destination, for all origins and destinations of the matrix;
- volumes by segment and line segment when these volumes are non-zero;
- the detailed paths for origin-destination requested in the input matrix.

6. <FILENAME>_OD.TXT

```
id;o;d;jour;heureo;heured;temps;tveh;tmap;tatt;tcorr;tatt1;cout;longueur;pole;volau;cveh;cwait;cmap;cboa;ctmap;tboa;njours;texte
1;13657;101;1.000;394.348;550.000;155.652;53.000;11.691;90.961;6.000;81.961;173.498;0.779;37802;103.22;1;1;1.5;2;1;2;0;MARCHE|ligne Citadine 1|ligne 54|ligne 63|MARCHE
2;2495;129;0.000;406.546;450.000;43.454;26.000;14.552;2.903;2.000;0.903;54.730;0.970;36881;100.58;1;1;1.5;2;1;2;0;MARCHE|ligne 319|MARCHE
3;1019;129;0.000;421.545;485.000;63.455;12.000;13.553;37.903;2.000;35.903;74.232;0.904;36828;75.41;1;1;1.5;2;1;2;0;MARCHE|ligne 319|MARCHE
4;183;129;0.000;650.998;665.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;63.59;1;1;1.5;2;1;2;0;MARCHE
5;1349;129;0.000;966.992;1085.000;118.008;59.000;6.105;52.903;2.000;50.903;125.061;0.407;36564;35.11;1;1;1.5;2;1;2;0;MARCHE|ligne 214|MARCHE
6;12660;129;0.000;1071.737;1170.000;98.263;62.000;9.361;26.903;6.000;19.903;114.943;0.624;35013;121.00;1;1;1.5;2;1;2;0;MARCHE|ligne 1|ligne 2|ligne 64|MARCHE
7;183;129;1.000;405.998;420.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;35.11;1;1;1.5;2;1;2;0;MARCHE
8;477;129;1.000;484.278;530.000;45.722;0.000;45.722;0.000;0.000;0.000;68.583;3.048;129;75.41;1;1;1.5;2;1;2;0;MARCHE
9;477;129;1.000;540.354;595.000;54.646;10.000;11.744;32.903;2.000;30.903;64.518;0.783;36772;63.59;1;1;1.5;2;1;2;0;MARCHE|ligne 63|MARCHE
10;183;129;1.000;590.998;605.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;35.11;1;1;1.5;2;1;2;0;MARCHE
11;477;129;1.000;559.278;605.000;45.722;0.000;45.722;0.000;0.000;0.000;68.583;3.048;129;35.11;1;1;1.5;2;1;2;0;MARCHE
12;8029;129;1.000;493.394;635.000;141.606;62.000;7.703;71.903;6.000;62.903;157.457;0.514;35978;142.66;1;1;1.5;2;1;2;0;MARCHE|ligne 73|ligne 2|ligne 64|MARCHE
13;1019;129;1.000;571.846;660.000;88.154;20.000;20.722;47.432;4.000;10.903;106.515;1.381;36701;96.80;1;1;1.5;2;1;2;0;MARCHE|ligne 63 R|MARCHE|ligne 64 R|MARCHE
14;1019;129;1.000;571.846;662.000;90.154;20.000;20.722;49.432;4.000;12.903;108.515;1.381;36701;96.80;1;1;1.5;2;1;2;0;MARCHE|ligne 63 R|MARCHE|ligne 64 R|MARCHE
15;183;129;1.000;660.998;675.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;35.11;1;1;1.5;2;1;2;0;MARCHE
16;4067;129;1.000;684.881;775.000;90.120;36.000;18.217;35.903;2.000;33.903;103.228;1.214;36752;100.55;1;1;1.5;2;1;2;0;MARCHE|ligne 64|MARCHE
17;1019;129;1.000;705.846;785.000;79.154;14.000;8.251;56.903;2.000;54.903;87.279;0.550;36701;63.59;1;1;1.5;2;1;2;0;MARCHE|ligne 63 R|MARCHE
18;477;129;1.000;849.278;895.000;45.722;0.000;45.722;0.000;0.000;0.000;68.583;3.048;129;63.59;1;1;1.5;2;1;2;0;MARCHE
19;183;129;1.000;915.998;930.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;35.11;1;1;1.5;2;1;2;0;MARCHE
20;183;129;1.000;960.998;975.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;188.00;1;1;1.5;2;1;2;0;MARCHE
21;183;129;1.000;985.998;1000.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;35.11;1;1;1.5;2;1;2;0;MARCHE
22;9748;129;1.000;813.300;1015.000;201.700;73.000;7.797;120.903;6.000;113.903;217.598;0.520;35027;75.41;1;1;1.5;2;1;2;0;MARCHE|ligne 2|ligne 1|ligne 63|MARCHE
23;183;129;1.000;1030.998;1045.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;35.11;1;1;1.5;2;1;2;0;MARCHE
24;4067;129;1.000;1095.881;1165.000;69.119;34.000;18.217;16.903;2.000;14.903;82.228;1.214;36752;100.55;1;1;1.5;2;1;2;0;MARCHE|ligne 64|MARCHE
25;183;129;1.000;1220.998;1235.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;35.11;1;1;1.5;2;1;2;0;MARCHE
26;183;129;1.000;1220.998;1235.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;63.59;1;1;1.5;2;1;2;0;MARCHE
27;183;129;1.000;1250.998;1265.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;188.00;1;1;1.5;2;1;2;0;MARCHE
28;183;129;2.000;340.998;355.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;63.59;1;1;1.5;2;1;2;0;MARCHE
29;1019;129;2.000;421.545;485.000;63.455;12.000;13.553;37.903;2.000;35.903;74.232;0.904;36828;188.00;1;1;1.5;2;1;2;0;MARCHE|ligne 319|MARCHE
30;477;129;2.000;476.354;505.000;28.646;5.000;18.994;4.653;2.000;2.653;42.143;1.266;36772;63.59;1;1;1.5;2;1;2;0;MARCHE|ligne 328|MARCHE
31;183;129;2.000;500.998;515.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;100.55;1;1;1.5;2;1;2;0;MARCHE
32;477;129;2.000;492.278;538.000;45.722;0.000;45.722;0.000;0.000;0.000;68.583;3.048;129;63.59;1;1;1.5;2;1;2;0;MARCHE
33;477;129;2.000;773.354;805.000;31.646;3.000;18.994;9.653;2.000;7.653;45.143;1.266;36772;63.59;1;1;1.5;2;1;2;0;MARCHE|ligne 214|MARCHE
34;183;129;2.000;805.998;820.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;63.59;1;1;1.5;2;1;2;0;MARCHE
35;183;129;2.000;925.998;940.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;63.59;1;1;1.5;2;1;2;0;MARCHE
36;183;129;2.000;1015.998;1030.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;63.59;1;1;1.5;2;1;2;0;MARCHE
37;183;129;2.000;1060.998;1075.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;63.59;1;1;1.5;2;1;2;0;MARCHE
38;183;129;2.000;1195.998;1210.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;35.11;1;1;1.5;2;1;2;0;MARCHE
39;477;129;3.000;459.354;490.000;30.646;5.000;18.994;6.653;2.000;4.653;44.143;1.266;36772;121.00;1;1;1.5;2;1;2;0;MARCHE|ligne 328|MARCHE
40;360;129;3.000;448.681;495.000;46.319;7.000;17.445;21.874;2.000;19.874;59.041;1.163;36660;188.00;1;1;1.5;2;1;2;0;MARCHE|ligne 214|MARCHE
41;686;129;3.000;448.839;505.000;56.161;22.000;20.287;13.874;4.000;9.874;74.305;1.352;36721;164.77;1;1;1.5;2;1;2;0;MARCHE|ligne 68|ligne 328|MARCHE
42;183;129;3.000;685.998;700.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;63.59;1;1;1.5;2;1;2;0;MARCHE
43;183;129;3.000;690.998;705.000;14.002;0.000;14.002;0.000;0.000;0.000;21.003;0.933;129;141.00;1;1;1.5;2;1;2;0;MARCHE
44;6204;129;3.000;640.131;785.000;144.869;62.000;25.966;56.903;2.000;54.903;161.852;1.731;35840;121.00;1;1;1.5;2;1;2;0;MARCHE|ligne 63 R|MARCHE
```

The results OD file is located in the directory that the user specified by clicking "OK". It is a "Delimited text" file with ";" as separator. It contains one line per line of the "matrix" file with the following fields:

- id: sequential number corresponding to the line number of the matrix file;
- o: origin node number of the trip;
- d: destination node number of the trip;
- jour: day of travel;
- heureo: departure time of the trip:
  - if time type = "d", this is the desired start time indicated in the matrix file;
  - if type of time = "a", it is the latest departure time to reach the destination before the arrival time;
- heured: arrival time: arrival time of the trip:
  - if type of time = "d", it is the earliest arrival time to reach the destination starting at the departure time;
  - if type of time = "a", this is the desired arrival time indicated in the matrix file;
- temps: travel time between the origin node and the destination time (minutes):

time = [arrival time] - [departure time];

- tveh: time spent into time based vehicles (minutes);
- tmap: walking or individual transport time (minutes);
- tatt: waiting time at a stop before boarding a time based vehicle (minutes);

• tcorr: time spent in transfers, this time is equal to the number of trips made multiplied by the time of transfers;

• ncorr: number of boardings. This number represents the number of boardings in a transit vehicle whose service is described by schedules;

• tatt1: waiting time before the first boarding (direction d) or waiting time between the actual arrival time and the desired arrival time (direction a);

• cout: generalized time minimized by the search algorithm for shortest paths.

[generalized time] = [TC weight] * [vehicle time] + [ATT weight] * [wait time] + [MAP weight] * [on time] + [number of TC trips] * [COR time];

• longueur: cumulative length;

• pole: node number of the first intermodal point (transition from the individual transport network to time based network). If the whole route is done on the individual network, the pole has the value of the node origin or destination according to the direction of calculation;

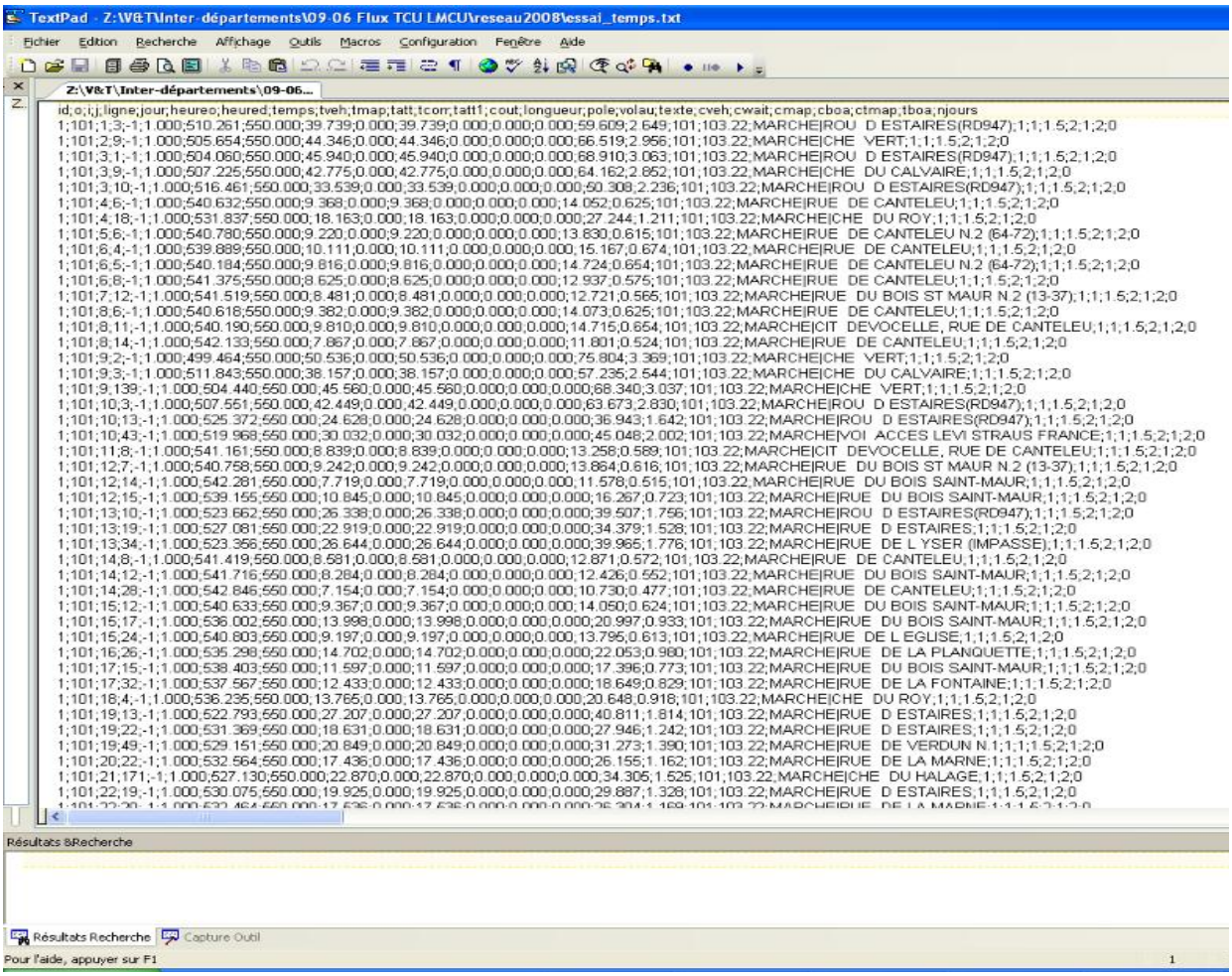
• volau: volume of demand on the last link of the route;

• texte: list of public transport lines used on the route. For this feature to provide suitable or actionable results, the names of the time based timetable links must start with an identifier (which may be of type characters) of the line followed by a "-". It is this identifier which will be taken up in the succession of the borrowed lines, otherwise, the identifier used will be the character string defined from the beginning of the name of the section until the first «-».

• nbpop: number of iterations when calculating the shortest paths. This indicator makes it possible to optimize the parameters of the algorithm for a minimum calculation time. For an equivalent destination-origin, a smaller nbpop value indicates a more optimal computation time;

• ttoll: cumulative toll along the route.

7. <FILENAME>_TEMPS.TXT



This result file will only be filled in if the “Sortie temps" box has been activated. The structure of this file is identical to that of the file origins-destinations except for the last field on borrowed lines which is absent.

The first line indicates the name of the fields.

Then, for each line of the matrix file, the file gives the set of sections of the network that are accessible with the generalized time calculation parameters defined in the dialog box described in 4.3. This file can therefore quickly become very large for large networks:

- id: sequential number corresponding to the row number of the matrix file;
- o: origin node number (direction d) or destination (direction a) of movement;
- i: origin node number of the accessible section;
- j: end node number of the accessible section;
- numtrc: number identifying the section
- jour: day of travel;
- heureo: departure time of the trip:
  - if time type = "d", this is the desired start time indicated in the matrix file;
  - if type of time = "a", it is the latest departure time to reach the destination before the arrival time;

• heured: arrival time of the trip:

◦ if type of time = "d", it is the earliest arrival time to reach the destination starting at the departure time;

◦ if type of time = "a", this is the desired arrival time indicated in the matrix file;

• temps: travel time between the origin node and the destination time (minutes). It is given by the formula time = [heured] - [heureo];

• tveh: time spent on public transport (minutes);

• tmap: walking or individual transport time (minutes);

• tatt: waiting time at a stop before boarding a public transport (minutes);

• tcorr: time spent in transfers, this time is equal to the number of trips made in TC multiplied by the time of matches;

• ncorr: number of transfers. This number corresponds to the number of rides in a transit vehicle whose service is described by schedules;

• tatt1: waiting time before getting on the first TC (direction d) or waiting time between the actual arrival time and the desired arrival time (direction a);

• cost: generalized time minimized by the search algorithm for shorter paths.

[generalized time] = [TC weight] * [vehicle time] + [ATT weight] * [wait time] + [MAP weight] * [on time] + [number of TC trips] * [COR time];

• length: cumulative length;

• pole: node number of the first intermodal point (transition from the individual transport network to the hourly network). If the whole route is done on the individual network, the pole has the value of the node origin or destination according to the direction of calculation;

• volau: volume of demand on the last link of the route;

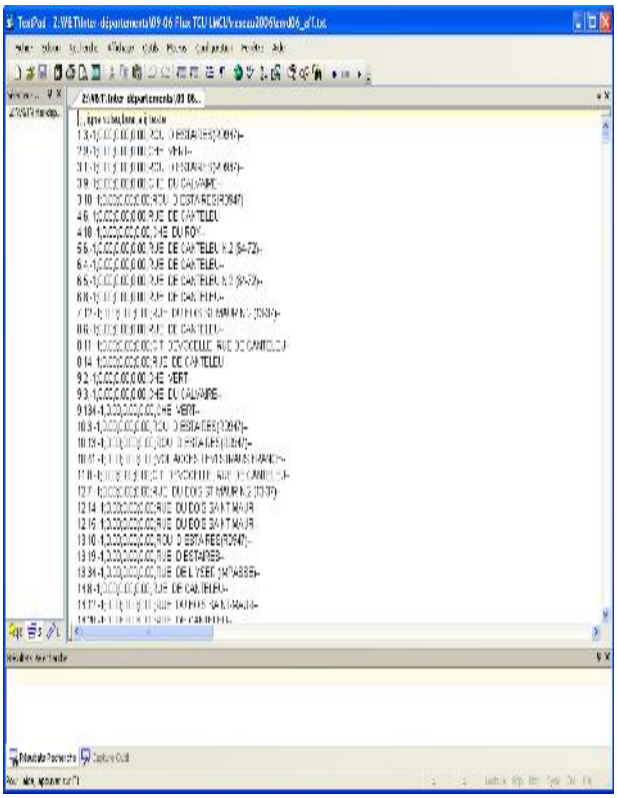
• previous: Number of the previous section. This field makes it possible to reconstitute the routes by going back successively the preceding sections

• type: link type indicator

• toll: Cumulative toll along the route

• ti: travel time between the origin node and the destination time (minutes) at the other end of the link. For example ti temps is relating to the time at the end node of the link, ti is relating yo the time a the start node of the link, and reciproqually.

8. <FILENAME>_AFF.txt

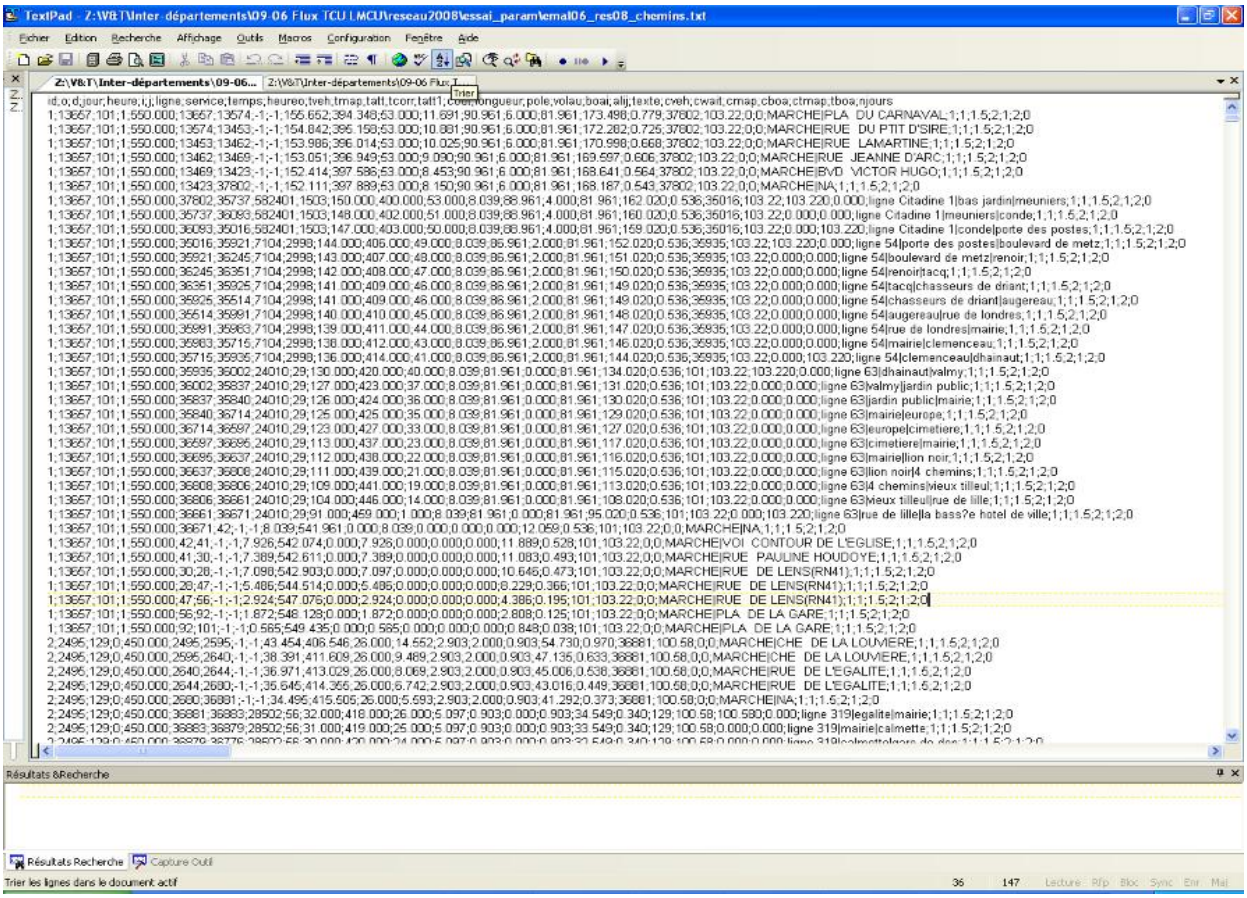


The assignment results file is a "Delimited text" file with ";" as separator. It contains a record by link and line segment and includes as fields:

- i: start node number of the section;
- j: end node number of the section;
- line :
  - TC line identifier if the mode is at times;
  - -1 if the mode is of the individual transport type;
- volau: volume assigned on the link;
- boai: number of boardings to the initial node;
- alij: number of alighting sat the end node;
- text: name of the section.
- type: type of section
- toll: toll of the section

9. <FILENAME>_CHEMINS.TXT

The detailed paths output file is "Delimited text" file with ";" as separator. It describes in detail the route of each record in the "matrix" file.



It contains a record by link and line segment describing the route and includes as fields:

- id: sequential number corresponding to the line number of the origin or OD identifier matrix file
- o: origin node number of the trip;
- d: destination node number of the trip;
- jour: day of travel;
- heure: desired departure or arrival time;
- i: start node number of the link;
- j: end node number of the link;
- ligne: TC line identifier if the mode is at times;
  - -1 if the mode is of the individual transport type;
- service:
  - TC service identifier if the mode is based on timetables;
  - -1 for individual transport type;

• temps: travel time between the origin node and the destination time (minutes):

time = [heured] - [heureo];

• heuro: time at the origin of the link if starting point or at arrival if point of arrival;

• tveh: time spent on public transport (minutes);

• tmap: walking or individual transport time (minutes);

• tatt: waiting time at a stop before boarding a public transport (minutes);

• tcorr: time spent in transfers, this time is equal to the number of trips made in public transport multiplied by the time of matches;

• ncorr: number of transfers. This number corresponds to the number of boarding in a transit vehicle whose service is described by schedules;

• tatt1: waiting time before getting on the first time based vehicle (direction d) or waiting time between the actual arrival time and the desired arrival time (direction a);

• cout: generalized time minimized by the search algorithm for shortest paths.

[generalized time] = [in vehicle weight] * [in vehicle time] + [wait weight] * [wait time] + [individual modes weight] * [individual modes time] + [number of boardings] * [transfer time];

• length: cumulative length;

• pole: node number of the first intermodal point (transition from the individual transport network to the time based network). If the whole route is done on the individual network, the pole has the value of the node origin or destination according to the direction of calculation;

• volau: volume of demand on the link for this specific trip;

• boai: number of boardings to the node origin of the link for the considered trip;

• alij: number of alightings at the destination node of the link for the considered trip;

• text: name of the link;

• type: type of link

• toll: detail of the cumulative toll along the way

By consulting this file and for all the lines having the same origin and destination nodes, it is possible to know the route taken on the chosen origin-destination, to know the volumes, boarding and alighting by service and to make detailed analyzes (trees, paths taken, ...)

10. <FILENAME>_NOEUDS.TXT

The output file by nodes is in the directory that the user specified by clicking "OK". It is a "Delimited text" file with ";" as separator. It contains one line per rline of the "matrix" file with fields as:

- id: sequential number corresponding to the line number of the matrix file or identifier of origin-destination;
- o: origin node number of the trip;
- d: destination node number of the trip;
- jour: day of travel;
- i: node identifier
- heureo: departure time of the trip:
  - if time type = "d", this is the desired start time indicated in the matrix file;
  - if type of time = "a", it is the latest departure time to reach the destination before the arrival time;
- heured: arrival time: arrival time of the trip:
  - if type of time = "d", it is the earliest arrival time to reach the destination starting at the departure time;
  - if type of time = "a", this is the desired arrival time indicated in the matrix file;
- temps: travel time between the origin node and the destination time (minutes):
  - time = [arrival time] - [departure time];
- tveh: time spent on public transport (minutes);
- tmap: walking or individual transport time (minutes);
- tatt: waiting time at a stop before boarding a public transport (minutes);
- tcorr: time spent in transfers, this time is equal to the number of boardings multiplied by the time of transfers;
- ncorr: number of transfers. This number represents the number of boardings in a transit vehicle whose service is described by schedules;
- tatt1: waiting time before getting on the first vehicle (direction d) or waiting time between the actual arrival time and the desired arrival time (direction a);
- cout: generalized time minimized by the search algorithm for shortest paths.

•[generalized time] = [in vehicle weight] * [in vehicle time] + [wait weight] * [wait time] + [individual modes weight] * [individual modes time] + [number of boardings] * [transfer time];

- • length: cumulative length;
- • pole: node number of the first intermodal point (transition from the individual transport network to the time based network). If the whole route is done on the individual network, the pole has the value of the node origin or destination according to the direction of calculation;
- • ttoll: cumulative toll along the route.

11. <NOM_FICHIER>_LOG.TXT

MUSLIW provides a small log file that summarizes information that can be useful when you want to find the files and parameters used or evaluate the differences in computation time corresponding to algorithm parameters. Here is an example of log file (in french) :

Version: 1.5.1.211
Début de la procédure: vendredi 04 mars 2011 11:39:16.968
Paramètres par défaut:

Temps correspondance minimum par défaut:2
Temps correspondance maximum par défaut:120
Pondération correspondance:5
Pondération attente:1
Pondération temps à horaires:1
Pondération temps individuel:1.5
Coefficient temps individuel:1
Temps individuel maximal:30
Pondération du péage:0
Nombre de jours:1
Interdiction demi-tours:False
Algorithme:0
Nombre d'intervalles:10000
Paramètre d'échelle de l'algorithme:15
Paramètre exposant de l'algorithme:2
Sortie chemins:True
Sortie temps:1
Noms fichiers sortie:C:\palmier\reseaux\tutu
Réseau:C:\palmier\reseaux\transpole2006_musliw.txt
Noeuds:32084
Liens:102580
Services horaires:359508
Matrice:C:\palmier\reseaux\matrice_dt_2.txt
Début Calcul: vendredi 04 mars 2011 11:39:23.906
Fin Calcul: vendredi 04 mars 2011 11:39:35.203
Temps Calcul:11.296875 sec

The log file summarizes:

- the version of Musliw
- the default calculation parameters (the specific parameters by origin-destination are in the result files);
- the paths of the input and output files;
- size elements of the network (nodes, links, turning movements and transfers, services);
- elements to evaluate the calculation time;
- non accessible origins and / or destinations.

12. <FILENAME>_SERVICES.TXT

The detailed services output file is a "Delimited text" file with ";” as separator. It describes in detail all the volumes of flows and boardings-alightings per service.

i;j;ligne;service;hd;hf;regime;volau;boia;alij;texte;type
908272;688583;1026;1026;1065;1145;OOOOOOO;1.00;0.00;0.00;ZRHDUSSwiss/Crossair;2
8000082;8000085;9510136;9510136;1291;1297;OOOOOOO;1.00;0.00;0.00; D�sseldorf Flughafen D�sseldorf Hbf;1
8000084;8003553;9510136;9510136;1359;1365;OOOOOOO;1.00;0.00;0.00; D�ren Langerwehe;1
8000085;8001584;9510136;9510136;1300;1305;OOOOOOO;1.00;0.00;0.00; D�sseldorf Hbf D�sseldorf-Benrath;1
8000178;8000084;9510136;9510136;1348;1358;OOOOOOO;1.00;0.00;0.00; Horrem D�ren;1
8000207;8000208;9510136;9510136;1335;1339;OOOOOOO;1.00;0.00;0.00; K�ln Hbf K�ln-Ehrenfeld;1
8000208;8000178;9510136;9510136;1340;1347;OOOOOOO;1.00;0.00;0.00; K�ln-Ehrenfeld Horrem;1
8000209;8003368;9510136;9510136;1322;1327;OOOOOOO;1.00;0.00;0.00; K�ln-M�lheim K�ln Messe/Deutz;1
8000348;8000406;9510136;9510136;1377;1382;OOOOOOO;1.00;0.00;0.00; Stolberg(Rheinl)Hbf Aachen-Rothe Erde;1
8000406;8000001;9510136;9510136;1383;1387;OOOOOOO;1.00;0.00;0.00; Aachen-Rothe Erde Aachen Hbf;1
8001584;8006713;9510136;9510136;1306;1314;OOOOOOO;1.00;0.00;0.00; D�sseldorf-Benrath Leverkusen Mitte;1
8001886;8000348;9510136;9510136;1373;1376;OOOOOOO;1.00;0.00;0.00; Eschweiler Hbf Stolberg(Rheinl)Hbf;1
8003368;8000207;9510136;9510136;1329;1332;OOOOOOO;1.00;0.00;0.00; K�ln Messe/Deutz K�ln Hbf;1
8003553;8001886;9510136;9510136;1366;1372;OOOOOOO;1.00;0.00;0.00; Langerwehe Eschweiler Hbf;1
8006713;8000209;9510136;9510136;1315;1321;OOOOOOO;1.00;0.00;0.00; Leverkusen Mitte K�ln-M�lheim;1

- i: origin node;
- j: destination node;
- ligne: line number;
- service: service number;
- hd: start time at node i;
- hf: arrival time at node j;
- regime: timetable for the circulation of the service;
- volau: volume of service flow;
- boai: number of boardings at node i;
- alij: number of alightings at node j;
- texte: description of the link;
- type: type of links
-
- Only services with non-zero flow are reported into the file.

13. <FILENAME>_TURNS.TXT

The turns output file is a "Delimited text" file with ";" as separator. It describes in detail all of the turning movements and transfers at a junction or stop.

j;i;lignei;textei;k;lignek;textek;volau;
8000082;8000085;9510115; D�sseldorf Hbf D�sseldorf Flughafen;688583;-1; DUS - Duesseldorf Flughafen ;1;a
688583;8000082;-1; Duesseldorf Flughafen - DUS ;908272;1026;ZRHDUSSwiss/Crossair;1;a
8000084;8003553;9510141; Langerwehe D�ren;8000178;9510136; Horrem D�ren;1;a
8000207;8000208;9511313; K�ln-Ehrenfeld K�ln Hbf;8003368;9510136; K�ln Messe/Deutz K�ln Hbf;1;a
8000406;8000001;9510139; Aachen Hbf Aachen-Rothe Erde;8000348;9510136; Stolberg(Rheinl)Hbf Aachen-Rothe Erde;1;a
8000085;8001584;9511014; D�sseldorf-Benrath D�sseldorf Hbf;8000082;9510136; D�sseldorf Flughafen D�sseldorf Hbf;1;a
8000178;8000084;9510139; D�ren Horrem;8000208;9510136; K�ln-Ehrenfeld Horrem;1;a
8003553;8001886;9510137; Eschweiler Hbf Langerwehe;8000084;9510136; D�ren Langerwehe;1;a
8001584;8006713;9510139; Leverkusen Mitte D�sseldorf-Benrath;8000085;9510136; D�sseldorf Hbf D�sseldorf-Benrath;1;a
8003368;8000207;9212850; K�ln Hbf K�ln Messe/Deutz;8000209;9510136; K�ln-M�lheim K�ln Messe/Deutz;1;a
8000208;8000178;9510139; Horrem K�ln-Ehrenfeld;8000207;9510136; K�ln Hbf K�ln-Ehrenfeld;1;a
8000209;8003368;9510137; K�ln Messe/Deutz K�ln-M�lheim;8006713;9510136; Leverkusen Mitte K�ln-M�lheim;1;a
8006713;8000209;9510135; K�ln-M�lheim Leverkusen Mitte;8001584;9510136; D�sseldorf-Benrath Leverkusen Mitte;1;a
8000348;8000406;9510139; Aachen-Rothe Erde Stolberg(Rheinl)Hbf;8001886;9510136; Eschweiler Hbf Stolberg(Rheinl)Hbf;1;a
8001886;8000348;9510139; Stolberg(Rheinl)Hbf Eschweiler Hbf;8003553;9510136; Langerwehe Eschweiler Hbf;1;a

- j: node corresponding to a junction or a stop;
- i: origin node;
- ligne i: line number of the incoming;
- textei: label of the origin link;
- k: destination node number;
- lignek: line number of the destination link;
- textek: label of the destination link;
- volau: volume of the flow of turning movements or line-to-line transfers.
-

•Only the turning movements or line-by-line transfers volumes, whose flow volume are greater than-zero, appear in the file.

14. Muslic — User guide: penalties, sub-paths and early/late weights

Documentation of the changes in the parallel version of Muslic (October 2026)

14.1. 1. Overview

Three optional features extend the shortest-path computation. If you do not use them, results are identical to the previous version.

Feature	What it is for	Where to code it
Extended movement penalties	Penalise or forbid the passage from one arc to another with a time, a toll and a cost	Penalties file
Early and late weights	Allow leaving earlier or arriving later than the requested time, at a cost per minute of deviation	Parameter file, or matrix columns
Sub-paths	Forbid or penalise a sequence of 3 arcs or more (or of 2 arcs with a cost)	Penalties file, lines starting with P

All three work in both computation directions: departure mode (earliest departure time) and arrival mode (latest arrival time).

14.2. 2. Movement penalties: time, toll and cost

14.2.1 2.1 Why they matter

A movement is the passage from one arc to the next at a node: a turn, a transfer, entering or leaving a site. Penalties represent turn bans, crossing times, tolls or costs specific to one movement. The nodes involved are detected automatically; nothing else has to be declared.

14.2.2 2.2 Classic format (6 or 7 columns)

One line per movement, fields separated by semicolons:

```
via_node;from_node;from_line;to_node;to_line;time;toll
```

Field	Meaning
via_node	Node where the movement takes place
from_node, from_line	Start node and line of the arc arriving at the via node
to_node, to_line	End node and line of the arc leaving the via node
time	Movement time in minutes. Negative value: movement forbidden. 0: no time penalty
toll	Optional (7th column). Toll of the movement, positive or negative

Line numbers are those of the network file, using the value your network uses for walking or road arcs (for example -1). Existing 6-column lines remain valid.

Effect of the time (strictly positive value). In departure mode, it replaces the minimum transfer time for that movement. In arrival mode, it is added to the minimum transfer time. This difference exists in the original program and was not changed.

Effect of the toll. It is multiplied by the toll weight of the type of the arc taken after the movement. The product is added to the path cost, and the toll itself is added to the toll column of the results. A negative toll acts as a bonus (see 2.5).

Prohibition. A negative time forbids the movement; any toll on that line is ignored.

14.2.3 2.3 Adding a cost: two-arc P line

The cost (in generalised cost units, like the cout column of the results) does not exist in the classic line. To code it, use a P line with two arcs:

```
P;time;cost;toll;node0;line1;node1;line2;node2
```

Example: a cost of 5 to go from arc 2→3 to arc 3→4 (walking arcs, line -1), with no time or toll:

```
P;0;5;0;2;-1;3;-1;4
```

If the same movement is defined several times (classic format or P lines), the first line read is kept.

14.2.4 2.4 Units

Value	Unit
time	minutes
cost	generalised cost units (minutes weighted by the computation parameters)
toll	monetary units, converted to cost by the toll weight of the arc type

14.2.5 2.5 Results and the limit on bonuses

The movement cost and toll are included in the cout and toll columns of the _od file and in the path files. Example: two routes of cost 6 (through the penalised movement) and 8. Toll weight = 1.

Movement toll	OD cost	Route chosen
none	6	direct
1	7 (toll = 1)	direct
3	8	the route of cost 8
-10	3 (toll = -10)	direct

The search algorithm assumes that cost never decreases along a path. A bonus (negative cost or toll) can therefore reduce the cost of an arc, but not bring it below the cost of the previous arc. In the example, the -10 bonus gives a cost of 3, not -4; the toll column does show -10.

14.3. 3. Early and late departure/arrival weights

14.3.1 3.1 Why they matter

By default, in departure mode the traveller leaves at the requested time and waits for the next service. In arrival mode, they take the last service arriving before the requested time. In reality a traveller may prefer to leave a little earlier (to avoid a long wait) or to arrive a little late (to avoid a much longer trip). The early and late weights give the cost per minute of that deviation from the desired time, and the program keeps the cheapest option.

14.3.2 3.2 Coding in the parameter file

Add two lines at the end of the file, after the output isolated links line:

```
False;output isolated links
0.5;early departure/arrival weight
1.0;late departure/arrival weight
```

Line	Meaning
1st line	Early weight: cost per minute of departure or arrival before the desired time
2nd line	Late weight: cost per minute of departure or arrival after the desired time

Rules: without these lines the feature is off. If only the first line is present, the late weight is 0. Use values greater than or equal to 0. The values are copied to the _param and _log files.

14.3.3 3.3 Coding row by row in the matrix

In advanced mode, a matrix row can carry its own weights in columns 26 (early) and 27 (late).

Columns	Content
1 to 7	origin, destination, demand, day, time, direction (d or a), label
8 to 18	coefficients: in-vehicle, wait, individual mode, boarding, individual travel time factor, min transfer, max transfer, number of days, max walk, toll, max cost
19 to 24	path output, time output, algorithm, algorithm parameter, max number of buckets, power
25	output types filter
26	early weight
27	late weight

Empty columns are dropped when the row is read, so columns 8 to 25 must all be filled, including the filter. To avoid filtering, list all arc types separated by | . A row without columns 26 and 27 uses the parameter file values; they do not carry over to the next rows. With column 26 alone, the late weight takes the file value.

14.3.4 3.4 How it is computed

The alternative applies only to the first boarding in departure mode and to the last alighting in arrival mode, including after an access or egress walk. It does not apply to intermediate transfers.

Departure mode. Let R be the arrival time at the boarding stop (requested time plus access walk) plus the minimum transfer time.

- Normal option: first service departing at R or later. Cost = in-vehicle time + wait × wait weight + late weight × (departure − R).
- Early option: last service departing before R, at most (max transfer time) before. Cost = in-vehicle time + early weight × (R − departure). There is no waiting cost.

Arrival mode. R is the deadline at the alighting node.

- Normal option: last service arriving at R or earlier. Cost = in-vehicle time + wait × wait weight + early weight × (R − arrival).
- Late option: first service arriving after R, at most (max transfer time) after. Cost = in-vehicle time + late weight × (arrival − R).

The late weight drives the choice between early and late, but not the choice among several services departing after R.

14.3.5 3.5 Example

One transit arc with two services: the first departs at 395 and arrives at 405, the second departs at 410 and arrives at 420. In-vehicle time of 10 minutes, in-vehicle and wait weights equal to 1.

Departure requested at 400

Early / Late	Service chosen	Cost	tatt	Reading
0.5 / 1	departs 395	12.5	-5	leaves 5 min early
10 / 1	departs 410	30	10	10 min wait
1 / 0	departs 395	15	-5	leaves early
3 / 0	departs 410	20	10	wait
0.2 / 5	departs 395	11	-5	leaves early

Arrival requested at 410

Early / Late	Service chosen	Cost	tatt	Reading
0.5 / 1	arrives 405	17.5	5	arrives 5 min early
10 / 1	arrives 420	20	-10	arrives 10 min late
1 / 0	arrives 420	10	-10	arrives late
0.2 / 5	arrives 405	16	5	arrives early

14.3.6 3.6 Reading the results

- **tatt and tatt1** (total wait and wait at the first boarding) become negative when the alternative is chosen. In departure mode, -5 means leaving 5 minutes earlier than planned. In arrival mode, -10 means arriving 10 minutes after the deadline.
- **temps** remains the difference between the arrival time and the requested time.
- **cout** includes the early and late penalties.
- The weights appear in _param and _log if they are in the parameter file. Weights coming from a matrix row are not copied there.

14.4. 4. Forbidden or penalised sub-paths

14.4.1 4.1 Why they matter

Some constraints apply to a sequence of arcs rather than to one movement: forbidding a chain of manoeuvres, preventing more than n consecutive segments on the same line, or favouring a sequence of segments with a bonus. The program keeps the best path that avoids the forbidden sequence, without blocking paths that only follow the beginning of it.

14.4.2 4.2 Coding

A line of the penalties file starting with P:

```
P;time;cost;toll;node0;line1;node1;line2;node2;line3;node3
```

Field	Meaning
P	Marks a sub-path line
time	Penalty time in minutes. Negative value: sub-path forbidden
cost	Penalty cost, positive or negative
toll	Penalty toll, positive or negative
node0, line1, node1, ...	The sequence of arcs. Each arc is described by its start node, its line and its end node; the end node of one arc is the start node of the next

At least two arcs are required. There is no upper limit. The arcs must exist in the network with that line. A line with an unknown node or arc is ignored and reported in the log.

Examples (walking arcs, line -1):

```
P;-1;0;0;1;-1;2;-1;3;-1;4          forbids the sequence 1→2, 2→3, 3→4
P;0;10;0;1;-1;2;-1;3;-1;4          cost of 10 for this sequence
P;0;0;1.5;1;-1;2;-1;3;-1;4         toll of 1.5 for this sequence
P;-1;0;0;1;-1;2;-1;3;-1;4;-1;7     forbids a sequence of 4 arcs
P;-1;0;0;1300;1;1400;1;1500;1;1600 forbids 3 consecutive segments of line 1
```

14.4.3 4.3 How the program works

For each arc, the program keeps several labels:

- the default label: the best path whose end does not resemble the beginning of any sub-path;
- one label per sub-path beginning currently being matched.

When a sequence is complete, the penalty applies to the path concerned, or the path is rejected if forbidden. The default label is untouched, so the best path outside the prohibition stays available. These extra labels are handled by copies of arcs, which are invisible in the results.

14.4.4 4.4 Combination rules

- If several sub-paths end at the same time, the time used is the largest, costs and tolls are added, and a prohibition prevails.
- A sub-path acts only once the sequence is complete. A path that follows only the beginning of it is not penalised.
- Classic movement penalties remain active inside a sub-path.
- A 2-arc P line works like a classic movement with an additional cost.
- The same file serves both computation directions.

14.4.5 4.5 Arc accessibility

An arc that is accessible without the prohibitions stays accessible if another path exists that uses no forbidden sequence. It becomes inaccessible only if all of its paths go through a forbidden sequence. On a sparse test network (111 arcs, 148 forbidden sub-paths), the program's result matched an exact search: among the arcs accessible without prohibitions, 30% kept the same cost, 36% stayed accessible at a higher cost and 34% became inaccessible.
If the destination of an OD is no longer accessible, no line is written for that OD. The usual caps (maximum cost, maximum walking time, maximum transfer time) remain active: a replacement path that is too long can fall outside the search perimeter.

14.4.6 4.6 Example

Walking network with two routes, nodes 1 to 7. Direct route: 1 → 2 → 3 → 4 (cost 3). Alternative route: 1 → 5 → 3 → 4 (cost 5). Node 6 follows node 3, node 7 follows node 4.

P line	OD 1→4	OD 1→6	OD 1→7
none	3	3	4
forbids 1→2→3→4	5	3	6
forbids 1→2→3→4→7	3	3	6
cost 1 on 1→2→3→4	4	3	5
cost 10 on 1→2→3→4	5	3	6
toll 1.5 (weight 1)	4.5	3	5.5
cost bonus -1	2	3	3
forbids 1→2→3 (2 arcs)	5	5	6

OD 1 → 6 goes through 1 → 2 → 3 without reaching 4: it is not affected by the ban on 1 → 2 → 3 → 4, because only the beginning of the sequence is followed.

14.4.7 4.7 Reading the results

- **Paths and times**: the arcs shown are always the original arcs. The _temps file shows, for each arc, the best of its labels.
- **cout, toll**: include the sub-path penalties.
- **Cumulative files** (_aff, _services, _transferts): volumes of the copies are brought back to the original arcs.
- **_log**: one line gives the number of sub-paths and copied arcs (Sub-path penalties: N sub-paths of 3 arcs or more, M arc copies). Ignored lines are reported with a Sub-path ignored message.

14.4.8 4.8 Limits

- A bonus cannot bring the cost of an arc below that of the previous arc (see 2.5).
- Each arc copy exists once per computation thread: many long sub-paths increase memory use. In a test with 80 sub-paths, the computation took about 4% longer.
- The feature was tested on walking arcs and one transit-line case, not on very large sets of sub-paths.
- Combined use with the early and late weights has not been tested.

14.5. 5. Summary of results

Item	Effect of the new features
cout	Includes tolls, penalty costs and early and late weights
toll	Accumulates the tolls of arcs and of movements or sub-paths
tatt, tatt1	Negative for an early departure (departure mode) or a late arrival (arrival mode)
temps	Arrival time minus requested time
_param	Copies the two weights from the parameter file, if present
_log	Weights, number of sub-paths and copied arcs, ignored lines
_temps	Best label of each arc
_transferts, _aff, _services	Volumes accumulated on the original arcs

14.6. 6. Other behaviour changes

- **Advanced matrix columns**: a row goes back to the parameter file settings before applying its own columns. It no longer inherits values from the previous row. A default value given alone (for example for the wait weight) now applies to all arc types.
- **Stops (sortie_stops)**: with a time output of 10 or more, the stops column is filled for every OD, and no longer only for the first one.
- **Reuse of computations**: an OD reuses the previous computation only if origin, day, time and direction are identical, the row has fewer than 13 columns and the parameters are identical.
- **Parallel computation**: all cores are used. The environment variable MUSLIC_THREADS sets the number of threads. Memory use grows with that number.
- **Corrected label**: individual travel time factor is now spelled correctly in _param.

14.7. 7. Troubleshooting

Symptom	Likely cause
A P line has no effect	Unknown node or arc (see the Sub-path ignored message in the log), or arc line different from the network
No line for an OD	The destination is not accessible (prohibitions, or cost or time caps)
Matrix weights ignored	Columns 8 to 25 incomplete: empty columns are dropped when read
Negative tatt	An early departure or late arrival was chosen: this is intended
A bonus seems too small	It is capped at the cost of the previous arc
Decimal value misread	The decimal separator follows the system culture; both forms

Symptom	Likely cause
	are accepted